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Antibodies against signal-regulatory protein alpha and methods of use

Abstract

Provided herein, inter alia, are isolated antibodies that bind an extracellular domain of a human SIRP- α v1 polypeptide (e.g., the D1 domain), an extracellular domain of a human SIRP- α v2 polypeptide, or both. In some embodiments, the antibodies also bind an extracellular domain of a monkey SIRP- α polypeptide, an extracellular domain of a mouse SIRP- α polypeptide, an extracellular domain of a human SIRP- β polypeptide, and/or an extracellular domain of a human SIRP- γ polypeptide. In some embodiments, the antibodies block or do not binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide, while in some embodiments, the antibodies reduce the affinity of a human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. Further provided herein are methods, polynucleotides, vectors, and host cells related thereto.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 15/710,798, filed Sep. 20, 2017, which claims the priority benefit of U.S. Provisional Application No. 62/397,752, filed Sep. 21, 2016, and U.S. Provisional Application No. 62/515,480, filed Jun. 5, 2017, each of which is hereby incorporated by reference in its entirety.

SUBMISSION OF SEQUENCE LISTING ON ASCII TEXT FILE

(1) The content of the following submission on ASCII text file is incorporated herein by reference in its entirety: a computer readable form (CRF) of the Sequence Listing (file name: 757972000110SEQLIST.txt, date recorded: Jun. 1, 2021, size: 411 KB).

FIELD

(2) The present disclosure relates to isolated antibodies that bind an extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, an extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both, as well as polynucleotides, vectors, host cells, and methods related thereto.

BACKGROUND

(3) Signal-regulatory protein alpha (SIRP- α) is part of a family of cell-surface receptors that plays critical roles in the regulation of the immune system (see, e.g., Barclay, A. N. and Brown, M. H.

(2006) *Nat. Rev. Immunol.* 6:457-64). SIRP- α is expressed on the surface of various cells, including leukocytes such as dendritic cells, eosinophils, neutrophils, and macrophages. SIRP- α includes an extracellular domain that interacts with external stimuli such as ligands and an intracellular domain that mediates a variety of intracellular signals.

(4) One of the major roles of SIRP- α is its regulation of the immune response through interactions with CD47. CD47 is expressed on the surface of a variety of cell types. When the IgSF domain of CD47 binds the extracellular domain (e.g., the D1 domain) of SIRP- α expressed on an immune cell (e.g., a macrophage), this transduces a SIRP- α -mediated signal in the immune cell that prevents phagocytosis of the CD47-expressing cell. Thus, CD47 serves to convey what has been termed a “don't eat me” signal to the immune system that prevents phagocytosis of healthy cells (see, e.g., WO2015/138600 and Weiskopf, K. et al. (2013) *Science* 341:88-91). However, CD47 has also been shown to be highly expressed by a variety of cancers, and its interaction with SIRP- α in this context is thought to allow tumors to mimic the healthy “don't eat me” signal in order to evade immune surveillance and phagocytosis by macrophages (see, e.g., Majeti, R. et al. (2009) *Cell* 138:286-99; Zhao, X. W. et al. (2011) *Proc. Natl. Acad. Sci.* 108:18342-7). As such, antibodies that block this interaction are highly desirable.

(5) SIRP- α is known to be a highly polymorphic protein in humans, monkeys, and mice. For example, 20 amino acid differences have been identified between SIRP- α proteins in the NOD and C57BL/6 mouse strains, and these polymorphisms lead to functional consequences related to CD47 binding and engraftment of human hematopoietic stem cells in these mouse strains. In humans, at least 10 distinct alleles of the SIRPA gene have been identified (Takenaka, K. et al. (2007) *Nat. Immunol.* 8:1313-23, Zhao, X. et al. (2011), *PNAS.* 108:18342-47; van der Heijden, J. (2014). Genetic variation in human Fc gamma receptors: Functional consequences of polymorphisms and copy number variation (Doctoral dissertation)).

(6) Due to the importance of the SIRP- α -CD47 interaction in normal immune function and tumorigenesis, as well as the polymorphic nature of SIRP- α and the existence of other SIRP family receptors, the identification of antibodies having different binding specificities with intra- and/or inter-species cross-reactivity is of great interest for development of clinical candidates that are effective across human populations and the characterization of these candidates in various animal models. Thus, a need exists for both research tools and potential clinical candidates that modulate SIRP- α function, e.g., its binding interaction with CD47. A need also exists for methods of isolating antibodies with a variety of SIRP- α binding specificities and effects on CD47-SIRP- α binding in order to understand and effectively target this critical interaction.

(7) All references cited herein, including patent applications, patent publications, non-patent literature, and UniProtKB/Swiss-Prot Accession numbers are herein incorporated by reference in their entirety, as if each individual reference were specifically and individually indicated to be incorporated by reference.

SUMMARY

(8) To meet these and other needs, provided herein, inter alia, are isolated antibodies that bind an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both. In some embodiments, the antibody binds an extracellular domain of a human SIRP- α v1 polypeptide comprising the amino acid sequence of

EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELIYNQKEGHFPR
VTTVSDLTKRNNMDFSIRIGNITPADAGTYCYVKFRKGGSPDDVEFKSGAGTELSVRAK PS
(SEQ ID NO:5). In some embodiments, the antibody binds an extracellular domain of a human SIRP- α v2 polypeptide comprising the amino acid sequence of
EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIYNQKEGHFPRV
TTVSESTKRENMDFSISISNITPADAGTYCYVKFRKGGSPDTEFKSGAGTELSVRAKPS (SEQ

ID NO:6). In some embodiments, the antibody binds an extracellular domain of a human SIRP- α v1 polypeptide comprising the amino acid sequence of
EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELIYNQKEGHFPR
VTTVSDLTKRNNMDFSIRIGNITPADAGTYCYVKFRKGGSPDDVEFKSGAGTELSVRAK PS
(SEQ ID NO:5) and binds an extracellular domain of a human SIRP- α v2 polypeptide comprising the amino acid sequence of
EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIYNQKEGHFPRV
TTVSESTKRENMDFSISISNITPADAGTYCYVKFRKGGSPDTEFKSGAGTELSVRAKPS (SEQ ID NO:6). In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of three, four, five, six, seven, eight, nine or ten different human SIRP- α variant polypeptides. In some embodiments, each of the three, four, five, six, seven, eight, nine or ten different human SIRP- α variant polypeptides comprises an extracellular domain comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 5, 6, and 76-83. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide. In some embodiments, the monkey SIRP- α polypeptide is a cynomolgus SIRP- α polypeptide. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of at least two different monkey SIRP- α variant polypeptides. In some embodiments, the antibody binds an extracellular domain of a cynomolgus SIRP- α polypeptide comprising the amino acid sequence of SEQ ID NO:11, an extracellular domain of a cynomolgus SIRP- α polypeptide comprising the amino acid sequence of SEQ ID NO:12, or both. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different murine SIRP- α variant polypeptides. In some embodiments, the antibody binds an extracellular domain of one or more murine SIRP- α polypeptides, and wherein the one or more murine SIRP- α polypeptides each comprise an amino acid sequence selected from the group consisting of SEQ ID NOs: 7-10. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide. In some embodiments, the antibody does not bind an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide. In some embodiments, the antibody does not bind an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide. In some embodiments, the antibody binds the extracellular domain of a human SIRP- β polypeptide comprising the amino acid sequence of SEQ ID NO:13, the extracellular domain of a human SIRP- β polypeptide comprising the amino acid sequence of SEQ ID NO:14, or both. In some embodiments, the antibody binds the extracellular domain of a human SIRP- γ polypeptide comprising the amino acid sequence of SEQ ID NO:15. In some embodiments, the antibody modulates SIRP- α signaling in a cell expressing a human SIRP- α polypeptide. In some embodiments, the cell is a leukocyte selected from the group consisting of a macrophage, a dendritic cell, a neutrophil, an eosinophil, and a myeloid-derived suppressor cell (MDSC). In some embodiments, the antibody inhibits SIRP- α signaling in a macrophage expressing a human SIRP- α polypeptide. In some embodiments, the antibody enhances phagocytosis by a macrophage expressing a human SIRP- α polypeptide. In some embodiments, the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, the antibody does not bind a complex comprising a SIRP- α D1 variant bound to an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human

CD47. In some embodiments, the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, the antibody binds a complex comprising a SIRP- α D1 variant bound to an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide increases $k_{\text{sub.off}}$ of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell increases $k_{\text{sub.off}}$ of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and wherein the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and wherein the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and wherein binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and wherein the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and wherein the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In

some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and wherein binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and wherein the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and wherein the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; wherein the antibody binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and wherein binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domain) of two or more different human SIRP- α variant polypeptides, a murine SIRP- α polypeptide, and a monkey SIRP- α polypeptide; wherein the antibody does not bind at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and wherein the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody comprises one, two, three, four, five, or six CDR sequences; a heavy chain variable domain sequence; and/or a light chain variable domain sequence from antibody S130. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a murine SIRP- α polypeptide, a monkey SIRP- α polypeptide, and at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and wherein the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody comprises one, two, three, four, five, or six CDR sequences; a heavy chain variable domain sequence; and/or a light chain variable domain sequence from an antibody selected from the group consisting of S8, S13, S14, and S121. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a murine SIRP- α polypeptide, and a monkey SIRP- α polypeptide; wherein the antibody does not bind, or binds with reduced affinity to, at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and wherein the antibody does not block binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody comprises one, two, three, four, five, or six CDR sequences; a heavy chain variable domain sequence; and/or a light chain variable domain sequence from antibody S137. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of

two or more different human SIRP- α variant polypeptides and a monkey SIRP- α polypeptide; wherein the antibody does not bind a murine SIRP- α polypeptide; wherein the antibody does not bind at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and wherein the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody comprises one, two, three, four, five, or six CDR sequences; a heavy chain variable domain sequence; and/or a light chain variable domain sequence from an antibody selected from the group consisting of S128. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a monkey SIRP- α polypeptide, and at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; wherein the antibody does not bind a murine SIRP- α polypeptide; and wherein the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody comprises one, two, three, four, five, or six CDR sequences; a heavy chain variable domain sequence; and/or a light chain variable domain sequence from an antibody selected from the group consisting of S9, S11, S119, S120, S122, and S135. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides and the extracellular domains (e.g., the D1 domains) of two or more different monkey SIRP- α variant polypeptides, and wherein binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody comprises one, two, three, four, five, or six CDR sequences; a heavy chain variable domain sequence; and/or a light chain variable domain sequence from an antibody selected from the group consisting of S115, S116, S117 and S118. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:120 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:97. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:127 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:104. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:133 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:134. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:135 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:136. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:137 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:138. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:139 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:140. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:141 and/or a VL domain comprising the amino acid sequence of SEQ ID NO:142.

(9) In some embodiments, the antibody comprises a VH domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to the VH domain of antibody 115 (e.g., as listed in Table 2) and/or a VL domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to the VL domain of antibody 115 (e.g., as listed in Table 2). In some embodiments, the antibody comprises a VH domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to the VH domain of antibody 116 (e.g., as listed in Table 2) and/or a VL domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to the VL domain of

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89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO:89. In some embodiments, the antibody comprises a VH domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO:113 and/or a VL domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO:90. In some embodiments, the antibody comprises a VH domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO:114 and/or a VL domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO:91. In some embodiments, the antibody comprises a VH domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO:115 and/or a VL domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO:92. In some embodiments, the antibody comprises a VH domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO:135, 137, 139, or 141 and/or a VL domain comprising an amino acid sequence that is 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO:136, 138, 140, or 142.

(10) In some embodiments, the antibody comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:227 or 230, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:228 or 231, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:229; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:232, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:233, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:234. In some embodiments, the antibody comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:219 or 235, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:236 or 238, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:237; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:239, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:240, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:241. In some embodiments, the antibody comprises (a) an HVR-H1 sequence comprising the amino acid sequence of GFSFSX.sub.1X.sub.2AMX.sub.3, wherein X.sub.1 is N or I; X.sub.2 is F or Y; and X.sub.3 is T or S (SEQ ID NO:185); (b) an HVR-H2 sequence comprising the amino acid sequence of TIGX.sub.4X.sub.5DTYYADSVKG, wherein X.sub.4 is S or A and X.sub.5 is G or D (SEQ ID NO:186); (c) an HVR-H3 sequence comprising the amino acid sequence of DSTVX.sub.6WSGDFFDY, wherein X.sub.6 is S or G (SEQ ID NO:187); (d) an HVR-L1 sequence comprising the amino acid sequence of RASQNVX.sub.7X.sub.8DX.sub.9A, wherein X.sub.7 is K or R; X.sub.8 is N or S; and X.sub.9 is L or I (SEQ ID NO:188); (e) an HVR-L2 sequence comprising the amino acid sequence of AAX.sub.10X.sub.11RX.sub.12T, wherein X.sub.10 is R or S; X.sub.11 is I or S; and X.sub.12 is E or D (SEQ ID NO:189); and (f) an HVR-L3 sequence comprising the amino acid sequence of QQYYDWPPFT (SEQ ID NO:148). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 119 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from the variable domain sequences of SEQ ID NOs:335 and 97 (e.g., one, two, or three heavy chain HVR sequences from the heavy chain variable domain sequence of SEQ ID NO:335 and/or one, two, or three light chain HVR sequences from the light chain variable domain sequence of SEQ ID NO:97). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 135 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from

the variable domain sequences of SEQ ID NOs:127 and 104 (e.g., one, two, or three heavy chain HVR sequences from the heavy chain variable domain sequence of SEQ ID NO:127 and/or one, two, or three light chain HVR sequences from the light chain variable domain sequence of SEQ ID NO:104). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from the variable domain sequences of SEQ ID NOs:97, 104, 120, 335, and 127 (e.g., one, two, or three heavy chain HVR sequences from the heavy chain variable domain sequence of SEQ ID NOs:335 and 127 and/or one, two, or three light chain HVR sequences from the light chain variable domain sequence of SEQ ID NOs:97 and 104). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:143-148 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:143-145 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:146-148). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:148-153 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:149-151 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:152, 153, and 148). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 136 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:155-160 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:155-157 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:158-160). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 21 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:161-166 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:161-163 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:164-166). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 25 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs: 161, 163, 168, and 170-172 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs: 161, 168, and 163 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:170-172). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 27 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs: 163, 173, 174, and 176-178 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:163, 173, and 174 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:176-178). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 66 (e.g., as listed in Table 2). In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:162, 163, 179, and 182-184 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs: 162, 163, and 179 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:182-184). In some embodiments, the antibody comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:143, 202, 204, or 205, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:144, 203, or 206, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:145 or 207; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:146 or 208, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:147 or 209, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:148 or 210. In some embodiments, the antibody comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:149, 211, 213, or 214, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:150, 212, or

215, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:151 or 216; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:152 or 217, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:153 or 218, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:148. In some embodiments, the antibody comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:155, 219, 221, or 222, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:156, 220, or 223, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:157 or 224; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:158 or 225, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:159 or 226, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:160. In some embodiments, the antibody comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:161, 191, or 194, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:162, 192, or 195, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:163 or 193; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166. In some embodiments, the antibody comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:161, 191, or 194, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:168, 196, or 197, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:163 or 193; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:170, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:171, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:172. In some embodiments, the antibody comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:173, 198, or 200, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:174, 199, or 201, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:163 or 193; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:176, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:177, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:178. In some embodiments, the antibody comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:179, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:162, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:163; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:182, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:183, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:184. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:135 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:137 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:170, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:171, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:172. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:139 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:176, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:177, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:178. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:141 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:182, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:183, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:184. In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 3 (e.g., as listed in Table 2). In some

embodiments, the antibody comprises (a) a VH domain comprising one, two, or three HVR sequences from SEQ ID NO:242; and/or (b) a VL domain comprising one, two, or three HVR sequences from SEQ ID NO:243. In some embodiments, the antibody comprises one, two, three, four, five, or six HVR sequences from antibody 45 (e.g., as listed in Table 2). In some embodiments, the antibody comprises (a) a VH domain comprising one, two, or three HVR sequences from SEQ ID NO:244; and/or (b) a VL domain comprising one, two, or three HVR sequences from SEQ ID NO:245. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:135 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:137 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:139 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, the antibody comprises a VH domain comprising the amino acid sequence of SEQ ID NO:141 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, one, two, three, four, five, or six of the HVR sequences are defined by Kabat. In some embodiments, one, two, three, four, five, or six of the HVR sequences are defined by Chothia. In some embodiments, one, two, three, four, five, or six of the HVR sequences are defined by IMGT. In some embodiments, the antibody comprises HVR sequences as defined by two or more of Kabat, Chothia, and IMGT (e.g., the antibody comprises one or more HVR sequences as defined by one delineation and one or more HVR sequences as defined by a different delineation).

(11) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody comprises: (a) a heavy chain variable (VH) domain comprising (i) an HVR-H1 sequence comprising the amino acid sequence of NFAMT (SEQ ID NO: 175), NFAVT (SEQ ID NO:204), or NFALT (SEQ ID NO:305), (ii) an HVR-H2 sequence comprising the amino acid sequence of TIGSGDTYYADSVKG (SEQ ID NO:144), and (iii) an HVR-H3 sequence comprising the amino acid sequence of DSTVSWSGDFFDY (SEQ ID NO:145); and/or (b) a light chain variable (VL) domain comprising (i) an HVR-L1 sequence comprising the amino acid sequence of RASQNVKNDLA (SEQ ID NO:146), (ii) an HVR-L2 sequence comprising the amino acid sequence of AARIRET (SEQ ID NO:147), and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDWPPFT (SEQ ID NO:148). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:120, 335, 246, 258, or 327; and/or the VL domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:97 or 312. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:246, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:258, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:335, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:327, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the

amino acid sequence of SEQ ID NO:246, and the VL domain comprises the amino acid sequence of SEQ ID NO:312; the VH domain comprises the amino acid sequence of SEQ ID NO:258, and the VL domain comprises the amino acid sequence of SEQ ID NO:312; the VH domain comprises the amino acid sequence of SEQ ID NO:335, and the VL domain comprises the amino acid sequence of SEQ ID NO:312; or the VH domain comprises the amino acid sequence of SEQ ID NO:327, and the VL domain comprises the amino acid sequence of SEQ ID NO:312.

(12) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody comprises: (a) a heavy chain variable (VH) domain comprising (i) an HVR-H1 sequence comprising the amino acid sequence of IYAMS (SEQ ID NO:269), IYAVS (SEQ ID NO:213), or IYALS (SEQ ID NO:306), (ii) an HVR-H2 sequence comprising the amino acid sequence of TIGADDTYYADSVKG (SEQ ID NO:150), and (iii) an HVR-H3 sequence comprising the amino acid sequence of DSTVGWSGDDFDY (SEQ ID NO:151); and/or (b) a light chain variable (VL) domain comprising (i) an HVR-L1 sequence comprising the amino acid sequence of RASQNVRSDIA (SEQ ID NO:152), (ii) an HVR-L2 sequence comprising the amino acid sequence of AASSRDT (SEQ ID NO:153), and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDWPPFT (SEQ ID NO:148). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:341, 247, 259, or 328; and/or the VL domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:104 or 248. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:127, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:247, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:259, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:328, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:127, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; the VH domain comprises the amino acid sequence of SEQ ID NO:247, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; the VH domain comprises the amino acid sequence of SEQ ID NO:259, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; or the VH domain comprises the amino acid sequence of SEQ ID NO:328, and the VL domain comprises the amino acid sequence of SEQ ID NO:248.

(13) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody comprises: (a) a heavy chain variable (VH) domain comprising: (i) an HVR-H1 sequence comprising the amino acid sequence of X.sub.1X.sub.2DX.sub.3N, wherein X.sub.1 is S or T; X.sub.2 is Y or S; and X.sub.3 is M, L, or V (SEQ ID NO:307); (ii) an HVR-H2 sequence comprising the amino acid sequence of LISGSGEIX.sub.1YYADSVKG, wherein X.sub.1 is I or T (SEQ ID NO:308); and (iii) an HVR-H3 sequence comprising the amino acid sequence of EX.sub.1X.sub.2X.sub.3YRFFDX.sub.4, wherein X.sub.1 is N or D; X.sub.2 is N or D; X.sub.3 is R or M; and X.sub.4 is D or Y (SEQ ID NO:309); and/or (b) a light chain variable (VL) domain comprising: (i) an HVR-L1 sequence comprising the amino acid sequence of RAX.sub.1QSVYX.sub.2YLA, wherein X.sub.1 is S or D; and X.sub.2 is T or S (SEQ ID NO:310); (ii) an HVR-L2 sequence comprising the amino acid sequence of X.sub.1AX.sub.2X.sub.3RAX.sub.4, wherein X.sub.1 is G, A, or D; X.sub.2 is S or R; X.sub.3 is S, N, or T; and X.sub.4 is T or A (SEQ ID NO:311); and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDRPPLT (SEQ ID NO:160). In some embodiments, the antibody comprises (a) a heavy chain variable (VH) domain comprising (i) an HVR-H1 sequence comprising the amino acid sequence of SYDMN (SEQ ID NO:270), SYDVN (SEQ ID NO:221), or SYDLN (SEQ ID NO:313), (ii) an HVR-H2 sequence comprising the amino acid sequence of LISGSGEIIYYADSVKG (SEQ ID NO:156), and (iii) an HVR-H3 sequence

comprising the amino acid sequence of ENNRYRFFDD (SEQ ID NO:157); and/or (b) a light chain variable (VL) domain comprising (i) an HVR-L1 sequence comprising the amino acid sequence of RASQSVYTYLA (SEQ ID NO:158), (ii) an HVR-L2 sequence comprising the amino acid sequence of GASSRAT (SEQ ID NO:159), and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDRPPLT (SEQ ID NO:160). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO: 249, 133, 260, or 329; and/or the VL domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:134, 250, or 251. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:251; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:251; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; or the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:251.

(14) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody comprises: a heavy chain variable (VH) domain comprising: an HVR-H1 sequence comprising the amino acid sequence of X.sub.1X.sub.2AX.sub.3S, wherein X.sub.1 is S or T; X.sub.2 is N, Y, H, or D; and X.sub.3 is M, L, or V (SEQ ID NO:297); an HVR-H2 sequence comprising the amino acid sequence of GISX.sub.1X.sub.2X.sub.3X.sub.4X.sub.5X.sub.6YYX.sub.7X.sub.8SX.sub.9KG, wherein X.sub.1 is A or S; X.sub.2 is G, S, or absent; X.sub.3 is S, D or G; X.sub.4 is G or S; X.sub.5 is D, S, or G; X.sub.6 is T or A; X.sub.7 is P, G, V, I, A, or S; X.sub.8 is A, D, or G; and X.sub.9 is V or M (SEQ ID NO:298); and an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193); and/or a light chain variable (VL) domain comprising: an HVR-L1 sequence comprising the amino acid sequence of SGGX.sub.1X.sub.2X.sub.3SX.sub.4YYX.sub.5, wherein X.sub.1 is D, G, S, I, or absent; X.sub.2 is S, W, G, Y, D, or absent; X.sub.3 is S, Y, T, or D; X.sub.4 is H, T, S, or Y; and X.sub.5 is G or A (SEQ ID NO:299); an HVR-L2 sequence comprising the amino acid sequence of SDX.sub.1X.sub.2RPX.sub.3, wherein X.sub.1 is D or N; X.sub.2 is E, K, or Q; and X.sub.3 is S or P (SEQ ID NO:300); and an HVR-L3 sequence comprising the amino acid sequence of X.sub.1X.sub.2YDX.sub.3X.sub.4X.sub.5YX.sub.6NX.sub.7, wherein X.sub.1 is G or A; X.sub.2 is G or A; X.sub.3 is G, Y, Q, S, or A; X.sub.4 is S, R, or T; X.sub.5 is T or S; X.sub.6 is A, I, V, L, or T; and X.sub.7 is T, A, D, or P (SEQ ID NO:301). In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody comprises: a heavy chain variable (VH) domain comprising: a heavy chain variable (VH) domain comprising: an HVR-H1 sequence comprising the amino acid sequence of SX.sub.1AX.sub.2S, wherein X.sub.1 is N or Y; and wherein X.sub.2 is M, L, or V (SEQ ID

NO:302); an HVR-H2 sequence comprising the amino acid sequence of GISX.sub.1GX.sub.2X.sub.3DTYYX.sub.4X.sub.5SVKG, wherein X.sub.1 is A or S; X.sub.2 is G or absent; X.sub.3 is S or G; X.sub.4 is P, G, or V; and X.sub.5 is A or D (SEQ ID NO:303); and an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193); and/or a light chain variable (VL) domain comprising: an HVR-L1 sequence comprising the amino acid sequence of SGGX.sub.1YSSYYA, wherein X.sub.1 is S or A (SEQ ID NO:304); an HVR-L2 sequence comprising the amino acid sequence of SDDKRPS (SEQ ID NO:336); and an HVR-L3 sequence comprising the amino acid sequence of GGYDQSSYTNP (SEQ ID NO:172). In some embodiments, the VH domain comprises (i) an HVR-H1 sequence comprising the amino acid sequence of SNAMS (SEQ ID NO:194), SNAVS (SEQ ID NO:271), or SNALS (SEQ ID NO:318), (ii) an HVR-H2 sequence comprising the amino acid sequence of GISAGGSDTYYPASVKG (SEQ ID NO:195), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:135, 263, 264, or 330. In some embodiments, the VH domain comprises (i) an HVR-H1 sequence comprising the amino acid sequence of SNAMS (SEQ ID NO:194), SNAVS (SEQ ID NO:271), or SNALS (SEQ ID NO:318), (ii) an HVR-H2 sequence comprising the amino acid sequence of GISSGSDTYYGDSVKG (SEQ ID NO:197), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:137, 265, 266, or 331. In some embodiments, the VH domain comprises (i) an HVR-H1 sequence comprising the amino acid sequence of SYAMS (SEQ ID NO:200), SYAVS (SEQ ID NO:272), or SYALS (SEQ ID NO:319), (ii) an HVR-H2 sequence comprising the amino acid sequence of GISSGGDTYYVDSVKG (SEQ ID NO:201), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:139, 267, 268, or 332. In some embodiments, the VL domain comprises the sequence FW1-HVR-L1-FW2-HVR-L2-FW3-HVR-L3-FW4 (N-terminus to C-terminus), wherein FW1 comprises the amino acid sequence SYELTQPPSVSVSPGQTARITC (SEQ ID NO:314), FW2 comprises the amino acid sequence WYQQKPGQAPVTLIY (SEQ ID NO:315), FW3 comprises the amino acid sequence NIPERFSGSSSGTTVTLTISGVQAEDEADYYC (SEQ ID NO:316), and FW4 comprises the amino acid sequence FGSGTKLTVL (SEQ ID NO:317). In some embodiments, the VL domain comprises (i) an HVR-L1 sequence comprising the amino acid sequence of SGGSYSSYYA (SEQ ID NO:170), (ii) an HVR-L2 sequence comprising the amino acid sequence of SDDKRPS (SEQ ID NO:336), and (iii) an HVR-L3 sequence comprising the amino acid sequence of GGYDQSSYTNP (SEQ ID NO:172). In some embodiments, the VL domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:252. In some embodiments, the VL domain comprises (i) an HVR-L1 sequence comprising the amino acid sequence of SGGAYSSYYA (SEQ ID NO:261), (ii) an HVR-L2 sequence comprising the amino acid sequence of SDDKRPS (SEQ ID NO:336), and (iii) an HVR-L3 sequence comprising the amino acid sequence of GGYDQSSYTNP (SEQ ID NO:172). In some embodiments, the VL domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:262. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:263, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:264, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:330, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:135, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:137, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the

VH domain comprises the amino acid sequence of SEQ ID NO:139, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:265, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:266, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:331, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:267, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:268, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:332, and the VL domain comprises the amino acid sequence of SEQ ID NO:252; the VH domain comprises the amino acid sequence of SEQ ID NO:263, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:264, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:330, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:265, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:266, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:331, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:267, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:268, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:332, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:135, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; the VH domain comprises the amino acid sequence of SEQ ID NO:137, and the VL domain comprises the amino acid sequence of SEQ ID NO:262; or the VH domain comprises the amino acid sequence of SEQ ID NO:139, and the VL domain comprises the amino acid sequence of SEQ ID NO:262.

(15) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody binds to a human SIRP- α v1 polypeptide at one or more amino acid positions selected from the group consisting of I31, V33, Q52, K53, T67, R69, N70, and K96, according to SEQ ID NO:296. In some embodiments, the antibody binds to the human SIRP- α v1 polypeptide at I31, V33, Q52, K53, T67, R69, N70, and K96, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at one or more amino acid positions selected from the group consisting of L30, P32, E54, T62, N71, M72, F74, and R95, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at L30, P32, E54, T62, N71, M72, F74, and R95, according to SEQ ID NO:296.

(16) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody binds to a human SIRP- α v1 polypeptide at one or more amino acid positions selected from the group consisting of 17, P9, D10, K11, S12, A42, A108, and E111, according to SEQ ID NO:296. In some embodiments, the antibody binds to the human SIRP- α v1 polypeptide at K11, A42, A108, and E111, according to SEQ ID NO:296. In some embodiments, the antibody binds to the human SIRP- α v1 polypeptide at 17, P9, D10, K11, S12, A108, and E111, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at one or more amino acid positions selected from the group consisting of L14, T26, T28, T88, Y90, S106, S113, and A116, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at L14, T88, Y90, S106, S113, and A116 of human SIRP- α v1, according to SEQ ID

NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at L14, T26, and T28, according to SEQ ID NO:296.

(17) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody binds to a human SIRP- α v1 polypeptide at one or more amino acid positions selected from the group consisting of E47, L48, P58, R59, T82, and A84, according to SEQ ID NO:296. In some embodiments, the antibody binds to the human SIRP- α v1 polypeptide at E47, L48, P58, R59, T82, and A84, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at one or more amino acid positions selected from the group consisting of A17, P44, G45, 149, E54, G55, H56, F57, and P83, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at A17, P44, G45, 149, E54, G55, H56, F57, and P83 of human SIRP- α v1, according to SEQ ID NO:296.

(18) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody binds the extracellular domain of a human SIRP- α v1 polypeptide with a dissociation constant ($K_{sub.D}$) of less than 100 nM, and wherein the antibody blocks binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain of a human SIRP- α v2 polypeptide with a dissociation constant ($K_{sub.D}$) of less than 100 nM. In some embodiments, the antibody binds the D1 domain of a human SIRP- α v1 polypeptide and the D1 domain of a human SIRP- α v2 polypeptide. In some embodiments, the antibody binds an extracellular domain of a cynomolgus SIRP- α polypeptide. In some embodiments, the antibody binds an extracellular domain of a human SIRP- β polypeptide. In some embodiments, the antibody binds an extracellular domain of a human SIRP- γ polypeptide. In some embodiments, the antibody binds an extracellular domain of a murine SIRP- α polypeptide.

(19) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody binds the D1 domain of a human SIRP- α polypeptide, and wherein the antibody does not block binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the D1 domain of a human SIRP- α with a dissociation constant ($K_{sub.D}$) of less than 100 nM. In some embodiments, the antibody binds the D1 domain of a human SIRP- α v1 polypeptide with a dissociation constant ($K_{sub.D}$) of less than 100 nM and/or binds the D1 domain of a human SIRP- α v2 polypeptide with a dissociation constant ($K_{sub.D}$) of less than 100 nM. In some embodiments, the antibody binds the D1 domain of a human SIRP- α v1 polypeptide and the D1 domain of a human SIRP- α v2 polypeptide. In some embodiments, the antibody binds an extracellular domain of a cynomolgus SIRP- α polypeptide. In some embodiments, the antibody binds an extracellular domain of a human SIRP- β polypeptide. In some embodiments, the antibody binds an extracellular domain of a murine SIRP- α polypeptide.

(20) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain and a light chain variable (VL) domain of an antibody selected from the group consisting of antibodies 119, 120, 121, 122, 21, 25, 27, 66, and 135. In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising: (a) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:120 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:97; (b) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:121 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:98; (c) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID

NO:130 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:107; (d) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:122 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:99; (e) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:135 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:136; (f) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:137 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:138; (g) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:139 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:140; (h) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:141 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:142; or (i) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:127 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:104.

(21) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain and a light chain variable (VL) domain of an antibody selected from the group consisting of antibodies 136 and 137. In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising: (a) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:133 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:134; or (b) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:128 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:105.

(22) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain and a light chain variable (VL) domain of an antibody selected from the group consisting of antibodies 3, 213, 173, and 209. In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising: (a) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:242 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:243; (b) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:275 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:276; (c) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:278 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:279; or (d) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:280 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:281.

(23) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain and a light chain variable (VL) domain of an antibody selected from the group consisting of antibodies 115, 116, 117, 118, and 132. In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody

comprising: (a) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:116 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:93; (b) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:117 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:94; (c) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:118 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:95; (d) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:119 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:96; or (e) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:282 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:283.

(24) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain and a light chain variable (VL) domain of an antibody selected from the group consisting of antibodies 218, 123, 149, 161, 162, and 194. In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising: (a) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:284 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:285; (b) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:123 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:100; (c) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:286 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:287; (d) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:288 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:289; (e) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:290 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:291; or (f) a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:292 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:293.

(25) In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain and a light chain variable (VL) domain of antibody 45. In other aspects, provided herein is an isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide (e.g., the D1 domain), wherein the antibody competes for binding the extracellular domain of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody comprising a heavy chain variable (VH) domain comprising the amino acid sequence of SEQ ID NO:244 and a light chain variable (VL) domain comprising the amino acid sequence of SEQ ID NO:245.

(26) In some embodiments of any of the above embodiments, the antibody enhances phagocytosis by a macrophage expressing a human SIRP- α polypeptide. In some embodiments, the antibody enhances activation of a dendritic cell expressing a human SIRP- α polypeptide. In some embodiments, the antibody inhibits in vivo growth of a tumor that expresses CD47. In some embodiments, the antibody does not prevent interactions between a CD47-expressing cell and a T cell.

(27) In some embodiments of any of the above embodiments, the antibody is a monoclonal antibody. In some embodiments, the antibody is a scFv-Fc, single domain antibody, single heavy

chain antibody, or single light chain antibody. In some embodiments, the antibody comprises a light chain constant region comprising the amino acid sequence of SEQ ID NO:325, 326, or 426. In some embodiments, the antibody comprises an amino acid sequence selected from the group consisting of SEQ ID NOs:320-324. In some embodiments, the antibody comprises an Fc region. In some embodiments, the Fc region is a human Fc region selected from the group consisting of an IgG1 Fc region, an IgG2 Fc region, and an IgG4 Fc region. In some embodiments, the Fc region comprises a human IgG1 Fc region comprising one or more mutations selected from the group consisting of L234A, L235A, L235E, G237A, and N297A, according to EU numbering. In some embodiments, the Fc region comprises a human IgG2 Fc region comprising one or more mutations selected from the group consisting of A330S, P331S and N297A, according to EU numbering. In some embodiments, the Fc region comprises a human IgG4 Fc region comprising one or more mutations selected from the group consisting of S228P, E233P, F234V, L235A, L235E, delG236, and N297A, according to EU numbering. In some embodiments, the antibody is an antibody fragment selected from the group consisting of a Fab, F(ab')₂, Fab'-SH, Fv, and scFv fragment. In some embodiments, the antibody is conjugated to a cytotoxic agent or label.

(28) In some embodiments, the antibody is a bispecific antibody. In some embodiments, the antibody comprises a first antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and a second antigen binding domain that binds an antigen expressed by a cancer cell. In some embodiments, the antigen expressed by the cancer cell is selected from the group consisting of CD19, CD20, CD22, CD30, CD33, CD38, CD52, CD56, CD70, CD74, CD79b, CD123, CD138, CS1/SLAMF7, Trop-2, 5T4, EphA4, BCMA, Mucin 1, Mucin 16, PD-L1, PTK7, STEAP1, Endothelin B Receptor, mesothelin, EGFRvIII, ENPP3, SLC44A4, GNMB, nectin 4, NaPi2b, LIV-1A, Guanylyl cyclase C, DLL3, EGFR, HER2, VEGF, VEGFR, integrin α V β 3, integrin α 5 β 1, MET, IGF1R, TRAILR1, TRAILR2, RANKL, FAP, Tenascin, Le.sup.y, EpCAM, CEA, gpA33, PSMA, TAG72, a mucin, CAIX, EPHA3, folate receptor α , GD2, GD3, and an MHC/peptide complex comprising a peptide from NY-ESO-1/LAGE, SSX-2, a MAGE family protein, MAGE-A3, gp100/pmel17, Melan-A/MART-1, gp75/TRP1, tyrosinase, TRP2, CEA, PSA, TAG-72, immature laminin receptor, MOK/RAGE-1, WT-1, SAP-1, BING-4, EpCAM, MUC1, PRAME, survivin, BRCA1, BRCA2, CDK4, CML66, MART-2, p53, Ras, β -catenin, TGF- β RII, HPV E6, or HPV E7. In some embodiments, the antibody is a chicken, humanized, chimeric, or human antibody. In some embodiments, the antibody is generated by or derived from a chicken.

(29) Further provided herein are polynucleotides comprising the antibody according to any one of the above embodiments. Further provided herein are vectors comprising the polynucleotide according to any one of the above embodiments. Further provided herein are host cells comprising the polynucleotide or vector according to any one of the above embodiments. Further provided herein are methods of producing an antibody, comprising culturing the host cell according to any one of the above embodiments such that the antibody is produced. In some embodiments, the methods further include recovering the antibody from the host cell.

(30) Further provided herein are methods of treating or delaying progression of cancer in an individual, the methods comprising administering to the individual an effective amount of the antibody according to any one of the above embodiments. In some embodiments, the methods further comprise administering to the individual an effective amount of a second antibody. In some embodiments, the second antibody binds an antigen expressed by a cancer cell. In some embodiments, the antigen expressed by the cancer cell is selected from the group consisting of CD19, CD20, CD22, CD30, CD33, CD38, CD52, CD56, CD70, CD74, CD79b, CD123, CD138, CS1/SLAMF7, Trop-2, 5T4, EphA4, BCMA, Mucin 1, Mucin 16, PTK7, STEAP1, Endothelin B Receptor, mesothelin, EGFRvIII, ENPP3, SLC44A4, GNMB, nectin 4, NaPi2b, LIV-1A, Guanylyl cyclase C, DLL3, EGFR, HER2, VEGF, VEGFR, integrin α V β 3, integrin α 5 β 1, MET, IGF1R, TRAILR1, TRAILR2, RANKL, FAP, Tenascin, Le.sup.y, EpCAM, CEA, gpA33, PSMA, TAG72, a

mucin, CAIX, EPHA3, folate receptor α , GD2, GD3, and an MHC/peptide complex comprising a peptide from NY-ESO-1/LAGE, SSX-2, a MAGE family protein, MAGE-A3, gp100/pmel17, Melan-A/MART-1, gp75/TRP1, tyrosinase, TRP2, CEA, PSA, TAG-72, immature laminin receptor, MOK/RAGE-1, WT-1, SAP-1, BING-4, EpCAM, MUC1, PRAME, survivin, BRCA1, BRCA2, CDK4, CML66, MART-2, p53, Ras, β -catenin, TGF- β RII, HPV E6, or HPV E7. In some embodiments, the methods further comprise administering to the individual an effective amount of an immunotherapeutic agent. In some embodiments, the immunotherapeutic agent comprises a second antibody. In some embodiments, the second antibody binds to an antigen selected from the group consisting of PD-1, PD-L1, OX40, CTLA-4, CD137/4-1BB, TNFR2, B7-H3, FZD7, CD27, CCR4, CSF1R, CSF, TIM-3, LAG-3, VISTA, ICOS, CCR2, IDO, A2R, CD39, CD73, TIGIT, CD80, CD47, arginase, TDO, and PVRIG. In some embodiments, the first antibody binds the extracellular domain of a human SIRP- α v1 polypeptide, the extracellular domain of a human SIRP- α v2 polypeptide, or the extracellular domains of both a human SIRP- α v1 polypeptide and a human SIRP- α v2 polypeptide with a dissociation constant ($K_{sub.D}$) of less than 100 nM, wherein the first antibody blocks binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide, and wherein the second antibody binds to PD-1. In some embodiments, the first antibody binds the D1 domain of a human SIRP- α polypeptide, wherein the first antibody does not block binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide, and wherein the second antibody binds to PD-1. In some embodiments, the first antibody binds the extracellular domain of a human SIRP- α v1 polypeptide with a dissociation constant ($K_{sub.D}$) of less than 100 nM, wherein the first antibody blocks binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide, and wherein the second antibody binds to PD-L1. In some embodiments, the first antibody binds the D1 domain of a human SIRP- α polypeptide, wherein the first antibody does not block binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide, and wherein the second antibody binds to PD-L1. In some embodiments, the individual is a human.

(31) Further provided herein are methods of treating or delaying progression of an autoimmune disease or an inflammatory disease in an individual, the methods comprising administering to the individual an effective amount of the antibody according to any one of the above embodiments. In some embodiments, the autoimmune disease or inflammatory disease is selected from the group consisting of multiple sclerosis, rheumatoid arthritis, a spondyloarthropathy, systemic lupus erythematosus, an antibody-mediated inflammatory or autoimmune disease, graft versus host disease, sepsis, diabetes, psoriasis, psoriatic arthritis, atherosclerosis, Sjogren's syndrome, progressive systemic sclerosis, scleroderma, acute coronary syndrome, ischemic reperfusion, Crohn's Disease, ulcerative colitis, endometriosis, glomerulonephritis, IgA nephropathy, polycystic kidney disease, myasthenia gravis, idiopathic pulmonary fibrosis, asthma, atopic dermatitis, acute respiratory distress syndrome (ARDS), vasculitis, and inflammatory autoimmune myositis. In some embodiments, the individual is a human.

(32) Further provided herein are methods of identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and does not block binding between human CD47 and the human SIRP- α polypeptide, the methods comprising (a) providing an antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide; (b) assembling a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; (c) contacting the antigen binding domain with the assembled complex; and (d) detecting binding of the antigen binding domain to the complex, wherein binding of the antigen binding domain to the complex indicates that the antigen

binding domain does not block binding between human CD47 and the human SIRP- α polypeptide. Further provided herein are methods of identifying an antibody or antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and does not block binding between human CD47 and the human SIRP- α polypeptide, the methods comprising contacting an antibody or antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide with a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; and detecting binding of the antigen binding domain to the complex, wherein binding of the antigen binding domain to the complex indicates that the antigen binding domain does not block binding between human CD47 and the human SIRP- α polypeptide. Further provided herein are methods of identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and blocks binding between human CD47 and the human SIRP- α polypeptide, the methods comprising (a) providing an antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide; (b) assembling a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; (c) contacting the antigen binding domain with the assembled complex; and (d) detecting binding of the antigen binding domain to the complex, wherein a lack of binding of the antigen binding domain to the complex indicates that the antigen binding domain blocks binding between human CD47 and the human SIRP- α polypeptide. Further provided herein are methods of identifying an antibody or antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and blocks binding between human CD47 and the human SIRP- α polypeptide, the methods comprising contacting an antibody or antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide with a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; and (d) detecting binding of the antigen binding domain to the complex, wherein a lack of binding of the antigen binding domain to the complex indicates that the antigen binding domain blocks binding between human CD47 and the human SIRP- α polypeptide. In some embodiments, the SIRP- α D1 variant comprises an amino acid sequence selected from the group consisting of SEQ ID NOs:17-52. In some embodiments, the IgSF domain of CD47 comprises the amino acid sequence of SEQ ID NO:16. In some embodiments, the polypeptide comprising the IgSF domain of CD47 comprises a human CD47 extracellular domain. In some embodiments, the polypeptide comprising the IgSF domain of CD47 further comprises an antibody Fc region.

(33) Further provided herein are methods of producing an anti-SIRP- α antibody that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide, the methods comprising: (a) immunizing a chicken with a peptide comprising at least a portion of a human SIRP- α extracellular domain (e.g., the D1 domain); (b) obtaining an antibody from an antibody-producing cell from the immunized chicken; and (c) detecting binding between the antibody obtained from the cell and the extracellular domains (e.g., the D1 domains) of a human SIRP- α polypeptide, wherein binding between the antibody and the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide indicates that the antibody is an anti-SIRP- α antibody that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α variant polypeptide.

In some embodiments, the antibody is a chicken, humanized, chimeric, or human antibody. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELIYNQKEGHFPR VTTVSDLTKRNNMDFSIRIGNITPADAGTY YCVKFRKGGSPDDVEFKSGAGTELSVRAK PS (SEQ ID NO:5). In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIYNQKEGHFPRV TTVSESTKRENMDFSISISNITPADAGTY YCVKFRKGGSPDTEFKSGAGTELSVRAKPS (SEQ ID NO:6). In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELIYNQKEGHFPR VTTVSDLTKRNNMDFSIRIGNITPADAGTY YCVKFRKGGSPDDVEFKSGAGTELSVRAK PS (SEQ ID NO:5) and binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIYNQKEGHFPRV TTVSESTKRENMDFSISISNITPADAGTY YCVKFRKGGSPDTEFKSGAGTELSVRAKPS (SEQ ID NO:6). In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of three, four, five, six, seven, eight, nine or ten different human SIRP- α variant polypeptides. In some embodiments, each of the three, four, five, six, seven, eight, nine or ten different human SIRP- α variant polypeptides comprises an extracellular domain (e.g., the D1 domain) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 5, 6, and 76-83. In some embodiments, the methods further comprise detecting binding between the antibody obtained from the cell and an extracellular domain (e.g., the D1 domain) of one or more SIRP- α polypeptides selected from the group consisting of a monkey SIRP- α polypeptide, a murine SIRP- α polypeptide, a human SIRP- β polypeptide, and a human SIRP- γ polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide. In some embodiments, the monkey SIRP- α polypeptide is a cynomolgus SIRP- α polypeptide. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of at least two different monkey SIRP- α variant polypeptides. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide comprising the amino acid sequence of SEQ ID NO:11, an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide comprising the amino acid sequence of SEQ ID NO:12, or both. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different murine SIRP- α variant polypeptides. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of one or more murine SIRP- α polypeptides, and wherein the one or more murine SIRP- α polypeptides each comprise an amino acid sequence selected from the group consisting of SEQ ID NOs: 7-10. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide. In some embodiments, the antibody does not bind an extracellular domain (e.g., the D1 domain) of a human SIRP-0 polypeptide. In some embodiments, the antibody does not bind an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP-0 polypeptide comprising the amino acid sequence of SEQ ID NO:13, the extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide comprising the amino acid sequence of SEQ ID NO:14, or both. In some embodiments, the antibody binds the

extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide comprising the amino acid sequence of SEQ ID NO:15. In some embodiments, the methods further comprise detecting binding or a lack of binding between the antibody obtained from the cell and a complex comprising a SIRP- α D1 variant bound to an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47. In some embodiments, the antibody binds a complex comprising a SIRP- α D1 variant bound to an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47. In some embodiments, the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, the antibody does not bind a complex comprising a SIRP- α D1 variant bound to an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47. In some embodiments, the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide increases $k_{\text{sub.off}}$ of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell increases $k_{\text{sub.off}}$ of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell. In some embodiments, the antibody modulates SIRP- α signaling in a cell expressing a human SIRP- α polypeptide. In some embodiments, the cell is a leukocyte selected from the group consisting of a macrophage, a dendritic cell, a neutrophil, an eosinophil, and a myeloid-derived suppressor cell (MDSC). In some embodiments, the antibody inhibits SIRP- α signaling in a macrophage expressing a human SIRP- α polypeptide. In some embodiments, the antibody enhances phagocytosis by a macrophage expressing a human SIRP- α polypeptide.

(34) It is to be understood that one, some, or all of the properties of the various embodiments described herein may be combined to form other embodiments of the present invention. These and other aspects of the invention will become apparent to one of skill in the art. These and other embodiments of the invention are further described by the detailed description that follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A shows an alignment among the D1 domains of 10 different human SIRP- α variant polypeptides. Sequences shown correspond to SEQ ID NOs: 5, 6, and 76-83 (from top to bottom). Amino acid differences are indicated by asterisks.
- (2) FIG. 1B shows an alignment between human v1, human v2, cynomolgus monkey, and 129 mouse SIRP- α D1 domains. Sequences shown correspond to SEQ ID NOs: 5, 6, 11, and 7 (from top to bottom). Amino acid differences are indicated by asterisks.
- (3) FIG. 1C shows alignments between various human and mouse SIRP- α D1 domains, with R1, R2 and R3 loops indicated. Shown is an alignment between human v1, human v2, 129 mouse, NOD mouse, C57BL/6 mouse, and BALB/c mouse SIRP- α D1 domains. Sequences shown correspond to SEQ ID NOs: 5-10 (from top to bottom). Amino acid differences are indicated by asterisks.
- (4) FIG. 2 shows an alignment between human v1, human v2, cynomolgus monkey, 129 mouse, and chicken SIRP- α D1 domains. Sequences shown correspond to SEQ ID NOs: 5, 6, 11, 7, and 84 (from top to bottom). Amino acid differences are indicated by asterisks.
- (5) FIGS. 3A & 3B show binding specificity of antibody clone S130 for a variety of SIRP peptides. FIG. 3A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP α D1 domains, as well as human SIRP γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants bound to the IgSF domain of CD47 (SEQ ID NO:16). The pre-formed complex was generated by mixing two high affinity human SIRP- α v1 and v2 polypeptides (SEQ ID NOs: 17 and 19) in a 1:1 ratio and combining the mixture with CD47 to generate the SIRP- α :CD47 complex. The pre-formed complex SIRP- α :CD47 complex for FIGS. 4A-9B are also prepared similarly. FIG. 3B summarizes the binding specificity of the clone against each of these targets (“+” indicates binding; “-” indicates non-binding).
- (6) FIGS. 4A & 4B show binding specificity of antibody clone S121 for a variety of SIRP peptides. FIG. 4A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP α D1 domains, as well as human SIRP γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants (SEQ ID NO:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO:16). FIG. 4B summarizes the binding specificity of the clone against each of these targets (“+” indicates binding; “-” indicates non-binding).
- (7) FIGS. 5A & 5B show binding specificity of antibody clone S137 for a variety of SIRP peptides. FIG. 5A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP α D1 domains, as well as human SIRP γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants (SEQ ID NO:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO:16). FIG. 5B summarizes the binding specificity of the clone against each of these targets (“+” indicates binding; “-” indicates non-binding).
- (8) FIGS. 6A & 6B show binding specificity of antibody clone S128 for a variety of SIRP peptides. FIG. 6A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP α D1 domains, as well as human SIRP γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants (SEQ ID NO:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO:16). FIG. 6B summarizes the binding specificity of the clone against each of these targets (“+” indicates binding; “-” indicates non-binding).
- (9) FIGS. 7A & 7B show binding specificity of antibody clone S135 for a variety of SIRP peptides. FIG. 7A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP α D1 domains, as well as human SIRP γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants (SEQ ID NO:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO:16). FIG. 7B summarizes the binding specificity of the clone against each of these targets (“+” indicates binding; “-” indicates non-binding).
- (10) FIGS. 8A & 8B show binding specificity of antibody clone S126 for a variety of SIRP

peptides. FIG. 8A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP α D1 domains, as well as human SIRP γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants (SEQ ID NO:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO:16). FIG. 8B summarizes the binding specificity of the clone against each of these targets (“+” indicates binding; “-” indicates non-binding).

(11) FIGS. 9A & 9B show binding specificity of antibody clone S138 for a variety of SIRP peptides. FIG. 9A shows the ELISA binding curves for the antibody against the human v1, human v2, murine, and cynomolgus SIRP α D1 domains, as well as human SIRP γ (SEQ ID NO:15) and a pre-formed complex of high-affinity SIRP- α variants (SEQ ID NO:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO:16). FIG. 9B summarizes the binding specificity of the clone against each of these targets (“+” indicates binding; “-” indicates non-binding).

(12) FIGS. 10A-10C show an alignment of VH and VL domains of the scFv-Fc clones obtained from a wild-type chicken. SEQ ID NOs:53-60 are shown (in order from top to bottom in the alignment). CDR and linker sequences are indicated by lines. Amino acid differences are indicated by asterisks.

(13) FIGS. 10D-10F show an alignment of VH and VL domains of the scFv-Fc clones obtained from a chicken that produces human antibodies. SEQ ID NOs:61-74 are shown (in order from top to bottom in the alignment). CDR and linker sequences are indicated by lines. Amino acid differences are indicated by asterisks.

(14) FIGS. 11A-11D show an alignment of VH and VL domains of Family 2 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 294, 139, 358, 362, 354, 380, 384, 350, 137, 374, 356, 352, 135, 348, 376, 346, 342, 344, 141, 360, 370, 382, 364, 366, 368, 372, and 378 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs 295, 363, 140, 359, 355, 351, 136, 349, 377, 138, 375, 357, 353, 381, 385, 345, 365, 367, 369, 347, 142, 343, 371, 379, 383, 361, and 373 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

(15) FIGS. 11E-11F show an alignment of VH and VL domains of Family 3 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 133, 128, 396, 386, 398, 402, 392, 388, 390, 394, and 400 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs 134, 105, 387, 389, 395, 397, 399, 403, 391, 393, and 401 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

(16) FIGS. 11G & 11H show an alignment of VH and VL domains of Family 4 clones. The amino acid sequences of the VH domains are SEQ ID NOs:116, 117, 118, 119, 282, 404, and 406 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs 93, 94, 95, 96, 283, 405, and 407 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

(17) FIG. 11I shows an alignment of VH and VL domains of Family 5, Bin 4 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 278 and 412 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs 279 and 413 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

(18) FIG. 11J shows an alignment of VH and VL domains of additional Family 5, Bin 4 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 275 and 414 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs: 276 and 415 (in order from top to bottom in the alignment). HVR and linker sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

(19) FIG. 11K shows the sequences of VH and VL domains (SEQ ID NOs 280 and 281,

respectively of Family 5, Bin 4 clone S209. The HVR sequences are underlined.

(20) FIG. 11L shows an alignment of VH and VL domains of Family 5, Bin 5 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 123 and 292 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs 100 and 293 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

(21) FIG. 11M shows an alignment of VH and VL domains of additional Family 5, Bin 5 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 288, 290, 408, and 410 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs: 289, 291, 409, and 411 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

(22) FIG. 11N shows the sequences of VH and VL domains of clone 149 (SEQ ID NOs 286 and 287, respectively) and clone 218 (SEQ ID NOs 284 and 285, respectively). HVR sequences are underlined.

(23) FIGS. 11O & 11P show an alignment of VH and VL domains of Family 1 clones. The amino acid sequences of the VH domains are SEQ ID NOs: 120, 121, 130, and 122 (in order from top to bottom in the alignment). The amino acid sequences of the VL domains are SEQ ID NOs: 97, 98, 107, and 99 (in order from top to bottom in the alignment). HVR sequences are indicated by lines. The HVRs are according to Kabat. Amino acid differences are indicated by asterisks.

(24) FIGS. 12A-12C show surface plasmon resonance (SPR) binding profiles of representative antibody clones binding a pre-formed complex of a high affinity SIRP- α variant (SEQ ID NO:18) mixed with increasing concentrations of the IgSF domain of CD47 (SEQ ID NO:16). FIG. 12A shows the binding curve of an antibody clone (S123) that does not block CD47 binding to SIRP- α (e.g., a non-blocking antibody). FIG. 12B shows the binding curve of an antibody clone (S119) that blocks CD47 binding to SIRP- α (e.g., a blocking antibody). FIG. 12C shows the binding curve of an antibody clone (S118) that binds to SIRP- α and reduces its affinity for binding CD47 (e.g., a “kick off” antibody).

(25) FIGS. 13A-13G show the results of in vitro tumor cell phagocytosis assays using macrophages treated with anti-SIRP- α antibody (at indicated series of concentrations), cetuximab or trastuzumab, anti-SIRP- α antibody plus cetuximab or trastuzumab, or control antibody (IgG1, κ), as indicated. Macrophages that had phagocytosed tumor cells were identified as cells positive for CD33, CD206, and CFSE by flow cytometry. Tumor cells assayed were DLD-1 (FIGS. 13A-13D & 13G) or OE19 cells (FIGS. 13E & 13F). Anti-SIRP- α antibodies tested were AB3a (FIGS. 13A & 13B), AB45a (FIGS. 13C & 13D), AB119a (FIG. 13E), AB135a (FIG. 13F), and AB136c (FIG. 13G).

(26) FIG. 14 shows the results of in vivo dendritic cell activation assays on dendritic cells isolated from the spleens of Balb/c mice treated with anti-SIRP- α antibody AB136b, control rat anti-mouse anti-SIRP- α antagonistic antibody (clone p84), rat IgG control, or mouse IgG control, as indicated. Mice were intravenously injected with the indicated antibody at 10 mg/kg, and spleens were harvested five hours after injection. Activation marker CD86 on dendritic cells was measured by flow cytometry.

(27) FIG. 15 shows the results of an in vivo syngeneic mouse colon carcinoma model to assess single agent activity. MC38 cells were implanted subcutaneously in C57BL/6 mice and randomized into groups (8 mice/group). Mice were treated with vehicle (PBS), CD47 blocking anti-SIRP- α antibody AB25b, CD47 blocking anti-SIRP- α antibody AB25c, CD47 blocking anti-SIRP- α antibody AB27b, CD47 non-blocking anti-SIRP- α antibody AB3b, or CD47 non-blocking anti-SIRP- α antibody 136b. Treatment was initiated when tumors were an average of 60 mm^{sup.3}, day 7 post implant. Mice were dosed intraperitoneally (IP) at 10 mg/kg twice a week for three weeks with anti-SIRP α antibodies. Animals were sacrificed when tumors reached a volume of ~2000 mm^{sup.3}.

(28) FIG. 16 shows the results of an in vivo syngeneic mouse colon carcinoma model to assess single agent activity. CT26 cells were implanted subcutaneously in BALB/c mice (8-9 mice were used per group) that were treated with AB136b or vehicle (PBS), as indicated. Treatment was initiated when tumors were an average of 80 mm^{sup.3}, day 7 post implant. Mice were dosed intraperitoneally (IP) at 3 mg/kg or 10 mg/kg twice a week for three weeks with anti-SIRP α antibodies. Animals were sacrificed when tumors reached a volume of ~2000 mm^{sup.3}.

(29) FIG. 17A shows a comparison of CD47 and anti-SIRP- α antibody clone 119 Fab binding to SIRP- α , as determined by X-ray crystallography.

(30) FIG. 17B shows the interaction site between anti-SIRP- α antibody clone 119 Fab and SIRP- α , as determined by buried surface area analyses. SIRP- α residues included in the antibody 119 Fab binding epitope are shaded according to buried surface area changes. Key antibody residues in the SIRP- α paratope are indicated: (H)=heavy chain residue; (L)=light chain residue.

(31) FIG. 18A shows a comparison of CD47 and anti-SIRP- α antibody clone 136 Fab binding to SIRP- α , as determined by X-ray crystallography.

(32) FIG. 18B shows the interaction site between anti-SIRP- α antibody clone 136 Fab and SIRP- α , as determined by buried surface area analyses. SIRP- α residues included in the antibody 136 Fab binding epitope are shaded according to buried surface area changes. Key antibody residues in the SIRP- α paratope are indicated: (H)=heavy chain residue; (L)=light chain residue.

(33) FIG. 19A shows a comparison of CD47 and anti-SIRP- α antibody clone 3 Fab binding to SIRP- α , as determined by X-ray crystallography.

(34) FIG. 19B shows the interaction site between anti-SIRP- α antibody clone 3 Fab and SIRP- α , as determined by buried surface area analyses. SIRP- α residues included in the antibody 3 Fab binding epitope are shaded according to buried surface area changes. Key antibody residues in the SIRP- α paratope are indicated: (H)=heavy chain residue; (L)=light chain residue.

(35) FIG. 19C shows a comparison of CD47 and anti-SIRP- α antibody clone 115 Fab binding to SIRP- α , as determined by X-ray crystallography.

(36) FIG. 19D shows the interaction site between anti-SIRP- α antibody clone 115 Fab and SIRP- α , as determined by buried surface area analyses. SIRP- α residues included in the antibody 115 Fab binding epitope are shaded according to buried surface area changes. Key antibody residues in the SIRP- α paratope are indicated: (H)=heavy chain residue; (L)=light chain residue.

(37) FIG. 20A shows a comparison of CD47, anti-SIRP- α antibody clone 119 Fab, anti-SIRP- α antibody clone 136 Fab, anti-SIRP- α antibody clone 3 Fab, anti-SIRP- α antibody clone 115 Fab binding to SIRP- α , as determined by X-ray crystallography.

(38) FIGS. 20B-20E show the epitopes for CD47, anti-SIRP- α antibody clone 119 Fab, anti-SIRP- α antibody clone 136 Fab, anti-SIRP- α antibody clone 3 Fab, and anti-SIRP- α antibody clone 115 Fab binding to SIRP- α , as determined by X-ray crystallography. Values indicate difference between surface accessible area of each residue atom in the Fab/CD47 when analyzed alone vs. when analyzed in complex with SIRP- α , expressed as buried surface area ($\text{\AA}^{\text{sup.2}}$). Residue numbering according to SEQ ID NO: 296.

(39) FIG. 21A shows a flowchart for epitope binning of anti-SIRP- α antibodies.

(40) FIG. 21B shows results of an exemplary assay for epitope binning of anti-SIRP- α antibodies A, B, C, D, E, and F.

(41) FIGS. 22A & 22B show the results of epitope binning of the indicated anti-SIRP- α antibodies. The clone number for the ligand (anti-SIRP α) bound to the chip is indicated as rows, and the clone number for the analytes (anti-SIRP α) injected over the chip is indicated as columns. White boxes indicate antibodies that form sandwiches (and are considered to bind different epitopes). Gray boxes indicate antibodies that did not form sandwiches (and are considered to bind the same epitope). "X" indicates scenarios where the data from one orientation disagrees with the other.

(42) FIG. 23 provides a model for anti-SIRP- α antibody and CD47 binding to the SIRP- α D1 domain based on epitope binning. Representative antibody clones for each bin are provided (and

labeled by number).

(43) FIG. 24A shows alignments between the parental 119 heavy chain (“119_VH_Wt”), the 119 variant heavy chain with 4 mutations (3 back-mutations to germline sequence in framework and one mutation in CDR-H1 removing a potential oxidation hot spot; “VH_MutALL”), the 119 variant heavy chain with 3 mutations (3 back-mutations to germline sequence in framework only; “VH_MutAll_V34M”), and the 119 variant heavy chain with 3 mutations and an M34L mutation (3 back-mutations to germline sequence in framework; “VH_MutAll_V34L”). Sequences depicted are: SEQ ID NO:335 for 119_VH_Wt, SEQ ID NO:246 for VH_MutALL, SEQ ID NO:258 for VH_MutAll_V34M, and SEQ ID NO:327 for VH_MutAll_V34L. CDR sequences are indicated with lines; amino acid differences are indicated by asterisks.

(44) FIG. 24B shows alignments between the parental 119 light chain (“119_VL_Wt”), and the 119 variant light chain with 4 mutations (4 back-mutations to germline sequence in framework; “VL_mutAll”). Sequences depicted are: SEQ ID NO:97 for 119_VL_Wt and SEQ ID NO:312 for VL_mutAll. CDR sequences are indicated with lines; amino acid differences are indicated by asterisks.

(45) FIG. 25A shows alignments between the parental 135 light chain (“VL_wt”) and the 135 variant light chain with 2 mutations (2 back-mutations to germline sequence in framework; “VL_mutALL”). Sequences depicted are: SEQ ID NO:104 for 135 VL_wt and SEQ ID NO:248 for 135 VL_MutALL. HVR sequences are indicated with lines; amino acid differences are indicated by asterisks.

(46) FIG. 25B shows alignments between the parental 135 heavy chain (“VH_wt”) the 135 variant heavy chain with 6 mutations (5 back-mutations to germline sequence in framework and one mutation in CDR-H1 removing a potential oxidation hot spot; “VH_MutAll”), the 135 variant heavy chain with 5 back-mutations to germline sequence in framework (“VH_MutAll_V34M”), and the 135 variant heavy chain with 5 back-mutations to germline sequence in framework and M34L mutation (“VH_MutAll_V34L”). Sequences depicted are: SEQ ID NO:341 for VH_wt, SEQ ID NO:247 for VH_MutAll, and SEQ ID NO:259 for VH_MutAll_V34M, and SEQ ID NO:328 for VH_MutAll_V34L. HVR sequences are indicated with lines; amino acid differences are indicated by asterisks.

(47) FIG. 26A shows alignments between the parental 136 light chain (“VL_wt”), the 136 variant light chain with 4 mutations (4 back-mutations to germline sequence in framework; “VL_mutaLL”), the 136 variant light chain with a single I2T back-mutation reverted to wild-type sequence in an otherwise “all mut” background (“VL_Mutall_I2T”). Sequences depicted are: SEQ ID NO: 134 for VL_wt, SEQ ID NO:250 for VL_mutaLL, and SEQ ID NO:251 for VL_Mutall_I2T. HVR sequences are indicated with lines; amino acid differences are indicated by asterisks.

(48) FIG. 26B shows alignments between the parental 136 heavy chain (“VH_wt”), the 136 variant heavy chain with 6 mutations (5 back-mutations to germline sequence in framework and one mutation in CDR-H1 removing a potential oxidation hot spot; “VH_mutall”), the 136 variant heavy chain with 5 back-mutations to germline sequence in framework (“VH_Mutall_V34M”), and the 136 variant heavy chain with 5 back-mutations to germline sequence in framework and M34L mutation (“VH_Mutall_V34L”). Sequences depicted are: SEQ ID NO:133 for VH_wt, SEQ ID NO:249 for VH_mutall, SEQ ID NO:260 for VH_Mutall_V34M, and SEQ ID NO:329 for VH_Mutall_V34L. HVR sequences are indicated with lines; amino acid differences are indicated by asterisks.

(49) FIG. 27A shows binding affinities of antibody 136 variants to six SIRP- α proteins: human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), cynomolgus SIRP- α (SEQ ID NO:11), NOD mouse SIRP- α (SEQ ID NO:8), BL/6 mouse SIRP- α (SEQ ID NO:9), and BALB/c mouse SIRP- α (SEQ ID NO:10). Antibody variants had mutant (“mut”) or parental (“wt”) light chains and mutant or parental heavy chains, as indicated in the order light chain/heavy chain. On

the graph, y-axis indicates the ratio of K.sub.D mut/K.sub.D wt. A ratio of 1 means antibody had equivalent K.sub.D to wt/wt antibody (indicated by dotted line); ratio of >1 indicates lower affinity than wt/wt; and ratio of <1 indicates higher affinity than wt/wt.

(50) FIG. 27B shows binding affinities of antibody 136 variants to seven SIRP- α proteins: BL/6 mouse SIRP- α (SEQ ID NO:9), NOD mouse SIRP- α (SEQ ID NO:8), BALB/c mouse SIRP- α (SEQ ID NO:10), human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), cynomolgus SIRP- α (SEQ ID NO:11), and human SIRP- γ v1 (SEQ ID NO:15). In addition to testing the wt 136 and all mutant 136, variants were constructed eliminating each individual mutation in an all mutant light chain background. On the graph, y-axis indicates the ratio of K.sub.D mut/K.sub.D wt. A ratio of 1 means antibody had equivalent K.sub.D to wt/wt antibody (indicated by dotted line); ratio of >1 indicates lower affinity than wt/wt; and ratio of <1 indicates higher affinity than wt/wt.

(51) FIG. 28 compares expression yield and binding affinity of antibodies having indicated human heavy chains and humanized light chains in FreeStyle™ 293-FS cells (Thermo Fisher).

(52) FIG. 29 shows an alignment between Hum1, Hum 8, and Hum9 VL domains. Hum8 was generated based on Hum1 but with 5 amino acid substitutions near or in HVR-L1 and -L2 that increase humanness. Hum9 was generated based on Hum1 but with 4 amino acid substitutions near or in HVR-L1 and -L2 that increase humanness. SEQ ID NOs: 252 (Hum1), 416 (Hum 8), and 262 (Hum9) are depicted. HVR sequences are indicated with lines; amino acid differences are indicated by asterisks.

(53) FIG. 30 shows alignments between the antibody 21 variant with germline back-mutations (“HC_MutAll”), the antibody 21 variant with germline back-mutations and mutation in CDR-H1 removing a potential oxidation hot spot M34V (“HC_MutAll_M34V”), the antibody 21 variant with germline back-mutations and M34L mutation (“HC_MutAll_M34L”), the antibody 25 variant with germline back-mutations (“HC_MutAll”), the antibody 25 variant with germline back-mutations and mutation in CDR-H1 removing a potential oxidation hot spot M34V (“HC_MutAll_M34V”), the antibody 25 variant with germline back-mutations and M34L mutation (“HC_MutAll_M34L”), the antibody 27 variant with germline back-mutations (“HC_MutAll”), the antibody 27 variant with germline back-mutations and mutation in CDR-H1 removing a potential oxidation hot spot M34V (“HC_MutAll_M34V”), and the antibody 27 variant with germline back-mutations and M34L mutation (“HC_MutAll_M34L”). Sequences depicted are SEQ ID NOs:263, 264, 330, 267, 268, 332, 265, 266, and 331 (top to bottom). HVR sequences are indicated with lines; amino acid differences are indicated by asterisks.

(54) FIG. 31 shows the results of in vitro phagocytosis assays using HER2(+) OE19 cells as the target and M2 macrophages as the phagocytosing cell. “Kick off” anti-SIRP- α antibodies were tested at the indicated concentrations in combination with the anti-HER2 antibody trastuzumab. Phagocytosis was measured by percentage of CFSE+ cells.

(55) FIG. 32 shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Non-blocking anti-SIRP- α antibodies were tested at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.

(56) FIGS. 33A-33C show the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Anti-SIRP- α antibodies were tested at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.

(57) FIG. 34 shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Non-blocking anti-SIRP- α antibodies 27 and 136 were each tested as a full-length antibody (with Fc region) or F(ab).sub.2 fragment at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.

- (58) FIG. 35 shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Blocking anti-SIRP- α antibody 119 variants were tested at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.
- (59) FIG. 36 shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Blocking anti-SIRP- α antibody 135 variants were tested at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.
- (60) FIG. 37 shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Non-blocking anti-SIRP- α antibodies were tested at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.
- (61) FIGS. 38A-38B show the results of in vivo dendritic cell activation assays with the indicated anti-SIRP- α antibodies. Mice were intravenously injected with the indicated antibody at 10 mg/kg, and spleens were harvested five hours after injection. Activation markers CD86, MHCII and CCR7 on CD4+ dendritic cells were measured by flow cytometry.
- (62) FIG. 39A shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. 218a and 218 variant anti-SIRP- α antibodies were tested at the indicated concentrations in combination with the anti-EGFR antibody cetuximab. Phagocytosis was measured by percentage of CFSE+ cells.
- (63) FIG. 39B shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Exemplary blocking anti-SIRP- α antibodies 119a, 120a, and 122a were tested at the indicated concentrations. Phagocytosis was measured by percentage of CFSE+ cells.
- (64) FIG. 39C shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Exemplary non-blocking anti-SIRP- α antibodies 136a and 137a were tested at the indicated concentrations. Phagocytosis was measured by percentage of CFSE+ cells.
- (65) FIG. 39D shows the results of in vitro phagocytosis assays using EGFR(+) DLD-1 cells as the target and M2 macrophages as the phagocytosing cell. Exemplary kick off anti-SIRP- α antibodies 115a, 116a, 117a, 118a, and 132a were tested at the indicated concentrations. Phagocytosis was measured by percentage of CFSE+ cells.
- (66) FIG. 40 shows the results of an in vivo syngeneic mouse colon carcinoma model to assess activity of combining anti-SIRP- α treatment with PD-L1/PD-1 pathway inhibition. CT26 cells were implanted subcutaneously in C57BL/6 mice and randomized into groups (8 mice/group). Mice were treated with vehicle (PBS), anti-PD-L1 antibody, CD47 blocking anti-SIRP- α antibody AB25b, or AB25b and PD-L1. Treatment was initiated when tumors were an average of 60 mm^{sup.3}, day 7 post implant. Mice were dosed intraperitoneally (IP) at 10 mg/kg twice a week for three weeks and sacrificed when tumors reach a volume of ~2000 mm^{sup.3}.
- (67) FIG. 41 shows the results of an in vivo syngeneic mouse colon carcinoma model to assess activity of combining anti-SIRP- α treatment with PD-L1/PD-1 pathway inhibition. MC38 cells were implanted subcutaneously in C57BL/6 mice and randomized into groups (8 mice/group). Mice were treated with vehicle (PBS), anti-PD-1 antibody, CD47 blocking anti-SIRP- α antibody AB25b, AB25b and anti-PD-1, CD47 non-blocking anti-SIRP- α antibody AB136b, or AB136b and anti-PD-1. Treatment was initiated when tumors were an average of 60 mm^{sup.3}, day 7 post implant. Mice were dosed intraperitoneally (IP) at 10 mg/kg twice a week for three weeks and sacrificed when tumors reach a volume of ~2000 mm^{sup.3}.

DETAILED DESCRIPTION

- (68) The present disclosure describes antibodies that bind the extracellular domains (e.g., the D1 domains) of one or more human SIRP- α polypeptides and have a variety of SIRP- α binding profiles

of potential interest. These unique SIRP- α binding profiles include one or more of the following binding capabilities, which are combined in a multifactorial manner to yield a multitude of unique specificities. For instance, the antibody can bind the extracellular domains (e.g., the D1 domains) of human SIRP- α v1, human SIRP- α v2, or both; the antibody can bind the extracellular domains (e.g., the D1 domains) of one or more monkey SIRP- α polypeptides, or it can lack binding thereto; the antibody can bind the extracellular domains (e.g., the D1 domains) of one or more murine SIRP- α polypeptides, or it can lack binding thereto; the antibody can bind the extracellular domain (e.g., the D1 domain) of a human SIRP β polypeptide, or it can lack binding thereto; and/or the antibody can bind the extracellular domain (e.g., the D1 domain) of a human SIRP γ polypeptide, or it can lack binding thereto. In addition, the present disclosure describes antibodies that block binding of CD47 to SIRP- α , antibodies that do not block binding of CD47 to SIRP- α , and antibodies that do not block binding of CD47 to SIRP- α but decrease SIRP- α 's affinity for CD47, leading to more rapid dissociation of the CD47/SIRP- α complex. In addition, the present disclosure describes anti-SIRP- α antibodies with one or more in vitro and/or in vivo biological properties of interest, such as the ability to enhance macrophage phagocytosis, enhance dendritic cell activation, inhibit in vivo growth of a tumor that expresses CD47, and/or the ability to accomplish one or more of these activities without preventing interactions between a CD47-expressing cell and a T cell.

(69) The methods described herein may be used to identify antibodies with unique combinations of the above binding specificities. Without wishing to be bound to theory, it is thought that the ability to identify unique anti-SIRP- α antibodies with different binding profiles as described above allows for the identification of antibodies (e.g., those described herein) with desirable clinical properties and advantages for pre-clinical research.

(70) In one aspect, provided herein are isolated antibodies that bind the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both. Further provided herein are polynucleotides and vectors encoding the antibodies of the present disclosure, as well as methods of antibody production related thereto.

(71) In another aspect, provided herein are methods for treating or delaying progression of cancer in an individual, comprising administering to the individual an effective amount of an antibody of the present disclosure.

(72) In another aspect, provided herein are methods for treating or delaying progression of an autoimmune or inflammatory disease in an individual, comprising administering to the individual an effective amount of an antibody of the present disclosure.

(73) In another aspect, provided herein are methods for identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and does not block binding between human CD47 and the human SIRP- α polypeptide. In some embodiments, the methods include (a) providing an antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide; (b) assembling a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; (c) contacting the antigen binding domain with the assembled complex; and (d) detecting binding of the antigen binding domain to the complex, wherein binding of the antigen binding domain to the complex indicates that the antigen binding domain does not block binding between human CD47 and the human SIRP- α polypeptide. In another aspect, provided herein are methods for identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and does not block binding between human CD47 and the human SIRP- α polypeptide. In some embodiments, the methods include contacting an antigen binding domain or antibody that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide with a

complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; and detecting binding of the antigen binding domain to the complex, wherein binding of the antigen binding domain to the complex indicates that the antigen binding domain does not block binding between human CD47 and the human SIRP- α polypeptide.

(74) In another aspect, provided herein are methods for identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and blocks binding between human CD47 and the human SIRP- α polypeptide. In some embodiments, the method includes (a) providing an antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide; (b) assembling a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; (c) contacting the antigen binding domain with the assembled complex; and (d) detecting binding of the antigen binding domain to the complex, wherein a lack of binding of the antigen binding domain to the complex indicates that the antigen binding domain blocks binding between human CD47 and the human SIRP- α polypeptide. In another aspect, provided herein are methods for identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and blocks binding between human CD47 and the human SIRP- α polypeptide. In some embodiments, the method includes contacting an antigen binding domain or antibody that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide with a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47, wherein the SIRP- α D1 variant is a non-naturally occurring high affinity SIRP- α D1 domain, and wherein the SIRP- α D1 variant binds to human CD47 with an affinity that is at least 10-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47; and detecting binding of the antigen binding domain to the complex, wherein a lack of binding of the antigen binding domain to the complex indicates that the antigen binding domain blocks binding between human CD47 and the human SIRP- α polypeptide.

(75) In another aspect, provided herein are methods for producing an anti-SIRP- α antibody that binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides. In some embodiments, the method includes (a) immunizing a chicken with a peptide comprising at least a portion of a human SIRP- α extracellular domain (e.g., the D1 domain); (b) obtaining an antibody from an antibody-producing cell from the immunized chicken; and (c) detecting binding between the antibody obtained from the cell and the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, wherein binding between the antibody and the extracellular domains (e.g., the D1 domains) of the two or more different human SIRP- α variant polypeptides indicates that the antibody is an anti-SIRP- α antibody that binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides.

(76) Definitions

(77) Before describing the disclosed embodiments in detail, it is to be understood that the present disclosure is not limited to particular compositions or biological systems, which can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting.

(78) As used in this specification and the appended claims, the singular forms “a”, “an” and “the” include plural referents unless the content clearly dictates otherwise. Thus, for example, reference to “a molecule” optionally includes a combination of two or more such molecules, and the like.

(79) The term “about” as used herein refers to the usual error range for the respective value readily known to the skilled person in this technical field. Reference to “about” a value or parameter herein includes (and describes) embodiments that are directed to that value or parameter per se.

(80) It is understood that aspects and embodiments of the present disclosure include “comprising,” “consisting,” and “consisting essentially of” aspects and embodiments.

(81) A “SIRP- α polypeptide” as used herein may refer to any endogenous or naturally occurring SIRP- α polypeptide encoded by a genome from any vertebrate, including mammals such as humans, monkeys, rodents (e.g., mouse or rat), and birds, such as chickens. The term also includes naturally occurring variants, e.g., alternatively spliced variants, allelic variants, or polymorphisms (e.g., those described herein). The term may further refer to full-length, unprocessed SIRP- α polypeptides as well as SIRP- α polypeptides that result from cellular processing, e.g., removal of a signal sequence, etc. Exemplary SIRP- α polypeptide sequences are described herein. In some embodiments, a human SIRP- α polypeptide is one encoded by a human SIRPA gene, e.g., as described by NCBI Gene ID No. 140885. As described herein, SIRP- α polypeptides are highly polymorphic within and among species. For example, at least 10 human variants with amino acid polymorphisms in the extracellular domain have been identified.

(82) SIRP- α polypeptides include an extracellular domain that binds ligands/partners, e.g., CD47. SIRP- α polypeptides comprise 3 highly homologous immunoglobulin (Ig)-like extracellular domains—D1, D2, and D3. The SIRP- α D1 domain (“D1 domain”) refers to the membrane distal, extracellular domain of SIRP- α and mediates binding of SIRP- α to CD47 (see, e.g., Hatherley, D. et al. (2008) *Mol. Cell* 31:266-77; Hatherley, D. et al. (2007) *J. Biol. Chem.* 282:14567-75; Hatherley, D. et al. (2009) *J. Biol. Chem.* 284:26613-9; and Lee, W. Y. et al. (2010) *J. Biol. Chem.* 285:37953-63). The extracellular domain generally refers to the entire extracellular portion of SIRP- α , e.g., as expressed on a cell surface, and may include distinct SIRP- α domains, such as the D1 domain. The D1 domain contains residues shown to be critical for mediating CD47 binding (see, e.g., Lee, W. Y. et al. (2007) *J. Immunol.* 179:7741-50). In some embodiments, an antibody that binds an extracellular domain of a SIRP- α polypeptide binds one or more residues of the D1 domain. Exemplary human SIRP- α D1 domain sequences are described throughout the present disclosure and include without limitation SEQ ID NOs:5, 6, and 76-83. Human SIRP- α D2 and D3 domain sequences are also known and include, without limitation, APVVSGPAARATPQHTVSFTCESHGFSPRDITLKWFKNGNELSDFQTNVDPVGESVSYSI HSTAKVVLTRDVDHSQVICEVAHVTLQGDPLRGTANLSETIR (SEQ ID NO:131) for the D2 domain and

VPPTLEVTTQQPVRAENQVNVTCQVRKFYPQRLQLTWLENGNVSRTETASTVTENKDGT YNWMSWLLVNVSAHRDDVKLTQCVEHDGQPAVSKSHDLKVS (SEQ ID NO:132) for the D3 domain.

(83) As used herein, “CD47” (also known as integrin associated protein (IAP), MER6, and OA3) refers to a polypeptide that, among other roles, serves as a binding partner for SIRP- α polypeptides. In some embodiments, CD47 refers to a human CD47 polypeptide, e.g., a polypeptide encoded by a human CD47 gene, such as that described by NCBI Ref Seq ID No. 961. Exemplary human CD47 amino acid sequences are known (see, e.g., NCBI Reference Sequence Accession No. NP_001768). In particular, the IgSF domain of CD47 refers to the N-terminal extracellular domain of CD47 that is known to be critical for SIRP- α binding (see, e.g., Barclay, A. N. and Brown, M. H. (2006) *Nat. Rev. Immunol.* 6:457-64 and Hatherley, D. et al. (2009) *J. Biol. Chem.* 284:26613-9). In some embodiments, an IgSF domain of CD47 comprises the amino acid sequence of

QLLFNKTKSVEFTFSNDTVVIPCFVTNMEAQNTTEVYVKWKFKGRDIYTFDGALNKSTV PTDFSSAKIEVSQLLKGDASLKMDKSDAVSHTGNYTCEVTELTREGETIIEELKYRVVS (SEQ ID NO:16). The term “CD47” may also include modified CD47 polypeptides that are able to bind SIRP- α , e.g., a polypeptide comprising an IgSF domain of CD47 conjugated to another polypeptide or other moiety, e.g., an Ig Fc region.

(84) As used herein, a “SIRP- α epitope” may refer to the amino acids of a SIRP- α polypeptide that form the binding site for an anti-SIRP- α antibody of the present disclosure and/or a SIRP- α binding partner, including without limitation CD47. Binding of an antibody or other polypeptide to an epitope can be characterized and/or mapped using a variety of assays known in the art, including without limitation a cross-blocking assay (see *Antibodies, A Laboratory Manual*, Cold Spring Harbor Laboratory, Ed Harlow and David Lane (1988)), epitope mapping (see Champe et al., *J. Biol. Chem.* 270:1388-1394 (1995)), X-ray co-crystallography, epitope binning, site-directed mutagenesis, oligo-peptide scanning, high-throughput mutagenesis mapping, hydrogen/deuterium exchange, limited proteolysis, and so forth.

(85) As used herein “modulating SIRP- α signaling” may refer to antagonizing, agonizing, or otherwise interfering with one or more aspects of SIRP- α signaling in a cell expressing a SIRP- α polypeptide. SIRP- α signaling may refer to one or more intracellular signaling events mediated by activation of a SIRP- α polypeptide, including without limitation tyrosine phosphorylation of the intracellular region of SIRP- α , phosphatase (e.g., SHIP1) binding, adaptor protein binding (e.g., SCAP2, FYB, and/or GRB2), cytokine production (e.g. IL-10, IL-1 β , IFN or TNF), and nitric oxide production; and/or one or more intercellular phenotypes, including without limitation macrophage phagocytosis and other activating or suppressive phenotypes of macrophages, eosinophils, neutrophils, dendritic cells, and myeloid-derived suppressor cells (MDSCs).

(86) As used herein, the term “antibody” may refer to intact antibodies; antibody fragments (including without limitation Fab, F(ab')₂, Fab'-SH, Fv, diabodies, scFv, scFv-Fc, single domain antibodies, single heavy chain antibodies, and single light chain antibodies), provided that they exhibit the desired biological activity (e.g. epitope binding); monoclonal antibodies; polyclonal antibodies; monospecific antibodies; multi-specific antibodies (e.g., bispecific antibodies); and antibody-like proteins.

(87) As used herein, the term “bispecific” when used in reference to an antibody or antibody fragment includes an antibody or antibody fragment that possesses two different binding specificities. For example, each binding specificity may recognize a different antigen, or each binding specificity may recognize the same antigen with different affinity and/or precise epitope. In some embodiments, each different binding specificity comprises one or more different antibody antigen binding domains (e.g., variable domains), such that the bispecific antibody or antibody fragment comprises at least a first antigen binding domain with a first binding specificity and a second antigen binding domain with a second binding specificity. A variety of exemplary bispecific antibody formats are described herein and known in the art.

(88) An “isolated” antibody may refer to an antibody that has been separated and/or recovered from a component of its natural environment, e.g., a host cell or organism. In some embodiments, an antibody is purified to a desired purity by weight (e.g., at least 95%); and/or homogeneity by SDS-PAGE using, for example, staining by silver, Coomassie, etc. In some embodiments, an isolated antibody is obtained following one or more purification steps.

(89) As is known in the art, “native” antibodies refer to typically heterotetrameric complexes including two identical light (L) chains and two identical heavy (H) chains. Variable numbers of disulfide bonds connect the two heavy chains, and one connects each light chain to a heavy chain, in addition to intrachain disulfide bridges. The heavy chains include a variable domain (VH) followed (N-terminus to C-terminus) by three or four constant domains. The light chains include a variable domain (VL) followed by a constant domain (CL). Typically, mammalian light chains fall into one of two categories based on amino acid sequence: kappa and lambda.

(90) A “constant domain” may refer to the more conserved portion of the antibody or fragment, e.g., outside the variable domains. The term may include the CL domain as well as heavy chain constant domains CH1, CH2, CH3 and optionally CH4.

(91) Constant domains of the heavy chain can be assigned to one of 5 major types: IgA, IgD, IgE, IgG, and IgM. Several subtypes exist for many of these major types. The subunit structures and

three-dimensional configurations of different classes of immunoglobulins are well known and described generally in, for example, Abbas et al. *Cellular and Mol. Immunology*, 4th ed. (W.B. Saunders, Co., 2000).

(92) As used herein, the term “antibody variable domain” refers to the portions of the light and heavy chains of an antibody that include the complementary determining regions (CDRs, e.g., CDR L1, CDR L2, CDR L3, CDR H1, CDR H2, and CDR H3) and framework regions (FRs).

(93) The term “variable” refers to the fact that subsequences of the variable domains differ substantially in sequence between antibodies and are critical to the binding specificity of a particular antibody for its antigen. Variability is concentrated in three hypervariable regions (HVRs) in both VH and VL domains. The more conserved portions of variable domains are called the framework regions (FR) in which the HVRs are interspersed. The variable domains of native heavy and light chains each comprise four FR regions connected by three HVRs that form loops (see Kabat et al., *Sequences of Proteins of Immunological Interest*, Fifth Edition, National Institute of Health, Bethesda, Md. (1991)).

(94) The term “hypervariable region (HVR)” may refer to the subregions of the VH and VL domains characterized by enhanced sequence variability and/or formation of defined loops. These include three HVRs in the VH domain (H1, H2, and H3) and three HVRs in the VL domain (L1, L2, and L3). H3 is believed to be critical in imparting fine binding specificity, with L3 and H3 showing the highest level of diversity. See Johnson and Wu, in *Methods in Molecular Biology* 248:1-25 (Lo, ed., Human Press, Totowa, N.J., 2003).

(95) A number of HVR delineations are known. The Kabat Complementarity Determining Regions (CDRs) are based on sequence variability and are the most commonly used (Kabat et al., *Sequences of Proteins of Immunological Interest*, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (1991)). Chothia refers instead to the location of the structural loops (Chothia and Lesk *J. Mol. Biol.* 196:901-917 (1987)). The AbM HVRs represent a compromise between the Kabat HVRs and Chothia structural loops, and are used by Oxford Molecular's AbM antibody modeling software. The “contact” HVRs are based on an analysis of the available complex crystal structures. The residues from each of these HVRs are noted below. “Framework” or “FR” residues are those variable domain residues other than the HVR residues.

(96) TABLE-US-00001 Loop Kabat AbM Chothia Contact L1 L24-L34 L24-L34 L26-L32 L30-L36 L2 L50-L56 L50-L56 L50-L52 L46-L55 L3 L89-L97 L89-L97 L91-L96 L89-L96 H1 H31-H35B H26-H35B H26-H32 H30-H35B (Kabat Numbering) H1 H31-H35 H26-H35 H26-H32 H30-H35 (Chothia Numbering) H2 H50-H65 H50-H58 H53-H55 H47-H58 H3 H95-H102 H95-H102 H96-H101 H93-H101

(97) “Extended” HVRs are also known: 24-36 or 24-34 (L1), 46-56 or 50-56 (L2) and 89-97 or 89-96 (L3) in the VL and 26-35 (H1), 50-65 or 49-65 (H2) and 93-102, 94-102, or 95-102 (H3) in the VH (Kabat numbering).

(98) “Numbering according to Kabat” may refer to the numbering system used for heavy chain variable domains or light chain variable domains of the compilation of antibodies in Kabat et al., supra. The actual linear amino acid sequence may contain fewer or additional amino acids corresponding to a shortening of, or insertion into, a FR or HVR of the variable domain. The Kabat numbering of residues may be determined for a given antibody by alignment at regions of homology of the sequence of the antibody with a “standard” Kabat numbered sequence. Typically, the Kabat numbering is used when referring to a residue in the variable domains (approximately residues 1-107 of the light chain and residues 1-113 of the heavy chain), whereas the EU numbering system or index (e.g., the EU index as in Kabat, numbering according to EU IgG1) is generally used when referring to a residue in the heavy chain constant region.

(99) “Full length” or “intact” antibodies typically include heavy chains with an Fc region, e.g., as opposed to an antibody fragment. Antigen-binding “Fab” fragments with a single antigen binding site may be released from the residual Fc fragment by papain digestion. F(ab')₂ fragments include

two antigen-binding sites produced by pepsin treatment of an antibody. Antibody fragments will, however, include one or more antibody variable regions.

(100) An “Fv” fragment contains a complete antigen-binding site. A single chain Fv (scFv) can include a VH and a VL domain linked by a peptide linker such that the VH and VL domains associate, e.g., as in an antibody or Fab fragment, such that the HVRs form an antigen binding site. See Pluckthün, in *The Pharmacology of Monoclonal Antibodies*, vol. 113, Rosenberg and Moore eds., (Springer-Verlag, New York, 1994), pp. 269-315. In some embodiments, the scFv is fused to an antibody Fc domain (e.g., scFv-Fc). While six HVRs typically comprise an antigen binding site, a single variable domain with three HVRs is still capable of binding an antigen, albeit at a lower affinity. See Hamers-Casterman et al., *Nature* 363:446-448 (1993); Sheriff et al., *Nature Struct. Biol.* 3:733-736 (1996). Single domain antibodies (e.g., camelid antibodies) typically include a single, monomeric variable domain for antigen binding. Single heavy chain (VHH) and single light chain antibodies are also known. A Fab’ fragment typically includes a few more residues at the C-terminal end than a Fab fragment. A Fab’-SH includes cysteine residues with a free thiol. Various chemical couplings of antibody fragments are known in the art.

(101) A “diabody” includes antibody fragments with two antigen-binding sites. These include a VH and VL domain connected by a linker, which is typically too short to facilitate pairing of domains in the same chain. Diabodies may be bivalent or bispecific. Tribodies and tetrabodies, or other numbers of VH/VL domains are known. See Hudson et al., *Nat. Med.* 9:129-134 (2003).

(102) As used herein, a “monoclonal” antibody refers to an antibody obtained from a population of substantially homogeneous antibodies, e.g., substantially identical but allowing for minor levels of background mutations and/or modifications. “Monoclonal” denotes the substantially homogeneous character of antibodies, and does not require production of the antibody by any particular method. In some embodiments, a monoclonal antibody is selected by its HVR, VH, and/or VL sequences and/or binding properties, e.g., selected from a pool of clones (e.g., recombinant, hybridoma, or phage-derived). A monoclonal antibody may be engineered to include one or more mutations, e.g., to affect binding affinity or other properties of the antibody, create a humanized or chimeric antibody, improve antibody production and/or homogeneity, engineer a multispecific antibody, resultant antibodies of which are still considered to be monoclonal in nature. A population of monoclonal antibodies may be distinguished from polyclonal antibodies as the individual monoclonal antibodies of the population recognize the same antigenic site. A variety of techniques for production of monoclonal antibodies are known; see, e.g., the hybridoma method (e.g., Kohler and Milstein, *Nature*, 256:495-97 (1975); Hongo et al., *Hybridoma*, 14 (3): 253-260 (1995), Harlow et al., *Antibodies: A Laboratory Manual*, (Cold Spring Harbor Laboratory Press, 2nd ed. 1988); Hammerling et al., in: *Monoclonal Antibodies and T-Cell Hybridomas* 563-681 (Elsevier, N.Y., 1981)), recombinant DNA methods (see, e.g., U.S. Pat. No. 4,816,567), phage-display technologies (see, e.g., Clackson et al., *Nature*, 352: 624-628 (1991); Marks et al., *J. Mol. Biol.* 222: 581-597 (1992); Sidhu et al., *J. Mol. Biol.* 338(2): 299-310 (2004); Lee et al., *J. Mol. Biol.* 340(5): 1073-1093 (2004); Fellouse, *Proc. Natl. Acad. Sci. USA* 101(34): 12467-12472 (2004); and Lee et al., *J. Immunol. Methods* 284(1-2): 119-132 (2004), and technologies for producing human or human-like antibodies in animals that have parts or all of the human immunoglobulin loci or genes encoding human immunoglobulin sequences (see, e.g., WO 1998/24893; WO 1996/34096; WO 1996/33735; WO 1991/10741; Jakobovits et al., *Proc. Natl. Acad. Sci. USA* 90: 2551 (1993); Jakobovits et al., *Nature* 362: 255-258 (1993); Bruggemann et al., *Year in Immunol.* 7:33 (1993); U.S. Pat. Nos. 5,545,807; 5,545,806; 5,569,825; 5,625,126; 5,633,425; and 5,661,016; Marks et al., *Bio/Technology* 10: 779-783 (1992); Lonberg et al., *Nature* 368: 856-859 (1994); Morrison, *Nature* 368: 812-813 (1994); Fishwild et al., *Nature Biotechnol.* 14: 845-851 (1996); Neuberger, *Nature Biotechnol.* 14: 826 (1996); and Lonberg and Huszar, *Intern. Rev. Immunol.* 13: 65-93 (1995).

(103) “Chimeric” antibodies may refer to an antibody with one portion of the heavy and/or light chain from a particular isotype, class, or organism and another portion from another isotype, class,

or organism. In some embodiments, the variable region will be from one source or organism, and the constant region will be from another.

(104) “Humanized antibodies” may refer to antibodies with predominantly human sequence and a minimal amount of non-human (e.g., mouse or chicken) sequence. In some embodiments, a humanized antibody has one or more HVR sequences (bearing a binding specificity of interest) from an antibody derived from a non-human (e.g., mouse or chicken) organism grafted onto a human recipient antibody framework (FR). In some embodiments, non-human residues are further grafted onto the human framework (not present in either source or recipient antibodies), e.g., to improve antibody properties. In general, a humanized antibody will comprise substantially all of at least one, and typically two, variable domains, in which all or substantially all of the hypervariable loops correspond to those of a non-human immunoglobulin, and all or substantially all of the FRs are those of a human immunoglobulin sequence. The humanized antibody optionally will also comprise at least a portion of an immunoglobulin constant region (Fc), typically that of a human immunoglobulin. See Jones et al., *Nature* 321:522-525 (1986); Riechmann et al., *Nature* 332:323-329 (1988); and Presta, *Curr. Op. Struct. Biol.* 2:593-596 (1992).

(105) A “human” antibody may refer to an antibody having an amino acid sequence which corresponds to that of an antibody produced by a human and/or has been made using any of the techniques for making human antibodies as disclosed herein. Human antibodies can be produced using various techniques known in the art, including phage-display libraries. Hoogenboom and Winter, *J. Mol. Biol.*, 227:381 (1991); Marks et al., *J. Mol. Biol.*, 222:581 (1991); preparation of human monoclonal antibodies as described in Cole et al., *Monoclonal Antibodies and Cancer Therapy*, Alan R. Liss, p. 77 (1985); Boerner et al., *J. Immunol.*, 147(1):86-95 (1991); and by administering the antigen to a transgenic animal that has been modified to produce such antibodies in response to antigenic challenge, but whose endogenous loci have been disabled, e.g., immunized xenomice (see, e.g., U.S. Pat. Nos. 6,075,181 and 6,150,584 regarding XENOMOUSE™ technology) or chickens with human immunoglobulin sequence(s) (see, e.g., WO2012162422, WO2011019844, and WO2013059159).

(106) As used herein, the term “linker” refers to a linkage between two elements, e.g., protein domains. In some embodiments, a linker can be a covalent bond or a spacer. The term “spacer” refers to a moiety (e.g., a polyethylene glycol (PEG) polymer) or an amino acid sequence (e.g., a 1-200 amino acid sequence) occurring between two polypeptides or polypeptide domains to provide space or flexibility (or both space and flexibility) between the two polypeptides or polypeptide domains. In some embodiments, an amino acid spacer is part of the primary sequence of a polypeptide (e.g., joined to the spaced polypeptides or polypeptide domains via the polypeptide backbone).

(107) The term “cytotoxic agent” as used herein may refer to any agent that inhibits cellular proliferation or induces cell death. Cytotoxic agents include, but are not limited to, chemotherapeutic agents; radioactive isotopes; growth inhibitory agents; and toxins such as small molecule toxins or enzymatically active toxins, including fragments and/or variants thereof. Exemplary cytotoxic agents include without limitation metabolic inhibitors, anti-microtubule agents, platinum containing compounds, alkylating agents, proteasome inhibitors, topoisomerase II inhibitors, antimetabolites, topoisomerase I inhibitors, signal transduction pathway inhibitors, non-receptor tyrosine kinase angiogenesis inhibitors, hormones and hormonal analogues, proapoptotic agents, inhibitors of LDH-A, cell cycle inhibitors, HDAC inhibitors, and antibiotic agents.

(108) As used herein, a “label” may include any moiety that serves as a detection agent, e.g., of binding between a labeled antibody of the present disclosure and a macromolecule or cell. Exemplary labels include without limitation fluorescent (e.g., compounds or proteins), radioactive, or enzymatic moieties, as well as affinity purification tags.

(109) As used herein, an antibody may be said to “bind” an antigen with an affinity sufficient to render the antibody useful for in vitro and/or in vivo manipulation of the antigen. In some

embodiments, an antibody that “binds” an antigen has a dissociation constant ($K_{sub.D}$) for the antigen that is less than or equal to 1 μ M at 25° C.

(110) As used herein, the term “affinity” or “binding affinity” refers to the strength of the binding interaction between two molecules. Generally, binding affinity refers to the strength of the sum total of non-covalent interactions between a molecule and its binding partner, such as a high affinity SIRP- α D1 variant and CD47. Unless indicated otherwise, binding affinity refers to intrinsic binding affinity, which reflects a 1:1 interaction between members of a binding pair. The binding affinity between two molecules is commonly described by the dissociation constant ($K_{sub.D}$) or the association constant ($K_{sub.A}$). Two molecules that have low binding affinity for each other generally bind slowly, tend to dissociate easily, and exhibit a large $K_{sub.D}$. Two molecules that have high affinity for each other generally bind readily, tend to remain bound longer, and exhibit a small $K_{sub.D}$. In some embodiments, the $K_{sub.D}$ of two interacting molecules is determined using known methods and techniques, e.g., surface plasmon resonance (SPR). $K_{sub.D}$ can be calculated as the ratio of $k_{sub.off}/k_{sub.on}$.

(111) As used herein, the term “ $K_{sub.D}$ less than” refers to a numerically smaller $K_{sub.D}$ value and an increasing binding affinity relative to the recited $K_{sub.D}$ value. As used herein, the term “ $K_{sub.D}$ greater than” refers to a numerically larger $K_{sub.D}$ value and a decreasing binding affinity relative to the recited $K_{sub.D}$ value.

(112) As used herein, “treatment” may refer to therapeutic administration of a molecule, compound, formulation, composition, etc. so as to alter one or more pathological symptoms in an individual or cell being treated. Desirable effects of treatment can include without limitation decelerating disease progression, ameliorating or palliating a pathological symptom or disease state, improving prognosis, and/or achieving disease remission. For example, an individual's cancer is successfully “treated” if one or more symptoms associated with cancer are mitigated or abolished, such as, without limitation, reducing the proliferation of cancer cells, eliminating cancer cells or tumor burden, decreasing symptoms resulting from the cancer, increasing the quality of life of the individual, lessening the dose of other medication(s), and/or prolonging survival of the individual. As another example, an autoimmune or inflammatory disease may be successfully “treated” if one or more symptoms associated with the autoimmune or inflammatory disease are mitigated or abolished, such as, without limitation, reducing autoreactive immune cells and/or inflammatory immune cells or cytokines, decreasing immune activation and/or inflammation, slowing or mitigating organ damage resulting from the disease, decreasing symptoms resulting from the disease, increasing the quality of life of the individual, lessening the dose of other medication(s), and/or prolonging survival of the individual.

(113) As used herein, “delaying progression” of a disease may refer to slowing, retarding, deferring, postponing development of, stabilizing, or otherwise hindering the pathological course of the disease. In some embodiments, the term may refer to a delay sufficient to effectively encompass prevention, e.g., in preventing the individual from developing the disease. In some embodiments, e.g., an advanced cancer, delaying progression may include delaying metastasis. One of skill in the art will appreciate that the precise length of delay may depend, e.g., upon the specific disease, condition of the individual, and the like.

(114) The terms “cancer” and “cancerous” may describe dysregulated or unregulated cell growth/proliferation by a cell or cells in a mammal. Any cancer type known in the art may be included, such as but not limited to carcinoma, sarcoma, lymphoma, leukemia, lymphoma, and blastoma. More particular examples of such cancers include, but are not limited to, lung cancer, squamous cell cancer, brain tumors, glioblastoma, head and neck cancer, hepatocellular cancer, colorectal cancer (e.g., colon or rectal cancers), liver cancer, bladder cancer, gastric or stomach cancer, pancreatic cancer, cervical cancer, ovarian cancer, cancer of the urinary tract, breast cancer, peritoneal cancer, uterine cancer, salivary gland cancer, kidney or renal cancer, prostate cancer, vulval cancer, thyroid cancer, anal carcinoma, penile carcinoma, melanoma, multiple myeloma and

B-cell lymphoma (including non-Hodgkin's lymphomas (NHL)); acute lymphoblastic leukemia (ALL); chronic lymphocytic leukemia (CLL); acute myeloid leukemia (AML); Merkel cell carcinoma; hairy cell leukemia; chronic myeloblastic leukemia (CML); and associated metastases. (115) As used herein, the term “effective amount” may refer to an amount of an antibody of the present disclosure or a pharmaceutical composition containing an antibody of the present disclosure that is sufficient and effective in achieving a desired therapeutic effect in treating or delaying progression of a patient having a disease, such as a cancer, e.g., solid tumor or hematological cancer. In some embodiments, a therapeutically effective amount will avoid adverse side effects, and/or such side effects will be outweighed by beneficial effects. An effective amount may depend upon the individual being treated, e.g., age, weight, sex, disease state, as well as the ability of the agent to produce a desired response. An effective amount can be administered in one or more administrations. As in the clinical context, an effective amount of a drug, compound, or pharmaceutical composition may or may not be achieved in conjunction with another drug, compound, or pharmaceutical composition, such as another therapeutic agent. Thus, an “effective amount” may also be considered in the context of administering one or more additional therapeutic agents, and a single agent may be considered to be given in an effective amount if, in conjunction with one or more other agents, a desirable result may be or is achieved.

(116) As used herein, the term “pharmaceutical composition” may refer to a medicinal or pharmaceutical formulation that includes an active ingredient as well as excipients or diluents (or both excipients and diluents) and enables the active ingredient to be administered by suitable methods of administration. In some embodiments, the pharmaceutical compositions disclosed herein include pharmaceutically acceptable components that are compatible with one or more antibodies of the present disclosure. In some embodiments, the pharmaceutical composition is in tablet or capsule form for oral administration or in aqueous form for intravenous or subcutaneous administration, for example by injection.

(117) As used herein, the terms “subject,” “individual,” and “patient” are used interchangeably to refer to a vertebrate, for example, a mammal. Mammals include, but are not limited to, murines, simians, humans, farm animals, sport animals, and pets.

(118) As used herein, “in conjunction with” or “in combination with” may refer to administration of one therapeutic in addition to (e.g., before, during, and/or after) another therapeutic.

(119) Antibodies

(120) Certain aspects of the present disclosure relate to antibodies that bind the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain of a human SIRP- α v2 polypeptide, or both. As demonstrated herein, antibodies have been characterized that specifically bind to v1 or v2, as well as antibodies that bind to both proteins. In humans, at least 10 distinct alleles of SIRPA have been identified (see FIG. 1A; see also Takenaka, K. et al. (2007) *Nat. Immunol.* 8:1313-23). Antibodies that bind one or more human SIRP- α polypeptides and possess one or more of the other binding specificities described herein are an advantageous discovery of the present disclosure. Further, the present disclosure demonstrates methods for producing and identifying antibodies representing a surprising diversity of novel SIRP- α binding specificity profiles.

(121) In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELIYNQKEGHFPRV TTVSDLTKRNNMDFSIRIGNITPADAGTYCYVKFRKGGSPDDVEFKSGAGTELSVRKPS (SEQ ID NO:5). In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide comprising the amino acid sequence of

EEELQVIQPDKSVSVAAGESAILHCTVTSIPVGPIQWFRGAGPARELIYNQKEGHFPRVT TVSESTKRENMDFSISISNITPADAGTYCYVKFRKGGSPDTEFKSGAGTELSVRKPS (SEQ ID

NO:6). In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide comprising the amino acid sequence of EEELQVIQPDKSVLVAAGETAWTLRCTATSLIPVGPIQWFRGAGPGRELIYNQKEGHFPRV TTVSDLTKRNNMDFSIRIGNITPADAGTYCYVKFRKGSPDDVEFKSGAGTELSVRAKPS (SEQ ID NO:5) and an extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide comprising the amino acid sequence of

(122) TABLE-US-00002 (SEQ ID NO: 6)

EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIY NQKEGHFPRVTTVSESTKRENMDFSISISNITPADAGTYCYVKFRKGSPD TEFKSGAGTELSVRAKPS.

(123) In some embodiments, an antibody of the present disclosure binds the extracellular domains (e.g., the D1 domains) of 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 different human SIRP- α variant polypeptides. As used herein, a “human SIRP- α variant polypeptide” may refer to a naturally occurring human SIRP- α variant polypeptide or polymorphism found expressed in a human, e.g., as opposed to a variant bearing one or more engineered mutations. For example, in some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of one or more human SIRP- α variant polypeptides comprising a sequence shown in the Table 1. In some embodiments, an antibody of the present disclosure binds to an extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide and/or an extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, and binds to an extracellular domain (e.g., the D1 domain) of one or more human SIRP- α polypeptides selected from v3, v4, v5, v6, v7, v8, v9, and v10.

(124) TABLE-US-00003 TABLE 1 Human SIRP- α variant polypeptide sequences corresponding to the D1 domain. Variant SEQ ID NO Sequence v1 5

EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRG AGPGRELIYNQKEGHFPRVTTVSDLTKRNNMDFSIRIGNIT PADAGTYCYVKFRKGSPDDVEFKSGAGTELSVRAKPS v2 6

EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRG AGPARELIYNQKEGHFPRVTTVSESTKRENMDFSISISNITP ADAGTYCYVKFRKGSPDTEFKSGAGTELSVRAKPS v3 76

EEELQVIQPDKSVSVAAGESAILLCTVTSLIPVGPIQWFRGAG PARELIYNQKEGHFPRVTTVSESTKRENMDFSISISNITPADA GTYYCYVKFRKGSPDTEFKSGAGTELSVRAKPS v4 77

EEGLQVIQPDKSVSVAAGESAILHCTATSLIPVGPIQWFRG AGPGRELIYNQKEGHFPRVTTVSDLTKRNNMDFSIRIGNIT PADAGTYCYVKFRKGSPDDVEFKSGAGTELSVRAKPS v5 78

EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRG AGPGRELIYNQKEGHFPRVTTVSDLTKRNNMDFSIRIGNIT PADAGTYCYVKFRKGSPDDVEFKSGAGTELSVRAKPS v6 79

EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRG AGPGRELIYNQKEGHFPRVTTVSDLTKRNNMDFPIRIGNIT PADAGTYCYVKFRKGSPDDVEFKSGAGTELSVRAKPS v7 80

EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRG AGPARELIYNQKEGHFPRVTTVSESTKRENMDFSISISNITP ADAGTYCYVKFRKGSPDTEFKSGAGTELSVRGKPS v8 81

EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRG AGPARELIYNQKEGHFPRVTTVSESTKRENMDFSISISNITP ADAGTYCYVKFRKGSPDTEFKSGAGTELSVRAKPS v9 82

EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRG AGPGRELIYNQKEGHFPRVTTVSDLTKRNNMDFSIRISNIT PADAGTYCYVKFRKGSPDDVEFKSGAGTELSVRAKPS v10 83

EEELQVIQPKSVSVAAGESAILHCTVTSLIPVGPIQWFRG
AGPARELIYNQKEGHFPRVTTVSESTKRENMDFSISISNITP
ADAGTYCYCVKFRKGSPDTEFKSGAGTELSVRAKPS

(125) In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide (e.g., the D1 domain of a monkey SIRP- α polypeptide). In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide (e.g., found in the organism *Macaca fascicularis*). In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of at least two different monkey SIRP- α variant polypeptides. In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of at least two different cynomolgus SIRP- α variant polypeptides. For example, in some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide comprising the amino acid sequence of

EEELQVIQPEKSVSVAAGESATLNCTATSLIPVGPIQWFRGVGPGRELIYHQKEGHFPRV
TPVSDPTKRNNMDFSIRISNITPADAGTYCYCVKFRKGSPDVELKSGAGTELSVRAKPS (SEQ
ID NO:11), an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide comprising the amino acid sequence of

EEELQVIQPEKSVSVAAGDSATLNCTVSSLIPVGPIQWFRGAGPGRELIYNLKEGHFPRVT
AVSDPTKRNNMDFSIRISNITPADAGTYCYCVKFRKGSPDVELKSGAGTELSVRAKPS (SEQ
ID NO:12), or both.

(126) In some embodiments, an antibody of the present disclosure binds an extracellular domain of a murine or mouse SIRP- α polypeptide (e.g., found in the organism *Mus musculus*; e.g., the D1 domain of a murine or mouse SIRP- α polypeptide). In some embodiments, the antibody binds the extracellular domains (e.g., the D1 domains) of two or more different murine SIRP- α variant polypeptides. A variety of murine SIRP- α variant polypeptides from different mouse strains are known. In some embodiments, the murine SIRP- α variant polypeptide comprises an amino acid sequence selected from

(127) TABLE-US-00004 (SEQ ID NO: 7; from 129 mouse strain)
KELKVTQPEKSVSVAAGDSTVLNCTLTSLIPVGPIKWYRGVVGQSRLLIYS
FTGEHFPRVTNVSDATKRNNMDFSIRISNVTPEDAGTYCYCVKFQKGPSEP
DTEIQSGGGTEVYVLAKPS, (SEQ ID NO: 8; from NOD mouse strain)
TEVKVIQPEKSVSVAAGDSTVLNCTLTSLIPVGPIRWYRGVVGQSRLLIYS
FTTEHFPRVTNVSDATKRNLDFSIRISNVTPEDAGTYCYCVKFQKGSPT
EIQSGGGTEVYVLAK, (SEQ ID NO: 9; from C57BL/6 mouse strain)
KELKVTQPEKSVSVAAGDSTVLNCTLTSLIPVGPIRWYRGVGPSRLLIYS
FAGEYVPRIRNVSDTTKRNNMDFSIRISNVTPADAGIYCYCVKFQKGSSEP
DTEIQSGGGTEVYVLAK, and (SEQ ID NO: 10; from BALB/c mouse strain)
TEVKVTQPEKSVSVAAGDSTILNCTVTSLIPVGPIRWYRGVVGQSRLLIYS
FTGEHFPRIRNVSDTTKRNNMDFSIRISNVTPEDAGTYCYCVKFQKGSSEP
DTEIQSGGGTEVYVLAK.

(128) In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a human SIRP family protein. In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide. In some embodiments, a human SIRP- β polypeptide refers to a polypeptide encoded by a human SIRPB1 gene, e.g., as described by NCBI Ref Seq ID No. 10326. In some embodiments, the extracellular domain (e.g., the D1 domain) of the human SIRP- β polypeptide comprises the amino acid sequence of

(129) TABLE-US-00005 (SEQ ID NO: 13)
EDELQVIQPEKSVSVAAGESATLRCAMTSLIPVGPIMWFRGAGAGRELIY
NQKEGHFPRVTTVSELTKRNNLDFSISISNITPADAGTYCYCVKFRKGSPD

DVEFKSGAGTELSVRAKPS or (SEQ ID NO: 14)
EEELQVIQPDKSISVAAGESATLHCTVTSLIPVGPIQWFRGAGPGRELIY
NQKEGHFPRVTTVSDLTKRNNMDFSIRISNITPADAGTYCYVKFRKGSPD
HVEFKSGAGTELSVRAKPS.

(130) In some embodiments, an antibody of the present disclosure binds an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide. In some embodiments, a human SIRP- γ polypeptide refers to a polypeptide encoded by a human SIRPG gene, e.g., as described by NCBI Ref Seq ID No. 55423. In some embodiments, the extracellular domain (e.g., the D1 domain) of the human SIRP- γ polypeptide comprises the amino acid sequence of

(131) TABLE-US-00006 (SEQ ID NO: 15)
EEELQMIQPEKLLLVTVGKTATLHCTVTSLLPVGPVLWFRGVGPGRELIY
NQKEGHFPRVTTVSDLTKRNNMDFSIRISSITPADVGTYCYVKFRKGSPE
NVEFKSGPGTEMALGAKPS.

(132) In addition to antibodies that bind one or more of the polypeptides described above, the present disclosure contemplates antibodies that do not bind one or more of the polypeptides described above. Stated another way, the binding profile of an antibody of the present disclosure may be characterized by positively or negatively reciting any of the binding specificities and/or properties described herein.

(133) In some embodiments, an antibody of the present disclosure modulates SIRP- α signaling in a cell expressing a human SIRP- α polypeptide. In some embodiments, an antibody of the present disclosure antagonizes SIRP- α signaling in a cell expressing a human SIRP- α polypeptide. In some embodiments, an antibody of the present disclosure interferes with SIRP- α signaling in a cell expressing a human SIRP- α polypeptide. In some embodiments, an antibody of the present disclosure agonizes SIRP- α signaling in a cell expressing a human SIRP- α polypeptide. In some embodiments, SIRP- α signaling includes one or more intracellular signaling events mediated by activation of a SIRP- α polypeptide, including without limitation tyrosine phosphorylation of the intracellular region of SIRP- α , phosphatase (e.g., SHIP1) binding, adaptor protein binding (e.g., SCAP2, FYB, and/or GRB2), and nitric oxide production. Various assays for measuring SIRP- α signaling in a cell include without limitation SIRP- α phosphorylation, SHP1 and SHP2 co-immunoprecipitation, PI3-kinase signaling, cytokine production (both inflammatory IL-12, IL-23, TNF α , IFN and suppressive cytokines IL-10, IL-4, IL-13, cell surface markers levels for M1 and M2 macrophage markers) or dendritic cell activation and function; Kharitonov, A. et al. (1997) *Nature* 386: 181-6; Ochi, F. et al. (1997) *Biochem. Biophys. Res. Commun.* 239:483-7; Kim, E. J. et al. (2013) *Inflammation Research* 62:377-86; Yi, T. et al. (2015) *Immunity* 43:764-75.

(134) In some embodiments, the cell expressing a human SIRP- α polypeptide is a leukocyte. In some embodiments, the cell is a macrophage, dendritic cell, neutrophil, eosinophil, or myeloid-derived suppressor cell (MDSC). In some embodiments, an antibody of the present disclosure decreases or antagonizes SIRP- α signaling in a cell expressing a human SIRP- α polypeptide by at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%, e.g., using one or more of the SIRP- α signaling assays described herein or otherwise known in the art. In some embodiments, an antibody of the present disclosure increases or agonizes SIRP- α signaling in a cell expressing a human SIRP- α polypeptide by at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%, e.g., using one or more of the SIRP- α signaling assays described herein or otherwise known in the art.

(135) In some embodiments, an antibody of the present disclosure modulates an intercellular phenotype mediated by SIRP- α . In some embodiments, an antibody of the present disclosure enhances phagocytosis by a macrophage expressing a human SIRP- α polypeptide. For example, phagocytic activity of a macrophage treated or contacted with an antibody of the present disclosure can be compared with phagocytic activity of a macrophage not treated or contacted with the antibody, or phagocytic activity of a macrophage that expresses a human SIRP- α polypeptide and is

treated or contacted with an antibody of the present disclosure can be compared with phagocytic activity of a macrophage that does not express a human SIRP- α polypeptide and is treated or contacted with the antibody. Exemplary phagocytosis assays may be found, e.g., in Wieskopf, K. et al (2013) *Science* 341: 88 and Willingham, S. B. et al. (2012) *Proc. Natl. Acad. Sci.* 109:6662-7. In some embodiments, an antibody of the present disclosure increases phagocytosis by a macrophage expressing a human SIRP- α polypeptide by at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%, e.g., using one or more of the phagocytosis assays described herein or otherwise known in the art.

(136) In some embodiments, an antibody of the present disclosure enhances activation of dendritic cell(s) expressing a human SIRP- α polypeptide (e.g., an increased level of activation of individual dendritic cells, or an increased proportion of dendritic cells that are activated within a sample population). For example, activation of dendritic cell(s) treated or contacted with an antibody of the present disclosure can be compared with activation of dendritic cell(s) not treated or contacted with the antibody, or activation of dendritic cell(s) that express a human SIRP- α polypeptide and are treated or contacted with an antibody of the present disclosure can be compared with activation of dendritic cell(s) that do not express a human SIRP- α polypeptide and are treated or contacted with the antibody. Exemplary dendritic cell activation assays are described herein. In some embodiments, an antibody of the present disclosure increases dendritic cell (e.g., expressing a human SIRP- α polypeptide) activation by at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%, e.g., using one or more of the dendritic cell activation assays described herein or otherwise known in the art.

(137) In some embodiments, an antibody of the present disclosure inhibits in vivo growth of a tumor that expresses CD47. For example, in vivo growth of a tumor that expresses CD47 and is treated with an antibody of the present disclosure can be compared against in vivo growth of a tumor that expresses CD47 and is not treated with an antibody of the present disclosure. Exemplary in vivo tumor growth assays are described herein. In some embodiments, an antibody of the present disclosure inhibits in vivo growth of a tumor that expresses CD47 by at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%, e.g., using one or more of the in vivo tumor growth assays described herein or otherwise known in the art.

(138) In some embodiments, an antibody of the present disclosure blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (e.g., a “blocking” antibody). For example, the antibody and the CD47 polypeptide may “compete” for the same SIRP- α epitope, and/or antibody binding to SIRP- α may be mutually exclusive with CD47 binding to SIRP- α . The binding interface between SIRP- α and CD47, as well as residues of both proteins that participate in binding, are known; see Hatherley, D. et al. (2007) *J. Biol. Chem.* 282:14567-75 and Nakaishi, A. et al. (2008) *J. Mol. Biol.* 375:650-60. Exemplary assays for determining whether an antibody blocks CD47 binding to SIRP- α are described herein. In some embodiments, an antibody of the present disclosure blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide in an in vitro assay, e.g., using purified SIRP- α and/or CD47 polypeptides. For example, in vitro ELISA and SPR assays are described herein, although this is not meant to be limiting, as other in vitro binding assays may also be used. In some embodiments, antibody binding to a complex comprising a SIRP- α D1 variant (e.g., a non-naturally occurring high affinity SIRP- α D1 domain binding to CD47 with higher affinity than one or more naturally occurring counterparts as described herein) bound to an IgSF domain of CD47 is used to screen for blocking, non-blocking, and/or kick off antibodies. In some embodiments, “blocking” and/or “non-blocking” antibodies can be tested via surface plasmon resonance (SPR; e.g., as described in Example 1). For example, a complex can be formed between an IgSF domain of CD47 and a SIRP- α D1 variant (e.g., a non-naturally occurring high affinity SIRP- α D1 domain binding to CD47 with higher affinity than one or more naturally occurring counterparts as described herein), then binding

of a test antibody to the complex can be measured. For an antibody that blocks binding of SIRP- α to CD47, at increasing concentrations of CD47, one would expect fewer molecules of SIRP- α to be available to bind to the antibody since the antibody competes for the same binding site as CD47 and most/all SIRP- α is complexed with CD47. Therefore, one would expect the resonance (RU) to decrease with increasing concentration of CD47 in the mixture. In some embodiments, a “blocking” anti-SIRP- α antibody of the present disclosure binds to the extracellular domain of a SIRP- α polypeptide (e.g., the D1 domain) at one or more residues of the binding interface between CD47 and SIRP- α , i.e., the blocking antibody and CD47 share partially or completely overlapping epitopes. In some embodiments, a “blocking” anti-SIRP- α antibody of the present disclosure binds to the extracellular domain of a SIRP- α polypeptide (e.g., the D1 domain) at one or more amino acid positions that are also bound by CD47 in the CD47:SIRP- α complex. The binding interfaces between SIRP- α and exemplary anti-SIRP- α antibodies or CD47 are described in Example 4. In some embodiments, an antibody of the present disclosure blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell, e.g., an in vivo binding assay between polypeptides expressed on the surface of cells. In some embodiments, the in vivo assay may assess binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell by assaying one or more aspects of SIRP- α signaling, e.g., one or more intracellular signaling events mediated by activation of a SIRP- α polypeptide, including without limitation tyrosine phosphorylation of the intracellular region of SIRP- α , phosphatase (e.g., SHIP1) binding, adaptor protein binding (e.g., SCAP2, FYB, and/or GRB2), cytokine production (e.g. IL-10, IL-1 β , IFN or TNF), and nitric oxide production; and/or one or more intercellular phenotypes, including without limitation macrophage phagocytosis and other activating or suppressive phenotypes of macrophages, neutrophils, dendritic cells, eosinophils, and myeloid-derived suppressor cells (MDSCs).

(139) In some embodiments, an antibody of the present disclosure does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (e.g., a “non-blocking” antibody). For example, the antibody and the CD47 polypeptide may bind distinct and/or non-overlapping epitopes of SIRP- α . In some embodiments, an antibody of the present disclosure does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide in an in vitro assay, e.g., using purified SIRP- α and/or CD47 polypeptides. For example, in vitro ELISA and SPR assays are described herein, although this is not meant to be limiting, as other in vitro binding assays may also be used. In some embodiments, antibody binding to a complex comprising a SIRP- α D1 variant (e.g., a non-naturally occurring high affinity SIRP- α D1 domain binding to CD47 with higher affinity than one or more naturally occurring counterparts as described herein) bound to an IgSF domain of CD47 is used to screen for blocking, non-blocking, and/or kick off antibodies. In some embodiments, “blocking” and/or “non-blocking” antibodies can be tested via surface plasmon resonance (SPR; e.g., as described in Example 1). For example, a complex can be formed between an IgSF domain of CD47 and a SIRP- α D1 variant (e.g., a non-naturally occurring high affinity SIRP- α D1 domain binding to CD47 with higher affinity than one or more naturally occurring counterparts as described herein), then binding of a test antibody to the complex can be measured. For an antibody that does not block binding of SIRP- α to CD47, the antibody would be expected to bind to SIRP- α /CD47 complex and form a sandwich. Therefore, at increasing concentrations of CD47, the resonance would increase accordingly due to increased sandwich formation. In some embodiments, a “non-blocking” anti-SIRP- α antibody of the present disclosure binds to the extracellular domain of a SIRP- α polypeptide (e.g., the D1 domain) at one or more residues that are distinct from the binding interface between CD47 and SIRP- α , i.e., the non-blocking antibody and CD47 share completely

non-overlapping epitopes. The binding interfaces between SIRP- α and exemplary anti-SIRP- α antibodies or CD47 are described in Example 4. In some embodiments, an antibody of the present disclosure does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell, e.g., an in vivo binding assay between polypeptides expressed on the surface of cells. In some embodiments, the in vivo assay may assess binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell by assaying one or more aspects of SIRP- α signaling, e.g., one or more intracellular signaling events mediated by activation of a SIRP- α polypeptide, including without limitation tyrosine phosphorylation of the intracellular region of SIRP- α , phosphatase (e.g., SHIP1) binding, adaptor protein binding (e.g., SCAP2, FYB, and/or GRB2), cytokine production (e.g. IL-10, IL-1 β , IFN or TNF), and nitric oxide production; and/or one or more intercellular phenotypes, including without limitation macrophage phagocytosis and other activating or suppressive phenotypes of macrophages, neutrophils, dendritic cells, eosinophils, and myeloid-derived suppressor cells (MDSCs). It is a surprising finding of the present disclosure that antibodies that do not block SIRP- α interaction with CD47 are able to increase phagocytosis and block in vivo tumor growth. Without wishing to be bound to theory, it is thought that non-blocking anti-SIRP- α antibodies can modulate one or more functions of SIRP- α independent of CD47 binding.

(140) In some embodiments, binding of an antibody of the present disclosure to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide (e.g., a “kick off” antibody). In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide increases $k_{\text{sub.off}}$ of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide. In some embodiments binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human high affinity SIRP- α polypeptide (e.g., as described herein) increases the $k_{\text{sub.off}}$ of the human high affinity SIRP- α polypeptide (e.g., as described herein) for binding an IgSF domain of a human CD47 polypeptide to greater than about $1 \times 10^{\text{sup.} -3}$ l/s. For example, the antibody and the CD47 polypeptide may have adjacent or partially overlapping SIRP- α epitopes, such that the antibody is able to bind SIRP- α when it is bound to CD47, but the antibody: SIRP- α promotes dissociation of the SIRP- α . CD47 complex. In some embodiments, an antibody of the present disclosure reduces affinity of an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide in an in vitro assay, e.g., using purified SIRP- α and/or CD47 polypeptides. In some embodiments, binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide increases $k_{\text{sub.off}}$ of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide in an in vitro assay, e.g., using purified SIRP- α and/or CD47 polypeptides. For example, in vitro ELISA and SPR assays are described herein, although this is not meant to be limiting, as other in vitro binding assays may also be used. In some embodiments, antibody binding to a complex comprising a SIRP- α D1 variant (e.g., a non-naturally occurring high affinity SIRP- α D1 domain binding to CD47 with higher affinity than one or more naturally occurring counterparts as described herein) bound to an IgSF domain of CD47 is used to screen for blocking, non-blocking, and/or kick off antibodies. In some embodiments, an antibody of the present disclosure reduces affinity of an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell for binding an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell, e.g., an in vivo binding assay between polypeptides expressed on the surface of cells. In some embodiments, a “kick off” anti-SIRP- α antibody of the present disclosure binds to the extracellular domain (e.g., the D1 domain) of a SIRP- α polypeptide at one or more residues of the binding interface between CD47

and SIRP- α , i.e., the kick off antibody and CD47 share partially overlapping epitopes. In some embodiments, a “kick off” anti-SIRP- α antibody of the present disclosure binds to the extracellular domain (e.g., the D1 domain) of a SIRP- α polypeptide at 1 or more residues that are also bound by CD47 in the CD47:SIRP- α complex. For example, a “kick off” anti-SIRP- α antibody can bind to the extracellular domain (e.g., the D1 domain) of a SIRP- α polypeptide at 2 or more residues that are also bound by CD47 in the CD47:SIRP- α complex and are at the periphery of the CD47 binding epitope of SIRP- α . The binding interfaces between SIRP- α and exemplary anti-SIRP- α antibodies or CD47 are described in Example 4. In some embodiments, binding of an antibody of the present disclosure to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell increases $k_{\text{sub.off}}$ of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell, e.g., an in vivo binding assay between polypeptides expressed on the surface of cells. In some embodiments, the in vivo assay may assess binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide expressed on the surface of a first cell and an IgSF domain of a human CD47 polypeptide expressed on the surface of a second cell by assaying one or more aspects of SIRP- α signaling, e.g., one or more intracellular signaling events mediated by activation of a SIRP- α polypeptide, including without limitation tyrosine phosphorylation of the intracellular region of SIRP- α , phosphatase (e.g., SHIP1) binding, adaptor protein binding (e.g., SCAP2, FYB, and/or GRB2), cytokine production (e.g. IL-10, IL-1 β , IFN or TNF), and nitric oxide production; and/or one or more intercellular phenotypes, including without limitation macrophage phagocytosis and other activating or suppressive phenotypes of macrophages, neutrophils, dendritic cells, eosinophils, and myeloid-derived suppressor cells (MDSCs).

(141) In some embodiments, an antibody of the present disclosure modulates one or more immune cell functions by binding to two or more (or all three) of SIRP- α , SIRP β , and SIRP γ .

(142) In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

(143) In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

(144) In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a monkey SIRP- α polypeptide; and binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide.

(145) In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

(146) In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1

domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

(147) In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide.

(148) In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a monkey (e.g., cynomolgus) SIRP- α polypeptide; binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and the antibody does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

(149) In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a monkey (e.g., cynomolgus) SIRP- α polypeptide; binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

(150) In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide, the extracellular domain (e.g., the D1 domain) of a human SIRP- α v2 polypeptide, or both; binds the extracellular domain (e.g., the D1 domain) of a monkey (e.g., cynomolgus) SIRP- α polypeptide; binds the extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide; and binding of the antibody to an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide reduces affinity of the human SIRP- α polypeptide for binding an IgSF domain of a human CD47 polypeptide.

(151) In some embodiments, an antibody of the present disclosure binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a murine SIRP- α polypeptide, and a monkey SIRP- α polypeptide; the antibody does not bind at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (see, e.g., FIGS. 3A & 3B).

(152) In some embodiments, an antibody of the present disclosure binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a murine SIRP- α polypeptide, a monkey SIRP- α polypeptide, and at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (see, e.g., FIGS. 4A & 4B).

(153) In some embodiments, an antibody of the present disclosure binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a murine SIRP- α polypeptide, and a monkey SIRP- α polypeptide; the antibody does not bind at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and the antibody does not block

binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (see, e.g., FIGS. 5A & 5B).

(154) In some embodiments, an antibody of the present disclosure binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides and a monkey SIRP- α polypeptide; the antibody does not bind a murine SIRP- α polypeptide; the antibody does not bind at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; and the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (see, e.g., FIGS. 6A & 6B).

(155) In some embodiments, an antibody of the present disclosure binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides, a monkey SIRP- α polypeptide, and at least one of an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide and an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide; the antibody does not bind a murine SIRP- α polypeptide; and the antibody blocks binding between the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide (see, e.g., FIGS. 7A & 7B).

(156) In some embodiments, an antibody of the present disclosure comprises three CDRs from a VH domain comprising a sequence set forth in Table 2 and/or three CDRs from a VL domain comprising a sequence set forth in Table 2 (VH and VL sequences with CDRs highlighted are provided in FIGS. 10A-10F & 11A-11P). For example, in some embodiments, an antibody of the present disclosure comprises three CDRs from a VH domain comprising a sequence selected from SEQ ID NOs: 294, 139, 358, 362, 354, 380, 384, 350, 137, 374, 356, 352, 135, 348, 376, 346, 342, 344, 141, 360, 370, 382, 364, 366, 368, 372, 378, 133, 128, 396, 386, 398, 402, 392, 388, 390, 394, 400, 116, 117, 118, 119, 282, 404, 406, 278, 412, 275, 414, 280, 123, 292, 288, 290, 408, 410, 286, 284, 120, 121, 130, and 122 and/or three CDRs from a VL domain comprising a sequence selected from SEQ ID NOs: 295, 363, 140, 359, 355, 351, 136, 349, 377, 138, 375, 357, 353, 381, 385, 345, 365, 367, 369, 347, 142, 343, 371, 379, 383, 361, 373, 134, 105, 387, 389, 395, 397, 399, 403, 391, 393, 401, 93, 94, 95, 96, 283, 405, 407, 279, 413, 276, 415, 281, 100, 293, 289, 291, 409, 411, 287, 285, 97, 98, 107, and 99. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising a sequence set forth in Table 2 and/or a VL domain comprising a sequence set forth in Table 2 (see FIGS. 10A-10F & 11A-11P). For example, in some embodiments, an antibody of the present disclosure comprises a VH domain comprising an amino acid sequence selected from SEQ ID NOs: 294, 139, 358, 362, 354, 380, 384, 350, 137, 374, 356, 352, 135, 348, 376, 346, 342, 344, 141, 360, 370, 382, 364, 366, 368, 372, 378, 133, 128, 396, 386, 398, 402, 392, 388, 390, 394, 400, 116, 117, 118, 119, 282, 404, 406, 278, 412, 275, 414, 280, 123, 292, 288, 290, 408, 410, 286, 284, 120, 121, 130, and 122 and/or a VL domain comprising an amino acid sequence selected from SEQ ID NOs: 295, 363, 140, 359, 355, 351, 136, 349, 377, 138, 375, 357, 353, 381, 385, 345, 365, 367, 369, 347, 142, 343, 371, 379, 383, 361, 373, 134, 105, 387, 389, 395, 397, 399, 403, 391, 393, 401, 93, 94, 95, 96, 283, 405, 407, 279, 413, 276, 415, 281, 100, 293, 289, 291, 409, 411, 287, 285, 97, 98, 107, and 99. In some embodiments, an antibody of the present disclosure comprises six CDR sequences from an antibody described in Table 2 (see FIGS. 10A-10F & 11A-11P). In some embodiments, an antibody of the present disclosure comprises a VL domain comprising the VL domain sequence of VL domains Hum1-Hum9 as described herein. In some embodiments, the antibody is a humanized antibody. In some embodiments, the antibody is a human antibody. In some embodiments, an antibody of the present disclosure comprises a VH domain and/or a VL domain from an antibody described in Table 2. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising a D or E residue followed by (e.g., in the direction of N-terminus to C-terminus) a VH domain sequence selected from SEQ ID NOs: 116-130.

(157) TABLE-US-00007 TABLE 2 Amino acid sequences of antibody clones

described - Chlebin/ SEQ Antibody Domain ID - NO Sequence S1 Chicken VL
85 ALTQPASVSANPGETVEITCSGGGSNNA YGWFQQKSPGSAPLTVIYDNGKRPSDIPS
RFSGSKSDSTGTLTITRVQAEDEAVYYC GSADNSGAGVFGAGTTTLTVL S2 Chicken VL 86
AVTQPASVSANPGETVRITCSGDSSSYYS WHQQKSPGSAPVSVIYSNTDRPSDIPSRF
SGSASGSTATLTITGVQAEDEAVYFCGA YDSSSDSDIFGAGTTTLTVL S8 Chicken VL 87
AVTQPSSVSANPGETVEITCSGSSTYYGW YQQKSPGSAPVTVIYDNDKRPSDIPSRFS
GSKSGSTHTLIITGVQVEDEAVYFCGNED NNYVAIFGAGTTTLTVL S9 Chicken VL 88
ALTQPSSVSANPGETVKITCSGDNSAHY YYGWYQQKSPGSAPVTVIYYNDKRPSGI
PSRFSGSASGSTATLIITGVQVEDEAVYF CGSADSSNPAIFGAGTTTLTVL S11 Chicken VL 89
AVTQPASVSANPGETVKITCSGSSSGSYG WYQQKSPGSAPVTLIYETNKRPSNIPSRF
SGSKSGSTATLTITGVQADDEAVYYCGS EDSSTYLSIFGAGTTTLTVL S12 Chicken VL 90
AVTQPASVSANPGETVKITCSGDSSYYG WYQQKSPGSAPVTVIYDDNKRPSNIPSRF
SGSKSGSTGTLTITGVQADDEAVYFCGN EDNSYVAIFGAGTTTLTVL S13 Chicken VL 91
AVTQPASVSANPGETVKITCSGSSSYYG WYRQKSPGSAPVTLIYDNDKRPSGIPSRF
SGSKSGSTNTLTITGVQADDEAVYYCGN EDNSYVGIFGAGTTTLTVL S14 Chicken VL 92
AVTQPASVSANLGETVKITCSGDSSYYG WYQQKAPGSAPVTLIYDNDKRPSNIPSRF
SGSKSGSTATLTITGVQADDEAVYYCGN EDMNYVGIFGAGTTTLTVL S115 Human VL 93
ETVLTQSPATLSVSPGERATLSCRASQTV GSKLAWHQKPGQAPRLLIYDATNRAT
GISDRFSGSGSGTDFTLTISLQTEDSAVY YCQQYYYWPPYRFGGGTKVEIK S116 Human
VL 94 ETVLTQSPATLSVSPGERATLSCRASQTV GSKLAWHQKPGQAPRLLIYDATNRAT
GISDRFSGSRSGTDFTLTISLQTEDSAVY YCQQYYYWPPYRFGGGTKVEIK S117 Human
VL 95 ETVLTQSPATLSVSPGERATLSCRASQTV GSKLAWHQKPGQAPRLLIYDATNRAT
GIPDRFSGSGSGTDFTLTISLQTEDSAVY YCQQYYYWPPYRFGGGTKVEIK S118 Human
VL 96 ETVLTQSPATLSVSPGERATLSCRASQTV GSKLAWHQKPGQAPRLLIYDASRRAT
GIPDRFSGSGSGTDFTLTISLQTEDSAVY YCQQYYYWPPYRFGGGTKVEIK S119 Human
VL 97 EIVLTQSPATLSVSPGERATFSCRASQNV KNDLAWYQQRPGQAPRLLIYAARIRETG
IPERFSGSGSGTEFTLTITSLQSEDFAVYY CQQYYDWPPFTFGGGTKVEIK S120 Human VL
98 EIVLTQSPATLSVSPGERATFSCRASQNV KNDLAWYQQRPGQAPRLLIYAARIRETG
IPERFSGSGSGTEFTLTITSLQSEDFAVYY CQQYYDWPPFTFGGGTKVEIK S122 Human VL
99 EIVLTQSPATLSVSPGERATFSCRASQNV KNDLAWYQQRPGQAPRLLIYAARIRETG
IPERFSGSGSGTEFTLTITSLQSEDFAVYY CQQYYDWPPFTFGGGTKVEIK S123 Human VL
100 EIVLTQSPGTLVSPGERVTLTCRASQGIA GKIAWYQQKPGQAPRLLIYDASSRATGIP
GRFSGSGSGTEFTLTITSLQSEDFAVYYC QQHYDWSPLTFGGGTKVEIK S126 Human VL
101 EIVLTQSPGTLTLSPGERATLSCRASQSIG SSYLAWYQQKPGQAPRLLIYDATNRATG
IPDRFSGSGSGTDFTLTISLQTEDSAVYY CQQYYYWPPYRFGGGTKVEIK S128 Human VL
102 ETVLTQSPATLSVSPGERATLSCRASQTV GSKLAWHQKPGQAPRLLIYDASNRAT
GIPDRFSGSGSGTDFTLTISLQTEDSAVY YCQQYYYWPPYRFGGGTKVEIK S130 Human
VL 103 EIVLTQSPGTLVSPGERATLSCRASQNV RSDLAWYQQKLGQAPRLLIYDANTRAT
DIPDRFSGSGSGTEFTLTISLQSEDFAVY YCQHYYDWPPVTFGGGTKVEIK S135 Human
VL 104 EIVLTQSPATLSVSPGERVTFSCRASQNV RSDIAWYQQKPGQAPRLLIYAASSRDTGI
PDRFSGSGSGTDFTLTISLQSEDFGVYY CQQYYDWPPFTFGGGTKVEIK S137 Human VL
105 ETVLTQSPGTLTLSPGERATLTCRASQSV YTYLAWYQEKPGQAPRLLIYGASSRATG
IPDRFSGSGSGTVFTLTISLQSEDFAVYY CQQYYDRPPLTFGGGTKVEIK S138 Human VL
106 EIVLTQSPGTLVSPGERVILTCRASQSD TYNLAWYQQKPGQAPRLLIYDLSTRATG
IPDRFSGSGSGTEFTLTINSLEPEDFAVYY CHQYYDWPPYTFGGGTKVEIK S121 Human VL
107 EIVLTQSPATLSVSPGERATFSCRASQNV KNDLAWYQQRPGQAPRLLIYAARIRETG
IPERFSGSGSGTEFTLTITSLQSEDFAVYY CQQYYDWPPFTFGGGTKVEIK S1 Chicken VH
108 AVTLDESGGGLQTPGGALSLVCKGSGFT FSSHAMNWVRQAPGKGLEWVAGISSDG
RFTYYGA AVQGRATISRDNQSTVRLQL NNLRAEDTATYYCTKNGGCGSGGDLDC
IDAWGHGTEVIVSS S2 Chicken VH 109 AVTLDESGGGLQTPGGGLSLVCKASGFD

FSNFNMAWVQPGLEWVQGRATISRDNGQSTVRLQLN
NLRAEDTGTTFCTRNFSSVSSIDAWGH GTEVIVSS S8 Chicken VH 110
AVTLDESGGGLQTPGGALSLVCKASGFT FSSYNMGWVRQAPGKGLEFVAGIYASG
SSTD TDTTYGPAVAGRATISRDNGQSTV RLQLNNLRAEDTGTYYCAKAAGGCSTH
TCTAYIADSIDAWGHGTEVIVSS S9 Chicken VH 111
AVTLDESGGGLQTPGRALSLVCRGSGFSI SSYNMGWVRQAPGKGLEFIASIGSDGSS
THYAPAVKGRATITRDVGQSTVRLQLNN LRAEDTGTTFCAKDAYQCSYATCNDYL
DTIDAWGHGTEVIVSS S11 Chicken VH 112 AVTLDESGGGLQTPGGALSLVCKASGFT
FSSFNMGWVRQAPGKGLEFVAIYSGNS AEYGA AVQGRATISRDNGQSTVRLQLN
NLRAEDTGIYFCAKDAGSGCYSGVCAGT SSIDAWGHGTEVIVSS S12 Chicken VH 113
AVTLDESGGGLQTPGGALSLVCKASGFT FSSYNMGWVRQAPGKGLEFVAGIYIASG
DLGTTYGA AVQGRATISRDNGQSTVRLQ LNNLRAEDTGTTFCAKSAGGCSAHSCDT
YIADSIDAWGHGTEVIVSS S13 Chicken VH 114 AVTLDESGGGLQTPGGALSLVCKASGFT
FSSYNMGWVRQAPDKGLEFVAGIYTGS DAGLSTTYGA AVQGRATISRDNGQSTVR
LQLNNLGAEDTGIYFCTKSAGGCSYDNC DAYIADSIDAWGHGTEVIVSS S14 Chicken VH
115 AVTLDESGGGLQTPGGALSLVCKASGFT FNSYNMGWVRQAPGKGLEFVAGIYSAG
GDTSTTYGA AVNGRATISRDNGQSTVRL QLNNLRAEDTGIYFCAKAAGGCTAHNC
DAYIADSIDAWGHGTEVIVSS S115 Human VH 116
VQLVESGGGVVRPGE SLRLSCAASGFSFS SYAMNWVRQAPGEGLEWVSRINSGGGG
TDYAESVKGRFTISRDNSENTLYLQMNS LRAEDTAVYYCAKQYDWN SFFDYWGL
GALVTVSS S116 Human VH 117 VQLVESGGGVVRPGE SLRLSCAASGFSFS
SYAMNWVRQAPGEGLEWVSRINSGGGG TDYAESVKGRFTISRDNSENTLYLQMNS
LRAEDTAVYYCAKQYDWN SFFDYWGL GALVTVSS S117 Human VH 118
VQLVESGGGVVRPGE SLRLSCAASGFSFS SYAMNWVRQAPGEGLEWVSRINSGGGG
TDYAESVKGRFTISRDNSENTLYLQMNS LRAEDTAVYYCAKQYDWN GFFDYWGL
GALVTVSS S118 Human VH 119 VQLVESGGGVVRPGE SLRLSCAASGFSFS
SYAMNWVRQAPGEGLEWVSRINSGGGG TDYAESVKGRFTISRDNSENTLYLQMNS
LRAEDTAVYYCAKQYDWN GFFDYWGL GALVTVSS S119 Human VH 120
VQLLESGGGVVQPGGSLRLSCAASGFSF SNFAMTWVRQAPGEGLEWVSTIGSGDT
YYADSVKGRFTISRDN SKNTLYLQMNSL RAEDTAVYYCAKDSTVSWSGDFFDYWG
LGTLVTVSS S120 Human VH 121 VQLVESGGGVVQPGGSLRLSCAASGFSF
SNFAMTWVRQAPGEGLEWVSTIGSGDT YYADSVKGRFTISRDN SKNTLYLQMNSL
RAEDTAVYYCAKDSTVSWSGDFFDYWG LGTRVTVSS S122 Human VH 122
VQLVESGGGVVRPGE SLRLSCAASGFRF SNFAMTWVRQAPGEGLEWVSTIGSGDT
YYADSVKGRFTISRDN SKNTLYLQMNSL RAEDTAVYYCAKDSTVSWSGDFFDYWG
LGTLVTVSS S123 Human VH 123 VQLVESGGGLVQPGGSLRLSCTASGFTF
RNYGMSWVRQAPGEGLEWVSASSGSGS TYYTDSVKGRFTISRDN SKNTLYLQMNS
LRAEDTAIYYCAKVTWNNFFDYWGLGT LTVTVSS S126 Human VH 124
VQLVESGGGVVRPGE SLRLSCAASGFTF SNYDMTWVRQAPGEGLEWVSGISGNGG
STYYADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCAMNRWWFDYWGLGT
LTVTVSS S128 Human VH 125 VQLVESGGGVVRPGE SLRLSCAASGFSF
RSYAMNWVRQAPGEGLEWVSRIDSGGG GTDYADSVKGRFTISRDN SKNTLYLQMN
SLRAEDTAVYYCAKQYDWN SFFDYWGL GAPVTVSS S130 Human VH 126
VQLVESGGGVVRPGE SLRLSCAASGFTF SNYAMSWVRQAPGEGLEWVSLITTNGD
GAYYADSVKGRFTISRDN SKNTLYLQM NSLRAEDTAIYYCAKDGA AHYYDIFFDY
WGLGTPVTVSS S135 Human VH 127 VQLVESGGGVVRPGE SLRLSCAASGFSFS
IYAMSWVRQAPGEGLEWVSTIGADDTY YADSVKGRFTISRDN SKNTLYLQMNSLR
AEDTAVYYCAKDSTVGWSGDFFDYWG LGTLVTVSS S137 Human VH 128
VQLVESGGGVVRPGE SLRLSCAASGFTF SSYDMNWVRQAPGEGLEWVSLISGSGEI
IYYADSVKGRFTISRDN SKNTLYLQMNS LRAEDTAVYYCAKENNRYRFFDDWGLG

TLTVSS S138 Human VH 129 VQLVESGGGVVRPGESESLRLSCAASGFTF
SNYAMNWVRQAPGEGLEWVSGISGRGG DTYYADSVKGRFTISRDN SKNTLYLQMN
SLRAEDTAIYYCAKGTWNYGSFDYWGL GTLTVTVSS S121 Human VH 130
VQLVESGGGVVQPGGSLRLSCAASGFSF SNFAMTWVRQAPGEGLEWVSTIGSGDT
YYADSVKGRFTISRDN SKNTLYLQMNSL RAEDTAVYYCAKDSTVSWSGDFFDYWG
LGTLTVTVSS AB136 Human VH 133 DVQLVESGGGVVRPGESESLRLSCAASGFT
FSSYDMNWVRQAPGEGLEWVSLISGSGE IYYADSVKGRFTISRDN SKNTLYLQMNS
LRAEDTAVYYCAKENNRYRFFDDWGLG TLTVTVSS AB136 Human VL 134
ETVLTQSPGTTLTSPGERATLTCRASQSV YTYLAWYQEKPGQAPRLLIYGASSRATG
IPDRFSGSGSGTEFTLTISSLQSEDFAVYY CQQYYDRPPLTFGGG TKVEIK AB21 Human VH
135 DVQLVESGGGVVRPGESESLRLSCAASGFT FSSNAMSWVRQAPGKGLEWLAGISAGG
SDTYYPASVKGRFTISRDN SKNTLYLQM NTLTAEDTAVYYCARETNH LFDYWGL
GTLTVTVSS AB21 Chicken VL 136 ALTQPASVSANPGETVKIACSGGDYYSY
YYGWYQQKAPGSALVTVIYSDDKRPSDI PSRFSGSASGSTATLTITGVRAEDEAVYY
CGGYDYSTYANAFGAGTTTLTVL AB25 Human VH 137
DVQLVESGGGVVRPGESESLRLSCEASGFT FSSNAMSWVRQAPGKGLEWVAGISSGS
DTYYGDSVKGRLTISRDN SKNLYLQMN SLTAEDTAVYYCARETNH LFDYWGLG
TLTVTVSS AB25 Chicken VL 138 ALTQPASVSANPGETVEITCSGGSYSSYY
YAWYQQKSPGSAPVTLIYSDDKRPSNIPS RFSGSASGSTATLTITGVRAEDEAVYFCG
GYDQSSYTNPFAGAGTTTLTVL AB27 Human VH 139
DVQLVESGGGVVRPGESESLRLSCAVSGFR FSSYAMSWVRQAPGKGLEWVSGISSGG
DTYYVDSVKGRFTISRDN SKNTLYLQVN SLTAEDTAIYYCARETNH LFDYWGLG
TLTVTVSS AB27 Chicken VL 140 ALTQPASVSADLGETVKITCSGGDSSSHY
YGWYQQKSPGSAPVTVIYSDDERPSDIPS RFSGSASGSTATLTITGVRVEDEAIYYCG
AYDGSTYANTFGAGTTTLTVL AB66 Human VH 141
DVQLVESGGGVVRPGESESLRLSCAASGFT FSSYAMSWVRQAPGKGLEWLAGISAGG
SDTYYIDSVKGRFTISRDN PKNSLYLQMS SLTAEDTAVYYCARETNH LFDYWGLG
TLTVTVSS AB66 Chicken VL 142 ALTQPASVSANPGETVKITCSGGDYYST
YYAWYQQKSPGSAPVTVIHSDDKRPSDI PSRFSGSASGSAATLIITGVRVEDEAVYY
CGGYDGRTYINTFGAGTTTLTVL AB119 Human HVR-H1 143 GFSFSNFAMT AB119 Human
HVR-H2 144 TIGSGDTYYADSVKG AB119 Human HVR-H3 145 DSTVSWSGDFFDY AB119
Human HVR-L1 146 RASQNVKNDLA AB119 Human HVR-L2 147 AARIRET AB119 Human
HVR-L3 148 QQYYDWPPFT AB135 Human HVR-H1 149 GFSFSIYAMS AB135 Human HVR-
H2 150 TIGADDTYYADSVKG AB135 Human HVR-H3 151 DSTVGWSGDFFDY AB135
Human HVR-L1 152 RASQNVRS DIA AB135 Human HVR-L2 153 AASSRDT AB135 Human
HVR-L3 148 QQYYDWPPFT AB136 Human HVR-H1 155 GFTFSSYDMN AB136 Human
HVR-H2 156 LISGSGEIYYADSVKG AB136 Human HVR-H3 157 ENNRYRFFDD AB136
Human HVR-L1 158 RASQSVYTYLA AB136 Human HVR-L2 159 GASSRAT AB136 Human
HVR-L3 160 QQYYDRPPLT AB21 Human HVR-H1 161 GFTFSSNA AB21 Human HVR-H2
162 ISAGGSDT AB21 Human HVR-H3 163 ARETNH LFDY AB21 Chicken HVR-L1 164
SGGDYYSYYYG AB21 Chicken HVR-L2 165 TVIYSDDKRPSD AB21 Chicken HVR-L3 166
GGYDYSTYANA AB25 Human HVR-H1 161 GFTFSSNA AB25 Human HVR-H2 168 ISSGSDT
AB25 Human HVR-H3 163 ARETNH LFDY AB25 Chicken HVR-L1 170 SGGSYSSYYA
AB25 Chicken HVR-L2 171 TLIYSDDKRPSN AB25 Chicken HVR-L3 172 GGYDQSSYTNP
AB27 Human HVR-H1 173 GFRFSSYA AB27 Human HVR-H2 174 ISSGGDT AB27 Human
HVR-H3 163 ARETNH LFDY AB27 Chicken HVR-L1 176 SGGDSSSHYYG AB27 Chicken
HVR-L2 177 TVIYSDDERPSD AB27 Chicken HVR-L3 178 GAYDGSTYANT AB66 Human
HVR-H1 179 GFTFSSYA AB66 Human HVR-H2 162 ISAGGSDT AB66 Human HVR-H3 163
ARETNH LFDY AB66 Chicken HVR-L1 182 SGGDYYSTYIA AB66 Chicken HVR-L2 183
TVIHSDDKRPSD AB66 Chicken HVR-L3 184 GGYDGRTYINT AB3 Chicken VH 242

AVTLDESGGLQTPGGALSLVCKASGFTF
YGA AVKGRATISRDNRQSTVKLQLNNLRA EDTGIYFCARDFGSGVGSIDAWGNGTEVIV
SS AB3 Chicken VL 243 ALTQPASVSANLGGTVKITCSGSRGRYGW
YQQRSPGSAPVTVIYRDNRPSNIPSRFSS TSGSTSTLTITGVQADDES VYFCGSYDGS
DIFGAGTTTLTVL AB45 Chicken VH 244 AVTLDESGGGLQTPGGALSLVCKASGFTF
SSYAMGWVRQAPGKGLEWVAGIDDDGST ANYGPAVKGRATISRDNGQSTVRLQLNNP
RAEDSGTYFCAKASVTGWSAHISGRDLTW GHGTEVIVSS AB45 Chicken VL 245
ALTQPASVSANPGETVKITCSGGGIYYYG WYQQKSPGSAPVTLIYENDKRPSDIPSRFS
GSTSGSTNTFTITGVQAEDEAVYYCGGYD SNTTSGIFGAGTTTLTVL AB119 Human with
VH 246 EVQLLES GGGVVP GGSLRLSCAASGFSFS mut D1E, E43K,
NFAVTWVRQAPGKGLEWVSTIGSGDTYY L112Q, and
ADSVKGRFTISRDN SKNTLYLQMNSLRAE M34V
DTAVYYCAKDSTVSWSGDFFDYWGQGTL mutations VTVSS AB135 Human with VH 247
EVQLVESGGGVVP GGSLRLSCAASGFSFS mut D1E, R13Q,
IYAVSWVRQAPGKGLEWVSTIGADDTYY E16G, E43K
ADSVKGRFTISRDN SKNTLYLQMNSLRAE L112Q, and
DTAVYYCAKDSTVGWSGDFFDYWGQGTL M34V VTVSS mutations AB136 Human with
VH 249 EVQLVESGGGVVP GRSLRLSCAASGFTFS mut all D1E, R13Q,
SYDVNWVRQAPGKGLEWVSLISGSGEIY E16R, E43K
YADSVKGRFTISRDN SKNTLYLQMNSLRA L111Q, and
EDTAVYYCAKENNRYRFFDDWGQGT LVT M34V VSS mutations AB136 Human with VL
250 EIVLTQSPGTL SLSPGERATL SCRASQSVYT mut all T2I, T12S,
YLAWYQQKPGQAPRL LIYGASSRATGIPD T22S, and
RFSGSGSGTEFTLTIS SLQSEDFAVYYCQQ E28Q YYDRPPLTFGGG TKVEIK mutations
AB136 Human with VL 251 ETVLTQSPGTL SLSPGERATL SCRASQSVY mut T12S, T22S
TYLAWYQQKPGQAPRL LIYGASSRATGIP all_I2T and E38Q
DRFSGSGSGTEFTLTIS SLQSEDFAVYYCQ mutations QYYDRPPLTFGGG TKVEIK Hum1
Humanized VL 252 SYELTQPPSVSVSPGQTARITC **SGGSYSSY**
YYAWYQQKPGQAPVT LIY SDDKRPS NIPE RFSGSSSGTTVTLTISGVQAEDEADYYC **GG**
YDQSSYTNP FGGG TKLTVL Hum2 Humanized VL 253
QSVLTQPPSVSAAPGQKVTISC **SGGSYSSY** **YYAWYQQ** LPGTAPKT LIY **SDDKRPS** NIPD
RFSGSKSGTSATLGITGLQTGDEADYYC **G** **YDQSSYTNP** FGTG TKVTVL Hum3
Humanized VL 254 SYELTQPPSVSVSPGQTARITC **SGGDYYST**
YYAWYQQKPGQAPVT VIH SDDKRPS DIPE RFSGSSSGTTVTLTISGVQAEDEADYYC **GG**
YDGRTYINT FGGG TKLTVL Hum4 Humanized VL 255
QSVLTQPPSVSAAPGQKVTISC **SGGDYYST** **YYAWYQQ** LPGTAPKT VIH **SDDKRPS** DIPD
RFSGSKSGTSATLGITGLQTGDEADYYC **G** **YDGRTYINT** FGTG TKVTVL Hum5
Humanized VL 256 QSALTQPASVSGSPGQSITISCTGTSSDV **GS**
YSSYY **YYAWYQQ** HPGKAPKT LIY **SDDKRPS** NVSNRFSGSKSGNTASLTISGLQAEDEADY
YC **GGYDQSSYTNP** FGGG TKLTVL Hum6 Humanized VL 257
QSVLTQPPSVSAAPGQKVTISC **SGGDYYSY** **YYGWYQQ** LPGTAPKT VIY **SDDKRPS** DIPD
RFSGSKSGTSATLGITGLQTGDEADYYC **G** **YDYSTYANA** FGTG TKVTVL 119 VH
Human with VH 258 EVQLLES GGGVVP GGSLRLSCAASGFSFS MutAll_V34M D1E,
E43K, NFAVTWVRQAPGKGLEWVSTIGSGDTYY and L112Q
ADSVKGRFTISRDN SKNTLYLQMNSLRAE mutations
DTAVYYCAKDSTVSWSGDFFDYWGQGTL VTVSS 135 VH Human with VH 259
EVQLVESGGGVVP GGSLRLSCAASGFSFS MutAll_V34M D1E, R13Q,
IYAMSWVRQAPGKGLEWVSTIGADDTYY E16G, E43K,
ADSVKGRFTISRDN SKNTLYLQMNSLRAE and L112Q
DTAVYYCAKDSTVGWSGDFFDYWGQGTL mutations VTVSS 136 VH Human with VH

260 EVQLVESGGGVVQPGGSLRLSCAASGFTFS MutAll_V34M D1E, R13Q,
SYDMNWVRQAPGKGLEWVSLISGSGEIIY E16R, E43K,
YADSVKGRFTISRDN SKNTLYLQMNSLRA and L111Q
EDTAVYYCAKENNRYRFFDDWGQGT LVT mutations VSS Hum 8 Humanized VL 416
SYELTQPPSVSVSPGQTARITCSGGAYSSY YYAWYQQKPGQAPVLVIYSDSKRPSGIPE
RFGSGSSGTTVTLTISGVQAEDEADYYCGG YDQSSYTNPFGGGTKLTVL Hum9 Humanized
HVR-L1 261 SGGAYSSYYA Hum9 Humanized VL 262
SYELTQPPSVSVSPGQTARITCSGGAYSSY YYAWYQQKPGQAPVLVIYSDDKRPSGIPE
RFGSGSSGTTVTLTISGVQAEDEADYYCGG YDQSSYTNPFGGGTKLTVL AB21 Human
with VH 263 EVQLVESGGGVVQPGGSLRLSCAASGFTF Mut All germline
SSNAMSWVRQAPGKGLEWVAGISAGGSD back-
TYYPASVKGRFTISRDN SKNTLYLQMNSL mutations
RAEDTAVYYCARETNH LFDYWGQGT L VTVSS AB21 Human with VH 264
EVQLVESGGGVVQPGGSLRLSCAASGFTF Mut All germline
SSNAVSWVRQAPGKGLEWVAGISAGGSD M34V back-
TYYPASVKGRFTISRDN SKNTLYLQMNSL mutations
RAEDTAVYYCARETNH LFDYWGQGT L and liability VTVSS mutation AB25 Human
with VH 265 EVQLVESGGGVVQPGGSLRLSCAASGFTF Mut All germline
SSNAMSWVRQAPGKGLEWVAGISSGSDT back-
YYGDSVKGRFTISRDN SKNTLYLQMNSLT mutations
AEDTAVYYCARETNH LFDYWGQGT LVT VSS AB25 Human with VH 266
EVQLVESGGGVVQPGGSLRLSCAASGFTF Mut All germline
SSNAVSWVRQAPGKGLEWVAGISSGSDTY M34V back-
YGDSVKGRFTISRDN SKNTLYLQMNSLTA mutations
EDTAVYYCARETNH LFDYWGQGT LVT V and liability SS mutation AB27 Human with
VH 267 EVQLVESGGGVVQPGGSLRLSCAASGFRF Mut All germline
SSYAMSWVRQAPGKGLEWVSGISSGGDT back-
YYVDSVKGRFTISRDN SKNTLYLQMNSLR mutations
AEDTAVYYCARETNH LFDYWGQGT LVT VSS AB27 Human with VH 268
EVQLVESGGGVVQPGGSLRLSCAASGFRF Mut All germline
SSYAVSWVRQAPGKGLEWVSGISSGGDTY M34V back-
YVDSVKGRFTISRDN SKNTLYLQMNSLRA mutations
EDTAVYYCARETNH LFDYWGQGT LVT V and liability SS mutation 135 Human with
HVR-H1 269 IYAMS V34M liability mutation 136 Human with HVR-H1 270 SYDMN V34M
liability mutation 21/25 Human HVR-H1 271 SNAVS M34V 27 Human HVR-H1 272 SYAVS
M34V 115 Human VH 273 DVQLVESGGGVVRPGE SLRLSCAASGFSFS
SYAMNWVRQAPGEGLWVSRINSGGGGT DYAESVKGRFTISRDNSENTLYLQMNSLR
AEDTAVYYCAKQYDWN SFFDYWGLGAL VTVSS 115 Human VL 274
ETVLTQSPATLSVSPGERATLSCRASQTVG SKLAWHQKPGQAPRL LIYDATNRATGIS
DRFSGSGSGTDFTLTIS SLQTEDSAVYYCQ QYYYWPPYRFGGGTKVEIK 213 Human VH
275 DVQLVESGGGVVRPGE SLRLSCEASGFTF RNY YMTWVRQAPGEGLWVSTISDTGDT
A Y YADSVKGRFTISRDN SKNSLYLQMNSL RADDTAIYYCAKSWIWWTFFDYWGLGTL
VTVSS 213 Chicken VL 276 ALTQPASVSANLGGTVEITCSGGNSNHYG
WYQQKSPGSAPVT LIYADTNRPSNIPSRFS GSTSGSTTTLTITGVQAEDEAVYYCGGSST
GDGIFGAGTTTLTVL 173 Human VH 278 DVQLVESGGGVVRPGE SLRLSCAASGFAF
SDHDMSWVRQGPGEGLEWVAGISLRGGV TWYADSVKGRFTISRDN SKNTLYLRLFSL
RTEDTAIYYCARESWNTFFDYWGLGTLVT VSS 173 Human VL 279
EIVLTQSPGTLSPGETATLSCRASQNVRS NLAWYQQKPGQAPRL LIYDASSRATGIPD
RFGSGSGTDFTLTIS SLQSEDFAVYYCQQ YGN GPPLTFGGGTKVEIK 209 Human VH 280
DVQLVESGGAVVRPGE SLRLSCKASGFTF TNFAMSWVRQAPGEGLWVSGISGSDDTT

YYADSVKSRDENSESTLYLQMNLSLR AEDTAVYYCVKDDSTVSNLTFDDYWGLGT
LVTVSS 209 Chicken VL 281 ALTQPASVSANLGGTVKITCSGGYGSDDG
SSSYYGWYQQKSPGSAPVILIYWDDKRPS DIPSRFSGSTSGSTTTTLTITGVQAEDEAVYF
CGTYDTSSGAIFGAGTTLTVL 132 Human VH 282
DVQLVESGGGVVRPGESELRRLSCAASGFSF RSYAMNWVRQAPGEGLEWVSRINSGGGG
TDYADSVK GRFTISRDN SKNTLYLQMNLSRAEDTAVY
YCAKQYDWN SFFDYWGLGALVTVSS 132 Human VL 283
ETVLTQSPATLSVSPGERATLSCRASQTVG SKLAWHQKPGQAPRLLIYDA
SNRATGIPDRFSGSGSGTDFTLTISSPQTED SAVYYCQQYYYPYRFGGGTKVEIK 218
Human VH 284 DVQLVESGGGVVRPGESELTLSCTASGFTFT
SSTMNWVRQAPGEGLDWVSSISTSGVITY YADSVKGRATISRDN SKNTLYLRFLSLRA
DDTAIYYCATDFTFDHWGPGLTVTVSS 218 Chicken VL 285
ALTQPASVSANPGETVKITCFGSSGNYGW FQQKSPGSAPVTVIHYNNKRPSDIPSRFSGS
KSGSTGTLTITGVRAEDEAVYFCGAWETG SATFGAGTTLTVL 149 Human VH 286
DVQLVESGGGVVRPGESELRRLSCAASGFTF SNFAMNWVRQAPGEGLEWVSLISVTATT
YYADSVKGRFTISRDN SKNTLYLQMNLSR AEDTAVYYCAKVTWNNLFDYWGLGTLV
TVSS 149 Human VL 287 EIVLTQSPGTL SLSPGERATLSCRASQPID
YLAWYQQKPGQAPRLLIYNTVTRATGIPD RFSGSGSGTDFTLTINRLEPEDFAVYYCQH
QYDWPPYIFGGGTKVEIK 161 Human VH 288 DVQLVESGGGVVRPGESELRRLSCAASGFTF
SNFAMTWVRQAPGKGPEWVSLVSVTATT YYADSVKGRFTISRDN SKSTLYLQMNLSR
AEDTAVYYCAKITWNNLFDYWGLGTLVT VSS 161 Human VL 289
EIVLTQSPGTL SLSPGERATLSCRASQTVGS KLAWEYQQKPGQAPRLLIYDSSSRASGIPDR
FSGSGSGTDFTLTISSLQSEDSAVYYCQQH NDWPPYTFGGGTKVEIK 162 Human VH 290
DVQLVESGGGLVRPGESELRRLSCAASGFTFT NYAVTWVRQAPGEGLEWVSLISVTGTTY
YADSVKGRFTISRDN SKSTLYLQMNLSR AEDTAVYYCAKVTWKNVFDYWGLGTLVT VSS
162 Human VL 291 EIVLTQSPGTL SVSPGERATFSCRASQTVGS
KLAWEYQQKPGQAPRLLIYDANTRATGIPA RFSGSGSGTEFTLTISSLQSEDFAVYYCQQ
HTDWPPYTFGGGTKVEIK 194 Human VH 292 DVQLVESGGGVVRPGESELRRLSCAASGFTF
RNYGMSWVRQAPGEGLEWVSASSGSGST YYADSVKGRFTISRDN SKNTLYLQMNLSR
AEDTAIYYCAKVTWNNFFDYWGLGTLVT VSS 194 Human VL 293
EIVLAQSPDTLSVSPGERATLTCRASQDVA GKLAWEYQQKPGQAPRLLIHATSSRADGIP
ARFSGSGSGTEFTLTITGLQSEDFAVYYCQ QHYDWSPLTFGGGTKVEIK 119 Human with
HVR-H1 305 NFALT M34L liability mutation 135 Human with HVR-H1 306 IYALS M34L
liability mutation 119 mut Human with VL 312 EIVLTQSPATLSVSPGERATLSCRASQNVK
all F21L, R39K, NDLAWEYQQKPGQAPRLLIYAA E60A, and
RIRETGIPARFSGSGSGTEFTLTISSLQSEDF T76S AVYYCQQYYDWPPFTFGGGTKVEIK
mutations 136 Human with HVR-H1 313 SYDLN M34L liability mutation 21/25 Human with
HVR-H1 318 SNALS M34L liability mutation 27 Human with HVR-H1 319 SYALS M34L
liability mutation 119_VH_MutAll_V34L Human with VH 327
EVQLLESGGGVVQPGGSLRLSCAASGFSFS germline NFALTWVRQAPGKGLEWVSTI back-
GSGDTYYADSVKGRFTISRDN SKNTLYLQ mutations
MNSLRAEDTAVYYCAKDSTVSWSGDFFD and liability YWGQGLTVTVSS mutation
135_VH_MutAll_V34L Human with VH 328 EVQLVESGGGVVQPGGSLRLSCAASGFSF
germline back- SIYALSWVRQAPGKGLEWVSTI mutations and
GADDTYYADSVKGRFTISRDN SKNTLYLQ liability
MNSLRAEDTAVYYCAKDSTVGWSGDFFD mutation YWGQGLTVTVSS
136_VH_Mutall_V34L Human with VH 329 EVQLVESGGGVVQPGRSLRLSCAASGFTF
germline back- SSYDLNWVRQAPGKGLEWVSLI mutations and
SGSGEIIYYADSVKGRFTISRDN SKNTLYL liability
QMNSLRAEDTAVYYCAKENNRYRFFDDW mutation GQGLTVTVSS

AB21_HC_MutAll_M34L Human with VH 330 EVQLVESGGGVVQPGGSLRLSCAASGFTF
germline back- SSNALSWVRQAPGKGLEWVAGISAGGSD mutations and
TYYPASVKGRFTISRDN SKNTLYLQMNSL liability
RAEDTAVYYCARETNH LFDYW GQGTL mutation VTVSS AB25_HC_MutAll_M34L
Human with VH 331 EVQLVESGGGVVQPGGSLRLSCAASGFTF germline back-
SSNALSWVRQAPGKGLEWVAGISSGSDTY mutations and
YGDSVKGRFTISRDN SKNTLYLQMNSLTA liability
EDTAVYYCARETNH LFDYW GQGTLVT mutation VSS AB27_HC_MutAll_M34L Human
with VH 332 EVQLVESGGGVVQPGGSLRLSCAASGFRF germline back-
SSYALSWVRQAPGKGLEWVSGISSGGDTY mutations and
YVDSVKGRFTISRDN SKNTLYLQMNSLRA liability
EDTAVYYCARETNH LFDYW GQGTLVT mutation VSS 218_Hum13 Humanized VL 333
QSALTQPASVSGSPGQSITISCFGSSGNYGL (218 VL
VSWYQQHPGKAPKLMIIYNNKRPSGVSN with
RFSGSKSGNTASLTISGLQAEDEADYYCG human AWETGSATFGGGTKLTVL IGLV2)
218_Hum14 Humanized VL 334 SYELTQPPSVSVSPGQTASITCFGSSGNYG (218 VL
WYQQKPGQSPVLVIYNNKRPSGIPERFSG with
SNSGNTATLTISGTQAMDEADYYCGAWE human TGSATFGGGTKLTVL IGLV3) 119 Human
VH 335 DVQLLESGGGVVQPGGSLRLSCAASGFSF
SNFAMTWVRQAPGEGLEWVSTIGSGDTY YADSVKGRFTISRDN SKNTLYLQMNSLRA
EDTAVYYCAKDSTVSWSGDFFDYWGLGT LVTVSS 135 Human VH 341
DVQLVESGGGVVRPGESLRLSCAASGFSFS IYAMSWVRQAPGEGLEWVSTIGADDTYY
ADSVKGRFTISRDN SKNTLYLQMNSLRAE DTAVYYCAKDSTVGWSGDFFDYWGLGTL
VTVSS 16 Human VH 294 DVQLVESGGGVVRPGESLRLSCAVSGFRFSSY
AMSWVRQAPGKGLEWVSGISSGGDTYYVDSV
KGRFTISRDN SKNTLYLQVNSLTAEDTAIYYCA RETWNH LFDYWGLGTLTVTSS 16
Chicken VL 295 ALTQPASVSANLGETVKITCSGGDSSSHYYGW
YQQKSPGSAPVTVIYSDDERPSDIPSRFSGSASG
STATLTITGVRVEDEAIYYCGAYDGSTYANTF GAGTTTLTVL 17 Human VH 342
DVQLVESGGAVVRPGESLRLSCAASGFTFSSY
AMSWVRQAPGKGLEWLAGISAGGSDTYYIDS
VKGRFTISRDN SENS LYLQMNSLTAEDTAVYY CARETNH LFDYWGLGTLTVTSS 17
Chicken VL 343 ALTQPASVSANPGETVKITCSGGDWYSTYYG
WYQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSA
SGSAATLTITGVRVEDEAVYYCAGYDGRTYIN TFGAGTTTLTVL 22 Human VH 344
DVQLVESGGGVVRPGESLRLSCAASGFTFSSY
AMSWVRQAPGKGLEWLAGISAGGSDTYYIDS
VKGRFTISR RQFQEQLSPNEPALTAEDTAVYY CARETNH LFDYWGLGTLTVTSS 22
Chicken VL 345 ALTQPASVSANPGETVKITCSGGDYYSTYYGW
YQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG
SAATLTIA GVRVEDEAVYFCGAYDGRTYINTF GAGTTTLTVL 23 Human VH 346
DVQLVESGGGVVRPGESLRLSCAASGFTFSSH
AMSWVRQAPGKGLEWLAGISAGGSDTYYIDS
VKGRFTISRDN SKSLYLRMNSLTAEDTAVYY CARETNH LFDYWGLGTLTVTSS 23
Chicken VL 347 ALTQPASVSANPGETVKITCSGGDYYSTYYAW
YQQKSPGSAPVTVIHSDDERPSDIPSRFSGSASG
SAATLIITGVRVEDEAVYFCGGYDGRTYINTFG AGTTTLTVL 24 Human VH 348
DVQLVESGGGVVRPGESLRLSCAASGFTFSSN
AMSWVRQAPGKGLEWLAGISAGGSDTYYPAS
VKGRFTISRDN PKNTLYLQMNTLTAEDTAVYY CARETNH LFDYWGLGTLTVTSS 24

Chicken VL 349 ALTQPASVSANLGGTVEITCSGGSSYYSSYYGW
YQQKSPGSALVTVIYSDDKRPSGIPSRFSGSAS
GSTATLTITGVRAEDEAVYYCGGYDYSSYTND FGAGTTTLTVL 26 Human VH 350
DVQLVESGGGVVRPGESELRRLSCAASGFTFSTY
AMSWVRQAPGKGLEWVSGISASGSGTYYGDS
VKGRFTMSRDNSKNTLYLQMNSLTAEDTAVY YCARETWNHLFDYWGLGTLVTVSS 26
Chicken VL 351 ALTQPASVSANLGGTVEITCSGGSSYYGWYQ
QKSPGSAPVTVIYSDNQRPSDIPSRFSGSASDST
ATLTITGVQVEDEAIYYCGGYDSSTYANTFGA GTTTLTVL 28 Human VH 352
DVQLVESGGGVVRPGESELRRLSCAASGFSFSSN
AMSWVRQAPGKGLEWVAGISASGDTYYSGSM
KGRFTISRDNNSKNTLYLQMNSLTAEDTAVYYC ARETWNHLFDYWGLGTLVTVSS 28
Chicken VL 353 ALTQPASVSANPGETVKITCSGGSDSYYYGWH
QQKSPGSAPVTVIYSDDQRPPDIPSRFSGSASGS
TTTLTITGVRAEDEAVYYCGGYDYSTYTNTFG AGTTLTVL 29 Human VH 354
DVQLVESGGGVVRPGESELRRLSCAVSGFRFSSY
AMSWVRQAPGKGLEWVSGISSDSDAYYVDSV
KGRFTISRDNNSKNTLYLQVNSLTAEDTAVYYC ARETWNHLFDYWGLGTMVTVSS 29
Chicken VL 355 ALTQPASVSANLGETVKITCSGGDSSSHYYGW
FQQKSPGSAPVTLIYSDDERPSDIPSRFSGSASG
STATLTITGVRVEDEAIYFCGAYDGSTYTNTFG AGTTLTVL 30 Human VH 356
DVQLVESGGGVVRPGESELRRLSCEASGFTFSSDA
MSWVRQAPGKGLEWVSGISSGSSTYYGGSVK
GRFTISRDNNSKNTLYLQMNSLTAEDTAVYYCA RETWNHLFDYWGLGTLVTVSS 30
Chicken VL 357 ALTQPASVSASPGETVEITCSGGSDSSYYYGW
YQQKSPGSAPVTVIYSDNKRPSNIPSRFSGSASG
STATLTITGVRVEDEAVYYCGGYDYSTYTNPFGAGTTLTVL 55 Human VH 358
DVQLVESGGGVVRPGESELRRLSCAVSGFRFSSY
AMSWVRQAPGKGLEWVSGISSGGDTYYVDSV
KGRFTISRDNNSKNTLYLQVNSLTAEDTAIYYCA RETWNHLFDYWGLGTLVTVSS 55
Chicken VL 359 ALTQPASVSANLGETVEITCSGGDSSSHYYGW
YQQKSPGSAPVTVIYSDDERPSDIPSRFSGSASG
STATLTITGVRVEDEAIYYCGAYDGSTYANTF GAGTTLTVL 56 Human VH 360
DVQLVESGGGVVRPGESELRRLSCAASGFTFSSY
AMSWVRQAPGKGLEWLAGISAGGSDTYIDS
VKGRFTISRDNNSKNSLYLQVNSLTAEDTAVYY CARETWNHLFDYWGLGTLVTVSS 56
Chicken VL 361 ALTQPASVSANPGETVKITCSGGGYSTYYGW
YQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG
SAATLTITGVRVEDEAVYYCAGYDGRTYINTF GAGTTLTVL 59 Human VH 362
DVQLVESGGGVVRPGESELRRLSCAVSGFRFSSH
AMSWVRQAPGKGLEWVSGISSGGDTYYVDSV
KGRFTISRDNNSKNTLYLQVNSLTAEDTAIYYCA RETWNHLFDYWGLGTLVTVSS 59
Chicken VL 363 ALTQPASVSANLGETVKITCSGGDSSSHYYGW
YQQKSPGSAPVTVIYSDDERPSDIPSRFSGSASG
STATLTITGVRVEDEAIYYCGAYDGSTYANTF GAGTTLTVL 60 Human VH 364
DVQLVDSGGGVVRPGESELRRLSCAASGFTFSSY
AMSWVRQAPGKGLEWLAGISAGGSDTYIDS
VKGRFTISRDNNSKNSLYLQMNSLTAEDTAVYY CARETWNHLFDYWGLGTLVTVSS 60
Chicken VL 365 ALTQPASVSANPGETVKITCSGGDYSTYYGW
YQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG

SAATLTITGVRVEDEAVYFCGAYDGRTYINTF GAGTTTLTVL 65 Human VH 366
 DVQLVESGGGVVVRPGESLRLSCAASGFTFSSY
 AMSWVRQAPGKGLEWLAGISAGGSDTYIDS
 VKGRFTISRDN SKNSLYLQMNSLTAEDTAVYY CARETNH LFDYWGLGTLVTVSS 65
 Chicken VL 367 ALTQPASVSANPGETVKITCSGGDYSTYYGW
 YQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG
 SAATLTITGVRVEDEAVYFCGAYDGRTYINTF GAGTTTLTVL 69 Human VH 368
 DLQLVESGGGVVVRPGESLRLSCAASGFTFSSYA
 MSWVRQAPGKGLEWLAGISAGGSDTYIDSV
 KGRFTISRDN SKNSLYLQMNSLTAEDTAVYYC ARETNH LFDYWGLGTLVTVSS 69
 Chicken VL 369 ALTQPASVSANPGETVKIICSGGDYSTYYGW
 YQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG
 SAATLTITGVRVEDEAVYFCGAYDGRTYINTF GAGTTTLTVL 70 Human VH 370
 DVQLVESGGGVVVRPGESLRLSCAASGFTFSSY
 AMSWVRQAPGKGLEWLAGISAGGSDAYIDS
 VKGRFTISRDN SKNSLYLQMNSLTAEDTAVYY CARETNH LFDYWGLGTLVTVSS 70
 Chicken VL 371 ALTQPASVSANPGETVKITCSGGDWYSTYYG
 WYQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSA
 SGSAATLTITGVRVEDEAVYYCAGYDGRTYIN TFGAGTTTLTVL 71 Human VH 372
 DVQLVESGGGVVVRPGESLRLSCAASGFTFSSY
 AMSWVRQAPGKGLEWLAGISAGGSDTYIDS
 VKGRFTISRDN SKNSLYLQMNSLTAEDTAVYY CARETNH LFDYWGLGTLVTVSS 71
 Chicken VL 373 ALTQPASVSANPGETVKITCSGGGYSTYYGW
 YQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSASG
 SAATLTITGVRVEDEAVYYCAGYDARTYINTF GAGTTTLTVL 73 Human VH 374
 DVQLVESGGGVVVRPGESLRLSCEASGFTFSSNA
 MSWARQAPGKGLEWVAGISSGSDTYYGDSVK
 GRLTISRDN SKNLYLQMNSLTAEDTAVYYCA RETNHLFDYWGLGTLVTVSS 73
 Chicken VL 375 ALTQPASVSANPGETVKITCSGGIYSSYYAW
 YQQKSPGSAPVTLIYSDDKRPSNIPSRFSGSASG
 STATLTITGVRAEDEAVYFCGGYDQSSYTNPF GAGTTTLTVL 74 Human VH 376
 DVQLVESGGGVVVRPGESLRLSCAASGFTFSSN
 AMSWVRQAPGKGLEWLAGISAGDSDTYYPAS
 VKGRFTISRDN PKNTLYLQMNLTAEDEAVYY CARETNH LFDYWGLGTLVTVSS 74
 Chicken VL 377 ALTQPASVSANPGETVKIACSGGSYYSYYGW
 YQQKSPGSALVTVIYSDDKRPSGIPSRFSGSAS
 GSTATLTITGVRAEDEAVYYCGGYDYSSYTND FGAGTTTLTVL 76 Human VH 378
 DVQLVESGGGVVVRPGESLRLSCAASGFTFSSY
 AMSWVRQAPGKGLEWLAGISAGGSDTYIDS
 VKGRFTISRDN SKNSLYLQMNSLTAEDTAVYY CARETNH LFDYWGLGTLVTVSS 76
 Chicken VL 379 ALTQPASVSANPGETVKITCSGGDWYSTYYG
 WYQQKSPGSAPVTVIHSDDKRPSDIPSRFSGSA
 SGSAATLTITGVRVEDEAVYYCAGYDGRTYIN TFGAGTTTLTVL 201 Human VH 380
 DVQLVESGGAVVRPGETLRLSCTASGFTFSSY
 AMSWVRQAPGKGLEWVSGISASGSDTYADS
 VKGRSTISRDN SKNTLYLRMSSLTAEDTAVYY CARETNH LFDYWGLGTLVTVSS 201
 Chicken VL 381 ALTQPASVSANPGETVKITCSGGGTSSYYGWY
 QQKSPGSAPVTLIHSDDKRPSDIPSRFSGSASGS
 TATLTITGVQVEDEAVYYCGGYDYTTYVNTFG AGTTTLTVL 202 Human VH 382
 DVQLVESGGGVVVRPGESLRLSCAASGFTFSSY

AMSWVRQAPGKLEWLAGISAGGSDTYIDS
VKGRFTISRDNSKNSLYLQMNSLTAEDTAVYY CARETWNHLFDYWGLGTLVTVSS 202
Chicken VL 383 ALTQPASVSANPGETVKITCSGGGYSTYYGW
YQQRSPGSAPVTVIHSDDKRPSDIPSRFSGSASG
SAATLTITGVRVEDEAVYYCAGYDGRTYLNTF GAGTTLTVL 206 Human VH 384
DVQLVESGGAVVRPGETLRLSCTASGFTFSSY
AMSWVRQAPGKGLEWVSGISASGSDTYADS
VKGRSTISRDNSKNTLYLRMSSLTAEDTAVYY CARETWNHLFDYWGLGTLVTLSS 206
Chicken VL 385 ALTQPASVSANPGETVKITCSGGGTSSYYGWY
QQKSPGSAPVTLIHSDDKRPSDIPSRFSGSASGS
TATLTITGVQAEDEAVYYCGGYDYTTYVNTFG AGTTLTVL 175 Human VH 386
DVQLVESGGGVVRPGESLRLSCAASGFTFSTD
MNWVRQAPGEGLEWVSLISGSGEITYYADSVK
GRFTISRDNSKNTLYLQMNSLRAEDTAVYYCA KENDRYRFFDYWGLGTLVTVSS 175
Human VL 387 ETVLTQSPGILSLSPGERATLTCRASQSVYTYL
AWYQEKPGQAPRLLIYGASSRAAGIPDRFSGS
GSGTEFTLTISLQSEDFAVYYCQQYYDRPPLT FGGGTKVEIK 177 Human VH 388
DVQLVESGGGVVRPGESLRLSCAASGFTFSSY
DMNWVRQAPGEGLEWVSLISGSGEIIYYADSV
KGRFTISRDNSKNTLYLQMNSLRAEDTAVYYC AKENNRYYRFFDYWGLGTLVTVSS 177
Human VL 389 KTVLTQSPGTLSLSPGERATLTCRADQSVYTYL
AWYQERPGQAPRLLIYDASSRATGIPDRFSGSG
SGTEFTLTISLQSEDFAVYYCQQYYDRPPLTF GGGGTKVEIK 178 Human VH 390
DVQLVESGGGVVRPGESLRLSCAASGFTFSSY
DMNWVRQAPGEGLEWVSLISGSGEIIYYADSV
KGRFTISRDNSKNTLYLQMNSLRAEDTAVYYC AKENNRYYRFFDYWGLGTLVTVSS 178
Human VL 391 ETVLTQSPGTTLTLSPGERATLSCRASQSVYTYL
AWYQEKPGQAPRLLIYGASSRATGVPDRFSGS
GSGTEFTLTISLQSEDFAVYYCQQYYDRPPLT FGGGTKVEIK 180 Human VH 392
DVQLVESGGGVVRPGESLRLSCAASGFTFSSY
DMNWVRQAPGEGLEWVSLISGSGEIIYYADSV
KGRFTISRDNSKNTLYLQMNSLRAEDTAVYYC AKENDRYRFFDYWGLGTLVTVSS 180
Human VL 393 ETVLTQSPGTTLTLSPGERATLTCRASQSVYTYL
AWYQEKPGQAPRLLIYGASSRATGIPDRFSGSG
SGTEFTLTISLQSEDFAVYYCQQYYDRPPLTF GGGGTKVEIK 184 Human VH 394
DVQLVESGGGVVRPGESLRLSCAASGFTFSSY
DMNWVRQAPGEGLEWVSLISGSGEIIYYADSV
KGRFTISRDNSKNTLYLQMNSLRAEDTAVYYC AKENNRYYRFFDYWGLGTLVTVSS 184
Human VL 395 ETVLTQSPGTLSLSPGERATLNCRASQSVYTYL
AWYQEKPGQAPRLLIYDASSRATGIPDRFSGSG
SGTEFTLTISLESEDFAVYYCQQYYDRPPLTF GGGGTKVEIK 185 Human VH 396
DVQLVESGGGVVRPGESLRLSCAASGFTFSSY
DMNWVRQAPGEGLEWVSLISGSGEIIYYADSV
KGRFTISRDNSKNTLYLQMNSLRAEDTAVYYC AKENNRYYRFFDDWGLGTLVTVSS 185
Human VL 397 ETVLTQSPGTLSLSPGERATLNCRASQSVYSYL
AWYQERPGQAPRLLIYGASTRATGIPDRFSGSG
SGTEFTLTISLESDDFAVYYCQQYYDRPPLTF GGGGTKVEIK 189 Human VH 398
DVQLVESGGGVVRPGESLRLSCAASGFTFSSSD
MNWVRQAPGEGLEWVSLISGSGEITYYADSVK
GRFTISRDNSKNTLYLQMNSLRAEDTAIYYCA KENNMYYRFFDYWGLGTLVTVSS 189

Human VL 399 ETVLTQSPGTLTSLSPGERATLSCRASQSVYTYL
AWYQQKPGQPPRLLIHAARNRAAGIPDRFSGS
GSGTEFTLTISLQSEDAVYYCQQYYDRPPLT FGGGTKVEIK 190 Human VH 400
DVQLVESGGGVVRPGESELRRLSCAASGFTFSSY
DMNWVRQAPGEGLEWVSLISGSGEIIYYADSV
KGRFTISRDN SKNTLYLQMNSLRAEDTAVYYC AKEDNRYRFFDYWGLGALVTVSS 190
Human VL 401 ETVLTQSPGTLTSLSPGERTTLTLCRASQSVYTYL
AWYQEKPGQAPRLLIYGASSRATGIPDRFSGSG
SGTEFTLTISLQSEDAVYYCQQYYDRPPLTF GGGTKVEIK 193 Human VH 402
DVQLVESGGGVVRPGESELRRLSCAASGFTFSSY
DMNWVRQAPGEGLEWVSLISGSGEITYYADSV
KGRFTISRDN SKNTLYLQMNSLRAEDTAVYYC AKENNRYRFFDYWGLGTLVTVSS 193
Human VL 403 ETVLTQSPGTLTSLSPGERATLSCRASQSVYTYL
AWYQEKPGQAPRLLIYAASTRATGIPDRFSGSG
SGTEFTLTISLQSEDAVYYCQQYYDRPPLTF GGGTKVEIK 191 Human VH 404
DVQLVESGGGVVRPGESELRRLSCAASGFSFSSY
AMNWVRQAPGEGLEWVSRINSGGGGTDYAES
VKGRFTISRDNSENTLYLQMNSLRAEDTAVYY CAKQYDWNGFFDYWGLGALVTVSS 191
Human VL 405 ETVLTQSPATLSVSPGERATLSCRASQTVGSKL AWHQQKPGQAPRLLIYD
ATNRATGIPDRFSGSGSGTDFTLTISGLQTEDSA VYYCQQYYYWPPYRFGGGTKVEIK 198
Human VH 406 DVQLVESGGGVVRPGESELRRLSCAASGFSFSSH
AMNWVRQAPGEGLEWVSRINSGGGGTDYAES
VKGRFTISRDNSENTLYLQMNSLRAEDTAVYY CAKQYDWNGFFDYWGLGALVTVSS 198
Human VL 407 ETVLTQSPATLSVSPGERATLSCRASQTVGSKL AWHQQKPGQAPRLLIYD
ATNRATGIPDRFSGSGSGTDFTLTISLQTEDSA VYYCQQYYYWPPYRFGGGTKVEIK 163
Human VH 408 DVQLVESGGGLVRPGESELRRLSCAASGFTFTNY
AVTWVRQAPGEGLEWVSLISVTGTTYADSV
KGRFTISRDN SKSTLYLQMNGLRAEDTAVYYC AKVTWKNVFDYWGLGTLVTVSS 163
Human VL 409 EIVLTQSPGTLTSLSPGERATFSCRASQTVGSKL
AWYQQKPGQAPRLLIYDANTRATGIPARFSGS
GSGTEFTLTISLQSEDAVYYCQQHTDWPPYT FGGGTKVEIK 164 Human VH 410
DVQLVESGGGLVRPGESELRRLSCAASGFTFTNY
AVTWVRQAPGEGLEWVSLISVTGTTYADSV
KGRFTISRDN SKSTLYLQMNGLRAEDTAVYYC AKVTWKNVFDYWGLGTLVTVSS 164
Human VL 411 EIVLTQSPGTLTSLSPGERATFSCRASQTVGSKL
AWYQQKPGQAPRLLIYDANTRATGIPARFSGS
RSGTEFTLTISLQSEDAVYYCQQHTDWPPYT FGGGTKVEIK 174 Human VH 412
DVQLVESGGGVVRPGESELRRLSCAASGFTFSDH
DMSWVRQGPGEGLWVAGISLRGGVTWYAD
SVKGRFTISRDN SKNTLYLRLFSLRTEDTAIYY CARESWNTFFDYWGLGTLVTVSS 174
Human VL 413 EIVLTQSPGTLTSLSPGETATLSCRASQNVRNL
AWYQQKPGQAPRLLIYDASSRATGIPDRFSGS
GSGTDFTLTISLQSEDAVYYCQQYGNGPPLT FGGGTKVEIK 214 Human VH 414
DVQLVESEGGVVRPGESELRRLSCEASGFTFRNSY
MTWVRQAPGEGLEWVSTISDTGDTAYYADSV
KGRFTISRDN SKNSLYLQMNSLRAEDTAIYYC AKSWIWWTFFDYWGLGTLVTVSS 214
Chicken VL 415 ALTQPASVSANLGGTVEITCSGGNSNHYGWYQ
QKSPGSAPVTLIYADTNRPSNIPSRFSGSTSGST
STLTITGVQAEDEAVYYCGGSSTGDGIFGAGTT LTVL Hum8 Humanized VL 416
SYELTQPPSVSVSPGQTARITCSGGAYSSYYA

WYQQKPGQAPVLIYSDSKRPSGIPERFSGSSS
 GTTVTLTISGVQAEDEADYYCGGYDQSSYTNP FGGGTKLTVL 136_MutAll(Light Human
 with VL 417 EIVLTQSPGTLTLSPGERATLSCRASQSVYTYL chain)_S12T D1E, R13Q,
 AWYQQKPGQAPRLLIYGA E16R, E43K, SSRATGIPDRFSGSGSGIEFTLTISLQSEDFAV
 and L111Q YYCQQYYDRPPLTFGGGTKVEIK mutations 136_MutAll(Light Human with VL
 418 EIVLTQSPGTLTLSPGERATLTLSCRASQSVYTYL chain)_S22T germline
 AWYQQKPGQAPRLLIYGA back- SSRATGIPDRFSGSGSGIEFTLTISLQSEDFAV mutations
 YYCQQYYDRPPLTFGGGTKVEIK 136_MutAll(Light Human with VL 419
 EIVLTQSPGTLTLSPGERATLSCRASQSVYTYL chain)_Q38E germline
 AWYQEKPQAPRLLIYGA back- SSRATGIPDRFSGSGSGIEFTLTISLQSEDFAV mutations
 YYCQQYYDRPPLTFGGGTKVEIK 119_wt_M34L Human with VH 420
 DVQLLES GGGVVP GGSLRLSCAASGFSFSNF liability
 ALTWVRQAPGEGLEWVSTIGSGDTYYADSVK mutation GRFTISR
 DNSKNTLYLQMNSLRAEDTAVYYCAKDSTVS WSGDFFDYWGLGTLTVSS
 119_wt_M34V Human with VH 421 DVQLLES GGGVVP GGSLRLSCAASGFSFSNF liability
 AVTWVRQAPGEGLEWVSTIGSGDTYYADSVK mutation GRFTISR
 DNSKNTLYLQMNSLRAEDTAVYYCAKDSTVS WSGDFFDYWGLGTLTVSS 135_wt_M34L
 Human with VH 422 DVQLVES GGGVVRP GESLRLSCAASGFSFSIYA liability
 LSWVRQAPGEGLEWVSTIGADDTYYADSVKG mutation RFTISR
 DNSKNTLYLQMNSLRAEDTAVYYCAKDSTVG WSGDFFDYWGLGTLTVSS
 135_wt_M34V Human with VH 423 DVQLVES GGGVVRP GESLRLSCAASGFSFSIYA
 liability VSWVRQAPGEGLEWVSTIGADDTYYADSVKG mutation RFTISR
 DNSKNTLYLQMNSLRAEDTAVYYCAKDSTVG WSGDFFDYWGLGTLTVSS
 136_wt_M34L Human with VH 424 DVQLVES GGGVVRP GESLRLSCAASGFTFSSY liability
 DLNWRQAPGEGLEWVSLISGSGEIYYADSV mutation KGRFTI
 SRDNSKNTLYLQMNSLRAEDTAVYYCAKENN RYRFFDDWGLGTLTVSS 136_wt_M34V
 Human with VH 425 DVQLVES GGGVVRP GESLRLSCAASGFTFSSY liability
 DVNWRQAPGEGLEWVSLISGSGEIYYADSV mutation KGRFTI
 SRDNSKNTLYLQMNSLRAEDTAVYYCAKENN RYRFFDDWGLGTLTVSS 119 Human
 HVR-H1 175 NFAMT HVR-H1

(158) In some embodiments, an antibody of the present disclosure binds to a human SIRP- α polypeptide at one or more residues. It is to be understood that residues of a SIRP- α polypeptide that are bound by an antibody of the present disclosure may be described with respect to a reference SIRP- α polypeptide, but this description is not limited to a single SIRP- α polypeptide (i.e., the reference SIRP- α polypeptide). Rather, specific amino acid residues of a reference SIRP- α polypeptide are described to identify corresponding amino acid positions that can be identified on various SIRP- α polypeptides. For example, specific residues that correspond to one or more amino acid positions of SEQ ID NO:296 can be identified for various human SIRP- α polypeptides, such as v1 and/or v2. Since the amino acid sequences of SEQ ID NO:296 and SEQ ID NO:5 differ only at the N80 position (excepting a small number of C-terminal residues of SEQ ID NO:296 useful for protein production and purification), one of skill in the art will appreciate that references herein to amino acid positions with respect to the amino acid sequence of SEQ ID NO:296 will correspond to the same positions in the amino acid sequence of SEQ ID NO:5. Techniques for determining the residues of a SIRP- α polypeptide bound by an antibody are known in the art; exemplary and non-limiting descriptions are provided in Example 4.

(159) TABLE-US-00008 SEQ ID NO: 296 EEELQVIQPD KSVLVAAGET
 ATLRCTATSL IPVGPIQWFR GAGPGRELIY NQKEGHFPRV TTVSDLTKRN
 NMDFSIRIGA ITPADAGTY C V KFRK GSPD DVEFKSGAGT ELSVRAKPST
 RHHHHHHH

(160) In some embodiments, an antibody of the present disclosure binds to a human SIRP- α v1

polypeptide at one or more amino acid positions selected from I31, V33, Q52, K53, T67, R69, N70, and K96, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at one or more amino acid positions selected from L30, P32, E54, T62, N71, M72, F74, and R95, according to SEQ ID NO:296.

(161) In some embodiments, an antibody of the present disclosure binds to a human SIRP- α v1 polypeptide at one or more amino acid positions selected from 17, P9, D10, K11, S12, A42, A108, and E111, according to SEQ ID NO:296. In some embodiments, the antibody binds to the human SIRP- α v1 polypeptide at K11, A42, A108, and E111, according to SEQ ID NO:296. In some embodiments, the antibody binds to the human SIRP- α v1 polypeptide at 17, P9, D10, K11, S12, A108, and E111, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at one or more amino acid positions selected from L14, T26, T28, T88, Y90, S106, S113, and A116, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at L14, T88, Y90, S106, S113, and A116 of human SIRP- α v1, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at L14, T26, and T28, according to SEQ ID NO:296.

(162) In some embodiments, an antibody of the present disclosure binds to a human SIRP- α v1 polypeptide at one or more amino acid positions selected from E47, L48, P58, R59, T82, and A84, according to SEQ ID NO:296. In some embodiments, the antibody further binds to the human SIRP- α v1 polypeptide at one or more amino acid positions selected from A17, P44, G45, 149, E54, G55, H56, F57, and P83, according to SEQ ID NO:296.

(163) In some embodiments, an antibody of the present disclosure binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide with a dissociation constant ($K_{sub.D}$) of less than 100 nM, less than 50 nM, or less than 30 nM. In some embodiments, the antibody blocks binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide, and the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide with a dissociation constant ($K_{sub.D}$) of less than 100 nM, less than 50 nM, or less than 30 nM. In some embodiments, the antibody binds the D1 domain of a human SIRP- α v1 polypeptide and the D1 domain of a human SIRP- α v2 polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- γ polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide.

(164) In some embodiments, an antibody of the present disclosure binds the D1 domain of a human SIRP- α polypeptide and does not block binding between an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide. In some embodiments, the antibody binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α v1 polypeptide with a dissociation constant ($K_{sub.D}$) of less than 100 nM, less than 50 nM, or less than 30 nM. In some embodiments, the antibody binds the D1 domain of a human SIRP- α v1 polypeptide. In some embodiments, the antibody binds the D1 domain of a human SIRP- α v1 polypeptide and the D1 domain of a human SIRP- α v2 polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a cynomolgus SIRP- α polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a human SIRP- β polypeptide. In some embodiments, the antibody binds an extracellular domain (e.g., the D1 domain) of a murine SIRP- α polypeptide.

(165) In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody. Techniques for determining whether an antibody competes for binding the

extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with a reference anti-SIRP- α antibody are known in the art; exemplary and non-limiting descriptions are provided in Example 5.

(166) In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more reference anti-SIRP- α antibodies selected from 119, 120, 121, 122, 21, 25, 27, 66, and 135. In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more of the following reference anti-SIRP- α antibodies: (a) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:120 and a VL domain comprising the amino acid sequence of SEQ ID NO:97, (b) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:121 and a VL domain comprising the amino acid sequence of SEQ ID NO:98, (c) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:130 and a VL domain comprising the amino acid sequence of SEQ ID NO:107, (d) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:122 and a VL domain comprising the amino acid sequence of SEQ ID NO:99, (e) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:135 and a VL domain comprising the amino acid sequence of SEQ ID NO:136, (f) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:137 and a VL domain comprising the amino acid sequence of SEQ ID NO:138, (g) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:139 and a VL domain comprising the amino acid sequence of SEQ ID NO:140, (h) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:141 and a VL domain comprising the amino acid sequence of SEQ ID NO:142, and (i) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:127 and a VL domain comprising the amino acid sequence of SEQ ID NO:104.

(167) In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more reference anti-SIRP- α antibodies selected from 136 and 137. In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more of the following reference anti-SIRP- α antibodies: (a) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:133 and a VL domain comprising the amino acid sequence of SEQ ID NO:134, and (b) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:128 and a VL domain comprising the amino acid sequence of SEQ ID NO:105.

(168) In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more reference anti-SIRP- α antibodies selected from 3, 213, 173, and 209. In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more of the following reference anti-SIRP- α antibodies: (a) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:242 and a VL domain comprising the amino acid sequence of SEQ ID NO:243, (b) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:275 and a VL domain comprising the amino acid sequence of SEQ ID NO:276, (c) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:278 and a VL domain comprising the amino acid sequence of SEQ ID NO:279, and (d) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:280 and a VL domain comprising the amino acid sequence of SEQ ID NO:281.

(169) In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more reference anti-SIRP- α antibodies selected from 115, 116, 117, 118, and 132. In some embodiments,

an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more of the following reference anti-SIRP- α antibodies: (a) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:116 and a VL domain comprising the amino acid sequence of SEQ ID NO:93, (b) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:117 and a VL domain comprising the amino acid sequence of SEQ ID NO:94, (c) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO: 118 and a VL domain comprising the amino acid sequence of SEQ ID NO:95, (d) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:119 and a VL domain comprising the amino acid sequence of SEQ ID NO:96, and (e) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:282 and a VL domain comprising the amino acid sequence of SEQ ID NO:283.

(170) In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more reference anti-SIRP- α antibodies selected from 218, 123, 149, 161, 162, and 194. In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with one or more of the following reference anti-SIRP- α antibodies: (a) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:284 and a VL domain comprising the amino acid sequence of SEQ ID NO:285, (b) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:123 and a VL domain comprising the amino acid sequence of SEQ ID NO:100, (c) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:286 and a VL domain comprising the amino acid sequence of SEQ ID NO:287, (d) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:288 and a VL domain comprising the amino acid sequence of SEQ ID NO:289, (e) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:290 and a VL domain comprising the amino acid sequence of SEQ ID NO:291, and (f) an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:292 and a VL domain comprising the amino acid sequence of SEQ ID NO:293.

(171) In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with reference anti-SIRP- α antibody 45. In some embodiments, an antibody of the present disclosure competes for binding the extracellular domain (e.g., the D1 domain) of the human SIRP- α polypeptide with an antibody comprising a VH domain comprising the amino acid sequence of SEQ ID NO:244 and a VL domain comprising the amino acid sequence of SEQ ID NO:245.

(172) The present disclosure provides multiple families of anti-SIRP- α antibodies, each family comprising multiple antibodies. As demonstrated herein, antibodies within a given family may share certain structural properties (e.g., similar or identical HVR or CDR sequences) as well as one or more functional properties, including but not limited to binding affinity to human, monkey, and/or mouse SIRP- α polypeptide(s), binding affinity to SIRP-0 polypeptides, binding affinity to SIRP- γ polypeptides, mode of binding to SIRP- α (e.g., CD47 blocking, CD47 non-blocking, or “kick off” binding), induction of phagocytosis (e.g., in an in vitro assay), activation of dendritic cells, anti-tumor efficacy, SIRP- α binding epitope residue(s) or “bin” (e.g., as determined by a binning assay), and the like (see, e.g., Tables P-T). Because of these shared properties, it is contemplated that HVR and/or VH or VL sequences of antibodies belonging to the same family can be interchanged or intermingled, such that an anti-SIRP- α antibody can comprise HVR and/or VH or VL derived from more than one specific an anti-SIRP- α antibody described herein. As discussed in greater detail herein, various methodologies for determining HVR or CDR sequences of an antibody variable domain are known in the art and can be used interchangeably herein, including without limitation the Kabat, Chothia, and IMGT definitions, as well as combinations thereof.

(173) In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs:120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs:120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs:120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs:97, 98, 107, 99, 104, and 312; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312; and/or an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312. In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs:120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; and an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328. In some embodiments, an antibody of the present disclosure comprises an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312; and an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312. In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs:120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 120, 246, 258, 327, 121, 130, 122, 127, 247, 259, 335, and 328; an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312; and an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 97, 98, 107, 99, 104, and 312.

(174) In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs:135, 137, 265, 266, 331, 139, 267, 332, 141, 263, 264, 268, 330, 294, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, and 384; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 135, 137, 265, 266, 331, 139, 267, 332, 141, 263, 264, 268, 330, 294, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, and 384; an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 135, 137, 265, 266, 331, 139, 267, 332, 141, 263, 264, 268, 330, 294, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, and 384; an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs:136, 138, 140, 142, 252, 254, 262, 416, 295, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, and 385; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 136, 138, 140, 142, 252, 254, 262, 416, 295, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, and 385; and/or an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 136, 138, 140, 142, 252, 254, 262, 416, 295, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, and 385. In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of

SEQ ID NOs: 135, 137, 265, 266, 331, 139, 267, 332, 141, 263, 264, 268, 330, 294, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, and 384; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 135, 137, 265, 266, 331, 139, 267, 332, 141, 263, 264, 268, 330, 294, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, and 384; and an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 135, 137, 265, 266, 331, 139, 267, 332, 141, 263, 264, 268, 330, 294, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, and 384. In some embodiments, an antibody of the present disclosure comprises an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 136, 138, 140, 142, 252, 254, 262, 416, 295, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, and 385; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 136, 138, 140, 142, 252, 254, 262, 416, 295, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, and 385; and an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 136, 138, 140, 142, 252, 254, 262, 416, 295, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, and 385. In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 135, 137, 265, 266, 331, 139, 267, 332, 141, 263, 264, 268, 330, 294, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, and 384; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 135, 137, 265, 266, 331, 139, 267, 332, 141, 263, 264, 268, 330, 294, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, and 384; an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 135, 137, 265, 266, 331, 139, 267, 332, 141, 263, 264, 268, 330, 294, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, and 384; an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 136, 138, 140, 142, 252, 254, 262, 416, 295, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, and 385; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 136, 138, 140, 142, 252, 254, 262, 416, 295, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, and 385; and an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 136, 138, 140, 142, 252, 254, 262, 416, 295, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, and 385.

(175) In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 133, 249, 260, 329, 128, 386, 388, 390, 392, 394, 396, 398, 400, and 402; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 133, 249, 260, 329, 128, 386, 388, 390, 392, 394, 396, 398, 400, and 402; an HVR-H3 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 133, 249, 260, 329, 128, 386, 388, 390, 392, 394, 396, 398, 400, and 402; an HVR-L1 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 134, 251, 105, 250, 417, 418, 419, 387, 389, 391, 393, 395, 397, 399, 401, and 403; an HVR-L2 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 134, 251, 105, 250, 417, 418, 419, 387, 389, 391, 393, 395, 397, 399, 401, and 403; and/or an HVR-L3 from a VL domain sequence selected from the amino acid sequences of SEQ ID NOs: 134, 251, 105, 250, 417, 418, 419, 387, 389, 391, 393, 395, 397, 399, 401, and 403. In some embodiments, an antibody of the present disclosure comprises an HVR-H1 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 133, 249, 260, 329, 128, 386, 388, 390, 392, 394, 396, 398, 400, and 402; an HVR-H2 from a VH domain sequence selected from the amino acid sequences of SEQ ID NOs: 133, 249, 260, 329, 128, 386, 388, 390, 392, 394,

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some embodiments, an antibody of the present disclosure comprises one, two, or three heavy chain HVR sequences from a VH domain comprising the amino acid sequence of SEQ ID NO:284 and/or one, two, or three light chain HVR sequences from a VL domain comprising the amino acid sequence of SEQ ID NOs:285, 333, or 334. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:284 and/or a VL domain comprising the amino acid sequence of SEQ ID NOs: 285, 333, or 334.

(188) In some embodiments, an antibody of the present disclosure comprises (a) a heavy chain variable (VH) domain comprising (i) an HVR-H1 sequence comprising the amino acid sequence of NFAMT (SEQ ID NO:175), NFAVT (SEQ ID NO:204), or NFALT (SEQ ID NO:305), (ii) an HVR-H2 sequence comprising the amino acid sequence of TIGSGDTYYADSVKG (SEQ ID NO:144), and (iii) an HVR-H3 sequence comprising the amino acid sequence of DSTVSWGDFFDY (SEQ ID NO:145); and/or (b) a light chain variable (VL) domain comprising (i) an HVR-L1 sequence comprising the amino acid sequence of RASQNVKNDLA (SEQ ID NO:146), (ii) an HVR-L2 sequence comprising the amino acid sequence of AARIRET (SEQ ID NO:147), and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDWPPFT (SEQ ID NO:148). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:120, 246, 258, or 327; and/or the VL domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:97 or 312. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:246, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:258, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:120, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:327, and the VL domain comprises the amino acid sequence of SEQ ID NO:97; the VH domain comprises the amino acid sequence of SEQ ID NO:246, and the VL domain comprises the amino acid sequence of SEQ ID NO:312; the VH domain comprises the amino acid sequence of SEQ ID NO:258, and the VL domain comprises the amino acid sequence of SEQ ID NO:312; the VH domain comprises the amino acid sequence of SEQ ID NO:120, and the VL domain comprises the amino acid sequence of SEQ ID NO:312; or the VH domain comprises the amino acid sequence of SEQ ID NO:327, and the VL domain comprises the amino acid sequence of SEQ ID NO:312.

(189) In some embodiments, an antibody of the present disclosure comprises (a) a heavy chain variable (VH) domain comprising (i) an HVR-H1 sequence comprising the amino acid sequence of IYAMS (SEQ ID NO:269), IYAVS (SEQ ID NO:213), or IYALS (SEQ ID NO:306), (ii) an HVR-H2 sequence comprising the amino acid sequence of TIGADDTYYADSVKG (SEQ ID NO:150), and (iii) an HVR-H3 sequence comprising the amino acid sequence of DSTVGWSGDFFDY (SEQ ID NO:151); and/or (b) a light chain variable (VL) domain comprising (i) an HVR-L1 sequence comprising the amino acid sequence of RASQNVRS DIA (SEQ ID NO:152), (ii) an HVR-L2 sequence comprising the amino acid sequence of AASSRDT (SEQ ID NO:153), and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDWPPFT (SEQ ID NO:148). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:341, 127, 247, 259, or 328; and/or the VL domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:104 or 248. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:127, and the VL

domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:341, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:247, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:259, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:328, and the VL domain comprises the amino acid sequence of SEQ ID NO:104; the VH domain comprises the amino acid sequence of SEQ ID NO:127, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; the VH domain comprises the amino acid sequence of SEQ ID NO:341, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; the VH domain comprises the amino acid sequence of SEQ ID NO:247, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; the VH domain comprises the amino acid sequence of SEQ ID NO:259, and the VL domain comprises the amino acid sequence of SEQ ID NO:248; or the VH domain comprises the amino acid sequence of SEQ ID NO:328, and the VL domain comprises the amino acid sequence of SEQ ID NO:248.

(190) In some embodiments, an antibody of the present disclosure comprises (a) a heavy chain variable (VH) domain comprising: (i) an HVR-H1 sequence comprising the amino acid sequence of X.sub.1X.sub.2DX.sub.3N, wherein X.sub.1 is S or T; X.sub.2 is Y or S; and X.sub.3 is M, L, or V (SEQ ID NO:307); (ii) an HVR-H2 sequence comprising the amino acid sequence of LISGSGEIX.sub.1YYADSVKG, wherein X.sub.1 is I or T (SEQ ID NO:308); and (iii) an HVR-H3 sequence comprising the amino acid sequence of EX.sub.1X.sub.2X.sub.3YRFFDX.sub.4, wherein X.sub.1 is N or D; X.sub.2 is N or D; X.sub.3 is R or M; and X.sub.4 is D or Y (SEQ ID NO:309); and/or (b) a light chain variable (VL) domain comprising: (i) an HVR-L1 sequence comprising the amino acid sequence of RAX.sub.1QSVYX.sub.2YLA, wherein X.sub.1 is S or D; and X.sub.2 is T or S (SEQ ID NO:310); (ii) an HVR-L2 sequence comprising the amino acid sequence of X.sub.1AX.sub.2X.sub.3RAX.sub.4, wherein X.sub.1 is G, A, or D; X.sub.2 is S or R; X.sub.3 is S, N, or T; and X.sub.4 is T or A (SEQ ID NO:311); and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDRPPLT (SEQ ID NO:160). In some embodiments, an antibody of the present disclosure comprises (a) a heavy chain variable (VH) domain comprising (i) an HVR-H1 sequence comprising the amino acid sequence of SYDMN (SEQ ID NO:270), SYDVN (SEQ ID NO:221), or SYDLN (SEQ ID NO:313), (ii) an HVR-H2 sequence comprising the amino acid sequence of LISGSGEIIYYADSVKG (SEQ ID NO:156), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ENNRYRFFDD (SEQ ID NO:157); and/or (b) a light chain variable (VL) domain comprising (i) an HVR-L1 sequence comprising the amino acid sequence of RASQSVYTYLA (SEQ ID NO:158), (ii) an HVR-L2 sequence comprising the amino acid sequence of GASSRAT (SEQ ID NO:159), and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDRPPLT (SEQ ID NO:160). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:133, 260, 329, or 249; and/or the VL domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:134, 250, 251, 417, 418, or 419. In some embodiments, the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises

the amino acid sequence of SEQ ID NO:417; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:417; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:417; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:417; the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:418; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:418; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:418; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:418; the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:419; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:419; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:419; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:419; the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; the VH domain comprises the amino acid sequence of SEQ ID NO:133, and the VL domain comprises the amino acid sequence of SEQ ID NO:251; the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:251; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; or the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:251.

(191) In some embodiments, an antibody of the present disclosure comprises (a) a heavy chain variable (VH) domain comprising: (i) an HVR-H1 sequence comprising the amino acid sequence of X.sub.1X.sub.2AX.sub.3S, wherein X.sub.1 is S or T; X.sub.2 is N, Y, H, or D; and X.sub.3 is M, L, or V (SEQ ID NO:297); (ii) an HVR-H2 sequence comprising the amino acid sequence of GISX.sub.1X.sub.2X.sub.3X.sub.4X.sub.5X.sub.6YYX.sub.7X.sub.8SX.sub.9KG, wherein X.sub.1 is A or S; X.sub.2 is G, S, or absent; X.sub.3 is S, D or G; X.sub.4 is G or S; X.sub.5 is D, S, or G; X.sub.6 is T or A; X.sub.7 is P, G, V, I, A, or S; X.sub.8 is A, D, or G; and X.sub.9 is V or M (SEQ ID NO:298); and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193); and/or (b) a light chain variable (VL) domain comprising: (i) an HVR-L1 sequence comprising the amino acid sequence of SGGX.sub.1X.sub.2X.sub.3SX.sub.4YYX.sub.5, wherein X.sub.1 is D, G, S, I, or absent; X.sub.2 is S, W, G, Y, D, or absent; X.sub.3 is S, Y, T, or D; X.sub.4 is H, T, S, or Y; and X.sub.5 is G or A (SEQ ID NO:299); (ii) an HVR-L2 sequence comprising the amino acid sequence of SDX.sub.1X.sub.2RPX.sub.3, wherein X.sub.1 is D or N; X.sub.2 is E, K, or Q; and X.sub.3 is S or P (SEQ ID NO:300); and (iii) an HVR-L3 sequence comprising the amino acid sequence of X.sub.1X.sub.2YDX.sub.3X.sub.4X.sub.5YX.sub.6NX.sub.7, wherein X.sub.1 is G or A; X.sub.2 is G or A; X.sub.3 is G, Y, Q, S, or A; X.sub.4 is S, R, or T; X.sub.5 is T or S; X.sub.6 is A, I, V, L, or T; and X.sub.7 is T, A, D, or P (SEQ ID NO:301). In some embodiments, an antibody of the present disclosure comprises (a) a heavy chain variable (VH) domain comprising: (i) an HVR-H1

sequence comprising the amino acid sequence of SX.sub.1AX.sub.2S, wherein X.sub.1 is N or Y; and wherein X.sub.2 is M, L, or V (SEQ ID NO:302); (ii) an HVR-H2 sequence comprising the amino acid sequence of GISX.sub.1GX.sub.2X.sub.3DTYYX.sub.4X.sub.5SVKG, wherein X.sub.1 is A or S; X.sub.2 is G or absent; X.sub.3 is S or G; X.sub.4 is P, G, or V; and X.sub.5 is A or D (SEQ ID NO:303); and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193); and/or (b) a light chain variable (VL) domain comprising: (i) an HVR-L1 sequence comprising the amino acid sequence of SGGX.sub.1YSSYYA, wherein X.sub.1 is S or A (SEQ ID NO:304); (ii) an HVR-L2 sequence comprising the amino acid sequence of SDDKRPS (SEQ ID NO:336); and (iii) an HVR-L3 sequence comprising the amino acid sequence of GGYDQSSYTNP (SEQ ID NO:172). In some embodiments, the VH domain comprises (i) an HVR-H1 sequence comprising the amino acid sequence of SNAMS (SEQ ID NO:194), SNAVS (SEQ ID NO:271), or SNALS (SEQ ID NO:318), (ii) an HVR-H2 sequence comprising the amino acid sequence of GISAGGSDTYYPASVKG (SEQ ID NO:195), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO: 135, 263, 264, or 330. In some embodiments, the VH domain comprises (i) an HVR-H1 sequence comprising the amino acid sequence of SNAMS (SEQ ID NO:194), SNAVS (SEQ ID NO:271), or SNALS (SEQ ID NO:318), (ii) an HVR-H2 sequence comprising the amino acid sequence of GISSGSDTYYGDSVKG (SEQ ID NO:197), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO: 137, 265, 266, or 331. In some embodiments, the VH domain comprises (i) an HVR-H1 sequence comprising the amino acid sequence of SYAMS (SEQ ID NO:200), SYAVS (SEQ ID NO:272), or SYALS (SEQ ID NO:319), (ii) an HVR-H2 sequence comprising the amino acid sequence of GISSGGDTYYVDSVKG (SEQ ID NO:201), and (iii) an HVR-H3 sequence comprising the amino acid sequence of ETWNHLFDY (SEQ ID NO:193). In some embodiments, the VH domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO: 139, 267, 268, or 332. In some embodiments, the VL domain comprises one or more human IGLV3 framework sequences. In some embodiments, the VL domain comprises four human IGLV3 framework sequences. In some embodiments, the VL domain comprises the sequence FW1-HVR-L1-FW2-HVR-L2-FW3-HVR-L3-FW4 (N-terminus to C-terminus), wherein FW1 comprises the amino acid sequence SYELTQPPSVSVSPGQTARITC (SEQ ID NO:314), FW2 comprises the amino acid sequence WYQQKPGQAPVTLIY (SEQ ID NO:315), FW3 comprises the amino acid sequence NIPERFSGSSSGTTVTLTISGVQAEDEADYYC (SEQ ID NO:316), and FW4 comprises the amino acid sequence FGGGTKLTVL (SEQ ID NO:317). In some embodiments, the VL domain comprises (i) an HVR-L1 sequence comprising the amino acid sequence of SGGSYSSYYA (SEQ ID NO:170), (ii) an HVR-L2 sequence comprising the amino acid sequence of SDDKRPS (SEQ ID NO:336), and (iii) an HVR-L3 sequence comprising the amino acid sequence of GGYDQSSYTNP (SEQ ID NO:172). In some embodiments, the VL domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:252. In some embodiments, the VL domain comprises an amino acid sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the amino acid sequence of SEQ ID NO:254. In some embodiments, the VL domain comprises an amino acid sequence that is at least 90%, at least

[illegible]

or Y; and X.sub.3 is T or S (SEQ ID NO:185); (b) an HVR-H2 sequence comprising the amino acid sequence of TIGX.sub.4X.sub.5DTYYADSVKG, wherein X.sub.4 is S or A and X.sub.5 is G or D (SEQ ID NO:186); (c) an HVR-H3 sequence comprising the amino acid sequence of DSTVX.sub.6WSGDFFDY, wherein X.sub.6 is S or G (SEQ ID NO:187); (d) an HVR-L1 sequence comprising the amino acid sequence of RASQNVX.sub.7X.sub.8DX.sub.9A, wherein X.sub.7 is K or R; X.sub.8 is N or S; and X.sub.9 is L or I (SEQ ID NO:188); (e) an HVR-L2 sequence comprising the amino acid sequence of AAX.sub.10X.sub.11RX.sub.12T, wherein X.sub.10 is R or S; X.sub.11 is I or S; and X.sub.12 is E or D (SEQ ID NO:189); and (f) an HVR-L3 sequence comprising the amino acid sequence of QQYYDWPPFT (SEQ ID NO:148).

(194) In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:143-148 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:143-145 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:146-148). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:148-153 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:149-151 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:152, 153, and 148). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:155-160 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:155-157 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:158-160). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:161-166 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:161-163 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:164-166). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:161-166 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:161-163 and/or one, two, or three light chain HVR sequences of a variable domain shown in Table 2). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:161, 163, 168, and 170-172 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs: 161, 168, and 163 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:170-172). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs: 161, 163, 168, and 170-172 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs: 161, 168, and 163 and/or one, two, or three light chain HVR sequences of a variable domain shown in Table 2). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs:163, 173, 174, and 176-178 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs:163, 173, and 174 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:176-178). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs: 162, 163, 179, and 182-184 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs: 162, 163, and 179 and/or one, two, or three light chain HVR sequences selected from SEQ ID NOs:182-184). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences having an amino acid sequence selected from SEQ ID NOs: 162, 163, 179, and 182-184 (e.g., one, two, or three heavy chain HVR sequences selected from SEQ ID NOs: 162, 163, and 179 and/or one, two, or three light chain HVR sequences of a variable domain shown in Table 2). In some embodiments, an antibody of the present disclosure comprises one, two, three,

four, five, or six HVR sequences from the variable domain sequences of SEQ ID NOs:120 and 97 (e.g., one, two, or three heavy chain HVR sequences from the heavy chain variable domain sequence of SEQ ID NO:120 and/or one, two, or three light chain HVR sequences from the light chain variable domain sequence of SEQ ID NO:97). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences from the variable domain sequences of SEQ ID NOs:127 and 104 (e.g., one, two, or three heavy chain HVR sequences from the heavy chain variable domain sequence of SEQ ID NO:127 and/or one, two, or three light chain HVR sequences from the light chain variable domain sequence of SEQ ID NO:104). In some embodiments, an antibody of the present disclosure comprises one, two, three, four, five, or six HVR sequences from the variable domain sequences of SEQ ID NOs:97, 104, 120, and 127 (e.g., one, two, or three heavy chain HVR sequences from the heavy chain variable domain sequence of SEQ ID NOs:120 and 127 and/or one, two, or three light chain HVR sequences from the light chain variable domain sequence of SEQ ID NOs:97 and 104).

(195) In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:143, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:144, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:145; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:146, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:147, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:148. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:149, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:150, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:151; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:152, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:153, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:148. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:143 or 149, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:144 or 150, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:145 or 151; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:146 or 152, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:147 or 153, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:148. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:155, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:156, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:157; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:158, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:159, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:160. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:161, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:162, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:163; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:161, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:168, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:163; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:170, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:171, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:172. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising an

comprising the amino acid sequence of SEQ ID NO:163; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:135 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:137 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:139 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising the amino acid sequence of SEQ ID NO:141 and/or a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, 170, 176, or 182; an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, 171, 177, or 183; and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:166, 172, 178, or 184. (198) In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising one, two, or three HVR sequences from SEQ ID NO:242; and/or (b) a VL domain comprising one, two, or three HVR sequences from SEQ ID NO:243. In some embodiments, an antibody of the present disclosure comprises (a) a VH domain comprising one, two, or three HVR sequences from SEQ ID NO:244; and/or (b) a VL domain comprising one, two, or three HVR sequences from SEQ ID NO:245.

(199) As described supra, various techniques for delineating hypervariable regions (HVRs) or complementarity determining regions (CDRs) are known in the art and can be applied to the variable domain sequences described herein. In some embodiments, an antibody of the present disclosure comprises HVRs as defined by Chothia, Kabat, IMGT, or a combination thereof (e.g., one or more HVRs as defined by one delineation and one or more HVRs as defined by a different delineation). HVR sequences of antibodies of the present disclosure delineated using three known delineation (Chothia, Kabat, and IMGT) are provided in Table 5. As used herein, unless otherwise specified, the numbering of HVR residues is defined by Kabat numbering.

(200) TABLE-US-00009 TABLE 5 HVR delineations. Hum 1 HVR-L1 170 SGGSYSSYYA (Chothia) HVR-L2 336 SDDKRPS HVR-L3 172 GGYDQSSYTNP Hum 1 HVR-L1 170 SGGSYSSYYA (Kabat) HVR-L2 336 SDDKRPS HVR-L3 172 GGYDQSSYTNP Hum 1 HVR-L1 337 GSYSS (IMGT) HVR-L2 338 IYS HVR-L3 339 GGYDQSSYT Hum 7 HVR-L1 164 SGGDYYSYYYG (Chothia) HVR-L2 336 SDDKRPS HVR-L3 166 GGYDYSTYANA Hum 7 HVR-L1 164 SGGDYYSYYYG (Kabat) HVR-L2 336 SDDKRPS HVR-L3 166 GGYDYSTYANA Hum 7 HVR-L1 340 GDYYS (IMGT) HVR-L2 338 IYS HVR-L3 190 GGYDYSTYA AB21 HVR-H1 191 GFTFSSN (Chothia) HVR-H2 192 SAGGSD HVR-H3 193 ETWNHLFDY AB21 HVR-H1 194 SNAMS (Kabat) HVR-H2 195 GISAGGSDTYYPASVKG HVR-H3 193 ETWNHLFDY AB21 HVR-H1 161 GFTFSSNA (IMGT) HVR-H2 162 ISAGGSDT HVR-H3 163 ARETWNHLFDY AB25 HVR-H1 191 GFTFSSN (Chothia) HVR-H2 196 SSGSD HVR-H3 193 ETWNHLFDY AB25 HVR-H1 194 SNAMS (Kabat) HVR-H2 197 GISSGSDTYYGDSVKG HVR-H3 193 ETWNHLFDY AB25 HVR-H1 161 GFTFSSNA (IMGT) HVR-H2 168 ISSGSDT HVR-H3 163 ARETWNHLFDY

AB27 HVR-H1 198 GFRFSSY (Chothia) HVR-H2 199 SSGGD HVR-H3 193 ETWNHLFDY
 AB27 HVR-H1 200 SYAMS (Kabat) HVR-H2 201 GISSGGDTYYVDSVKG HVR-H3 193
 ETWNHLFDY AB27 HVR-H1 173 GFRFSSYA (IMGT) HVR-H2 174 ISSGGDT HVR-H3 163
 ARETWNHLFDY AB119 HVR-H1 202 GFSFSNF (Chothia) HVR-H2 203 GSGD HVR-H3 145
 DSTVSWSGDFFDY AB119 HVR-H1 204 NFAVT (Kabat) HVR-H2 144 TIGSGDTYYADSVKG
 HVR-H3 145 DSTVSWSGDFFDY AB119 HVR-H1 205 GFSFSNFA (IMGT) HVR-H2 206
 IGSGDT HVR-H3 207 AKDSTVSWSGDFFDY AB119 HVR-L1 146 RASQNVKNDLA
 (Chothia) HVR-L2 147 AARIRET HVR-L3 148 QQYYDWPPFT AB119 HVR-L1 146
 RASQNVKNDLA (Kabat) HVR-L2 147 AARIRET HVR-L3 148 QQYYDWPPFT AB119 HVR-
 L1 208 QNVKND (IMGT) HVR-L2 209 AAR HVR-L3 210 QQYYDWP AB135 HVR-H1 211
 GFSFSIY (Chothia) HVR-H2 212 GADD HVR-H3 151 DSTVGWSGDFFDY AB135 HVR-H1
 213 IYAVS (Kabat) HVR-H2 150 TIGADDTYYADSVKG HVR-H3 151 DSTVGWSGDFFDY
 AB135 HVR-H1 214 GFSFSIYA (IMGT) HVR-H2 215 IGADDT HVR-H3 216
 AKDSTVGWSGDFFDY AB135 HVR-L1 152 RASQNVRSIDIA (Chothia) HVR-L2 153
 AASSRDT HVR-L3 148 QQYYDWPPFT AB135 HVR-L1 152 RASQNVRSIDIA (Kabat) HVR-
 L2 153 AASSRDT HVR-L3 148 QQYYDWPPFT AB135 HVR-L1 217 QNVRSID (IMGT) HVR-
 L2 218 AAS HVR-L3 148 QQYYDWPPFT AB136 HVR-H1 219 GFTFSSY (Chothia) HVR-H2
 220 SGSGEI HVR-H3 157 ENNRYRFFDD AB136 HVR-H1 221 SYDVN (Kabat) HVR-H2 156
 LISGSGEIYYADSVKG HVR-H3 157 ENNRYRFFDD AB136 HVR-H1 222 GFTFSSYD
 (IMGT) HVR-H2 223 ISGSGEI HVR-H3 224 AKENNRYRFFDD AB136 HVR-L1 158
 RASQSVYTYLA (Chothia) HVR-L2 159 GASSRAT HVR-L3 160 QQYYDRPPLT AB136 HVR-
 L1 158 RASQSVYTYLA (Kabat) HVR-L2 159 GASSRAT HVR-L3 160 QQYYDRPPLT AB136
 HVR-L1 225 QSVYTY (IMGT) HVR-L2 226 GAS HVR-L3 160 QQYYDRPPLT AB3 HVR-H1
 227 DYGMN (Kabat) HVR-H2 228 QITSGSRTYYGA AVKG HVR-H3 229 DFGSGVGSIDA
 AB3 HVR-H1 230 GFIFSDY (CHOTHIA) HVR-H2 231 TSGSR HVR-H3 229 DFGSGVGSIDA
 AB3 HVR-L1 232 SGSRGYRG (Chothia) HVR-L2 233 RDNQRPS HVR-L3 234 GSYDGSIDI
 AB3 HVR-L1 232 SGSRGYRG (Kabat) HVR-L2 233 RDNQRPS HVR-L3 234 GSYDGSIDI
 AB45 HVR-H1 235 SYAMG (Kabat) HVR-H2 236 GIDDDGSTANYGPAVKG HVR-H3 237
 ASVTGWSAHISGRLDT AB45 HVR-H1 219 GFTFSSY (CHOTHIA) HVR-H2 238 DDGST
 HVR-H3 237 ASVTGWSAHISGRLDT AB45 HVR-L1 239 SGGGIYYYG (Chothia) HVR-L2
 240 ENDKRPS HVR-L3 241 GGYDSNTTSGI AB45 HVR-L1 239 SGGGIYYYG (Kabat) HVR-
 L2 240 ENDKRPS HVR-L3 241 GGYDSNTTSGI

(201) In some embodiments, an antibody of the present disclosure comprises 1, 2, 3, 4, 5, or 6 HVRs listed in Table 5 (e.g., a VL domain comprising 1, 2, or 3 light chain HVRs listed in Table 5 and/or a VH domain comprising 1, 2, or 3 heavy chain HVRs listed in Table 5).

(202) In some embodiments, an antibody of the present disclosure comprises a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:191, an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 192, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO: 193. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:191, an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 196, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO: 193. In some embodiments, an antibody of the present disclosure comprises a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO: 198, an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 199, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO: 193. In some embodiments, an antibody of the present disclosure comprises a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO:164, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:165, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO: 166 and/or a VH domain comprising an HVR-H1 comprising the amino acid sequence of SEQ ID NO:191 or 198, an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 192, 196, or

HVR-H1 comprising the amino acid sequence of SEQ ID NO: 222, an HVR-H2 comprising the amino acid sequence of SEQ ID NO: 223, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO: 224; and/or (b) a VL domain comprising an HVR-L1 comprising the amino acid sequence of SEQ ID NO: 225, an HVR-L2 comprising the amino acid sequence of SEQ ID NO: 226, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO: 160.

(208) In some embodiments of any of the above embodiments, the antibody enhances phagocytosis by a macrophage expressing a human SIRP- α polypeptide, enhances activation of a dendritic cell expressing a human SIRP- α polypeptide, inhibits in vivo growth of a tumor or tumor cell(s) that expresses CD47, and/or does not prevent interactions between a CD47-expressing cell and a T cell. Exemplary assays for measuring phagocytosis, dendritic cell activation, tumor growth inhibition, and interactions between CD47-expressing cells and T cells (e.g., adhesion assays) are described herein and known in the art.

(209) Antibody Production and Other Antibody Properties

(210) An antibody of the present disclosure may be produced by any means known in the art. Exemplary techniques for antibody production are described below; however these exemplary techniques are provided for illustrative purposes only and are not intended to be limiting. In addition, exemplary antibody properties contemplated for use with the antibodies described herein are further described.

(211) In some embodiments, an antibody that “binds” an antigen has a dissociation constant ($K_{sub.D}$) for the antigen that is less than or equal to 1 μ M at 25° C. In some embodiments, an antibody of the present disclosure has a dissociation constant ($K_{sub.D}$) for human v1 and/or v2 SIRP- α polypeptides that is less than or equal to 1 μ M at 25° C., less than or equal to 500 nM at 25° C., less than or equal to 400 nM at 25° C., less than or equal to 300 nM at 25° C., less than or equal to 250 nM at 25° C., less than or equal to 200 nM at 25° C., less than or equal to 200 nM at 25° C., less than or equal to 100 nM at 25° C., or less than or equal to 50 nM at 25° C. In some embodiments, an antibody that binds a human SIRP- α polypeptide and one or more non-human SIRP- α polypeptides binds the human SIRP- α polypeptide at a higher affinity (e.g., 10-fold or 100-fold higher) than the non-human SIRP- α polypeptide, though it still considered to “bind” both polypeptides. In some embodiments, an antibody that binds a non-human SIRP- α polypeptide and one or more human SIRP- α polypeptides binds the non-human SIRP- α polypeptide at a higher affinity (e.g., 10-fold or 100-fold higher) than the human SIRP- α polypeptide, though it still considered to “bind” both polypeptides. Assays for determining binding affinity are known in the art and include without limitation surface plasmon resonance (SPR), e.g., as described herein; radiolabeled antigen binding assay (RIA), e.g., using a Fab version of an antibody and its antigen; and the like. Other exemplary binding assays are described herein.

(212) To prepare an antigen, the antigen may be purified or otherwise obtained from a natural source, or it may be expressed using recombinant techniques. In some embodiments, the antigen may be used as a soluble protein. In some embodiments, the antigen may be conjugate to another polypeptide or other moiety, e.g., to increase its immunogenicity. For example, an antigen described herein may be coupled with an Fc region. In some embodiments, a cell expressing the antigen on its cell surface may be used as the antigen.

(213) Polyclonal antibodies can be raised in an animal by multiple subcutaneous (sc) or intraperitoneal (ip) injections of the antigen and an adjuvant. For example, descriptions of chicken immunization are described herein. In some embodiments, the antigen is conjugated with an immunogenic protein, e.g., keyhole limpet hemocyanin, serum albumin, bovine thyroglobulin, or soybean trypsin inhibitor using a bifunctional or derivatizing agent. Exemplary methods for immunization of chickens are provided herein. Relevant methods suitable for a variety of other organisms, such as mammals, are well known in the art.

(214) As described supra, monoclonal antibodies may be produced by a variety of methods. In some embodiments, a monoclonal antibody of the present disclosure is made using the hybridoma

method first described by Kohler et al., *Nature*, 256:495 (1975), and further described in Hongo et al., *Hybridoma*, 14 (3): 253-260 (1995); Harlow et al., *Antibodies: A Laboratory Manual*, (Cold Spring Harbor Laboratory Press, 2nd ed. 1988); and Hammerling et al., in: *Monoclonal Antibodies and T-Cell Hybridomas* 563-681 (Elsevier, N.Y., 1981). Human hybridoma technology (Trioma technology) is described in Vollmers and Brandlein, *Histology and Histopathology*, 20(3):927-937 (2005) and Vollmers and Brandlein, *Methods and Findings in Experimental and Clinical Pharmacology*, 27(3):185-91 (2005). A culture medium in which hybridoma cells are grown may be screened for the presence of an antibody of interest, e.g., by in vitro binding assay, immunoprecipitation, ELISA, RIA, etc.; and the binding affinity may be determined, e.g., by Scatchard analysis. A hybridoma that produces an antibody with desired binding properties can be subcloned and grown using known culture techniques, grown in vivo as ascites tumors in an animal, and the like.

(215) In some embodiments, a monoclonal antibody is made using a library method, such as a phage display library. See, e.g., Hoogenboom et al. in *Methods in Molecular Biology* 178:1-37 (O'Brien et al., ed., Human Press, Totowa, NJ, 2001). In some embodiments, repertoires of VH and VL genes are cloned by polymerase chain reaction (PCR) and recombined randomly in phage libraries, which are then screened for antigen-binding phage, e.g., as described in Winter et al., *Ann. Rev. Immunol.*, 12: 433-455 (1994). Phage typically display antibody fragments, either as single-chain Fv (scFv) fragments or as Fab fragments. Alternatively, the naive repertoire can be cloned (e.g., from human) to provide a single source of antibodies to a wide range of non-self and also self-antigens without any immunization as described by Griffiths et al., *EMBO J*, 12: 725-734 (1993). Finally, naive libraries can also be made synthetically by cloning unrearranged V-gene segments from stem cells, and using PCR primers containing random sequence to encode the highly variable CDR3 regions and to accomplish rearrangement in vitro, as described by Hoogenboom and Winter, *J. Mol. Biol.*, 227: 381-388 (1992).

(216) In some embodiments, an antibody of the present disclosure is a chicken antibody. Chicken antibodies can be produced using various techniques known in the art; see, e.g., U.S. Pat. Nos. 6,143,559; 8,592,644; and 9,380,769.

(217) In some embodiments, an antibody of the present disclosure is a chimeric antibody. See, e.g., U.S. Pat. No. 4,816,567 and Morrison et al., *Proc. Natl. Acad. Sci. USA*, 81:6851-6855 (1984). In some embodiments, a chimeric antibody comprises a non-human variable region (e.g., a variable region derived from a chicken, mouse, rat, hamster, rabbit, or non-human primate, such as a monkey) and a human constant region. In some embodiments, a chimeric antibody is a "class switched" antibody in which the class or subclass has been changed from that of the parent antibody. Chimeric antibodies include antigen-binding fragments thereof.

(218) In some embodiments, a chimeric antibody is a humanized antibody. A non-human antibody can be humanized to reduce immunogenicity to humans, while retaining the specificity and affinity of the parental non-human antibody. Generally, a humanized antibody comprises one or more variable domains in which HVRs, e.g., CDRs, (or portions thereof) are derived from a non-human antibody (e.g., a chicken antibody), and FRs (or portions thereof) are derived from human antibody sequences. A humanized antibody optionally will also comprise at least a portion of a human constant region. In some embodiments, some FR residues in a humanized antibody are substituted with corresponding residues from a non-human antibody (e.g., the antibody from which the HVR residues are derived), e.g., to restore or improve antibody specificity or affinity. Humanized antibodies and methods of making them are reviewed, e.g., in Almagro and Fransson, *Front. Biosci.* 13:1619-1633 (2008). Methods of humanizing a chicken antibody have also been described, e.g., in WO2005014653.

(219) Human framework regions useful for humanization include but are not limited to: framework regions selected using the "best-fit" method; framework regions derived from the consensus sequence of human antibodies of a particular subgroup of light or heavy chain variable regions;

human somatically mutated framework regions or human germline framework regions; and framework regions derived from screening FR libraries. See, e.g., Sims et al. *J. Immunol.* 151:2296 (1993); Carter et al. *Proc. Natl. Acad. Sci. USA*, 89:4285 (1992); Presta et al. *J. Immunol.*, 151:2623 (1993); Almagro and Fransson, *Front. Biosci.* 13:1619-1633 (2008); and Baca et al., *J. Biol. Chem.* 272:10678-10684 (1997).

(220) In some embodiments, an antibody of the present disclosure is a human antibody. Human antibodies can be produced using various techniques known in the art. In some embodiments, the human antibody is produced by a non-human animal, such as the genetically engineered chickens (see, e.g., U.S. Pat. Nos. 8,592,644; and 9,380,769) and/or mice described herein. Human antibodies are described generally in Lonberg, *Curr. Opin. Immunol.* 20:450-459 (2008).

(221) In some embodiments, an antibody of the present disclosure is generated by or derived from a chicken, e.g., using the methods described herein.

(222) In some embodiments, an antibody of the present disclosure is an antibody fragment, including without limitation a Fab, F(ab')₂, Fab'-SH, Fv, or scFv fragment, or a single domain, single heavy chain, or single light chain antibody. Antibody fragments can be generated, e.g., by enzymatic digestion or by recombinant techniques. In some embodiments, Proteolytic digestion of an intact antibody is used to generate an antibody fragment, e.g., as described in Morimoto et al., *Journal of Biochemical and Biophysical Methods* 24:107-117 (1992) and Brennan et al., *Science*, 229:81 (1985). In some embodiments, an antibody fragment is produced by a recombinant host cell. For example, Fab, Fv and ScFv antibody fragments are expressed by and secreted from *E. coli*. Antibody fragments can alternatively be isolated from an antibody phage library.

(223) Fab'-SH fragments can be directly recovered from *E. coli* and chemically coupled to form F(ab')₂ fragments. See Carter et al., *Bio Technology* 10:163-167 (1992). F(ab')₂ fragments can also be isolated directly from a recombinant host cell culture. Fab and F(ab')₂ fragment with increased in vivo half-life comprising salvage receptor binding epitope residues are described in U.S. Pat. No. 5,869,046.

(224) In some embodiments, an antibody is a single chain Fv fragment (scFv). See WO 93/16185 and U.S. Pat. Nos. 5,571,894 and 5,587,458. scFv fusion proteins can be constructed to produce a fusion of an effector protein at either the amino or the carboxy terminus of an scFv. The antibody fragment may also be a "linear antibody", e.g., as described in U.S. Pat. No. 5,641,870, for example. Such linear antibodies may be monospecific or bispecific.

(225) In some embodiments, an antibody of the present disclosure is a multispecific antibody. Multispecific antibodies possess binding specificities against more than one antigen (e.g., having two, three, or more binding specificities). In some embodiments, the antibody is a bispecific antibody. In some embodiments, a bispecific antibody comprises two different binding specificities for the same antigen (e.g., having different binding affinity and/or specific epitope of the same antigen). In some embodiments, a bispecific antibody comprises binding specificities for two distinct antigens. In some embodiments, the bispecific antibody is a full-length or intact antibody. In some embodiments, the bispecific antibody is an antibody fragment of the present disclosure.

(226) Bispecific or multispecific antibodies with a variety of combinations of binding specificities are contemplated herein. In some embodiments, the bispecific antibody has a first binding specificity for one or more SIRP- α polypeptides as described herein. In some embodiments, the bispecific antibody has a second binding specificity for an antigen expressed by a cancer cell, e.g., on the cell surface. Exemplary such antigens include without limitation CD19, CD20, CD22, CD30, CD33, CD38, CD52, CD56, CD70, CD74, CD79b, CD123, CD138, CS1/SLAMF7, Trop-2, 5T4, EphA4, BCMA, Mucin 1, Mucin 16, PTK7, PD-L1, STEAP1, Endothelin B Receptor, mesothelin, EGFRvIII, ENPP3, SLC44A4, GNMB, nectin 4, NaPi2b, LIV-1A, Guanylyl cyclase C, DLL3, EGFR, HER2, VEGF, VEGFR, integrin α V β 3, integrin α 5 β 1, MET, IGF1R, TRAILR1, TRAILR2, RANKL, FAP, Tenascin, Le.sup.y, EpCAM, CEA, gpA33, PSMA, TAG72, a mucin, CAIX, EPHA3, folate receptor α , GD2, GD3, and an MHC/peptide complex comprising a peptide

from NY-ESO-1/LAGE, SSX-2, a MAGE family protein, MAGE-A3, gp100/pmel17, Melan-A/MART-1, gp75/TRP1, tyrosinase, TRP2, CEA, PSA, TAG-72, immature laminin receptor, MOK/RAGE-1, WT-1, SAP-1, BING-4, EpCAM, MUC1, PRAME, survivin, BRCA1, BRCA2, CDK4, CML66, MART-2, p53, Ras, β -catenin, TGF- β RII, HPV E6, or HPV E7. Without wishing to be bound to theory, it is thought that combining such a binding specificity with a binding specificity against a SIRP- α is particularly advantageous, e.g., to direct FcR-expressing leukocytes to target a tumor cell with the second binding specificity while also inhibiting the responsiveness of SIRP- α expressed by the leukocyte to any CD47 expressed by the tumor cell with the first binding specificity.

(227) Various methods are known in the art for generating and purifying a bispecific antibody. Numerous approaches have been described. One approach is the “knobs-into-holes” or “protuberance-into-cavity” approach (see, e.g., U.S. Pat. No. 5,731,168). In some embodiments, heterodimerization of Fc domain monomers is promoted by introducing different, but compatible, substitutions in the two Fc domain monomers, such as “knob-into-hole” residue pairs and charge residue pairs. The knob and hole interaction favors heterodimer formation, whereas the knob-knob and the hole-hole interaction hinder homodimer formation due to steric clash and deletion of favorable interactions. A hole refers to a void that is created when an original amino acid in a protein is replaced with a different amino acid having a smaller side-chain volume. A knob refers to a bump that is created when an original amino acid in a protein is replaced with a different amino acid having a larger side-chain volume. For example, in some embodiments, an amino acid being replaced is in the CH3 antibody constant domain of an Fc domain monomer and involved in the dimerization of two Fc domain monomers. In some embodiments, a hole in one CH3 antibody constant domain is created to accommodate a knob in another CH3 antibody constant domain, such that the knob and hole amino acids act to promote or favor the heterodimerization of the two Fc domain monomers. In some embodiments, a hole in one CH3 antibody constant domain is created to better accommodate an original amino acid in another CH3 antibody constant domain. In some embodiments, a knob in one CH3 antibody constant domain is created to form additional interactions with original amino acids in another CH3 antibody constant domain.

(228) In some embodiments, a hole is constructed by replacing amino acids having larger side chains such as tyrosine or tryptophan with amino acids having smaller side chains such as alanine, valine, or threonine, for example a Y407V mutation in the CH3 antibody constant domain. Similarly, in some embodiments, a knob is constructed by replacing amino acids having smaller side chains with amino acids having larger side chains, for example a T366W mutation in the CH3 antibody constant domain. In some embodiments, one Fc domain monomer includes the knob mutation T366W and the other Fc domain monomer includes hole mutations T366S, L358A, and Y407V. In some embodiments, a polypeptide of the disclosure including a high affinity SIRP- α D1 variant is fused to an Fc domain monomer including the knob mutation T366W to limit unwanted knob-knob homodimer formation. Examples of knob-into-hole amino acid pairs are included, without limitation, in Table 3.

(229) TABLE-US-00010 TABLE 3 Knob-into-hole amino acid pairs. Fc Y407A Y407A F405A T394S T366S T394W T394S T366W domain L358A Y407T Y407A T394S monomer 1 Y407V Fc T366Y T366W T394W F405W T366W T366Y T366W F405W domain F405A F405W Y407A monomer 2

(230) Another approach uses antibody variable domains with the desired binding specificities (antibody-antigen combining sites) fused to immunoglobulin constant domain sequences, e.g., with an immunoglobulin heavy chain constant domain, comprising at least part of the hinge, CH2, and CH3 regions. In some embodiments, the bispecific antibody has a hybrid immunoglobulin heavy chain with a first binding specificity in one arm and a hybrid immunoglobulin heavy chain-light chain pair (providing a second binding specificity) in the other arm. See WO 94/04690. Another approach uses cross-linking (see, e.g., U.S. Pat. No. 4,676,980) to produce a heteroconjugate

antibody. In some embodiments, bispecific antibodies can be prepared using chemical linkage (see, e.g., Brennan et al., *Science*, 229: 81 (1985)) to proteolytically cleave an intact antibody into F(ab')₂ fragments that are reduced in the presence of a dithiol complexing agent and converted to thionitrobenzoate (TNB) derivatives, one of which is reconverted to the Fab'-thiol by reduction and mixed with the other Fab'-TNB derivative to form the bispecific antibody. In some embodiments, Fab'-SH fragments are chemically coupled. In some embodiments, bispecific antibody fragments are produced in cell culture using leucine zippers, as in Kostelny et al., *J. Immunol.*, 148(5):1547-1553 (1992). For other bispecific antibody formats, see, e.g., Spiess, C. et al. (2015) *Mol. Immunol.* 67:95-106.

(231) In some embodiments, an antibody of the present disclosure is a diabody. See, e.g., Hollinger et al., *Proc. Natl. Acad. Sci. USA*, 90:6444-6448 (1993). In a diabody, the VH and VL domains of one fragment pair with complementary VL and VH domains of another fragment, thus forming two antigen-binding sites. Another strategy for making bispecific antibody fragments by the use of single-chain Fv (sFv) dimers has also been reported. See Gruber et al, *J. Immunol*, 152:5368 (1994).

(232) In some embodiments, an antibody of the present disclosure is a single-domain antibody. A single-domain antibody refers to a single polypeptide chain comprising all or a portion of the heavy chain variable domain or all or a portion of the light chain variable domain of an antibody. In certain embodiments, a single-domain antibody is a human single-domain antibody (see, e.g., U.S. Pat. No. 6,248,516 B1). In one embodiment, a single-domain antibody includes all or a portion of the heavy chain variable domain of an antibody. Camelid antibodies are also known.

(233) Antibodies can be produced using recombinant methods. For recombinant production of an anti-antigen antibody, nucleic acid encoding the antibody is isolated and inserted into a replicable vector for further cloning (amplification of the DNA) or for expression. DNA encoding the antibody may be readily isolated and sequenced using conventional procedures (e.g., by using oligonucleotide probes that are capable of binding specifically to genes encoding the heavy and light chains of the antibody). Many vectors are available. The vector components generally include, but are not limited to, one or more of the following: a signal sequence, an origin of replication, one or more marker genes, an enhancer element, a promoter, and a transcription termination sequence.

(234) An antibody of the present disclosure can be produced recombinantly as a fusion polypeptide with a heterologous polypeptide, e.g., a signal sequence or other polypeptide having a specific cleavage site at the N-terminus of the mature protein or polypeptide. The heterologous signal sequence selected can be one that is recognized and processed (e.g., cleaved by a signal peptidase) by the host cell. For prokaryotic host cells that do not recognize and process a native antibody signal sequence, the signal sequence is substituted by a prokaryotic signal sequence selected, for example, from alkaline phosphatase, penicillinase, lpp, or heat-stable enterotoxin II leaders. For yeast secretion the native signal sequence may be substituted by, e.g., the yeast invertase leader, a factor leader (including *Saccharomyces* and *Kluyveromyces* α -factor leaders), or acid phosphatase leader, the *C. albicans* glucoamylase leader, etc. In mammalian cell expression, mammalian signal sequences as well as viral secretory leaders, for example, the herpes simplex gD signal, are available.

(235) Both expression and cloning vectors contain a nucleic acid sequence that enables the vector to replicate in one or more selected host cells, e.g., to allow the vector to replicate independently of the host chromosomal DNA. This sequence can include origins of replication or autonomously replicating sequences. Such sequences are well known for a variety of bacteria, yeast, and viruses. Generally, the origin of replication component is not needed for mammalian expression vectors (the SV40 origin may be used because it contains the early promoter).

(236) Expression and cloning vectors can contain a selection gene or selectable marker. Typical selection genes encode proteins that (a) confer resistance to antibiotics or other toxins, e.g., ampicillin, neomycin, methotrexate, or tetracycline, (b) complement auxotrophic deficiencies, or

(c) supply critical nutrients not available from complex media. Examples of dominant selection use the drugs neomycin, mycophenolic acid and hygromycin. Another example of suitable selectable markers for mammalian cells are those that enable the identification of cells competent to take up antibody-encoding nucleic acid, such as DHFR, glutamine synthetase (GS), thymidine kinase, metallothionein-I and -II, preferably primate metallothionein genes, adenosine deaminase, ornithine decarboxylase, and the like. For example, a Chinese hamster ovary (CHO) cell line deficient in endogenous DHFR activity transformed with the DHFR gene is identified by culturing the transformants in a culture medium containing methotrexate (Mtx), a competitive antagonist of DHFR.

(237) Alternatively, host cells (particularly wild-type hosts that contain endogenous DHFR) transformed or co-transformed with DNA sequences encoding an antibody of interest, wild-type DHFR gene, and another selectable marker such as aminoglycoside 3'-phosphotransferase (APH) can be selected by cell growth in medium containing a selection agent for the selectable marker such as an aminoglycosidic antibiotic, e.g., kanamycin, neomycin, or G418.

(238) Expression and cloning vectors generally contain a promoter that is recognized by the host organism and is operably linked to nucleic acid encoding an antibody. Promoters suitable for use with prokaryotic hosts include the *phoA* promoter, β -lactamase and lactose promoter systems, alkaline phosphatase promoter, a tryptophan (*trp*) promoter system, and hybrid promoters such as the *tac* promoter. However, other known bacterial promoters are suitable. Promoter sequences are known for eukaryotes. Yeast promoters are well known in the art and can include inducible promoters/enhancers regulated by growth conditions. Virtually all eukaryotic genes have an AT-rich region located approximately 25 to 30 bases upstream from the site where transcription is initiated. Examples include without limitation the promoters for 3-phosphoglycerate kinase or other glycolytic enzymes, such as enolase, glyceraldehyde-3-phosphate dehydrogenase, hexokinase, pyruvate decarboxylase, phosphofructokinase, glucose-6-phosphate isomerase, 3-phosphoglycerate mutase, pyruvate kinase, triosephosphate isomerase, phosphoglucose isomerase, and glucokinase. Antibody transcription from vectors in mammalian host cells can be controlled, for example, by promoters obtained from the genomes of viruses. The early and late promoters of the SV40 virus are conveniently obtained as an SV40 restriction fragment that also contains the SV40 viral origin of replication. The immediate early promoter of the human cytomegalovirus is conveniently obtained as a HindIII E restriction fragment. Alternatively, the Rous Sarcoma Virus long terminal repeat can be used as the promoter.

(239) Transcription of a DNA encoding an antibody of this invention by higher eukaryotes is often increased by inserting an enhancer sequence into the vector. Many enhancer sequences are now known from mammalian genes (globin, elastase, albumin, α -fetoprotein, and insulin). Typically, however, one will use an enhancer from a eukaryotic cell virus.

(240) Expression vectors used in eukaryotic host cells (yeast, fungi, insect, plant, animal, human, or nucleated cells from other multicellular organisms) will also contain sequences necessary for the termination of transcription and for stabilizing the mRNA.

(241) Suitable host cells for cloning or expressing the DNA in the vectors herein are the prokaryote, yeast, or higher eukaryote cells described above. Suitable prokaryotes for this purpose include eubacteria, such as Gram-negative or Gram-positive organisms, for example, Enterobacteriaceae such as *Escherichia*, e.g., *E. coli*, *Enterobacter*, *Erwinia*, *Klebsiella*, *Proteus*, *Salmonella*, e.g., *Salmonella typhimurium*, *Serratia*, e.g., *Serratia marcescans*, and *Shigella*, etc. In addition to prokaryotes, eukaryotic microbes such as filamentous fungi or yeast are suitable cloning or expression hosts for antibody-encoding vectors. *Saccharomyces cerevisiae*, or common baker's yeast, is the most commonly used among lower eukaryotic host microorganisms. Certain fungi and yeast strains may be selected in which glycosylation pathways have been "humanized," resulting in the production of an antibody with a partially or fully human glycosylation pattern. See, e.g., Li et al., *Nat. Biotech.* 24:210-215 (2006).

(242) Plant cell cultures of cotton, corn, potato, soybean, petunia, tomato, duckweed (Lemnaceae), alfalfa (*M. truncatula*), and tobacco can also be utilized as hosts.

(243) Suitable host cells for the expression of glycosylated antibody are also derived from multicellular organisms (invertebrates and vertebrates). Examples of invertebrate cells include plant and insect cells. Numerous baculoviral strains and variants and corresponding permissive insect host cells from hosts such as *Spodoptera frugiperda* (caterpillar), *Aedes aegypti* (mosquito), *Aedes albopictus* (mosquito), *Drosophila melanogaster* (fruitfly), and *Bombyx mori* have been identified.

(244) Vertebrate cells may be used as hosts, and propagation of vertebrate cells in culture (tissue culture) has become a routine procedure. Examples of useful mammalian host cell lines are monkey kidney CV1 line transformed by SV40 (COS-7, ATCC CRL 1651); human embryonic kidney line (293 or 293 cells subcloned for growth in suspension culture, Graham et al., *J. Gen Virol.* 36:59 (1977)); baby hamster kidney cells (BHK, ATCC CCL 10); mouse sertoli cells (TM4, Mather, *Biol. Reprod.* 23:243-251 (1980)); monkey kidney cells (CV1 ATCC CCL 70); African green monkey kidney cells (VERO-76, ATCC CRL-1587); human cervical carcinoma cells (HELA, ATCC CCL 2); canine kidney cells (MDCK, ATCC CCL 34); buffalo rat liver cells (BRL 3A, ATCC CRL 1442); human lung cells (W138, ATCC CCL 75); human liver cells (Hep G2, HB 8065); mouse mammary tumor (MMT 060562, ATCC CCL51); TRI cells (Mather et al., *Annals N.Y. Acad. Sci.* 383:44-68 (1982)); MRC 5 cells; FS4 cells; and a human hepatoma line (Hep G2). Other useful mammalian host cell lines include Chinese hamster ovary (CHO) cells, including DHFR-CHO cells (Urlaub et al., *Proc. Natl. Acad. Sci. USA* 77:4216 (1980)); and myeloma cell lines such as NSO and Sp2/0. For a review of certain mammalian host cell lines suitable for antibody production, see, e.g., Yazaki and Wu, *Methods in Molecular Biology*, Vol. 248 (B. K. C. Lo, ed., Humana Press, Totowa, N.J., 2003), pp. 255-268.

(245) The host cells of the present disclosure may be cultured in a variety of media. Commercially available media such as Ham's F10 (Sigma), Minimal Essential Medium ((MEM), (Sigma), RPMI-1640 (Sigma), and Dulbecco's Modified Eagle's Medium ((DMEM), Sigma) are suitable for culturing the host cells. In addition, any of the media described in Ham et al., *Meth. Enz.* 58:44 (1979), Barnes et al., *Anal. Biochem.* 102:255 (1980), U.S. Pat. Nos. 4,767,704; 4,657,866; 4,927,762; 4,560,655; or 5,122,469; WO 90/03430; WO 87/00195; or U.S. Pat. Re. 30,985 may be used as culture media for the host cells. Any of these media may be supplemented as necessary with hormones and/or other growth factors (such as insulin, transferrin, or epidermal growth factor), salts (such as sodium chloride, calcium, magnesium, and phosphate), buffers (such as HEPES), nucleotides (such as adenosine and thymidine), antibiotics (such as GENTAMYCINTM drug), trace elements (defined as inorganic compounds usually present at final concentrations in the micromolar range), and glucose or an equivalent energy source. Any other necessary supplements may also be included at appropriate concentrations that would be known to those skilled in the art. The culture conditions, such as temperature, pH, and the like, are those previously used with the host cell selected for expression, and will be apparent to one of skill in the art.

(246) When using recombinant techniques, the antibody can be produced intracellularly, in the periplasmic space, or directly secreted into the medium. If the antibody is produced intracellularly, as a first step, the particulate debris, either host cells or lysed fragments, are removed, for example, by centrifugation or ultrafiltration. Carter et al., *Bio Technology* 10:163-167 (1992) describe a procedure for isolating antibodies which are secreted to the periplasmic space of *E. coli*.

(247) The antibody composition prepared from the cells can be purified using, for example, hydroxylapatite chromatography, hydrophobic interaction chromatography, gel electrophoresis, dialysis, and affinity chromatography, with affinity chromatography being among one of the typically preferred purification steps.

(248) In some embodiments, an antibody of the present disclosure comprises a kappa or lambda light chain constant region. In some embodiments, an antibody of the present disclosure comprises

a light chain constant region comprising the amino acid sequence of SEQ ID NO:325, 326, or 426. Exemplary and non-limiting light chain constant region sequences are provided in Table 6. In some embodiments, an antibody of the present disclosure comprises an IGLC3 lambda light chain constant region or an IGLC7 constant region.

(249) In some embodiments, an antibody of the present disclosure includes an Fc region. For example, in some embodiments, the Fc region is a human Fc region, e.g., IgG1, IgG2, or IgG4 and subtypes thereof. Exemplary and non-limiting Fc regions are provided within the amino acid sequences of SEQ ID NOs:320-324 shown in Table 6. In some embodiments, an Fc region within one or more of the amino acid sequences of SEQ ID NOs:320-324 comprises one or more of the mutations described herein, e.g., infra.

(250) TABLE-US-00011 TABLE 6 Exemplary constant region sequences SEQ ID Name
NO Sequence IgG1 wildtype 320

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS GALTSGVHTFPAVLQSS
GLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC DKTHHTCPPCPAPELLGG
PSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVD GVEVHNAKTKPREEQYN
STYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSREE
MTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLD SDGSFFLYSKLTVDKSRW
QQGNVFSCSVMHEALHNHYTQKSLSLSPGK IgG1_AAA_N297A 321

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS
GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVN
HKPSNTKVDKKVEPKSCDKTHHTCPPCPAPEAAGAPSVFLFPPK
PKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYKCKVSNKAL
PAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKS
RWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK IgG2 322

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNS GALTSGVHTFPAVLQSSG
LYSLSSVVTVPSSNFGTQTYTCNVDHKPSNTKVDKTKVERKCC VECPPCPAPPVAGPSVFL
FPPKPKDTLMISRTPEVTCVVVDVSHEDPEVQFNWYVDGVEV HNAKTKPREEQFNSTFRV
VSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQPR EPQVYTLPPSREEMTKNQ
VSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPMLDSDGSFF LYSKLTVDKSRWQQGNV
FSCSVMHEALHNHYTQKSLSLSPGK IgG2Da 323

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNS
GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSNFGTQTYTCNVD
HKPSNTKVDKTKVERKCCVECPPCPAPPVAGPSVFLFPPKPKDT
LMISRTPEVTCVVVDVSHEDPEVQFNWYVDGVEVHNAKTKP
REEQFNSTFRVVSVLTVVHQDWLNGKEYKCKVSNKGLPSSIE
KTISKTKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDI
AVEWESNGQPENNYKTTTPMLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGK IgG4_S228P 324

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNS
GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYTCNVD
HKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLFPPKPKDT
LMISRTPEVTCVVVDVSDQEDPEVQFNWYVDGVEVHNAKTKP
REEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIE
KTISKAKGQPREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSD
IAVEWESNGQPENNYKTTTPVLDSDGSFFLYSRLTVDKSRWQ
EGNVFSCSVMHEALHNHYTQKSLSLSLGK Human Kappa 325
RTVAAPSFIFFPSDEQLKSGTASVCLLNNFYFPREAKVQWKV
DNALQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVY

ACEVTHQGLSSPVTKNGEC Human Lambda IGLC1 326
GQPKANPTVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWK
ADGSPVKAGVETTKPSKQSNNKYAASSYLSLTPEQWKSHRSY
SCQVTHEGSTVEKTVAPTECS Human Lambda IGLC2 426
GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWK
ADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKSHRSY
SCQVTHEGSTVEKTVAPTECS

(251) In some embodiments, the Fc region includes one or more mutations that influence one or more antibody properties, such as stability, pattern of glycosylation or other modifications, effector cell function, pharmacokinetics, and so forth. In some embodiments, an antibody of the present disclosure has reduced or minimal glycosylation. In some embodiments, an antibody of the present disclosure has ablated or reduced effector function. Exemplary Fc mutations include without limitation (i) a human IgG1 Fc region mutations L234A, L235A, G237A, and N297A; (ii) a human IgG2 Fc region mutations A330S, P331S and N297A; and (iii) a human IgG4 Fc region mutations S228P, E233P, F234V, L235A, delG236, and N297A (EU numbering). In some embodiments, the human IgG2 Fc region comprises A330S and P331S mutations. In some embodiments, the human IgG4 Fc region comprises an S288P mutation. In some embodiments, the human IgG4 Fc region comprises S288P and L235E mutations.

(252) Antibodies that target cell surface antigens can trigger immunostimulatory and effector functions that are associated with Fc receptor (FcR) engagement on immune cells. There are a number of Fc receptors that are specific for particular classes of antibodies, including IgG (gamma receptors), IgE (eta receptors), IgA (alpha receptors) and IgM (mu receptors). Binding of the Fc region to Fc receptors on cell surfaces can trigger a number of biological responses including phagocytosis of antibody-coated particles (antibody-dependent cell-mediated phagocytosis, or ADCP), clearance of immune complexes, lysis of antibody-coated cells by killer cells (antibody-dependent cell-mediated cytotoxicity, or ADCC) and, release of inflammatory mediators, placental transfer, and control of immunoglobulin production. Additionally, binding of the C1 component of complement to antibodies can activate the complement system. Activation of complement can be important for the lysis of cellular pathogens. However, the activation of complement can also stimulate the inflammatory response and can also be involved in autoimmune hypersensitivity or other immunological disorders. Variant Fc regions with reduced or ablated ability to bind certain Fc receptors are useful for developing therapeutic antibodies and Fc-fusion polypeptide constructs which act by targeting, activating, or neutralizing ligand functions while not damaging or destroying local cells or tissues.

(253) In some embodiments, a Fc domain monomer refers to a polypeptide chain that includes second and third antibody constant domains (e.g., CH2 and CH3). In some embodiments, an Fc domain monomer also includes a hinge domain. In some embodiments, the Fc domain monomer is of any immunoglobulin antibody isotype, including IgG, IgE, IgM, IgA, and IgD. Additionally, in some embodiments, an Fc domain monomer is of any IgG subtype (e.g., IgG1, IgG2, IgG2a, IgG2b, IgG2c, IgG3, and IgG4). In some embodiments, Fc domain monomers include as many as ten changes from a wild-type Fc domain monomer sequence (e.g., 1-10, 1-8, 1-6, 1-4 amino acid substitutions, additions or insertions, deletions, or combinations thereof) that alter the interaction between an Fc domain and an Fc receptor.

(254) In some embodiments, an Fc domain monomer of an immunoglobulin or a fragment of an Fc domain monomer is capable of forming an Fc domain with another Fc domain monomer. In some embodiments, an Fc domain monomer of an immunoglobulin or a fragment of an Fc domain monomer is not capable of forming an Fc domain with another Fc domain monomer. In some embodiments, an Fc domain monomer or a fragment of an Fc domain is fused to a polypeptide of the disclosure to increase serum half-life of the polypeptide. In some embodiments, an Fc domain monomer or a fragment of an Fc domain monomer fused to a polypeptide of the disclosure

dimerizes with a second Fc domain monomer to form an Fc domain which binds an Fc receptor, or alternatively, an Fc domain monomer binds to an Fc receptor. In some embodiments, an Fc domain or a fragment of the Fc domain fused to a polypeptide to increase serum half-life of the polypeptide does not induce any immune system-related response. An Fc domain includes two Fc domain monomers that are dimerized by the interaction between the CH3 antibody constant domains.

(255) A wild-type Fc domain forms the minimum structure that binds to an Fc receptor, e.g., FcγRI, FcγRIIa, FcγRIIb, FcγRIIIa, FcγRIIIb, and FcγRIV. In some embodiments, the Fc domain in an antibody of the present disclosure comprises one or more amino acid substitutions, additions or insertions, deletions, or any combinations thereof that lead to decreased effector function such as decreased antibody-dependent cell-mediated cytotoxicity (ADCC), decreased complement-dependent cytotoxicity (CDC), decreased antibody-dependent cell-mediated phagocytosis (ADCP), or any combinations thereof. For example, an antibody of the present disclosure can exhibit decreased binding (e.g., minimal binding or absence of binding) to a human Fc receptor and decreased binding (e.g., minimal binding or absence of binding) to complement protein C1q; decreased binding (e.g., minimal binding or absence of binding) to human FcγRI, FcγRIIA, FcγRIIB, FcγRIIIB, FcγRIIIB, or any combinations thereof, and C1q; altered or reduced antibody-dependent effector function, such as ADCC, CDC, ADCP, or any combinations thereof, and so forth. Exemplary mutations include without limitation one or more amino acid substitutions at E233, L234, L235, G236, G237, D265, D270, N297, E318, K320, K322, A327, A330, P331, or P329 (numbering according to the EU index of Kabat (Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD (1991))).

(256) In some embodiments, an antibody of the present disclosure has reduced or ablated binding to CD16a, CD32a, CD32b, CD32c, and CD64 Fcγ receptors. In some embodiments, an antibody with a non-native Fc region described herein exhibits at least a 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or greater reduction in C1q binding compared to an antibody comprising a wild-type Fc region. In some embodiments, an antibody with a non-native Fc region as described herein exhibit at least a 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or greater reduction in CDC compared to an antibody comprising a wild-type Fc region.

(257) In some embodiments, the Fc variants herein are minimally glycosylated or have reduced glycosylation relative to a wild-type sequence. In some embodiments, deglycosylation is accomplished with a mutation of N297A, or by mutating N297 to any amino acid which is not N.

(258) In some embodiments, variants of antibody IgG constant regions (e.g., Fc variants) possess a reduced capacity to specifically bind Fcγ receptors or have a reduced capacity to induce phagocytosis. In some embodiments, variants of antibody IgG constant regions (e.g., Fc variants) possess a reduced capacity to specifically bind Fcγ receptors and have a reduced capacity to induce phagocytosis. For example, in some embodiments, an Fc domain is mutated to lack effector functions, typical of a “dead” Fc domain. For example, in some embodiments, an Fc domain includes specific amino acid substitutions that are known to minimize the interaction between the Fc domain and an Fcγ receptor. In some embodiments, an Fc domain monomer is from an IgG1 antibody and includes one or more of amino acid substitutions L234A, L235A, G237A, and N297A (as designated according to the EU numbering system per Kabat et al., 1991). In some embodiments, one or more additional mutations are included in such IgG1 Fc variant. Non-limiting examples of such additional mutations for human IgG1 Fc variants include E318A and K322A. In some instances, a human IgG1 Fc variant has up to 12, 11, 10, 9, 8, 7, 6, 5 or 4 or fewer mutations in total as compared to wild-type human IgG1 sequence. In some embodiments, one or more additional deletions are included in such IgG1 Fc variant. For example, in some embodiments, the C-terminal lysine of the Fc IgG1 heavy chain constant region is deleted, for example to increase the homogeneity of the polypeptide when the polypeptide is produced in bacterial or mammalian cells. In some instances, a human IgG1 Fc variant has up to 12, 11, 10, 9, 8, 7, 6, 5 or 4 or fewer deletions in total as compared to wild-type human IgG1 sequence.

(259) In some embodiments, an Fc domain monomer is from an IgG2 antibody and includes amino acid substitutions A330S, P331S, or both A330S and P331S. The aforementioned amino acid positions are defined according to Kabat, et al. (1991). The Kabat numbering of amino acid residues can be determined for a given antibody by alignment at regions of homology of the sequence of the antibody with a “standard” Kabat numbered sequence. In some embodiments, the Fc variant comprises a human IgG2 Fc sequence comprising one or more of A330S, P331S and N297A amino acid substitutions (as designated according to the EU numbering system per Kabat, et al. (1991)). In some embodiments, one or more additional mutations are included in such IgG2 Fc variants. Non-limiting examples of such additional mutations for human IgG2 Fc variant include V234A, G237A, P238S, V309L and H268A (as designated according to the EU numbering system per Kabat et al. (1991)). In some instances, a human IgG2 Fc variant has up to 12, 11, 10, 9, 8, 7, 6, 5, 4, 3 or fewer mutations in total as compared to wild-type human IgG2 sequence. In some embodiments, one or more additional deletions are included in such IgG2 Fc variant.

(260) When the Fc variant is an IgG4 Fc variant, in some embodiments, such Fc variant comprises a S228P, E233P, F234V, L235A, L235E, or delG236 mutation (as designated according to Kabat, et al. (1991)). In some instances, a human IgG4 Fc variant has up to 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2 or 1 mutation(s) in total as compared to wild-type human IgG4 sequence.

(261) In some embodiments, the Fc variant exhibits reduced binding to an Fc receptor of the subject compared to the wild-type human IgG Fc region. In some embodiments, the Fc variant exhibits ablated binding to an Fc receptor of the subject compared to the wild-type human IgG Fc region. In some embodiments, the Fc variant exhibits a reduction of phagocytosis compared to the wild-type human IgG Fc region. In some embodiments, the Fc variant exhibits ablated phagocytosis compared to the wild-type human IgG Fc region.

(262) Antibody-dependent cell-mediated cytotoxicity, which is also referred to herein as ADCC, refers to a form of cytotoxicity in which secreted Ig bound onto Fc receptors (FcRs) present on certain cytotoxic cells (e.g., Natural Killer (NK) cells and neutrophils) enabling these cytotoxic effector cells to bind specifically to an antigen-bearing target cell and subsequently kill the target cell. Antibody-dependent cell-mediated phagocytosis, which is also referred to herein as ADCP, refers to a form of cytotoxicity in which secreted Ig bound onto Fc receptors (FcRs) present on certain phagocytic cells (e.g., macrophages) enabling these phagocytic effector cells to bind specifically to an antigen-bearing target cell and subsequently engulf and digest the target cell. Ligand-specific high-affinity IgG antibodies directed to the surface of target cells can stimulate the cytotoxic or phagocytic cells and can be used for such killing. In some embodiments, polypeptide constructs comprising an Fc variant as described herein exhibit reduced ADCC or ADCP as compared to a polypeptide construct comprising a wild-type Fc region. In some embodiments, polypeptide constructs comprising an Fc variant as described herein exhibit at least a 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or greater reduction in ADCC or ADCP compared to a polypeptide construct comprising a wild-type Fc region. In some embodiments, antibodies comprising an Fc variant as described herein exhibit ablated ADCC or ADCP as compared to a polypeptide construct comprising a wild-type Fc region.

(263) Complement-directed cytotoxicity, which is also referred to herein as CDC, refers to a form of cytotoxicity in which the complement cascade is activated by the complement component C1q binding to antibody Fc. In some embodiments, polypeptide constructs comprising an Fc variant as described herein exhibit at least a 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or greater reduction in C1q binding compared to a polypeptide construct comprising a wild-type Fc region. In some cases, polypeptide constructs comprising an Fc variant as described herein exhibit reduced CDC as compared to a polypeptide construct comprising a wild-type Fc region. In some embodiments, polypeptide constructs comprising an Fc variant as described herein exhibit at least a 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or greater reduction in CDC compared to a polypeptide construct comprising a wild-type Fc region. In some cases, antibodies

comprising an Fc variant as described herein exhibit negligible CDC as compared to a polypeptide construct comprising a wild-type Fc region.

(264) Fc variants herein include those that exhibit reduced binding to an Fcγ receptor compared to the wild-type human IgG Fc region. For example, in some embodiments, an Fc variant exhibits binding to an Fcγ receptor that is less than the binding exhibited by a wild-type human IgG Fc region to an Fcγ receptor. In some instances, an Fc variant has reduced binding to an Fcγ receptor by a factor of 10%, 20% 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, 99% or 100% (fully ablated effector function). In some embodiments, the reduced binding is for any one or more Fcγ receptors, e.g., CD16a, CD32a, CD32b, CD32c, or CD64.

(265) In some instances, the Fc variants disclosed herein exhibit a reduction of phagocytosis compared to its wild-type human IgG Fc region. Such Fc variants exhibit a reduction in phagocytosis compared to its wild-type human IgG Fc region, wherein the reduction of phagocytosis activity is e.g., by a factor of 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, 99% or 100%. In some instances, an Fc variant exhibits ablated phagocytosis compared to its wild-type human IgG Fc region.

(266) In some embodiments, the Fc variants disclosed herein are coupled to one or more fusion partners. In some cases the fusion partner is a therapeutic moiety, such as a cytotoxic agent of the present disclosure. In some cases, the fusion partner is selected to enable targeting of an expressed protein, purification, screening, display, and the like. In some embodiments, the fusion partner also affects the degree of binding to Fc receptors or the degree of phagocytosis reduction.

(267) In some embodiments, fusion partners are linked to the Fc variant sequence via a linker sequence. In some embodiments, the linker sequence generally comprises a small number of amino acids, such as less than ten amino acids, although longer linkers are also utilized. In some cases, the linker has a length less than 10, 9, 8, 7, 6, or 5 amino acids or shorter. In some cases, the linker has a length of at least 10, 11, 12, 13, 14, 15, 20, 25, 30, or 35 amino acids or longer. Optionally, in some embodiments, a cleavable linker is employed.

(268) In some embodiments, a fusion partner is a targeting or signal sequence that directs an Fc variant protein and any associated fusion partners to a desired cellular location or to the extracellular media. In some embodiments, certain signaling sequences target a protein to be either secreted into the growth media, or into the periplasmic space, located between the inner and outer membrane of the cell. In some embodiments, a fusion partner is a sequence that encodes a peptide or protein that enables purification or screening. Such fusion partners include, but are not limited to, polyhistidine tags (His-tags) (for example His6 and His10) or other tags for use with Immobilized Metal Affinity Chromatography (IMAC) systems (e.g., Ni+2 affinity columns), GST fusions, MBP fusions, Strep-tag, the BSP biotinylation target sequence of the bacterial enzyme BirA, and epitope tags which are targeted by antibodies (for example c-myc tags, flag-tags, and the like).

(269) In some embodiments, such tags are useful for purification, for screening, or both. For example, in some embodiments, an Fc variant is purified using a His-tag by immobilizing it to a Ni+2 affinity column, and then after purification the same His-tag is used to immobilize the antibody to a Ni+2 coated plate to perform an ELISA or other binding assay.

(270) Various fusion partners that enable a variety of selection methods are available. For example, by fusing the members of an Fc variant library to the gene III protein, phage display can be employed. In some embodiments, fusion partners enable Fc variants to be labeled. Alternatively, in some embodiments, a fusion partner binds to a specific sequence on the expression vector, enabling the fusion partner and associated Fc variant to be linked covalently or noncovalently with the nucleic acid that encodes them.

(271) In some embodiments, when a fusion partner is a therapeutic moiety, the therapeutic moiety is, e.g., a cytotoxic agent, a peptide, a protein, an antibody, a siRNA, or a small molecule.

(272) In some embodiments, an antibody of the present disclosure is bound to various carriers or

labels and used to detect the presence of specific antigen expressing cells. Examples of carriers include glass, polystyrene, polypropylene, polyethylene, dextran, nylon, amylases, natural and modified celluloses, polyacrylamides, agaroses, and magnetite. The nature of the carrier can be either soluble or insoluble. Various different labels and methods of labeling are known. Examples of labels include enzymes, radioisotopes, fluorescent compounds, colloidal metals, chemiluminescent compounds, and bio-luminescent compounds. Various techniques for binding labels to antibodies disclosed herein are available. In some embodiments, the antibodies are coupled to low molecular weight haptens. These haptens are then specifically detected by means of a second reaction. For example, in some embodiments, the hapten biotin is used with avidin or the haptens dinitrophenol, pyridoxal, or fluorescein are detected with specific anti-hapten antibodies (e.g., anti-dinitrophenol antibodies, anti-pyridoxal antibodies, and anti-fluorescein antibodies respectively). In some embodiments, the antibodies described herein are utilized in vitro for binding assays, such as immune assays. For example, in some embodiments, the antibodies are utilized in liquid phase or bound to a solid phase carrier. In some embodiments, antibodies utilized for immunoassays are detectably labeled in various ways.

(273) Methods of Identifying and/or Generating Antibodies

(274) Certain aspects of the present disclosure relate to methods of identifying an antigen binding domain that binds an extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide. Advantageously, the methods described herein can be used to identify antigen binding domains that block binding between human CD47 and a human SIRP- α polypeptide, do not block binding between human CD47 and a human SIRP- α polypeptide, or reduce affinity of a human SIRP- α polypeptide for human CD47.

(275) In some embodiments, the methods include providing an antigen binding domain that binds the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide. Exemplary antigen binding domains and antibodies that bind the extracellular domain (e.g., the D1 domain) of a human SIRP- α polypeptide, and/or other SIRP polypeptides, are described herein, as are exemplary methods for identifying such antigen binding domains/antibodies. Exemplary methods are described in greater detail in the Examples *infra*.

(276) In some embodiments, the methods include assembling a complex comprising a SIRP- α D1 variant bound to a polypeptide comprising an IgSF domain of CD47. In some embodiments, “assembling” the complex includes providing a solution containing both a SIRP- α D1 variant and a polypeptide comprising an IgSF domain of CD47. In some embodiments, the SIRP- α D1 variant is a non-naturally occurring variant, e.g., that binds to human CD47 with an affinity that is at least 10-fold, at least 100-fold, or at least 100-fold greater than the affinity of a naturally occurring SIRP- α D1 domain binding to human CD47. Advantageously, this facilitates antibody screening, as a natural SIRP- α :CD47 interaction may be too weak for use in binding and screening assays. Exemplary variants are described in greater detail *infra*.

(277) In some embodiments, the methods include contacting the antigen binding domain with the assembled complex. In some embodiments, binding, or a lack or deficiency thereof, of the antigen binding domain to the complex is detected. Various detection techniques are described herein. In some embodiments, SPR or ELISA is used. Detectable binding of the antigen binding domain to the complex indicates that the antigen binding domain does not block binding between human CD47 and the human SIRP- α polypeptide. A lack of binding of the antigen binding domain to the complex indicates that the antigen binding domain blocks binding between human CD47 and the human SIRP- α polypeptide. In addition, SPR is used to distinguish blocking, non-blocking and kick-off antibodies. For example, in some embodiments, a non-blocking antibody increases RUs when injected on top of pre-formed SIRP α :CD47 complex, a kick-off antibody increases K_{sub.off} of a pre-formed SIRP α :CD47 complex, and blocking antibody does not change the RUs or K_{sub.off} of a pre-formed SIRP α :CD47 complex.

(278) In some embodiments, the IgSF domain of CD47 is a human IgSF domain. In some

embodiments, the polypeptide comprising the IgSF domain of CD47 comprises a human CD47 extracellular domain. In some embodiments, the IgSF domain of CD47 comprises the amino acid sequence of

QLLFNKTKSVEFTFSNDTVVIPCFVTNMEAQNTTEVYVKWKFKGRDIYTFDGALNKSTVPTDFSSAKIEVSQLLKGDASLKMDKSDAVSHTGNYTCEVTELTREGETIIEELKYRVVS (SEQ ID NO:16). In some embodiments, the polypeptide comprising the IgSF domain of CD47 is conjugated to another polypeptide or other moiety, e.g., an Ig Fc region.

(279) High Affinity SIRP- α D1 Domain Variants

(280) A variety of high affinity SIRP- α D1 variants are contemplated for use herein. For example, in certain embodiments, the SIRP- α D1 variant comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 17-52 (see Table 4). Further descriptions of SIRP- α D1 variants follow.

(281) TABLE-US-00012 TABLE 4 Exemplary SIRP- α D1 variant amino acid sequences. SEQ ID NO Sequence

17 EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRELIYNQREG
PFPRVTTVSDTTKRNNMDFSIRIGAITPADAGTY YCVKFRKKGSPDDVEFKSGAG
TELSVRAKPS 18

19 EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQRQG
PFPRVTTVSETTKRENMDFSISISNITPADAGTY YCIKFRKKGSPDTEFKSGAGTE LSVRAKPS
20 EEELQVIQPDKSVSVAAGESAILHCTVTSLFPVGPIQWFRGAGPARELIYNQRQ
GPFPRVTTVSESTKRENMDFSISISNITPADAGTY YCVKFRKKGSPDTEFKSGAGT
ELSVRAKPS 20

21 EEELQVIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQRQG
PFPRVTTVSETTKRENMDFSISISNITPADAGTY YCIKFRKKGSPDTEFKSGAGTE LSVRAKPS
22 EEELQIIQPDKSVSVAAGESAILHCTITSLIPVGPIQWFRGAGPARVLIYNQRQGP
FPRVTTVSETTKRENMDFSISISNITPADAGTY YCIKFRKKGSPDTEFKSGAGTEL SVRAKPS

23 EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARELIYNQRQG
PFPRVTTVSETTKRENMDFSISISNITPADAGTY YCIKFRKKGSPDTEFKSGAGTE LSVRAKPS
24 EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQKQG
PFPRVTTVSETTKRENMDFSISISNITPADAGTY YCIKFRKKGSPDTEFKSGAGTE LSVRAKPS

25 EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQREG
PFPRVTTVSETTKRENMDFSISISNITPADAGTY YCIKFRKKGSPDTEFKSGAGTE LSVRAKPS
26 EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQRQG
HFPRVTTVSETTKRENMDFSISISNITPADAGTY YCIKFRKKGSPDTEFKSGAGTE LSVRAKPS

27 EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQRQG
PFPRVTTVSESTKRENMDFSISISNITPADAGTY YCIKFRKKGSPDTEFKSGAGTEL SVRAKPS
28 EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARVLIYNQRQG
PFPRVTTVSETTKRENMDFSISISNITPADAGTY YCVKFRKKGSPDTEFKSGAGTE

LSVRAKPS 29
EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIYNQREG
PFPRVTTVSESTKRENMDFSISISNITPADAGTY YCVKFRKKGSPDTEFKSGAGTE

LSVRAKPS 30
EEELQVIQPDKSVSVAAGESAILHCTVTSLFPVGPIQWFRGAGPARELIYNQRE
GPFPRVTTVSESTKRENMDFSISISNITPADAGTY YCVKFRKKGSPDTEFKSGAGT

ELSVRAKPS 31
EEELQVIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARELIYNQREG
PFPRVTTVSESTKRENMDFSISISNITPADAGTY YCVKFRKKGSPDTEFKSGAGTE

LSVRAKPS 32

EEELQVIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARELIYNQREG
PFPRVTTVSETTKRENMDFSISISNITPADAGTYYCVKFRKGSPDTEFKSGAGTE
LSVRAKPS 33
EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARELIYNQREGP
FPRVTTVSESTKRENMDFSISISNITPADAGTYYCVKFRKGSPDTEFKSGAGTEL SVRAKPS
34 EEELQVIQPDKSVSVAAGESAILHCTITSLIPVGPIQWFRGAGPARELIYNQREG
PFPRVTTVSESTKRENMDFSISISNITPADAGTYYCVKFRKGSPDTEFKSGAGTE
LSVRAKPS 35
EEELQIIQPDKSVSVAAGESAILHCTITSLFPVGPIQWFRGAGPARELIYNQREGP
FPRVTTVSETTKRENMDFSISISNITPADAGTYYCVKFRKGSPDTEFKSGAGTEL SVRAKPS
36 EEELQVIQPDKSVLVAAGETATLRCTATSLFPVGPIQWFRGAGPGRELIYNQRQ
GPFPRVTTVSDLTKRNNMDFSIRIGNITPADAGTYYCVKFRKGSPDDVEFKSGA
GTELSVRAKPS 37
EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRELIYNQREG
PFPRVTTVSDLTKRNNMDFSIRIGNITPADAGTYYCVKFRKGSPDDVEFKSGAG
TELSVRAKPS 38
EEELQVIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRELIYNQRE
GPFPRVTTVSDTTKRNNMDFSIRIGNITPADAGTYYCVKFRKGSPDDVEFKSGA
GTELSVRAKPS 39
EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRELIYNQREG
PFPRVTTVSDTTKRNNMDFSIRIGNITPADAGTYYCVKFRKGSPDDVEFKSGAG
TELSVRAKPS 40
EEELQVIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRELIYNQRE
GPFPRVTTVSDLTKRNNMDFSIRIGNITPADAGTYYCVKFRKGSPDDVEFKSGA
GTELSVRAKPS 41
EEELQVIQPDKSVLVAAGETATLRCTATSLFPVGPIQWFRGAGPGRELIYNQRE
GPFPRVTTVSDLTKRNNMDFSIRIGNITPADAGTYYCVKFRKGSPDDVEFKSGA
GTELSVRAKPS 42
EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRVLIYNQRQG
PFPRVTTVSDTTKRNNMDFSIRIGNITPADAGTYYCIKFRKGSPDDVEFKSGAG
TELSVRAKPS 43
EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRVLIYNQRQG
PFPRVTTVSDTTKRNNMDFSIRIGAITPADAGTYYCIKFRKGSPDDVEFKSGAG
TELSVRAKPS 44
EEELQVIQPDKSVLVAAGETATLRCTATSLFPVGPIQWFRGAGPGRELIYNQRQ
GPFPRVTTVSDLTKRNNMDFSIRIGAITPADAGTYYCVKFRKGSPDDVEFKSGA
GTELSVRAKPS 45
EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRVLIYNQREG
PFPRVTTVSDTTKRNNMDFSIRIGAITPADAGTYYCIKFRKGSPDDVEFKSGAG
TELSVRAKPS 46
EEELQVIQPDKSVLVAAGETATLRCTATSLFPVGPIQWFRGAGPGRELIYNQRE
GPFPRVTTVSDLTKRNNMDFSIRIGAITPADAGTYYCVKFRKGSPDDVEFKSGA
GTELSVRAKPS 47
EEELQVIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRELIYNQRE
GPFPRVTTVSDLTKRNNMDFSIRIGAITPADAGTYYCVKFRKGSPDDVEFKSGA
GTELSVRAKPS 48
EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRELIYNQREG
PFPRVTTVSDLTKRNNMDFSIRIGAITPADAGTYYCVKFRKGSPDDVEFKSGAG
TELSVRAKPS 49
EEELQVIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRELIYNQRE

GPFPRVTTVSDTTKRNNMDFSIRIGAITPADAGTYCYVKFRKGSPDDVEFKSGA
 GTELSVRAKPS 50
 EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELIYNQKE
 GHFPRVTTVSDLTNRNNMDFSIRIGNITPADAGTYCYVKFRKGSPDDVEFKSG
 AGTELSVRAKPS 51
 EEELQIIQPDKSVLVAAGETATLRCTITSLFPVGPIQWFRGAGPGRELIYNQREG
 PFPRVTTVSDTTKRNNMDFSIRIGAITPADAGTYCYVKFRKGSPDDVEFKSGAG
 TELSVAKPS 52
 EEELQVIQPDKSVLVAAGETATLRCTATSLFPVGPIQWFRGAGPGRELIYNQR
 QGPFPRVTTVSDLTNRNNMDFSIRIGNITPADAGTYCYVKFRKGSPDDVEFK
 SGAGTELSVRAKPS

(282) In some embodiments, a SIRP- α D1 variant polypeptide or fragment thereof binds to CD47 with a $K_{sub.D}$ less than $1 \times 10^{sup.-8}$ M, less than $5 \times 10^{sup.-9}$ M, less than $1 \times 10^{sup.-9}$ M, less than $5 \times 10^{sup.-10}$ M, less than $1 \times 10^{sup.-10}$ M or less than $1 \times 10^{sup.-11}$ M. In some embodiments, a SIRP- α D1 variant polypeptide or fragment thereof binds to CD47 with a $K_{sub.D}$ between about 500 nM and 100 nM, between about 100 nM and 50 nM, between about 50 nM and 10 nM, between about 10 nM and 5 nM, between about 5 nM and 1 nM, between about 1 nM and 500 pM, between about 500 pM and 100 pM, between about 100 pM and 50 pM, or between about 50 pM and 10 pM.

(283) In some embodiments, fragments include polypeptides of less than 10 amino acids in length, about 10 amino acids in length, about 20 amino acids in length, about 30 amino acids in length, about 40 amino acids in length, about 50 amino acids in length, about 60 amino acids in length, about 70 amino acids in length, about 80 amino acids in length, about 90 amino acids in length, about 100 amino acids in length, or more than about 100 amino acids in length. Fragments retain the ability to bind to CD47. Preferably, SIRP- α D1 variant polypeptides and fragments thereof bind to CD47 with a higher affinity than a SIRP- α polypeptide binds to CD47.

(284) In some embodiments, the above-mentioned SIRP- α D1 variant polypeptides are attached or fused to a second polypeptide. In some embodiments, the second polypeptide includes, without limitation, an Fc polypeptide, an Fc variant, an HSA polypeptide, an albumin peptide, a PEG polymer or a fragment of the foregoing.

(285) In some embodiments, the polypeptide includes a high affinity SIRP- α D1 domain that has at least 85% sequence identity (e.g., at least 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity) to any variant provided in Table 4.

(286) Antibody Generation in Chicken

(287) Certain aspects of the present disclosure relate to methods of producing an anti-SIRP- α antibody that binds the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides. Without wishing to be bound to theory, it is thought that the use of chicken is particularly advantageous because of the greater diversity between chicken SIRP- α and mammalian (e.g., human, monkey, mouse, etc.) SIRP- α . Further, the phylogenetic distance between chickens and mammals allows for the identification of antibodies that cross-react against, e.g., human and mouse SIRP- α polypeptides, which can be difficult to achieve by producing antibodies in mouse due to self-tolerance.

(288) In some embodiments, the methods include immunizing a chicken with a peptide comprising at least a portion of a human SIRP- α extracellular domain (e.g., the D1 domain). An exemplary immunization schedule is described infra. Methods for chicken immunization are described, e.g., in Mettler Izquierdo, S. et al. (2016) *Microscopy (Oxf)* 1-16. In some embodiments, the methods include obtaining an antibody from an antibody-producing cell from the immunized chicken.

(289) In some embodiments, the methods include detecting binding between the antibody obtained from the cell and the extracellular domains (e.g., the D1 domains) of two or more different human SIRP- α variant polypeptides. For example, in some embodiments, human SIRP- α v1 and v2, e.g.,

as described herein, are used. Exemplary detection techniques are described herein and include without limitation the GEM assay (see, e.g., WO2009111014 and Mettler Izquierdo, S. et al. (2016) *Microscopy (Oxf)* 1-16), SPR, and ELISA.

(290) Methods of Treatment

(291) Certain aspects of the present disclosure relate to treating a disease or disorder using an antibody described herein. In some embodiments, the disease is cancer. In some embodiments, the disease is an autoimmune or inflammatory disease.

(292) For example, provided herein are methods of treating or delaying progression of cancer in an individual by administering an effective amount of an antibody of the present disclosure. Without wishing to be bound to theory, it is thought that the antibodies described herein may be useful in the treatment of cancer, e.g., by abrogating the cancer's ability to inhibit phagocytosis and immune surveillance through the CD47: SIRP- α signaling axis, or by otherwise enhancing activation of the immune system (such as by activation of dendritic cells).

(293) In some embodiments, an antibody of the present disclosure is administered in combination with a chemotherapeutic agent.

(294) In some embodiments, an antibody of the present disclosure is administered in combination with a second antibody, e.g., an antibody that binds an antigen expressed by the cancer. Exemplary antigens expressed by cancers are known in the art and include without limitation CD19, CD20, CD22, CD30, CD33, CD38, CD52, CD56, CD70, CD74, CD79b, CD123, CD138, CS1/SLAMF7, Trop-2, 5T4, EphA4, BCMA, Mucin 1, Mucin 16, PTK7, PD-L1, STEAP1, Endothelin B Receptor, mesothelin, EGFRvIII, ENPP3, SLC44A4, GNMB, nectin 4, NaPi2b, LIV-1A, Guanylyl cyclase C, DLL3, EGFR, HER2, VEGF, VEGFR, integrin α V β 3, integrin α 5 β 1, MET, IGF1R, TRAILR1, TRAILR2, RANKL, FAP, Tenascin, Le.sup.y, EpCAM, CEA, gpA33, PSMA, TAG72, a mucin, CAIX, EPHA3, folate receptor α , GD2, GD3, and an MHC/peptide complex comprising a peptide from NY-ESO-1/LAGE, SSX-2, a MAGE family protein, MAGE-A3, gp100/pmel17, Melan-A/MART-1, gp75/TRP1, tyrosinase, TRP2, CEA, PSA, TAG-72, immature laminin receptor, MOK/RAGE-1, WT-1, SAP-1, BING-4, EpCAM, MUC1, PRAME, survivin, BRCA1, BRCA2, CDK4, CML66, MART-2, p53, Ras, β -catenin, TGF- β RII, HPV E6, or HPV E7. For example, in some embodiments, an antibody of the present disclosure is administered in combination with a monoclonal antibody that binds CD123 (also known as IL-3 receptor alpha), such as talacotuzumab (also known as CSL362 and JNJ-56022473). In some embodiments, an antibody of the present disclosure is administered in combination with a monoclonal antibody that binds EGFR (such as cetuximab). In some embodiments, the second antibody includes one or more effector functions, e.g., effector functions that are associated with Fc receptor (FcR) engagement on immune cells including without limitation ADCC or ADCP, and/or complement-dependent cytotoxicity (CDC). Without wishing to be bound to theory, it is thought that combining such an antibody with an antibody of the present disclosure is particularly advantageous, e.g., to direct FcR-expressing leukocytes to target a tumor cell to which the second antibody is bound while also inhibiting the responsiveness of SIRP- α expressed by the leukocyte to any CD47 expressed by the tumor cell with the SIRP- α antibody.

(295) In some embodiments, an antibody of the present disclosure is administered in combination with an immunotherapeutic agent. An immunotherapeutic agent may refer to any therapeutic that targets the immune system and promotes a therapeutic redirection of the immune system, such as a modulator of a costimulatory pathway, cancer vaccine, recombinantly modified immune cell, etc. In some embodiments, the immunotherapeutic agent comprises an antibody. Exemplary antigens of immunotherapeutic antibodies are known in the art and include without limitation PD-1, PD-L1, OX40, CTLA-4, CD137/4-1BB, B7-H3, FZD7, CD27, TNFR2, CCR4, CSF1R, CSF, TIM-3, LAG-3, VISTA, ICOS, CCR2, IDO, A2R, CD39, CD73, TIGIT, CD80, CD47, arginase, TDO, and PVRIG. Immunotherapeutic agents that are approved or in late-stage clinical testing include, without limitation, ipilimumab, pembrolizumab, nivolumab, atezolizumab, avelumab, durvalumab,

and the like. Without wishing to be bound to theory, it is thought that the antibodies of the present disclosure are suitable for use with immunotherapeutic agents due to complementary mechanisms of action, e.g., in activating both macrophages and T.sub.effector cells to target tumor cells. In certain embodiments, an antibody of the present disclosure is administered in combination with an inhibitor of the PD-L1/PD-1 pathway, e.g., an anti-PD-L1 or anti-PD-1 antibody. As demonstrated herein, combined administration of an anti-SIRP- α antibody of the present disclosure and an inhibitor of the PD-L1/PD-1 pathway can result in synergistic anti-tumor activity. For example, in some embodiments, a blocking anti-SIRP- α antibody of the present disclosure is administered in combination with an anti-PD-1 antibody. In some embodiments, a non-blocking anti-SIRP- α antibody of the present disclosure is administered in combination with an anti-PD-1 antibody. In some embodiments, a blocking anti-SIRP- α antibody of the present disclosure is administered in combination with an anti-PD-L1 antibody. In some embodiments, a non-blocking anti-SIRP- α antibody of the present disclosure is administered in combination with an anti-PD-L1 antibody.

(296) Any cancer type known in the art may be included, such as but not limited to carcinoma, sarcoma, lymphoma, leukemia, lymphoma, and blastoma. More particular examples of such cancers include, but are not limited to, lung cancer, squamous cell cancer, brain tumors, glioblastoma, head and neck cancer, hepatocellular cancer, colorectal cancer (e.g., colon or rectal cancers), liver cancer, bladder cancer, gastric or stomach cancer, pancreatic cancer, cervical cancer, ovarian cancer, cancer of the urinary tract, breast cancer, peritoneal cancer, uterine cancer, salivary gland cancer, kidney or renal cancer, prostate cancer, vulval cancer, thyroid cancer, anal carcinoma, penile carcinoma, melanoma, multiple myeloma and B-cell lymphoma (including non-Hodgkin's lymphomas (NHL)); acute lymphoblastic leukemia (ALL); chronic lymphocytic leukemia (CLL); acute myeloid leukemia (AML); Merkel cell carcinoma; hairy cell leukemia; chronic myeloblastic leukemia (CML); and associated metastases.

(297) In addition to cancer therapies, the antibodies provided herein are useful in therapies in which monoclonal antibodies are administered for the purpose of depleting cells, e.g., in the treatment of inflammatory diseases by depletion immune cells. For such purposes the an antibody provided herein is administered in combination with a second therapeutic antibody, e.g. with rituximab for depletion of B cells in inflammatory diseases and autoimmune conditions; alemtuzumab for multiple sclerosis; OKT3 for immunosuppression; others for bone marrow transplant conditioning; and the like.

(298) Further provided herein are methods of treating or delaying progression of an autoimmune disease or an inflammatory disease in an individual by administering an effective amount of an antibody of the present disclosure. Autoimmune diseases and inflammatory diseases amenable to treatment according to the disclosure include, but are not limited to, multiple sclerosis, rheumatoid arthritis, a spondyloarthropathy, systemic lupus erythematosus, an antibody-mediated inflammatory or autoimmune disease, graft versus host disease, sepsis, diabetes, psoriasis, atherosclerosis, Sjogren's syndrome, progressive systemic sclerosis, scleroderma, acute coronary syndrome, ischemic reperfusion, Crohn's Disease, endometriosis, glomerulonephritis, myasthenia gravis, idiopathic pulmonary fibrosis, asthma, acute respiratory distress syndrome (ARDS), vasculitis, and inflammatory autoimmune myositis. In some embodiments, an antibody of the present disclosure is administered in combination with a therapeutic agent, such as an immunosuppressive, anti-inflammatory, or immunomodulatory agent. In some embodiments, an antibody provided herein is used in the treatment of an autoimmune disease or an inflammatory disease, e.g., multiple sclerosis, rheumatoid arthritis, a spondyloarthropathy, systemic lupus erythematosus, an antibody-mediated inflammatory or autoimmune disease, graft versus host disease, sepsis, diabetes, psoriasis, psoriatic arthritis, atherosclerosis, Sjogren's syndrome, progressive systemic sclerosis, scleroderma, acute coronary syndrome, ischemic reperfusion, Crohn's Disease, ulcerative colitis, endometriosis, glomerulonephritis, IgA nephropathy, polycystic kidney disease, myasthenia gravis, idiopathic pulmonary fibrosis, asthma, atopic dermatitis, acute respiratory distress syndrome (ARDS),

vasculitis, or inflammatory autoimmune myositis.

(299) In some embodiments, an antibody of the present disclosure is part of a pharmaceutical formulation, e.g., including the antibody and one or more pharmaceutically acceptable carriers. Pharmaceutical compositions and formulations as described herein can be prepared by mixing the active ingredients (such as an antibody or a polypeptide) having the desired degree of purity with one or more optional pharmaceutically acceptable carriers (*Remington's Pharmaceutical Sciences* 16th edition, Osol, A. Ed. (1980)), in the form of lyophilized formulations or aqueous solutions. Pharmaceutically acceptable carriers are generally nontoxic to recipients at the dosages and concentrations employed, and include, but are not limited to: buffers such as phosphate, citrate, and other organic acids; antioxidants including ascorbic acid and methionine; preservatives; low molecular weight (less than about 10 residues) polypeptides; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; sugars such as sucrose, mannitol, trehalose or sorbitol; salt-forming counter-ions such as sodium; metal complexes (e.g. Zn-protein complexes); and/or non-ionic surfactants such as polyethylene glycol (PEG). In some embodiments, an antibody of the present disclosure is lyophilized.

EXAMPLES

(300) The present disclosure will be more fully understood by reference to the following examples. The examples should not, however, be construed as limiting the scope of the present disclosure. It is understood that the examples and embodiments described herein are for illustrative purposes only and that various modifications or changes in light thereof will be suggested to persons skilled in the art and are to be included within the spirit and purview of this application and scope of the appended claims.

Example 1: Identification of Antibodies with Novel Binding Specificities to SIRP- α Proteins

(301) Methods

(302) Antibody Production

(303) The following proteins were used for immunization. Each includes a human or mouse SIRP- α peptide fused to a modified Fc region (either an S228P human IgG4 Fc or an L234A/L235A/G237A/N297A human IgG1 Fc designated as IgG1_AAA_N297A) for increased immunogenicity.

(304) TABLE-US-00013 TABLE A Immunogen sequences. Description SEQ ID NO
Sequence Human sirpa v1 1 EEELQVIQPDKSVLVAAGETATLRCTATSLIPV (Fusion
with Fc of IgG4_S228P) GPIQWFRGAGPGRELIYNQKEGHFPRVTTVSD

LTKRNNMDFSIRIGNITPADAGTY YCVKFRKG
SPDDVEFKSGAGTELSVRAKPSESKYGPPCPP
CPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTC
VVVDVSQEDPEVQFNWYVDGVEVHNAKTKP
REEQFNSTYRVVSVLTVLHQDWLNGKEYKCK
VSNKGLPSSIEKTISKAKGQPREPQVYTLPPSQ
EEMTKNQVSLTCLVKGFYPSDIAVEWESNGQ
PENNYKTTTPVLDSDGSFFLYSRLTVDKSRWQ
EGNVFSCSVMHEALHNHYTQKSLSLGLK

Human sirpa v2 2
EEELQVIQPDKSVSVAAGESAILHCTVTSLIPV (Fusion with Fc of IgG4_S228P)

GPIQWFRGAGPARELIYNQKEGHFPRVTTVSE
STKRENMDFSISISNITPADAGTY YCVKFRKGS
PDTEFKSGAGTELSVRAKPSESKYGPPCPPCPA
PEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVV
DVSQEDPEVQFNWYVDGVEVHNAKTKPREE
QFNSTYRVVSVLTVLHQDWLNGKEYKCKVS

NKGLPSSIEKTISKAKGQPREPQVYTLPPSQEE
 MTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
 NNYKTTTPPVLDSDGSFFLYSRLTVDKSRWQE GNVFSCSVMHEALHNHYTQKSLSLGLK
 Mouse 129 sirpa 3 KELKVTQPEKSVSVAAGDSTVLNCTLTSLLPV (Fusion with Fc of
 GPIKWYRGVVGQSRLLIYSFTGEHFPRVTNVSD IgG1_AAA_N297A)
 ATKRNNDMDFSIRISNVTPEAGTYCYVKFQKG
 PSEPDTEIQSGGGTEVYVLAKPSDKTHTCPPCP
 APEAAGAPSVFLFPPKPKDTLMISRTPEVTCVV
 VDVSHEDPEVKFNWYVDGVEVHNAKTKPRE
 EQYASTYRVVSVLTVLHQDWLNGKEYKCKV
 SNKALPAPIEKTISKAKGQPREPQVYTLPPSRE
 EMTKNQVSLTCLVKGFYPSDIAVEWESNGQP
 ENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQ QGNVFSCSVMHEALHNHYTQKSLSLSPGK
 Mouse NOD sirpa 4 TEVKVIQPEKSVSVAAGDSTVLNCTLTSLLPV (Fusion with Fc
 of GPIRWYRGVVGQSRQUYSFTTEHFPRVTNVSD IgG1_AAA_N297A)
 ATKRSNLDIFSIRISNVTPEAGTYCYVKFQRG
 SPDTEIQSGGGTEVYVLAKDKTHTCPPCPAPE
 AAGAPSVFLFPPKPKDTLMISRTPEVTCVVVD
 VSHEDPEVKFNWYVDGVEVHNAKTKPREEQ
 YASTYRVVSVLTVLHQDWLNGKEYKCKVSN
 KALPAPIEKTISKAKGQPREPQVYTLPPSREEM
 TKNQVSLTCLVKGFYPSDIAVEWESNGQPEN
 NYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQG NVFSCSVMHEALHNHYTQKSLSLSPGK

(305) The above proteins were used to immunize wild-type chickens, SynVH chickens which are transgenic chickens containing VH from human and VL from chicken, or chickens with fully human “HuMAB” immunoglobulin loci (Crystal Bioscience; see, e.g., WO2012162422, WO2011019844, and WO2013059159). Chickens were immunized with varied schedules having alternating doses of antigen. An exemplary immunization schedule is as follows: initial immunization with 100 µg dose of antigen having the sequence of SEQ ID NO:1 at week 1, boost of 100 µg of antigen having the sequence of SEQ ID NO: 2 at week 3, draw at week 4, boost with 50 µg dose of antigen having the sequence of SEQ ID NO:1 at week 5, draw at week 6, boost with 50 µg of antigen having the sequence of SEQ ID NO: 2 at week 7, and draw at week 8. Additional descriptions of chicken immunization may be found, e.g., in Mettler Izquierdo, S. et al. (2016) *Microscopy (Oxf)* 1-16.

(306) Screening and Expression of Antibody Clones

(307) Clones were generated and screened according to the GEM assay (see Mettler Izquierdo, S. et al. (2016) *Microscopy (Oxf)* 1-16). Clones were tested in a scFv-Fc format in which the light chain is fused via linker with the heavy chain. The scFv was fused to the N-terminus of human Fc-IgG1. Clones were expressed in FreeStyle™ 293-FS cells (Thermo Fisher) and secreted media was used for ELISA and SPR binding characterization.

(308) SPR

(309) Binding of the antibody clones to various SIRP proteins was determined using surface plasmon resonance (SPR) detection on a ProteOn XPR36 instrument (Bio-Rad, Hercules, CA) using phosphate buffered saline (PBS, pH 7.4) supplemented with 0.01% Tween-20 (PBST) as running buffer. The pre-filtered media containing the secreted antibodies was used directly for the assay. First, anti-Human IgG Fc (BR-1008-39, GE Healthcare) was amine-coupled onto a GLC sensor chip to generate the capture surfaces for the antibodies. About 4000 RU per flow cell of immobilized anti-human IgG Fc is achieved. Each clone is screened using the same method as follows. The SIRP analytes used for the screen are listed in Table C. (1) ~5-10 uL of pre-filtered media in 10 mM sodium acetate buffer (pH4.5) was injected for 2 mins at 30 ul/min; (2) buffer

flow for 1 min at 100 uL/min; (3) SIRP analyte (100 nM) injected for 1 min at 100 uL/min, followed by a dissociation cycle of 10 mins; (4) regeneration of chip surface by flowing 3M Magnesium Chloride for 1 min at 25 uL/min in both orientation; and (5) buffer flow for 1 min at 100 uL/min.

(310) Biosensor data were double-referenced by subtracting the interspot data (containing no immobilized anti-human IgG Fc) from the reaction spot data (immobilized anti-human IgG Fc) and then subtracting the response of a buffer “blank” analyte injection from that of an analyte injection. Binding was fitted using a 1:1 Langmuir and $K_{sub.off}$ (1/S) values calculated. All SPR assays were performed at 25° C.

(311) ELISA

(312) ELISA assays were carried out to screen binding of antibody clones to SIRP analytes and SIRP- α :CD47 complex. Briefly, 96-well flat-bottom, high binding plates (Greiner Bio-One #655061) were coated with the following proteins in separate ELISA experiments: Avidin (Sigma A9275) (2 ug/ml) followed by biotinylated human SIRP α V1 (0.5 ug/ml), Avidin (2 ug/ml) followed by human SIRP α V2 biotin (0.5 ug/ml), mouse NOD SIRP α (2 ug/ml), human SIRP γ (2 ug/ml), CD47 (2 ug/ml) followed by high affinity human SIRP α V1 and V2 (at 2 ug/ml each) or anti-hFc (2 ug/ml Rockland 609-4103). Plates were blocked with PBSTM (Phosphate Buffered Saline pH 7.4, 0.05% Tween®20 polysorbate, 3% milk), 50 ul of supernatant containing the secreted scFv-Fc are added for 1 hr at room temp. Plates were washed with PBST. 50 ul of anti-hFc-HRP (1:5000 Rockland 609-4303) added for 1 hr at room temp. Plates were washed with PBST. TMB was developed for 5 min and stopped with 1N HCl. ELISA results were read using BioTek Synergy H1 Hybrid Reader. The corresponding proteins used for the assay and described here are: human SIRP α V1 (SEQ ID NO: 5); human SIRP α V2 (SEQ ID NO:6); cynomolgus SIRP α (SEQ ID NO:11); mouse NOD SIRP α (SEQ ID NO: 8); human SIRP γ (SEQ ID NO: 15); CD47 (SEQ ID NO: 16); high affinity SIRP α V1 (SEQ ID NO: 42); high affinity SIRP α V2 (SEQ ID NO:17).

(313) Results

(314) SIRP- α is a highly polymorphic protein in humans, monkeys, and mice. For example, 20 amino acid differences have been identified between SIRP- α proteins in the NOD and C57BL/6 mouse strains, and these polymorphisms lead to functional consequences related to CD47 binding and engraftment of human hematopoietic stem cells in these mouse strains. In humans, at least 10 distinct alleles of SIRPA have been identified, and amino acid variations that distinguish the alleles are found in predicted CD47-binding residues (Takenaka, K. et al. (2007) *Nat. Immunol.* 8:1313-23). An alignment of 10 human variant SIRP- α protein sequences is provided in FIG. 1A. The identification of antibodies having different binding specificities with intra- and/or inter-species cross-reactivity is of great interest for development of clinical candidates that are effective across human populations and the characterization of these candidates in various animal models.

(315) The D1 domains of SIRP- α proteins from human (v1 and v2 variants), cynomolgus monkey, and mouse 129 were aligned to identify conserved amino acids (FIG. 1B). While the cynomolgus monkey and mouse 129 sequences are much more divergent from human v1 and v2 than human variants v3-v10 as shown in FIG. 1A, these alignments demonstrate some degree of conservation among all four proteins, suggesting that cross-reactive antibodies could perhaps be identified. However, each protein also shows unique polymorphisms, suggesting that specific antibodies were also possible.

(316) As shown in FIG. 1C, SIRP- α protein sequences representing multiple human variants and mouse strains (e.g., 129, NOD, C57BL/6, and BALB/c) were also aligned. R1, R2, R3 delineate residues located around binding sites of SIRP α to CD47. These alignments demonstrated that the R2 and R3 are considerably more divergent among and between human and mouse sequences than the R1. Without wishing to be bound to theory, it is thought that anti-SIRP- α antibodies that bind to specific murine SIRP- α proteins may be useful, e.g., for pharmacokinetic studies (such as those

using CD-1 mice), development of transgenic mice (such as those in a C57BL/6 background), and/or characterization in SCID (e.g., in a NOD background) or syngeneic (e.g., in a BALB/c or C57BL/6 background) mouse models.

(317) Mammalian SIRP- α D1 domains described above were also aligned with the D1 domain of chicken SIRP- α comprising the sequence

DFKLQQPQSSVVVIKGDTLTLNCTASGSGPIGAVKWVKGWGSNDQTVYEHKGSFPRVM
RAVPDPTNDFTRISNVSLEDAGTYCYVKLRKGIVDDVVFTRGGGTEVSVHA (SEQ ID
NO:84) (FIG. 2). Compared with the mammalian SIRP- α sequences, the sequence of chicken
SIRP α was found to be significantly more divergent. Pairwise comparisons of the percentage of
sequence identity between various SIRP- α proteins is shown in Table B.

(318) TABLE-US-00014 TABLE B Pairwise sequence identities (%) between SIRP- α proteins.

Human v1 Human v2 Mouse Cyno Chicken (SEQ ID (SEQ ID (SEQ ID (SEQ ID (SEQ ID NO: 5)
NO: 6) NO: 7) NO: 11) NO: 84) Human v1 — 89 66 91 46 (SEQ ID NO: 5) Human v2 89 — 68 88
41 (SEQ ID NO: 6) Mouse 66 68 — 72 47 (SEQ ID NO: 7) Cyno 91 88 72 — 47 (SEQ ID NO: 11)
Chicken 46 41 47 47 — SEQ ID NO: 84)

(319) Without wishing to be bound to theory, it was thought that this divergence would provide
unique opportunities to generate antibodies that cross-react across multiple mammalian SIRP- α
proteins. For example, it may be difficult to generate anti-SIRP- α antibodies that cross-react with
the murine sequence from a mouse host due to immune tolerance. Moreover, the greater divergence
between the chicken and mammalian immune systems may lead to a greater diversity in antibody
production.

(320) In order to identify antibodies with novel binding specificities to SIRP- α proteins, antibody
clones were characterized in the scFv-Fc format as described above. Each antibody clone was
tested at concentrations between 0.008 and 1.0 μ g/mL for binding using ELISA to the following
targets: the D1 domain of human SIRP- α v1 (sequence according to SEQ ID NO:5), the D1 domain
of human SIRP- α v2 (sequence according to SEQ ID NO:6), the D1 domain of a cynomolgus
SIRP- α variant (sequence according to SEQ ID NO:11), the D1 domain of mouse 129 SIRP- α
(sequence according to SEQ ID NO:7), and a human SIRP γ isoform (sequence according to SEQ
ID NO:15). As used hereinafter, antibody clones are referred to by clone ID number. In addition,
the notation “S[clone number]” refers to an sc-Fv-Fc format; the notation “AB[clone number]”
refers to a full IgG antibody format; the notation “AB[clone number]a” refers to a human IgG1
with L234A, L235A, G237A, and N297A mutations; “AB[clone number]b” refers to a mouse IgG1
N297A format; “AB[clone number]c” refers to a mouse IgG2a format; and “[clone number] Fab”
refers to a Fab fragment format.

(321) In addition, the binding of each antibody clone to a pre-complex prepared using a 1:1 mix of
two high affinity SIRP- α variants (SEQ ID NO:42 and SEQ ID NO:18) bound with the IgSF
domain of CD47 (sequence according to SEQ ID NO:16) was characterized. Advantageously, since
the affinity between the wild-type SIRP- α D1 domain and the IgSF domain of CD47 is relatively
low, the use of a complex comprising a high affinity SIRP- α variant allows the identification of
antibodies that bind to SIRP- α while it is complexed with CD47 (e.g., a non-blocking antibody). It
also allows the identification of antibodies that are unable to bind SIRP- α while it is complexed
with CD47, suggesting that the antibody and CD47 compete for the same binding interface on
SIRP- α (e.g., a blocking antibody). The two SIRP α variants used (SEQ ID NO:17 and SEQ ID
NO:19) correspond to high affinity SIRP α D1 domain engineered using human SIRP α polypeptide
variant 1 and variant 2, respectively.

(322) FIG. 3A shows the ELISA binding curve for clone S130 (SEQ ID NO:71). This clone
demonstrated cross-reactivity across mammalian SIRP- α proteins, with binding to multiple human
variants as well as cynomolgus and murine proteins (FIG. 3B). However, its binding was also
isoform-specific, since no binding was observed with human SIRP γ . This clone was also identified
as a blocker of the interaction between SIRP- α and CD47, as no binding to a pre-formed complex

containing CD47 bound to a high-affinity SIRP- α variant was detected.

(323) FIG. 4A shows the ELISA binding curve for clone S121 (SEQ ID NO:75). This clone also demonstrated cross-reactivity across mammalian SIRP- α proteins, but it also bound to human SIRP γ , indicating pan-isoformic binding (FIG. 4B). This clone was also identified as a blocker of the interaction between SIRP- α and CD47.

(324) FIG. 5A shows the ELISA binding curve for clone S137 (SEQ ID NO:73). Like the clone shown in FIGS. 3A & 3B, this clone demonstrated cross-reactivity across mammalian SIRP- α proteins with isoform-specific binding, as it did not bind to human SIRP γ (FIG. 5B). However, this clone was identified as a non-blocker of the interaction between SIRP- α and CD47, since it bound to the pre-formed complex.

(325) FIG. 6A shows the ELISA binding curve for clone S128 (SEQ ID NO:70). This clone demonstrated isoform-specific cross-reactivity across primate SIRP- α proteins, but it did not bind to the murine protein (FIG. 6B). This clone was also identified as a blocker of the interaction between SIRP- α and CD47.

(326) FIG. 7A shows the ELISA binding curve for clone S135 (SEQ ID NO:72). Similar to the clone shown in FIGS. 6A & 6B, this blocking clone demonstrated cross-reactivity across primate SIRP- α proteins, but it did not bind to the murine protein (FIG. 7B). Unlike clone S128, this clone was found to cross-react with human SIRP γ .

(327) FIG. 8A shows the ELISA binding curve for clone S126 (SEQ ID NO:69). This clone was found to be human-specific, binding to both human SIRP- α variants but none of the other peptides (FIG. 8B). This clone was identified as a blocker of the interaction between SIRP- α and CD47.

(328) FIG. 9A shows the ELISA binding curve for clone S138 (SEQ ID NO:74). This clone only bound human SIRP- α variant 1, demonstrating a high degree of intra- and inter-species binding specificity (FIG. 9B). This clone was identified as a blocker of the interaction between SIRP- α and CD47.

(329) In conclusion, novel antibodies with a number of unique binding specificities to SIRP- α proteins were characterized. FIGS. 10A-10C provide an alignment showing scFv-Fc variable domain sequences obtained from a chicken that produces chicken antibodies (CDRs are indicated). FIGS. 10D-10F provide an alignment showing scFv-Fc variable domain sequences obtained from a chicken that produces human antibodies (CDRs are indicated). These studies highlight the utility of the chicken as a source of antibody diversity for the identification of anti-SIRP- α antibodies.

(330) To assess if antibodies can block binding of SIRP α to CD47, an SPR screen was also carried out, in addition to screening by ELISA. Antibody capture was carried out using an anti-human IgG-Fc immobilized GLC surface prepared as described above. A SIRP α variant (SEQ ID NO:17) engineered to bind CD47 (SEQ ID NO:16) with high nM affinity was used for the screen rather than a wildtype SIRP- α . This is because the wildtype SIRP- α variant has low μ M binding affinity to CD47, which does not allow stable complex interaction to assess sandwich formation. Phosphate buffered saline (PBS, pH 7.4) supplemented with 0.01% Tween-20 (PBST) as running buffer.

(331) First, approximately 5-10 μ L of pre-filtered media containing the antibodies in 10 mM sodium acetate buffer (pH4.5) was injected for 2 mins at 30 μ L/min and captured over the anti-human IgG-Fc immobilized GLC surface followed by a brief buffer flow of 1 min at 100 μ L/min. Next, 100 nM of a high affinity SIRP- α variant (SEQ ID NO:17) pre-mixed with CD47 (SEQ ID NO:16) at different concentrations of 0, 20, 55, 500, or 1500 nM was injected separately for a minute at 100 μ L/min with a dissociation time for 10 mins. For the description below, SIRP- α refers to SEQ ID NO:17.

(332) For an antibody that does not block binding of SIRP- α to CD47, it would be expected to bind to SIRP- α /CD47 complex and form a sandwich. Therefore, at increasing concentration of CD47, the resonance increased accordingly due to increased sandwich formation. Antibody clone S123 was found to match this profile. An example of the profile for S123 is shown in FIG. 12A.

(333) For an antibody that blocks binding of SIRP- α to CD47, the profile will be different than a

non-blocking antibody. At increasing concentration of CD47, one would expect fewer molecules of SIRP- α to be available to bind to the antibody since the antibody competes for the same binding site as CD47 and most/all SIRP- α is complexed with CD47. Therefore, one would expect the resonance (RU) to decrease with increasing concentration of CD47 in the mixture. Antibody clone S119 was found to match this profile. An example of the profile for S119 is shown in FIG. 12B. (334) In addition to blockers and non-blockers, a third category of antibodies was isolated that have a “kick off” profile. Antibody clone S118 was found to match this profile. An example of the profile for S118 is shown in FIG. 12C. At higher concentration of CD47 (e.g 500 nM and 1500 nM CD47), a transient sandwich was formed between the antibody, SIRP- α and CD47 as indicated by the higher resonance of 300 RU which then decayed over time. This decay of resonance units indicates that the antibody was able to bind to SIRP- α in the complex and “kick SIRP- α off” from binding to CD47.

(335) The $k_{sub.off}$ rates of binding of each clone to each SIRP analyte were determined using SPR (Table C). The SPR screening conditions have been described herein, and the K_{off} values were determined using Langmuir kinetic fittings. The CD47 blocking properties (block, non-block, kick-off) are described in the last column of the Table C. Each antibody is identified according to its corresponding SEQ ID NO. SIRP analyte sequences are as follows: CV1-3, SEQ ID NO:18; v1, SEQ ID NO:5; v2, SEQ ID NO:6; cyno1, SEQ ID NO:11; cyno2, SEQ ID NO:12; m129, SEQ ID NO:7; NOD, SEQ ID NO:8; BL6, SEQ ID NO: 9; sirpb1, SEQ ID NO:13; sirpg, SEQ ID NO:15.

(336) TABLE-US-00015 TABLE C Langmuir kinetic fittings ($k_{sub.off}$ 1/s) SEQ ID CV1- cd47
Clone NO 3 v1 v2 cyno1 cyno2 m129 NOD BL6 sirpb1 sirpg blocking S1 53 2.19E-03 2.33E-03
NB 3.48E-02 4.06E-03 2.12E-03 7.89E-03 1.70E-03 NB 2.19E-03 Block S2 54 1.49E-01
2.52E-03 NB LB 5.59E-03 1.42E-02 NB 8.85E-03 — — Block S8 55 LB 1.14E-04 1.42E-04
4.15E-03 6.12E-04 1.29E-02 1.79E-03 6.96E-03 9.30E-04 3.23E-04 Block S9 56 1.68E-04
1.29E-04 1.40E-04 3.38E-04 2.70E-04 NB 1.81E-02 NB 1.03E-04 7.10E-03 Block S11 57
2.69E-03 3.35E-05 1.86E-04 1.11E-03 3.92E-03 NB LB LB 1.59E-04 3.86E-04 Block S12 58
5.15E-02 1.01E-04 1.68E-04 1.17E-03 5.24E-04 NB 2.60E-02 1.98E-03 1.73E-04 3.07E-04
Block S13 59 LB 2.07E-04 2.91E-04 1.84E-03 5.01E-04 4.93E-02 3.83E-03 1.03E-02
1.35E-04 4.02E-04 Block S14 60 3.33E-03 5.42E-05 7.64E-05 6.49E-04 4.52E-04 3.73E-02
4.98E-03 1.81E-02 6.01E-05 1.26E-04 Block S115 61 4.84E-04 7.86E-06 1.95E-05 2.22E-05
6.91E-05 NB NB NB — 3.34E-05 Kick off S116 62 4.80E-04 2.84E-05 5.07E-05 7.01E-05
1.20E-04 NB NB NB — 8.25E-05 Kick off S117 63 2.75E-04 1.20E-05 3.40E-05 1.83E-05
5.92E-05 NB NB NB — 4.30E-06 Kick off S118 64 2.47E-04 9.17E-06 4.12E-05 6.97E-05
8.06E-05 NB NB NB 1.31E-05 1.13E-05 Kick off S119 65 5.65E-04 4.15E-04 1.48E-04
2.34E-04 3.10E-04 NB LB NB 4.28E-04 3.95E-04 Block S120 66 6.26E-04 4.04E-04
1.49E-04 2.36E-04 3.14E-04 NB LB NB 4.25E-04 3.93E-04 Block S122 67 6.63E-04
4.98E-04 2.26E-04 2.66E-04 3.27E-04 NB LB NB 3.52E-04 3.61E-04 Block S123 68
1.31E-03 1.55E-03 NB 1.57E-03 1.54E-03 NB NB NB NB/LB 1.43E-03 Non-block S126 69
NB NB NB NB NB NB NB NB NB NB — S128 70 NB 4.59E-03 2.11E-03 1.52E-02 1.47E-02
NB NB NB 4.21E-03 NB — S130 71 3.73E-03 2.82E-03 2.08E-02 LB LB LB NB NB LB
5.06E-03 Block S135 72 3.63E-03 6.58E-04 8.78E-05 4.73E-04 3.88E-04 NB NB NB
5.21E-04 1.46E-03 Block S137 73 1.37E-03 1.65E-03 1.88E-03 2.04E-03 1.79E-03 5.15E-04
4.43E-04 4.63E-03 3.76E-03 1.48E-02 Non-block S138 74 LB NB NB NB NB NB NB NB NB
NB — NB = No binding LB = very low binding ($K_{sub.off}$ cannot be calculated by Langmuir
fittings) — = not screened

(337) These data indicate that antibody clones with a variety of binding specificities for SIRP- α proteins across different species and intra-species variants were identified. The binding specificities for selected antibodies were further characterized using ELISA as described above by generating binding curves against the human v1 (SEQ ID NO: 5), human v2 (SEQ ID NO: 6), NOD mouse (SEQ ID NO:8), and cynomolgus SIRP- α D1 domains (SEQ ID NO:11), as well as human SIRPy

(SEQ ID NO: 15) and a pre-formed complex of a 1:1 mixture of two high-affinity SIRP- α variants (SEQ ID NOs:17 and 19) bound to the IgSF domain of CD47 (SEQ ID NO: 16).

(338) Sequences of scFv-Fc clones tested are provided in Table D.

(339) TABLE-US-00016 TABLE D scFv-Fc sequences. Framework/ SEQ ID Clone CDR NO
Sequence S1 Chicken 53 ALTQPASVSANPGETVEITCSGGGSNNAYGWFQQ

KSPGSAPLTVIYDNGKRPSDIPSRFSGSKSDSTGTLT
ITRVQAEDEAVYYCGSADNSGAGVFGAGTTTLTVL
GQSSRSSGGGGSSGGGGSSMAAVTLDESGGGLQTP
GGALSLVCKGSGFTFSSHAMNWVRQAPGKGLEW
VAGISSDGRFTYYGA AVQGRATISRDNGQSTVRLQ
LNNLRAEDTATYYCTKNGGCGSGGDLDCIDAWG
HGTEVIVSSSLDPKSSDKTHTCPPCPAPELLGGPSV
FLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKF
NWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQ
PREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIA
VEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVD
KSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK V S2 Chicken 54

AVTQPASVSANPGETVRITCSGDSSSYYSWHQQKS
PGSAPVSVIYSNTDRPSDIPSRFSGSASGSTATLTIT
GVQAEDEAVYFCGAYDSSSDSDIFGAGTTTLTVLGQ
SSRSSGGGGSSGGGGSSMAAVTLDESGGGLQTPGG
GLSLVCKASGFDNFNMAWVRQGPCKGLEEYVA
EISDTGSTPYYGSAVQGRATISRDNGQSTVRLQLN
NLRAEDTGTYFCTRNFSSVSSIDAWGHGTEVIVS
SSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKPK
DTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGV
EVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLN
GKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYT
LPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGKV S8 Chicken 55

AVTQPSSVSANPGETVEITCSGSSTYYGWYQQKSP
GSAPVTVIYDNDKRPSDIPSRFSGSKSGSTHTLIITG
VQVEDEAVYFCGNEDNNYVAIFGAGTTTLTVLGQS
SRSSGGGGSSGGGGSSMAAVTLDESGGGLQTPGGA
LSLVCKASGFTFSSYNMGWVRQAPGKGLEFVAGI
YASGSSTDTDTTYGPAVAGRATISRDNGQSTVRLQ
LNNLRAEDTGTYCAKAAGGCSTHTCTAYIADSID
AWGHGTEVIVSSSLDPKSSDKTHTCPPCPAPELLG
GPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPE
VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSK
LTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSL SPGKV S9 Chicken 56
ALTQPSSVSANPGETVKITCSGDNSAHYYYGWYQ
QKSPGSAPVTVIYYNDKRPSGIPSRFSGSASGSTAT
LIITGVQVEDEAVYFCGSADSSNPAIFGAGTTTLTVL
GQSSRSSGGGGSSGGGGSSMAAVTLDESGGGLQTP

GRALSLVCRGSISSYNNMGWVRQAPGKGLEFIA
SIGSDGSSTHYAPAVKGRATITRDVGQSTVRLQLN
NLRAEDTGTYFCAKDAYQCSYATCNDYLDTIDAW
GHGTEVIVSSSLDPKSSDKTHTCPPCPAPELLGGPS
VFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVK
FNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLT
VLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKG
QPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDI
AVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTV
DKSRWQQGNVFCSCSVMHEALHNHYTQKSLSLSPG KV S11 Chicken 57

AVTQPASVSANPGETVKITCSGSSSGSYGWYQQKS
PGSAPVTLIYETNKRPSNIPSRFSGSKSGSTATLTIT
GVQADDEAVYYCGSEDSSTYLSIFGAGTTTLTVLGQ
SSRSSGGGGSSGGGGSSMAAVTLDES GGGLQTPGG
ALSLVCKASGFTFSSFNMGWVRQAPGKGLEFVAA
IYSGNSAEYGA AVQGRATISRDNGQSTVRLQLNNL
RAEDTGTYFCAKDAGSGCYSGVCAGTSSIDAWGH
GTEVIVSSSLDPKSSDKTHTCPPCPAPELLGGPSVFL
FPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNW
YVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQ
DWLNGEYKCKVSNKALPAPIEKTISKAKGQPREP
QVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEW
ESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSR
WQQGNVFCSCSVMHEALHNHYTQKSLSLSPGKV S12 Chicken 58

AVTQPASVSANPGETVKITCSGDSSYYGWYQQKS
PGSAPVTVIYDDNKRPSNIPSRFSGSKSGSTGTLTIT
GVQADDEAVYFCGNEDNSYVAIFGAGTTTLTVLGQ
SSRSSGGGGSSGGGGSSMAAVTLDES GGGLQTPGG
ALSLVCKASGFTFSSYNMGWVRQAPGKGLEFVAG
IYIASGDLGTTYGA AVQGRATISRDDGQSTVRLQL
NNLRAEDTGTYFCAKSAGGCSAHSCDTYIADSIDA
WGHGTEVIVSSSLDPKSSDKTHTCPPCPAPELLGGP
SVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEV
KFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVL
TVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAK
GQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPS
DIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKL
TVDKSRWQQGNVFCSCSVMHEALHNHYTQKSLSLS PGKV S13 Chicken 59

AVTQPASVSANPGETVKITCSGSSSYYGWYRQKSP
GSAPVTLIYDNDKRPSGIPSRFSGSKSGSTNTLTITG
VQADDEAVYYCGNEDNSYVGIFGAGTTTLTVLGQS
SRSSGGGGSSGGGGSSMAAVTLDES GGGLQTPGGA
LSLVCKASGFTFSSYNMGWVRQAPDKGLEFVAGI
YTGSDAGLSTTYGA AVQGRATISRDNGQSTVRLQ
LNNLGAEDTGTYFCTKSAGGCSDYNCDAYIADSID
AWGHGTEVIVSSSLDPKSSDKTHTCPPCPAPELLG
GPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPE
VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYP

SDIAVEWESNGQPNYKTTTPVLDSDGSFFLYSK
LTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSL SPGKV S14 Chicken 60
AVTQPASVSANLGETVKITCSGDSSYYGWYQQKA
PGSAPVTLIYDNDKRPSNIPSRFSGSKSGSTATLTIT
GVQADDEAVYYCGNEDMNYVGIFGAGTTTLTVLG
QSSRSSGGGGSSGGGGGMAAVTLDESGGGLQTPG
GALSLVCKASGFTFNSYNMGWVRQAPGKGLEFV
AGIYSAGGDTSTTYGAAVNGRATISRDNGQSTVRL
QLNNLRAEDTGIYFCAKAAGGCTAHNCDAYIADSI
DAWGHGTEVIVSSSLDPKSSDKTHTCPPCPAPELL
GGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDP
EVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVS
VLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISK
AKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFY
PSDIAVEWESNGQPNYKTTTPVLDSDGSFFLYS
KLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSL SPGKV S115 Human 61

ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAW
HQQKPGQAPRLLIYDATNRATGISDRFSGSGSGTD
FTLTISSLQTEDSAVYYCQQYYYWPPYRFGGGTKV
EIKGQSSRSSGGGGSSGGGGSDVQLVESGGGVVVRP
GESLRLSCAASGFSFSSYAMNWVRQAPGEGLEWV
SRINSGGGGTDYAESVKGRFTISRDNSENTLYLQM
NSLRAEDTAVYYCAKQYDWNSSFDDYWGLGALVT
VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKP
KDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDG
VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL
NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPNYKTTTPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSL SPGKV S116 Human 62

ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAW
HQQKPGQAPRLLIYDATNRATGISDRFSGSRSGTD
FTLTISSLQTEDSAVYYCQQYYYWPPYRFGGGTKV
EIKGQSSRSSGGGGSSGGGGSDVQLVESGGGVVVRP
GESLRLSCAASGFSFSSYAMNWVRQAPGEGLEWV
SRINSGGGGTDYAESVKGRFTISRDNSENTLYLQM
NSLRAEDTAVYYCAKQYDWNSSFDDYWGLGALVT
VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKP
KDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDG
VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL
NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPNYKTTTPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSL SPGKV S117 Human 63

ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAW
HQQKPGQAPRLLIYDATNRATGIPDRFSGSGSGTD
FTLTISSLQTEDSAVYYCQQYYYWPPYRFGGGTKV
EIKGQSSRSSGGGGSSRGGGSDVQLVESGGGVVVRP
GESLRLSCAASGFSFSSYAMNWVRQAPGEGLEWV
SRINSGGGGTDYAESVKGRFTISRDNSENTLYLQM

NSLRAEDTAVYYCAKQYDWNNGFFDYWGLGALVT
VSSSLDPKSSDKTHTCPPCPVPELLGGPSVFLFPPKP
KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG
VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL
NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGKV S118 Human 64

ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAW
HQQKPGQAPRLLIYDASRRATGIPDRFSGSGSGTDF
TLTISSLQTEDSAVYYCQQYYYWPPYRFGGGTKV
EIKGQSSRSSGGGGSSGGGGSDVQLVESGGGVVVRP
GESLRLSCAASGFSFSSYAMNWVRQAPGEGLEWV
SRINSGGGGTDYAESVKGRFTISRDNSENTLYLQM
NSLRAEDTAVYYCAKQYDWNNGFFDYWGLGALVT
VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKP
KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG
VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL
NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGKV S119 Human 65

EIVLTQSPATLSVSPGERATFSCRASQNVKNDLAW
YQQRPGQAPRLLIYAARIRETGIPERFSGSGSGTEFT
LTITSLQSEDAVYYCQQYYDWPPFTFGGGTKVEI
KGQSSRSSGGGGSSGGGGSDVQLLESAGGGVVQPG
GSLRLSCAASGFSFSNFAMTWVRQAPGEGLEWVS
TIGSGDTYYADSVKGRFTISRDN SKNTLYLQMNSL
RAEDTAVYYCAKDSTVSWSGDFFDYWGLGTLVT
VSSSLDPKSSDKPHTCPPCPAPELLGGPSVFLFPPKP
KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG
VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL
NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGKV S120 Human 66

EIVLTQSPATLSVSPGERATFSCRASQNVKNDLAW
YQQRPGQAPRLLIYAARIRETGIPERFSGSGSGTEFT
LTITSLQSEDAVYYCQQYYDWPPFTFGGGTKVEI
KGQSSRSSGGGGSSGGGGSDVQLVESGGGVVQPG
GSLRLSCAASGFSFSNFAMTWVRQAPGEGLEWVS
TIGSGDTYYADSVKGRFTISRDN SKNTLYLQMNSL
RAEDTAVYYCAKDSTVSWSGDFFDYWGLGTRVT
VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKP
KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG
VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL
NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGKV S122 Human 67

EIVLTQSPATLSVSPGERATFSCRASQNVKNDLAW
YQQRPGQAPRLLIYAARIRETGIPERFSGSGSGTEFT
LTITSLQSEDAVYYCQQYYDWPPFTFGGGTKVEI
KGQSSRSSGGGGSSGGGGSDVQLVESGGGVVRPG
ESLRLSCAASGFRFSNFAMTWVRQAPGEGLEWVS
TIGSGDTYYADSVKGRFTISRDN SKNTLYLQMNSL
RAEDTAVYYCAKDSTVSWSGDFFDYWGLGTLVT
VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKP
KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG
VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL
NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGKV S123 Human 68

EIVLTQSPGTLSPGERVTLTCRASQGIAGKIAWY
QQKPGQAPRLLIYDASSRATGIPGRFSGSGSGTEFT
LTITSLQSEDAVYYCQQHYDWSPLTFGGGTKVEI
KGQSSRSSGGGGSSGGGGSDVQLVESGGGLVQPG
GSLRLSCTASGFTFRNYGMSWVRQAPGEGLEWVS
ASSGSGSTYYTDSVKGRFTISRDN SKNTLYLQMNS
LRAEDTAIYYCAKVTWNNFFDYWGLGTLVTVSS
LDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKPKDT
LMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEV
HNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGK
EYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPP
SRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
NNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNV FSCSVMHEALHNHYTQKSLSLSPGKV
S126 Human 69 EIVLTQSPGTLTLSPGERATLSCRASQSIGSSYLAW

YQQKPGQAPRLLIYDATNRATGIPDRFSGSGSGTD
FTLTISLQTEDSAVYYCQQYYYWPPYRFGGGTKV
EIKGQSSRSSGGGGSSGGGGSDVQLVESGGGVVRP
GESLRLSCAASGFTFSNYDMTWVRQAPGEGLEWV
SGISGNGGSTYYADSVKGRFTISRDN SKNTLYLQM
NSLRAEDTAVYYCAMNRWWFDYWGLGTLVTVSS
SLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKPKD
TLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVE
VHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNG
KEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTL
PPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQ
PENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQG
NVFSCSVMHEALHNHYTQKSLSLSPGKV S128 Human 70

ETVLTQSPATLSVSPGERATLSCRASQTVGSKLAW
HQQKPGQAPRLLIYDASN RATGIPDRFSGSGSGTD
FTLTISLQTEDSAVYYCQQYYYWPPYRFGGGTKV
EIKGQSSRSSGGGGSSGGGGSDVQLVESGGGVVRP
GESLRLSCAASGFSFRSYAMNWVRQAPGEGLEWV
SRIDSGGGGTDYADSVKGRFTISRDN SKNTLYLQM
NSLRAEDTAVYYCAKQYDWNSFFDYWGLGAPVT
VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKP
KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG

VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL
NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGKV S130 Human 71
EIVLTQSPGTLSVSPGERATLSCRASQNVRSDLAW
YQQKLGQAPRLLIYDANTRATDIPDRFSGSGSGTE
FTLTISSLQSEDFAVYYCQHYYDWPPVTFGGGTKV
EIKGQSSRSSGGGGSSGGGGSDVQLVESGGGVVVRP
GESLRLSCAASGFTFSNYAMSWVRQAPGEGLEWV
SLITNGDGAYYADSVKGRFTISRDN SKNTLYLQM
NSLRAEDTAIYYCAKDGA AHYYDIFFDYWGLGTP
VTVSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPP
KPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYV
DGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD
WLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQ
VYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRW

QQGNVFSCSVMHEALHNHYTQKSLSLSPGKV S135 Human 72
EIVLTQSPATLSVSPGERVTFSCRASQNVRSDI AWY
QQKPGQAPRLLIYAASSRDTGIPDRFSGSGSGTDFT
LTISSLQSEDFGVYYCQQYYDWPPFTFGGGTKVEI
KGQSSRSSGGGGSSGGGGSDVQLVESGGGVVRPG
ESLRLSCAASGFSFSIYAMSWVRQAPGEGLEWVST
IGADDTYYADSVKGRFTISRDN SKNTLYLQMNSLR
AEDTAVYYCAKDSTVGWSGDFFDYWGLGTLVTV
SSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKPK
DTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGV
EVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLN
GKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYT
LPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ

GNVFSCSVMHEALHNHYTQKSLSLSPGKV S137 Human 73
ETVLTQSPGTLTLSVSPGERATLTCRASQSVYTYLAW
YQEKPGQAPRLLIYGASSRATGIPDRFSGSGSGTVF
TLTISSLQSEDFAVYYCQQYYDRPPLTFGGGTKVEI
KGQSSRSSGGGGSSGGGGSDVQLVESGGGVVRPG
ESLRLSCAASGFTFSSYDMNWVRQAPGEGLEWVS
LISGSGEIIYYADSVKGRFTISRDN SKNTLYLQMNS
LRAEDTAVYYCAKENNRYRFFDDWGLGTLVTVSS
SLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKPKD
TLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVE
VHNAKTKPREEQYNSTYRVVSVLTVLHQDWLN
GKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYT
LPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQ
PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQG

NVFSCSVMHEALHNHYTQKSLSLSPGKV S138 Human 74
EIVLTQSPGTLSVSPGERVILT CRASQSVDTYNLAW
YQQKPGQAPRLLIYDLSTRATGIPDRFSGSGSGTEF
TLTINSLEPEDFAVYYCHQYYDWPPYTFGGGTKVE

IKGQSSRSSGGSSGGSDVQLVESGGGVVVRPG
 ESLRLSCAASGFTFSNYAMNWVRQAPGEGLEWVS
 GISGRGGDTYYADSVKGRFTISRDN SKNTLYLQM
 NSLRAEDTAIYYCAKGTWNYGSFDYWGLGTLVT
 VSSSLDPKSSDKTDTCPPCPAPELLGGPSVFLFPPKP
 KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG V S121 Human 75
 EIVLTQSPATLSVSPGERATFSCRASQNVKNDLAW
 YQQRPGQAPRLLIYAARIRETGIPERFSGSGSGTEFT
 LTITSLQSEDEAVYYCQQYYDWPPFTFGGGTKVEI
 KGQSSRSSGGGGSSGGGGSDVQLVESGGGVVQPG
 GSLRLSCAASGFSFSNFAMTWVRQAPGEGLEWVS
 TIGSGDTYYADSVKGRFTISRDN SKNTLYLQMNSL
 RAEDTAVYYCAKDSTVSWSGDFFDYWGLGTLVT
 VSSSLDPKSSDKTHTCPPCPAPELLGGPSVFLFPPKP
 KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG
 VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL
 NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
 TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
 GQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ
 QGNVFSCSVMEALHNHYTQKSLSLSPGKV

(340) Without wishing to be bound to theory, it is thought that antibodies with cross-reactive binding among human, cynomologus, and/or murine proteins may allow for characterization of antibodies in both animal models and clinical testing. Antibodies with isoform- and/or variant-specific binding may be useful for personalized medicine approaches to specific human populations and/or studies on specific variants of interest. In addition to profiling intra- and inter-species binding specificities, the methods described herein also allow for identification of antibodies that block or do not block binding between SIRP- α and CD47, as well as antibodies that “kick SIRP α -off” from binding to CD47.

Example 2: Identification of Additional Antibodies with Novel Binding Specificities to SIRP- α Proteins

(341) Methods

(342) Determination of K.sub.D

(343) The interactions of anti-SIRP α antibodies with SIRP α from various species (human v1, human v2, cynomolgus, mouse 129, BL6, BALBc, NOD), SIRP β and SIRP γ were analyzed using two methods, direct immobilization of the antibodies (via GLC chip) or capture via biotinylated Protein A (via NLC chip), according to the following protocols. All experiments were performed at 25° C. using a SPR-based ProteOn XPR36 biosensor (BioRad, Inc, Hercules, CA) equipped with GLC or NLC sensor chips. Antibodies were expressed using FreeStyle™ 293-FS cells (Thermo Fisher) as described in Example 1. Purification was carried out by standard Protein A affinity column chromatography and eluted antibodies were stored in PBS buffer.

(344) The running buffer was PBS pH 7.4 with 0.01% Tween-20 (PBST+). All analytes were used at their nominal concentrations as determined by A280 Absorbance and using their molar calculated extinction coefficient. Analytes were injected in a “one-shot” kinetic mode as described elsewhere (see, e.g., Bravman, T. et al. (2006) *Anal. Biochem.* 358:281-288).

(345) For the method using GLC chip, the analytes were injected and flowed over anti-SIRP α antibodies immobilized (~1000 RUs) on GLC chips using Proteon™ Amine Coupling Kit. For the immobilization step, GLC chip was activated with EDAC/Sulpho-NHS 1:1 (Biorad) diluted 1/100 for 300 s at 25 μ L/min. Anti-SIRP α antibodies were diluted to 80 nM concentration in 10 mM sodium acetate buffer pH 4.5 and immobilized to the chip at 30 μ L/min for 50 s. Chip was inactivated with ethanolamine for 300 s at 25 μ L/min. The analytes (e.g., SIRP- α from different

species, SIRP- β , SIRP- γ) were injected in a “one-shot” kinetic mode at nominal concentrations of 100, 33, 11, 3.7, 1.2 and 0 nM. Association times were monitored for 90 s at 100 μ L/min, and dissociation times were monitored for 1200 s. The surfaces were regenerated with a 2:1 v/v blend of Pierce IgG elution buffer/4M NaCl.

(346) Alternatively, K.sub.D determination was performed using antibody capture via an NLC chip. In this case, 15 μ g/mL biotinylated protein A (ThermoFisher) was injected at 30 μ L/min for 120 s over the NLC chip to obtain an immobilization response of ~1000-1200RUs. Next, anti-SIRP α antibodies (~160 nM) were injected for 80 s at 30 μ L/min. The analytes (SIRP α from different species, SIRP- β and SIRP- γ) were subsequently injected in a “one-shot” kinetic mode at nominal concentrations of 100, 33, 11, 3.7, 1.2 and 0 nM. Association times were monitored for 60 s at 25 μ L/min, and dissociation times were monitored for 120 s. The surfaces were regenerated with a 2:1 v/v blend of Pierce IgG elution buffer/4M NaCl.

(347) Biosensor data were double-referenced by subtracting the interspot data (containing no immobilized protein) from the reaction spot data (immobilized protein) and then subtracting the response of a buffer “blank” analyte injection from that of an analyte injection. Double-referenced data were fit globally to a simple Langmuir model and the K.sub.D value was calculated from the ratio of the apparent kinetic rate constants (K.sub.D=k.sub.d/k.sub.a).

(348) Results

(349) Three representative antibodies (a blocking, non-blocking, and “kick off” antibody) were further characterized for binding to various SIRP- α , - β , - γ proteins as described above. The three antibodies contain human sequences and were derived from the HuMab chicken immunization experiments. These antibodies were tested as full-length human IgG1 antibodies with L234A, L235A, G237A, and N297A mutations. SIRP- α proteins examined in these experiments corresponded to human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), cynomolgus SIRP- α (SEQ ID NO:11), human SIRP β isoform 1 (SEQ ID NO:13), human SIRP β isoform 2 (SEQ ID NO:14), and human SIRP γ isoform 1 (SEQ ID NO:15). Human SIRP β isoform 1 (SEQ ID NO:13) is also known in the art as SIRP β 1 isoform 1.

(350) K.sub.D values obtained for the three antibodies are shown in Table E below.

(351) TABLE-US-00017 TABLE E Summary of K.sub.D values (M) beta beta gamma v1 v2 cyno isoform 1 isoform 2 isoform 1 Kick-off AB132a 4.26×10^{-10} 1.86×10^{-9} 2.41×10^{-9} 1.31×10^{-10} 1.64×10^{-10} 5.19×10^{-10} Blocker AB119a 1.98×10^{-10} 7.99×10^{-11} 1.45×10^{-10} 3.42×10^{-10} 3.76×10^{-10} 8.86×10^{-10} Non- AB136a 6.88×10^{-10} 3.22×10^{-9} 2.65×10^{-9} 4.35×10^{-9} 2.27×10^{-9} 1.20×10^{-7} blocker

(352) Next, the binding kinetics of various antibody clones to selected mouse SIRP- α proteins were determined as described above. Mouse proteins that were tested include BALBc (SEQ ID NO:10), BL6 (SEQ ID NO:9), NOD (SEQ ID NO:8), and m129 (SEQ ID NO:7) SIRP- α proteins. The results are summarized in Tables F-I below. Antibody clones labeled as “c” were tested as full-length mouse IgG2a antibodies; antibody clones labeled as “a” were tested as full-length human IgG1 antibodies with L234A, L235A, G237A, and N297A mutations.

(353) TABLE-US-00018 TABLE F Summary of kinetics for binding of selected antibodies to BALBc mouse SIRP- α protein (SEQ ID NO: 10) Antibody K.sub.on (1/Ms) K.sub.off (1/s) K.sub.D (M) AB136a 8.27×10^{-5} 2.73×10^{-4} 3.31×10^{-10} AB3c 7.76×10^{-5} 4.12×10^{-3} 5.31×10^{-9} AB21c 4.62×10^{-5} 6.18×10^{-4} 1.34×10^{-9} AB25c 3.03×10^{-5} 2.92×10^{-3} 9.64×10^{-9} AB27c 1.50×10^{-5} 2.26×10^{-3} 1.50×10^{-8} AB66c 2.24×10^{-5} 7.62×10^{-4} 3.41×10^{-9}

(354) TABLE-US-00019 TABLE G Summary of kinetics for binding of selected antibodies to BL6 mouse SIRP- α protein (SEQ ID NO: 9) Antibody K.sub.on (1/Ms) K.sub.off (1/s) K.sub.D (M) AB136a 6.24×10^{-5} 5.85×10^{-3} 9.37×10^{-9} AB3c 4.12×10^{-5} 6.19×10^{-3} 1.50×10^{-8} AB21c 2.76×10^{-5} 2.41×10^{-4} 8.76×10^{-10} AB25c $1.42 \times$

10.sup.5 3.99 × 10.sup.-4 2.81 × 10.sup.-9 AB27c 1.47 × 10.sup.5 1.40 × 10.sup.-3 9.49 × 10.sup.-9 AB66c 1.07 × 10.sup.5 6.19 × 10.sup.-4 5.80 × 10.sup.-9

(355) TABLE-US-00020 TABLE H Summary of kinetics for binding of selected antibodies to NOD mouse SIRP-α protein (SEQ ID NO: 8) Antibody K.sub.on (1/Ms) K.sub.off (1/s) K.sub.D (M) AB136a 8.41 × 10.sup.5 4.48 × 10.sup.-4 5.33 × 10.sup.-10 AB3c 9.99 × 10.sup.5 1.43 × 10.sup.-3 1.43 × 10.sup.-9 AB21c 7.49 × 10.sup.5 4.79 × 10.sup.-4 6.40 × 10.sup.-10 AB25c 3.66 × 10.sup.5 1.43 × 10.sup.-3 3.90 × 10.sup.-9 AB27c 2.96 × 10.sup.5 1.01 × 10.sup.-3 3.42 × 10.sup.-9 AB66c 4.12 × 10.sup.5 2.32 × 10.sup.-4 5.64 × 10.sup.-10

(356) TABLE-US-00021 TABLE I Summary of kinetics for binding of selected antibodies to m129 mouse SIRP-α protein (SEQ ID NO: 7) Antibody K.sub.on (1/Ms) K.sub.off (1/s) K.sub.D (M) AB136a 6.88 × 10.sup.5 5.97 × 10.sup.-4 8.67 × 10.sup.-10 AB3c 5.34 × 10.sup.5 1.18 × 10.sup.-2 2.20 × 10.sup.-8 AB21c 5.63 × 10.sup.5 3.31 × 10.sup.-5 5.88 × 10.sup.-11 AB25c 3.52 × 10.sup.5 2.07 × 10.sup.-5 5.87 × 10.sup.-11 AB27c 2.07 × 10.sup.5 1.01 × 10.sup.-4 4.87 × 10.sup.-10 AB66c 2.07 × 10.sup.5 1.98 × 10.sup.-5 9.55 × 10.sup.-11

(357) For the above antibodies, VH and VL domain sequences (respectively) were as follows: AB136a: SEQ ID NO: 133 and 134; AB3c: SEQ ID NO: 242 and 243; AB21c: SEQ ID NO: 135 and 136; AB25c: SEQ ID NO: 137 and 138; AB27c: SEQ ID NO: 139 and 140; AB66c: SEQ ID NO: 141 and 142.

(358) These results demonstrate that multiple antibody clones bound to all four mouse SIRP-α proteins, making these antibodies suitable for characterization in in vivo mouse models.

Example 3: Functional Properties of Anti-SIRP-α Antibodies

(359) The previous Examples describe the identification and characterization of anti-SIRP-α antibodies with a diverse array of properties, such as binding specificity for various human, cynomolgus, and/or murine SIRP-α, SIRP-β, and SIRP-γ proteins, and whether or not the antibodies block binding between SIRP-α and CD47. Antibodies representing particular categories of interest were next examined in a variety of in vitro and in vivo functional assays aimed at characterizing their effects on SIRP-α's biological functions. In particular, “blocking” (i.e., antibodies that block binding between SIRP-α and CD47) and “non-blocking” (i.e., antibodies that do not block binding between SIRP-α and CD47) were characterized for activities in various in vitro and in vivo assays.

(360) As noted above, antibody clones labeled as “a” were tested as full-length human IgG1 antibodies with L234A, L235A, G237A, and N297A mutations. Antibody clones labeled as “b” were tested as full-length mouse IgG1 antibodies with an N297A mutation. Antibody clones labeled as “c” were tested as full-length mouse IgG2a antibodies.

(361) Methods

(362) Tumor Cell Line Culturing

(363) DLD-1 (human colorectal adenocarcinoma) and OE19 (human esophageal carcinoma) cells were maintained in growth medium comprised of RPMI (Gibco) supplemented with 10 percent heat-inactivated Fetal Bovine Serum (Gibco), one percent penicillin/streptomycin (Gibco), and one percent Glutamax™ glutamine supplement (Gibco).

(364) Derivation and Culture of Human Monocyte-Derived Macrophages

(365) Trima residuals were received from Blood Centers of the Pacific and diluted 1:4 with Phosphate Buffered Saline (PBS, Gibco). Diluted blood was split into four tubes and underlaid with 20 ml Ficoll-Paque® Plus (GE Healthcare) lymphocyte medium. Tubes were centrifuged for 30 minutes at 400×g. PBMCs were collected from the interface and resuspended in FACS buffer (PBS with 0.5 percent Bovine Serum Albumin (Gibco)). CD14.sup.+ monocytes were purified by negative selection using the Monocyte Isolation Kit II (Miltenyi Biotec) and LS columns (Miltenyi Biotec) according to the manufacturer's protocol.

(366) For nonpolarized macrophages, CD14.sup.+ monocytes were seeded into 15 cm tissue culture plates (Corning) at 10 million cells per dish in 25 ml IMDM (Gibco) supplemented with 10

percent human AB serum (Corning), one percent penicillin/streptomycin, and one percent Glutamax™ glutamine supplement. Cells were cultured for seven to ten days.

(367) For M2 polarized macrophages, CD14.sup.+ monocytes were seeded into 15 cm tissue culture plates (Corning) at 6 million cells per dish in 25 ml RPMI (Gibco) supplemented with 10 fetal bovine serum (Thermo Fisher), one percent penicillin/streptomycin, and one percent Glutamax, and 50 ng/ml M-CSF (Miltenyi). Cells were cultured for seven to ten days.

(368) In Vitro Phagocytosis Assays

(369) DLD-1 and OE19 cells were detached from culture plates by washing twice with 20 ml PBS and incubation in 10 ml TrypLE™ Select (Gibco) adherent cell dissociation enzyme for 10 minutes at 37° C. Cells were centrifuged, washed in PBS, and resuspended in medium. Cells were labeled with the Celltrace™ CFSE Cell Proliferation kit (Thermo Fisher) according to the manufacturer's instructions and resuspended in IMDM. Macrophages were detached from culture plates by washing twice with 20 ml PBS and incubation in 10 ml TrypLE™ Select adherent cell dissociation enzyme for 20 minutes at 37° C. Cells were removed with a cell scraper (Corning), washed in PBS, and resuspended in IMDM.

(370) Phagocytosis assays were assembled in ultra-low attachment U-bottom 96 well plates (Corning) containing 100,000 DLD-1 or OE19 cells, 50,000 macrophages, five-fold serial dilutions of anti-SIRP- α antibody from 100 nM to 6.4 pM and cetuximab (Absolute Antibody) at 1 or 0.01 ug/ml, trastuzumab at 0.01 ug/ml, or control antibody of the same isotype (Southern Biotech). All anti-SIRP- α antibodies tested had a human IgG1 with L234A, L235A, G237A, and N297A mutations except AB136c, which had a mouse IgG2a. Plates were incubated two hours at 37° C. in a humidified incubator with 5 percent carbon dioxide. Cells were pelleted by centrifugation for five minutes at 400 $\times$ g and washed in 250 μ l FACS buffer. Macrophages were stained on ice for 15 minutes in 50 μ l FACS buffer containing 10 μ l human FcR Blocking Reagent (Miltenyi Biotec), 0.5 μ l anti-CD33 BV421 (Biolegend), and 0.5 μ l anti-CD206 APC-Cy7 (Biolegend). Cells were washed in 200 μ l FACS buffer, washed in 250 μ l PBS, and stained on ice for 30 minutes in 50 μ l Fixable Viability Dye eFluor™ 506 (ebioscience) dye diluted 1:1000 in PBS. Cells were washed twice in 250 μ l FACS buffer and fixed for 30 minutes on ice in 75 μ l Cytofix™ (BD Biosciences) fixation solution. Cells were washed in 175 μ l FACS buffer and resuspended in 75 μ l FACS buffer. Cells were analyzed on a FACS Canto™ II (BD Biosciences) flow cytometer, with subsequent data analysis by Flowjo 10.7 (Treestar). Dead cells were excluded by gating on the e506-negative population. Macrophages that had phagocytosed tumor cells were identified as cells positive for CD33, CD206, and CFSE.

(371) In Vivo Dendritic Cell Activation

(372) Balb/c mice (n=3/group) were intravenously injected with a control rat anti-mouse anti-SIRP- α antagonistic antibody (clone p84), rat IgG control, AB136b (AB136 with a mouse IgG1 Fc region bearing an N297A mutation), mouse IgG control, or vehicle (PBS) at 10 mg/kg. Five hours post injection, spleens were harvested and processed into single cell suspension by mechanical dissociation through a 70 μ m cell strainer. Cells were washed two with PBS and red blood cell lysis was performed. Subsequently, cells were washed for an additional two times with PBS/10% FBS and prepared for cell staining. Cells were stained with fluorochrome conjugated MHC-II, CD8, 33D1, CD4, CD11c, CD86, CCR7 and viability dye at 4 degrees for one hour. Cells were washed two times and analyzed using a BD Canto™ II flow cytometer and data were processed using Flowjo.

(373) In Vivo Anti-Tumor Activity

(374) For the CT26 syngeneic mouse colon carcinoma model, CT26 cells were implanted subcutaneously in BALB/c mice and randomized into groups (8-9 mice/group). Treatment groups included vehicle (PBS) and AB136b. AB136 anti-SIRP- α had a mouse IgG1 Fc region bearing an N297A mutation. Treatment was initiated when tumors were an average of 75-80 mm.sup.3, day 7 or 8 post implant. Mice were dosed intraperitoneally (IP) at 3 mg/kg twice a week for three weeks

AB136b. Animals were sacrificed when tumors reached a volume of ~2000 mm³. (375) For the MC38 syngeneic mouse colon carcinoma model, MC38 cells were implanted subcutaneously in C57BL/6 mice and randomized into groups (8-10 mice/group). Treatment groups included vehicle (PBS), AB25b, AB25c, AB27b, AB3b, and AB136b. All anti-SIRP α antibodies had a mouse IgG1 Fc region bearing an N297A mutation except for AB25c, which had a mouse IgG2a. Treatment was initiated when tumors were an average of 60-65 mm³, day 7 post implant. Mice were dosed intraperitoneally (IP) at 10 mg/kg twice a week for three weeks for anti-SIRP α . Animals were sacrificed when tumors reached a volume of ~2000 mm³.

(376) Results

(377) Various anti-SIRP- α antibodies were examined for induction of phagocytosis in in vitro phagocytosis assays using polarized and non-polarized macrophages.

(378) Additional anti-SIRP- α antibodies AB3 and AB45 were tested in the experiments described below. Both are non-blocking antibodies. The k_{sub.off} rates of binding of each clone (in scFv-Fc format) to each SIRP analyte were determined using SPR (Table J1). The SPR screening conditions have been described herein, and the K_{off} values were determined using Langmuir kinetic fittings. SIRP analyte sequences are as follows: CV1-3, SEQ ID NO:18; v1, SEQ ID NO:5; v2, SEQ ID NO:6; cyno1, SEQ ID NO: 11; cyno2, SEQ ID NO:12; m129, SEQ ID NO:7; NOD, SEQ ID NO:8; BL6, SEQ ID NO:9; sirpb1, SEQ ID NO: 13; sirpg, SEQ ID NO:15.

(379) TABLE-US-00022 TABLE J1 Langmuir kinetic fittings K_{sub.D} (M) for AB3 and AB45

SEQ ID NO:	5	6	11	12	8	9	10	15	13	14	V1	v2	cyno1	Cyno2	NOD	BL6	BALBc	sirp-g	Sirpb1	Sirpb2	AB3*
	1.62E-10	7.67E-11	2.29E-09	2.85E-09	1.63E-09	3.65E-09	1.16E-09	8.36E-08	1.63E-09	6.78E-11	AB45**	6.63E-11	1.34E-10	NB	NB	NB	NB	NB	2.71E-08	1.06E-08	2.53E-11

NB—No binding. *See SEQ ID NOs: 242 and 243 for VH and VL domains, respectively. **See SEQ ID NOs: 244 and 245 for VH and VL domains, respectively.

(380) As shown in FIGS. 13A & 13B, non-blocking anti-SIRP- α antibody AB3a was found to induce phagocytosis of DLD-1 tumor cells in M2 polarized macrophages. In particular, treatment of macrophages with cetuximab (anti-EGFR antibody) and AB3a led to robust induction of tumor cell phagocytosis. Similarly, non-blocking anti-SIRP- α antibody AB45a was found to induce phagocytosis of DLD-1 tumor cells M2 polarized macrophages, and treatment of macrophages with cetuximab (anti-EGFR antibody) and AB45a led to robust induction of tumor cell phagocytosis (FIGS. 13C & 13D). Blocking anti-SIRP- α antibodies AB119a (FIG. 13E) and AB135a (FIG. 13F) were also found to induce phagocytosis of OE19 tumor cells when co-administered with trastuzumab (anti-HER2 antibody). Non-blocking anti-SIRP- α antibody AB136c was also found to induce phagocytosis of DLD-1 tumor cells (FIG. 13G).

(381) A dendritic cell activation assay was used to characterize the in vivo effects of non-blocking anti-SIRP- α antibody AB136b (SEQ ID NOs:133 and 134 for VH and VL domain sequences, respectively). Failure to engage mouse SIRP- α receptor on splenic dendritic cells via CD47 binding leads to splenic dendritic cell activation. Control anti-SIRP- α antagonist antibody p84 activated splenic dendritic cells when injected intravenously into mice (FIG. 14). Non-blocking anti-SIRP- α antibody AB136b was tested in vivo to determine if it leads to dendritic cell activation. Interestingly, AB136b treatment led to activation of splenic dendritic cells at a similar level as p84 (FIG. 14).

(382) Next, the in vivo anti-tumor effects of various anti-SIRP- α antibodies were assayed in two syngeneic mouse colon carcinoma models. The anti-tumor activities of blocking anti-SIRP- α antibodies AB25b, AB25c and AB27b and non-blocking anti-SIRP- α antibodies AB3b and AB136b were examined in an MC38 syngeneic mouse colon carcinoma model to assess their single agent activities. Both blocking (AB25b, AB25c and AB27b) and non-blocking (AB3b and AB136b) anti-SIRP- α antibodies delayed tumor formation at 10 mg/kg as compared to vehicle alone (FIG. 15) in the MC38 syngeneic mouse model. On day 25, groups treated with anti-SIRP α antibodies had three mice below 600 mm³ for AB25b and AB27b, four mice below 600 mm³ for AB25c and

AB3b and five mice below 600 mm.sup.3 for AB136b, while the vehicle-treated group had only two mice below 600 mm.sup.3.

(383) The anti-tumor activity of non-blocking AB136b was next evaluated in a CT26 syngeneic mouse colon carcinoma model to assess single agent activity. These results confirmed that AB136b treatment delayed tumor formation at 3 mg/kg as compared to vehicle alone in the CT26 syngeneic mouse model (FIG. 16).

(384) Taken together, these results demonstrate the efficacy of anti-SIRP- α antibody treatment in inducing tumor cell phagocytosis by macrophages, activating dendritic cells, and inhibiting tumor growth in vivo. Multiple blocking anti-SIRP- α antibodies were found to promote tumor cell phagocytosis, activate dendritic cells and block in vivo tumor growth. Surprisingly, however, treatment with non-blocking anti-SIRP- α antibody was also found to activate splenic dendritic cells and inhibit tumor growth in vivo in two different syngeneic mouse tumor models. The finding that non-blocking anti-SIRP- α antibodies were able to increase phagocytosis and block in vivo tumor growth was both surprising and unexpected. Previous work has suggested that only blocking anti-SIRP- α antibodies would be able to inhibit in vivo tumor growth (see Yanagita, T. et al. (2017) *JCI Insight* 2:e89140).

Example 4: Structural Analysis of Anti-SIRP- α Antibody Epitopes on SIRP- α

(385) As described in Examples 1-3 above, anti-SIRP- α antibodies have been generated with a variety of specificities and modes of binding to SIRP- α , e.g., antibodies that block CD47 binding to SIRP- α , antibodies that do not block CD47 binding to SIRP- α , and antibodies that bind to SIRP- α and reduce its affinity for binding CD47 ("kick off" antibodies). Structural analyses were undertaken in order to understand how these types of antibodies bind to the D1 domain of SIRP- α as compared to CD47 and characterize the epitopes of selected anti-SIRP- α antibodies.

(386) Methods

(387) Crystallography and Structural Analysis

(388) Expression of the Fabs and SIRP α was similar to previously established protocols and involved traditional methods of affinity chromatography and size exclusion for protein purification. A human SIRP- α v1 mutant bearing an N80A mutation as compared to SEQ ID NO:5 was used for ease of protein production in Expi293 (SEQ ID NO:296). The goal of using an N80A SIRP α was to produce a homogenous, non-glycosylated form of SIRP α that would be most amenable for crystallization. The final purification buffer is minimal with only 10 mM Tris pH 7.5 and 50 mM NaCl. The purified complex sample is stable at 4° C. and was concentrated to 10-12 mg/mL in preparation for crystallization experiments and eventual structure determination.

(389) The methods for the Fab:SIRP α complex project followed well established principles or crystallography starting with the sparse matrix technique, which led to custom optimized conditions, establishing a routine protocol. Crystallization was carried out utilizing the sitting drop vapor diffusion technique. For the initial sparse matrix screening, condition kits commercially available through Qiagen® were utilized (Table J2). These crystallization experiments were set in drops with varying ratios of protein to crystallant condition. The ratios set were in the range of 1:1, 2:1, and 3:1 protein to condition in total volume of 1 μ L in the subwells of the 96 well (8 $\times$ 12) tray. 100 μ L of crystallization condition was placed in the well reservoir. These drops were set by utilizing the Mosquito drop setting robot and the completed plates were sealed and stored in a 12° C. incubator. The experiment was monitored by viewing the plates/drops under the microscope to see developments that included drop precipitation, aggregation, phase separation, amorphous formation, as well as initial protein crystals.

(390) TABLE-US-00023 TABLE J2 Kits used for initial sparse matrix screening. Kit Cat. No./ID Classics Suite 130901 Classics II Suite 130923 Classics L Suite 130902 PEGs Suite 130904 PEGs II Suite 130916 PACT Suite 130918 ProComplex Suite 130915 AmSO.sub.4 Suite 130905
Crystallization Summary and Crystal Harvesting

(391) Crystallization of the 4 complexes was achieved with derivatives of two main conditions as

shown in Table J3). Crystal harvesting was done on optimal crystal forms so that manipulation and cryo-freezing in liquid nitrogen would not jeopardize the integrity of the crystal prior to X-ray diffraction screening and possible data collection. To prevent icing, a cryo-protectant was implemented during freezing. The typical cryo-protectant included an addition of 20% glycerol to the crystallization condition that formed the crystal. When crystals formed in conditions with 30% PEG 4000 or above, the addition of glycerol was not necessary. The high percentage of PEG 4000 behaved as a viable cryo-protectant. Crystals of complexes were manipulated with cryo loops that are either nylon or Mitigen® crystal mount style. Single crystals were isolated and excised out of the drop in which they formed and transferred into cryo-protectant for a short period before plunging into liquid nitrogen to flash freeze.

(392) TABLE-US-00024 TABLE J3 Crystallization conditions for forming Fab:SIRP- α complex crystals. Fab Antibody Buffer Salt Precipitant 119 0.1M Tris-HCl pH 8.5 0.2M MgCl₂ 15% (w/v) PEG 4000 136 0.1M Tris-HCl pH 8.5 0.2M MgCl₂ 30% (w/v) PEG 4000 3 0.1M Sodium 0.45M Ammonium 35% (w/v) Acetate pH 4.6 Sulfate PEG 4000 115 0.1M Tris pH 7.2 0.2M MgCl₂ 18% (w/v) PEG 4000

Data Collection and Processing

(393) Data were collected on crystals flash frozen in liquid nitrogen as described above. These samples remained in the cryo stream when screened for protein lattice diffraction prior to subsequent dataset collection via the oscillation method. Collection occurred at either the Advanced Photon Source or Diamond Light Source. Datasets were reduced using the xia2 suite. xia2 is a wrapper script that allows for automated reduction of macromolecular crystallographic data. The program is able to utilize multiple data reducing programs such as XDS, DIALLS, Mosflm, and Aimless. These programs allow for the diffraction data to be indexed into the appropriate space group and unit cell, integration of intensities, and scaling to produce an estimate of intensity of each unique reflection. For the initial Complex 1 dataset, phases were calculated via molecular replacement (MR) using Phaser MR of the CCP4 Suite utilizing homolog models of SIRP α , as well as Fab (see PDB CODE: 2UV3 and PDB CODE: 4NM4 respectively). For subsequent datasets of Complex 2, 3, and 4, the completed structure model coordinates of Complex 1 was used as the search model for the phase calculation and the initial model build.

(394) Structure Building and Refinement

(395) Model building utilized the Coot program. As an example, Complex 1 was calculated to have 4 molecules in the ASU, therefore, 4 pairs of AB119f complexed with SIRP α are to be built to complete the structure model. The strategy of structure building was to build amino acid residues into electron density following the known sequence of the target proteins. Quality of data directly correlates to the fit of the structure model into the observed crystal dataset. Therefore, building the tertiary structure of the Complex followed established protocol. Initially, the peptide bone of the residue was built. This was followed by the placement of the correct residue side chain if the electron density map permits. In the instance an original amino acid in the protein sequence was not modeled, it was due to lack of density for the respective side chain, thus only the peptide backbone of the amino acid can be modeled and an alanine residue side chain is built in place. Stretches of the sequence may be a disordered region and cannot be built due to lack of density even for placement of the amino acid backbone. The builds were followed by subsequent rounds of refinement through the Refmac program, a part of the CCP4 Suite. The refinement program is utilized to minimize coordinate parameters through the Maximum Likelihood residual. Refinement was completed when the model parameters fell into favorable statistical tolerances for parameters including: the Rvalue, bond length and angles, and Ramachandran fit. In addition, to check physical tolerances of the model, Molprobity was also utilized to ensure complete and proper structure determination.

(396) Epitope Mapping and Superimposition Analysis

(397) Buried surface area of the antigen for the epitope was calculated as the difference between

the solvent-accessible surface area of the antigen alone and antigen in complex with Fab fragment of the antibody. Conversely, buried surface area of the antibody heavy and light chains for the paratope was calculated as the difference between the solvent-accessible surface area of the Fab fragment alone and in complex with its antigen. The surface accessible area was calculated by the rolling ball method with probe radius of 1.4 Å. Buried surface area is reported in Å². All antibody: SIRPα complexes were superimposed by selecting SIRPα from each structure and superimposing its carbon atoms. CD47: SIRPα, complex (V1 variant) structure used for the analysis is PDB:4CMM. The superimposition and the RMSD calculation was performed using PYMOL.

(398) Results

(399) First, the structure of blocking antibody 119 Fab bound to SIRP-α was determined and compared to CD47 binding to SIRP-α. As shown in FIG. 17A, antibody 119 and CD47 bound to a similar epitope of the SIRP-α D1 domain. 56% of the SIRP-α residues in the antibody 119 epitope were also found in the CD47 epitope, while 75% of the SIRP-α residues in the CD47 epitope were also found in the antibody 119 epitope. The SIRP-α residues participating in the interaction with antibody 119 (as determined by buried surface area changes, described supra) are shaded and shown as space-filled models in FIG. 17B. Key residues from the antibody 119 Fab paratope are also shown (black sticks), including heavy chain residues N31, S53, D55, Y57, T99, S101, and W102 (N31 is not visible in the structure orientation shown in FIG. 17B) and light chain residues Y92 and W94 (based on the heavy and light chain variable domain sequences according to SEQ ID NOs:335 and 97, respectively).

(400) FIG. 18A shows a comparison between CD47 binding to the SIRP-α D1 domain and non-blocking antibody 136 Fab binding to the SIRP-α D1 domain. The SIRP-α binding epitopes of antibody 136 and CD47 were found to be completely non-overlapping. The SIRP-α residues participating in the interaction with antibody 136 are shaded and shown as space-filled models in FIG. 18B. Key residues from the antibody 136 Fab paratope are also shown (black sticks), including heavy chain residues E56, Y59, and R102, and light chain residues Y92 and R94 (based on the heavy and light chain variable domain sequences according to SEQ ID NOs:133 and 134, respectively).

(401) FIG. 19A shows a comparison between CD47 binding to the SIRP-α D1 domain and non-blocking antibody 3 Fab binding to the SIRP-α D1 domain. The SIRP-α binding epitopes of antibody 3 and CD47 were also found to be completely non-overlapping. The SIRP-α residues participating in the interaction with antibody 3 are shaded and shown as space-filled models in FIG. 19B. Key residues from the antibody 3 Fab paratope are also shown (black sticks), including heavy chain residues R56 and G100 and light chain residues R24, R26, Y86, and G88 (based on the heavy and light chain variable domain sequences according to SEQ ID NOs:242 and 243, respectively).

(402) FIG. 19C shows a comparison between CD47 binding to the SIRP-α D1 domain and “kick off” antibody 115 Fab binding to the SIRP-α D1 domain. The SIRP-α binding epitope of antibody 115 was found to be adjacent to the epitope of CD47. While mAb 115 was found to bind an epitope of the SIRP-α D1 domain adjacent to that of CD47, parts of the 115 epitope likely overlap with CD47 itself. Comparing both epitopes, there are 2 identical residues, suggesting that the interactions from the non-overlapping portions of each epitope allow the 115 Fab and CD47 to bind SIRP-α simultaneously. Kinetic analyses described herein demonstrate that antibody 115 forms a transient complex with SIRP-α and CD47 but reduces the affinity of CD47 for the SIRP-α D1 domain. The SIRP-α residues participating in the interaction with antibody 115 are shaded in FIG. 19D. Key residues from the antibody 115 Fab paratope are also shown, including heavy chain residues S31, Y32, N52, Y100, and W102 and light chain residues K32 and Y92 (based on the heavy and light chain variable domain sequences according to SEQ ID NOs:273 and 274, respectively).

(403) FIG. 20A illustrates the binding of antibodies 119, 136, 3, and 115 to SIRP-α, as compared to

CD47 binding to SIRP- α . Blocking antibody 119 was found to bind a completely non-overlapping SIRP- α epitope, as compared with the SIRP- α epitopes of both non-blocking antibodies 136 and 3. SIRP- α and antibody residues that participate in these interactions are summarized in Table K1. SIRP- α residues determined to interact with CD47 were determined as follows: 29, 30, 31, 33, 34, 35, 36, 37, 50, 51, 52, 53, 54, 66, 67, 68, 69, 74, 93, 96, 97, 98, 99, and 101 (according to SEQ ID NO:5). FIGS. 20B-20E summarize the epitopes for CD47 and anti-SIRP- α Fabs 119, 136, 3, and 115 binding to SIRP- α (residues according to SEQ ID NO:296) and lists buried surface area changes. As described herein, the numbering of amino acid residues of an antibody used to indicate the paratope is based on numbering according to the amino acid sequence of the heavy or light chain, and not, e.g., the Kabat or Chothia numbering of antibody residues.

(404) TABLE-US-00025 TABLE K1 SIRP- α epitopes and antibody paratopes for selected anti-SIRP- α antibodies. Antibody Heavy chain antibody Light chain antibody clone SIRP- α epitope residues paratope residues paratope residues 119 3, 28, 29, 30, 31, 32, 33, 35, 36, 30, 31, 32, 33, 50, 52, 53, 27, 29, 30, 32, 49, 53, 38, 50, 52, 53, 54, 62, 64, 65, 66, 54, 55, 56, 57, 96, 98, 99, 55, 56, 91, 92, 94, and 95 67, 68, 69, 70, 71, 72, 73, 74, 75, 100, 101, 102, 103, 104, (SEQ ID NO: 97) 76, 95, 96, 97, and 98 105, 106, and 108 (SEQ ID NO: 296) (SEQ ID NO: 335) 136 6, 8, 9, 10, 11, 12, 14, 42, 43, 33, 52, 54, 56, 57, 59, 99, 27, 30, 32, 92, 93, 94, 44, 46, 87, 88, 90, 103, 104, 105, 100, 101, 102, 103, and 104 and 95 106, 107, 108, 109, 111, 112, (SEQ ID NO: 133) (SEQ ID NO: 134) 113, 115, and 116 (SEQ ID NO: 296) 3 5, 6, 7, 8, 9, 10, 11, 12, 14, 24, 31, 50, 52, 53, 54, 56, 24, 25, 26, 45, 86, 87, 26, 28, 72, 88, 90, 108, 109, 111, 58, 99, 100, 101, and 102 88, and 89 113, and 115 (SEQ ID NO: 242) (SEQ ID NO: 243) (SEQ ID NO: 296) 115 17, 39, 40, 44, 45, 46, 47, 48, 28, 30, 31, 32, 33, 50, 52, 32, 91, 92, 93, and 94 49, 51, 54, 55, 56, 57, 58, 59, 80, 53, 54, 56, 57, 75, 100, (SEQ ID NO: 274) 82, 83, 84, 85, 89, 100 101, 102 (SEQ ID NO: 296) (SEQ ID NO: 273)

(405) TABLE-US-00026 TABLE K2 Anti-SIRP- α antibody paratopes and buried surface area calculations. Residue Residue Change in name number buried area ($\text{\AA}^2$) 119 Heavy Chain TRP 102 -228 SER 101 -97.2 THR 99 -70.4 SER 53 -60.9 ASP 55 -59.2 ASN 31 -58.2 TYR 57 -53.5 PHE 32 -43.1 SER 103 -30.7 SER 98 -24.7 SER 30 -19.3 GLY 52 -18.4 VAL 100 -18.4 ASP 105 -17.2 ALA 33 -16.3 GLY 54 -7.8 ASP 108 -6.2 PHE 106 -5.7 GLY 104 -5.6 LYS 96 -5.3 THR 56 -2.9 THR 50 -1.7 119 Light Chain TYR 92 -78 TRP 94 -57.5 ILE 53 -31.5 TYR 49 -30.2 PRO 95 -20.4 GLU 55 -16.6 VAL 29 -12.3 ALA 30 -8.8 ASP 32 -8 TYR 91 -3.5 THR 56 -1.5 GLN 27 -3.4 136 Heavy Chain ARG 102 -114.9 GLU 56 -67.6 TYR 59 -57.3 ARG 104 -46.7 ILE 57 -30.8 ASN 101 -12.6 ASP 33 -12 TYR 103 -10.9 SER 52 -5.1 SER 54 -4 GLU 99 -3.3 ASN 100 -1.1 136 Light Chain ARG 94 -97.6 TYR 92 -95.2 TYR 30 -89 GLN 27 -46.6 TYR 32 -46.5 ASP 93 -19.1 PRO 95 -7.3 3 Heavy Chain ARG 56 -107.9 GLY 100 -62 ASP 31 -34.9 SER 53 -27.1 THR 52 -25.2 SER 101 -22.7 GLY 54 -15.1 TYR 58 -12.2 GLN 50 -7.3 GLY 102 -5.1 PHE 99 -4.5 3 Light Chain ARG 24 -123.6 TYR 86 -62.5 ARG 26 -56.4 GLY 88 -49.1 ASP 87 -22.1 GLY 25 -13.1 ARG 45 -4.8 SER 89 -1.7 115 Light Chain LYS 32 -70.2 TYR 91 -2.9 TYR 92 -85 TYR 93 -22.3 TRP 94 -21.4 115 Heavy Chain SER 28 -35.3 SER 30 -17.2 SER 31 -70.6 TYR 32 -50.2 ALA 33 -1.3 ARG 50 -9.5 ASN 52 -54.4 SER 53 -39.8 GLY 54 -1.4 GLY 56 -34 GLY 57 -11.2 SER 75 -13.3 TYR 100 -53.6 ASP 101 -45.8 TRP 102 -136.4

(406) These results elucidate the binding interfaces between the SIRP- α D1 domain and various anti-SIRP- α antibodies, as well as the binding interface between CD47 and SIRP- α . They also provide a structural basis for the observations that antibody 119 blocks binding between CD47 and SIRP- α (since they share overlapping SIRP- α epitopes), that antibodies 136 and 3 do not block binding (as their epitopes do not overlap with that of CD47), and that antibody 115 “kicks CD47 off” from binding SIRP- α (since they share only slightly overlapping SIRP- α epitopes). These studies further highlight specific antibody and SIRP- α residues that participate in each interaction. (407) With the SIRP α binding epitopes of non-blockers (3, 136), blockers (119) and kick-off (115) antibodies defined by crystallography, the binding locations for additional anti-SIRP α antibodies can be characterized by carrying out epitope binning described in Example 5. For instance, for any

anti-SIRP α antibodies that compete with non-blocker 136 for binding to SIRP α , we can predict that these antibodies would share similar binding epitopes as 136 as defined by crystallography.

Example 5: Epitope Binning of Anti-SIRP- α Antibodies

(408) Anti-SIRP- α antibodies described above were next assayed in epitope binning experiments with SIRP- α in order to categorize antibodies having shared and distinct epitopes.

(409) Methods

(410) Epitope Binning

(411) Briefly, epitope binning was conducted as shown in FIG. 21A. A first anti-SIRP- α antibody was immobilized on a chip, then human SIRP- α v1 (SEQ ID NO:5) was injected. A second anti-SIRP- α antibody was then injected. If the second antibody was able to bind the complex formed between the first anti-SIRP- α antibody and SIRP- α , the first and second antibodies were determined to bind different epitopes. If the second antibody was not able to bind, the first and second antibodies were determined to share an epitope.

(412) Exemplary results of a binning assay using anti-SIRP- α antibody (A) immobilized to the chip and injecting anti-SIRP- α antibodies (B-F) are shown in FIG. 21B. Anti-SIRP α antibody (B), (E) and (F) form sandwiches with the complex, and this is indicated by the increasing RU at time 60 s (upon injection of respective anti-SIRP α antibodies). As such, they are determined to bind different epitopes than anti-SIRP α (A) and are indicated by white boxes in the binning plot (FIGS. 22A & 22B). For anti-SIRP α antibody (C) and (D), they did not form a sandwich, and this is indicated by a steady RU after injection of respective anti-SIRP α antibodies. As such, they are determined to bind the same epitope as anti-SIRP α (A) and are shaded gray in the binning plot (FIGS. 22A & 22B). The clone number for the ligand (anti-SIRP α) bound to the chip is indicated as rows, and the clone number for the analytes (anti-SIRP α) injected over the chip is indicated as columns in FIGS. 22A & 22B. "X" indicates scenarios where the data from one orientation disagreed with the other.

(413) First, anti-SIRP α antibodies were immobilized on GLC chips using Proteon™ Amine Coupling Kit as described before. Briefly, for the immobilization step, GLC chip was activated with EDAC/Sulpho-NHS 1:1 (Biorad) diluted 1/80 for 300 s at 25 μ L/min. Anti-SIRP α antibodies were diluted to 80 nM concentration in 10 mM sodium acetate buffer pH 4.5 and immobilized to the chip at 30 μ L/min for 50 s. Chip was inactivated with ethanolamine for 300 s at 25 μ L/min. Afterwhich, SIRP- α v1 (SEQ ID NO:5) (100 nM) was first injected at 100 μ L/min for 60 s followed by injecting the anti-SIRP α antibodies in testing (100-150 nM) at 100 μ L/min for 60 s. The surfaces were regenerated with a 2:1 v/v blend of Pierce IgG elution buffer/4M NaCl. The resultant sensograms were used to score and group the antibodies into different bins according to their binding profiles.

(414) Results

(415) The following anti-SIRP- α antibody clones were binned: 3, 21, 25, 27, 45, 66, 115, 116, 117, 118, 119, 120, 121, 122, 123, 132, 135, 136, 137, 149, 161, 162, 173, 194, 209, 213, and 218. As shown in FIGS. 22A & 22B, the results demonstrated that each mode (B: blocking, NB: non-blocking, and KO: "kick off") of anti-SIRP- α antibody binding to human SIRP- α v1 binned separately. For the non-blockers, the anti-SIRP α antibodies could be further separated into 4 bins based on their binding profiles. For instance, anti-SIRP α antibodies 123, 149, 161, 162, 194 and 218 were grouped as Bin5. Anti-SIRP α antibodies 3, 173, 209 and 213 were grouped as Bin 4. Anti-SIRP α antibodies 136 and 137 were grouped as Bin 2. Anti-SIRP α antibody 45 has a unique binding profile and is grouped as Bin 6.

(416) Combined with the structural analyses described in Example 4, these results led to the model of antibody and CD47 binding to the SIRP- α D1 domain shown in FIG. 23. For instance, various anti-SIRP α antibodies in Bins 1, 2, 3 and 4 are postulated to share similar binding epitopes with anti-SIRP α 119, 136, 115 and 3 which were categorized in the respective bins. The binding epitopes for anti-SIRP α 119, 136, 115 and 3 have been obtained from crystal solutions as described in Example 4.

(417) In addition, anti-SIRP α antibodies in Bin 5 are postulated to bind adjacent to anti-SIRP α 136 (Bin 2) and 115 (Bin 3) respectively and share overlapping epitopes. This is based on binning data showing anti-SIRP α antibodies in Bin 5 competed with antibodies in Bins 2 and 3 for binding to SIRP α V1, and they did not compete with antibodies in Bins 1, 4 and 6 (FIG. 22B).

(418) Anti-SIRP α antibody 45 (Bin 6) is postulated to bind adjacent to anti-SIRP α 3 (Bin 4) and 136 (Bin 2) respectively and share overlapping epitopes. This is based on binning data showing anti-SIRP α 45 competed with antibodies in Bins 2 and 4 for binding to SIRP α V1, and anti-SIRP α 45 did not compete with antibodies in Bins 1, 3 and 5 (FIG. 22B).

(419) The model in FIG. 23 illustrates the epitope of each antibody bin based on structural and binning analyses, and demonstrates the overlap between various antibody bins using exemplary antibody clones. Additional anti-SIRP- α antibodies isolated herein are grouped into families and bins based on their epitope mapping profiles, homologies in their VH/VL domain sequences, binding specificities to SIRP proteins and the sources of the antibodies. For instance, anti-SIRP- α antibodies in family 1/Bin 1 are CD47 blockers that share high sequence homologies in their VH and VL domains (Table P and FIGS. 11O & 11P). The VH and VL are fully human sequences. Anti-SIRP- α antibodies in family 2/Bin 1 are also CD47 blockers. Their VH and VL share high homologies and are from human and chicken sources respectively (Table P, FIGS. 11A-11D). The anti-SIRP- α antibodies in Family 3/Bin 2 include CD47 non-blockers, and their highly homologous VH/VL are fully human sequences (Table P, FIGS. 11E-11F). The anti-SIRP- α antibodies in Family 4/Bin 3 include kick-off antibodies, and their highly homologous VH/VL are fully human sequences (Table P, FIGS. 11G & 11H). Family 5 includes CD47 non-blockers that are separately mapped into Bins 4, 5, and 6. The sequence alignments of these antibodies and their binding profiles are shown in FIGS. 11I-11N and Table P, respectively. The K.sub.off binding values to various SIRP- α , SIRP- β , and SIRP- γ polypeptides of the corresponding anti-SIRP α antibodies in Families 1-5 are presented in Table T.

Example 6: Germline and Liability Mutation Variants of Anti-SIRP- α Antibodies

(420) Some anti-SIRP- α antibodies described above are fully human antibodies generated in a chicken (e.g., antibodies 119, 135, and 136). As such, some of these antibodies may contain mutations in the variable domain framework sequences, as compared to wild-type human germline sequence, by virtue of generating these antibodies in chicken B cells. Therefore, it is desirable to “back-mutate” these respective residues to match that of human germline sequence with the goal of limiting immunogenicity when these anti-SIRP α antibodies are tested in humans as potential therapeutics. In addition to germline back-mutations, the CDRs and framework region of antibodies 119, 135 and 136 were analyzed for liability hot spots. These analyses identified sites where engineering may be desired to limit risk due to modifications that may occur during manufacturing, storage and/or drug development of anti-SIRP α antibodies.

(421) Methods

(422) Germline/Liability Mutations

(423) The K.sub.D for respective germline and liability mutants binding to SIRP were determined using direct immobilization using GLC chip as described supra.

(424) The wildtype and mutant antibodies were expressed in Expi293 and purified by Protein A affinity column chromatography as described earlier. All antibodies were expressed as human L234A/L235A/G237A/N297A IgG1 Fc antibodies. Mutagenesis was carried out using QuikChange Lightning Site Directed Mutagenesis kit according to manufacturer's instructions (Agilent Catalog #210518).

(425) Results

(426) Antibodies 119, 135, and 136 were examined. Selected antibody sequences were aligned with available human germline sequences using IgBlast (NCBI). For instance, a total of 7 sites on the heavy and light chains of 119 were identified. As shown in FIG. 24A, residues that were not commonly occurring in human germline sequence of 119 VH (e.g., D1, E43, and L112) were back-

mutated to match human germline sequence (e.g., D1E, E43K, and L112Q) while keeping CDR sequences intact. As for 119 VL (FIG. 24B), residues that were not commonly occurring in human germline sequence of 119 VL (e.g. F21, R39, E60, and T76) were back-mutated to match human germline sequence (e.g. F21L, R39K, E60A, and T76S) while keeping CDR sequences intact. As used herein, the terms “all mut” and “mut” refer to variable domains containing all of the germline mutations described herein for a particular antibody variable domain. The amino acid numberings used to describe the germline and liability mutations are based on sequential numbering accordingly to respective SEQ IDs.

(427) In addition, antibody sequences were also analyzed for “liability” hot spots, including residues that may be susceptible to oxidation, deamidation, isomerization, hydrolysis, and N-linked glycosylation. Potential hot spots are shown in Table L. In particular, M34V and M34L variants of HVR-H1 were generated for the VH domain of multiple antibodies.

(428) TABLE-US-00027 TABLE L Potential and known hot spots Oxidation M Deamidation NG NS NT Isomerization DG DS DT Hydrolysis DP N-linked glycosylation site(s) NXS/T

(429) Variants of antibody 119 were generated using heavy and/or light chain variable domains bearing germline and liability back-mutations. A 119 mutant (“mut”) VH domain was generated with the germline back mutations D1E, E43K, and L112Q, as well as the M34V mutations in CDR-H1 that remove a methionine residue that could potentially be oxidized (see SEQ ID NOs:246 for VH sequences). Another variant was generated with the germline back mutations D1E, E43K, and L112Q (see SEQ ID NO:258 for VH sequence). Alignments between the parental and variant sequences are shown in FIG. 24A. A 119 mutant (“mut”) light chain was also generated with the germline back mutations F21L, R39K, E60A, and T76S; an alignment between the parental and variant sequence is shown in FIG. 24B.

(430) Antibody 119 variants with the mutant heavy and/or light chain were compared with the parental 119 antibody with an IgG1 Fc region bearing L234A, L235A, G237A, and N297A mutations (EU numbering), and a parental 119 antibody with an IgG4 Fc region bearing an S228P mutation (EU numbering), for binding affinity to human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), and cynomolgus SIRP- α (SEQ ID NO:11). The 119 mutant heavy and light chains were both found to cause slight reductions in binding affinity to all three SIRP- α proteins. However, the 119 antibody variant with mutated heavy and light chains still displayed strong binding to both human SIRP- α proteins, with a K_{sub}.D of approximately 30 nM (Table M). Compared with the parental antibody, yield of the 119 antibody variant with mutated heavy and light chains also decreased by approximately 4.5-fold (Table M).

(431) To investigate effect of methionine in HVR-H1 on 119 SIRP α binding, M34V and M34L single mutations were generated in the 119 VH wildtype background and combined with 119 wildtype light chain. Both 119 wt/wt_M34V and 119 wt/wt_M34L had comparable affinities (K_{sub}.D, M) to human SIRP α v1 and v2 as compared with 119 wt/wt. This indicates that residue M34 is not critical for SIRP- α binding and can be substituted with M34L or M34V mutations. The corresponding VH sequences for M34V and M34L single mutations generated in the 119 VH wildtype background are SEQ ID NO: 421 and 420, respectively.

(432) TABLE-US-00028 TABLE M Binding affinities (K_{sub}.D, M) of 119 variant antibodies to human and cyno SIRP- α proteins K_{sub}.D for K_{sub}.D for K_{sub}.D for Yield Antibody VL/VH
human v1 human v2 cyno (mg/ml) 119 Mut/mut 3.17E-10 8.75E-11 1.95E-10 1.575 119 wt/mut 2.54E-10 6.94E-11 1.55E-10 5.377 119 mut/wt 2.14E-10 8.64E-11 1.38E-10 3.527 119 wt/wt 1.83E-10 6.82E-11 1.12E-10 7.192 119 Mut/ 2.15E-10 6.88E-11 1.18E-10 NT mut_V34M
AB119 Wt/wt with 1.74E-10 5.98E-11 1.16E-10 0.659 hIgG4 NT = not tested

(433) Next, similar variants of antibody 135 were generated. A 135 mutant (mut) heavy chain was generated with the germline back mutations D1E, R13Q, E16G, E43K, and L12Q, as well as the M34V mutation in CDR-H1 that removes a methionine residue that could potentially be oxidized (see SEQ ID NO:247 for VH sequence). A similar variant was constructed without the M34V

mutation in CDR-H1 (see SEQ ID NO:259 for VH sequence). A 135 mutant (mut) light chain was generated with the germline back mutations F21L and D60A (see SEQ ID NO:248 for VL sequence). Alignments between the parental and variant sequences are shown in FIGS. 25A & 25B. (434) Antibody 135 variants with the mutant heavy and/or light chain were compared with the parental 135 antibody for binding affinity to human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), cynomolgus SIRP- α (SEQ ID NO:11), and human SIRP- γ v1 (SEQ ID NO:15). The 135 mutant heavy and light chains had comparable binding affinity to all four SIRP- α proteins, as well as comparable yields (Table N1).

(435) To investigate the effect of methionine in HVR-H1 of 135 on SIRP- α binding, M34V and M34L single mutations were generated in the 135 VH wildtype background and combined with a 135 wildtype light chain. Both 135 wt/wt_M34V and 135 wt/wt_M34L had comparable affinities (K.sub.D, M) to human SIRP α v1 and v2 as compared with 135 wt/wt. This indicates that residue M34 is not critical for SIRP- α binding and can be substituted with an M34L or M34V mutation. The corresponding VH sequences for M34V and M34L single mutations generated in the 135 VH wildtype background are SEQ ID NO: 423 and 422, respectively.

(436) TABLE-US-00029 TABLE N1 Binding affinities (K.sub.D, M) of 135 variant antibodies to human and cyno SIRP- α proteins and human SIRP- γ protein K.sub.D for K.sub.D for K.sub.D
K.sub.D for Anti- VL/ human human for human Yield body VH α v1 α v2 cyno γ v1 (mg/mL) 135
mut/ 1.92E-10 1.75E-11 1.27E-10 7.33E-10 3.189 mut 135 wt/ 1.84E-10 1.97E-11 1.25E-10
7.79E-10 2.885 mut 135 mut/ 1.49E-10 2.66E-11 1.02E-10 5.22E-10 2.689 wt 135 wt/ 1.51E-10
2.90E-11 9.69E-11 5.39E-10 3.264 wt 135 wt/ 1.50E-10 1.61E-11 7.94E-11 NT NT mut V34M
NT = not tested

(437) Similar variants of antibody 136 were generated. A 136 mutant (mut) heavy chain was generated with the germline back mutations D1E, R13Q, E16R, E43K, and L111Q, as well as the M34V mutation in CDR-H1 that removes a methionine residue that could potentially be oxidized (see SEQ ID NO:249 for VH sequence). A similar variant was constructed without the M34V mutation in CDR-H1 (see SEQ ID NO:260 for VH sequence). A 136 mutant (mut) light chain was generated with the germline back mutations T2I, T12S, T22S, and E38Q (see SEQ ID NO:250 for VL). Alignments between the parental and variant sequences are shown in FIGS. 26A & 26B.

(438) As shown in FIG. 27A, antibody 136 variants with the mutant heavy and/or light chain were compared with the parental ("wt") 136 antibody as IgG1_AAA_N297A for binding affinity to human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), cynomolgus SIRP- α (SEQ ID NO:11), NOD mouse SIRP- α (SEQ ID NO:8), BL/6 mouse SIRP- α (SEQ ID NO:9), and BALB/c mouse SIRP- α (SEQ ID NO:10). In FIG. 27A, the Y-axis shows the ratio of K.sub.D mut/K.sub.D wt binding to various SIRP. If ratio=1 indicated no change in K.sub.D compare to parental antibody binding. Ratio=>1 and <1 indicated decrease and increase affinity binding to SIRP α (s) compare to parental antibody. While the 136 wt (VL)/mut (VH) antibody behaved similar to wt (VL)/wt(VH), both variants with the mutated light chain had poorer binding affinities, indicating that some VL mutations were not tolerated.

(439) In order to analyze the effect of each light chain back mutation on binding affinity, additional antibody 136 variants were constructed and characterized for binding affinity to BL/6 mouse SIRP- α (SEQ ID NO:9), NOD mouse SIRP- α (SEQ ID NO:8), BALB/c mouse SIRP- α (SEQ ID NO:10), human SIRP- α v1 (SEQ ID NO:5), human SIRP- α v2 (SEQ ID NO:6), cynomolgus SIRP- α (SEQ ID NO:11), and human SIRP- γ v1 (SEQ ID NO:15). Starting with the "all mutant" background, each individual mutation was reversed. I2T, S12T, S22T, and Q38E mutations were individually tested in otherwise "all mutant" light chains, as shown in FIG. 27B. The I2T mutation in an otherwise all mut background showed consistently similar binding affinity, as compared with the parental wt/wt antibody (see SEQ ID NO:251 for I2T in all mut background and FIG. 27A for alignment with parental and mutant 136 antibodies). However, the other three reverse mutations (S12T, S22T, and Q38E) consistently showed binding affinities more similar to 136 mut/mut,

indicating that the T2I mutation is responsible for reduced binding affinity to various SIRP proteins. Additional data from these experiments are provided in Table Q infra.

(440) TABLE-US-00030 TABLE N2 Binding affinities (K.sub.D, M) of 136 variant antibodies to human, mouse, and cyno SIRP- α proteins and human SIRP- γ protein V1 V2 NOD BL6 BALBc Cyno Sirpg Light chain/ SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID Heavy chain NO: 5 NO: 6 NO: 8 NO: 9 NO: 10 NO: 11 NO: 15 136 MutRv_I2T/mut 7.17E-10 1.86E-09 6.80E-10 1.53E-08 4.22E-10 2.33E-09 4.00E-08 136 MutRv_S12T/mt 4.88E-09 1.05E-08 2.85E-09 1.63E-08 2.09E-09 7.99E-09 6.67E-08 136 MutRv_S22T/mut 4.99E-09 8.28E-09 2.44E-09 1.33E-08 1.64E-09 6.84E-09 6.17E-08 136 MutRv_Q38E/mut 6.18E-09 1.25E-08 5.98E-09 3.31E-08 2.98E-09 1.04E-08 1.93E-08 136 MutRv_I2T/mut_V34M 5.51E-10 1.74E-09 5.96E-10 1.38E-08 3.61E-10 2.11E-09 3.35E-08 136 Mut/mut 7.32E-09 1.27E-08 5.42E-09 2.16E-08 2.67E-09 9.48E-09 2.88E-08 136 Wt/wt 9.72E-10 2.54E-09 8.21E-10 1.64E-08 4.58E-10 2.83E-09 3.36E-08

Example 7: Humanization of Anti-SIRP- α Antibodies

(441) The Examples above describe the generation of anti-SIRP- α antibodies having a fully human heavy chain and a chicken light chain. In order to humanize the chicken-derived light chains, chicken HVRs of these antibodies were grafted onto various human lambda light chain frameworks.

(442) Methods

(443) Humanization

(444) Antibodies were humanized using standard techniques. For measuring production yield, equal volume of Expi293 cultures expressing anti-SIRP α antibodies were purified by Protein A affinity chromatography. After buffer exchange into PBS, the protein concentration was determined by A280 and expressed in mg/mL.

(445) Results

(446) In order to design humanized light chains, each chicken light chain sequence was aligned to the closest human germline framework by IgBLAST (NCBI). Using this analysis, the closest match to the chicken lambda light chain framework is human IGLV3 (see SEQ ID NOs:314-317).

(447) In another approach, a literature search was undertaken to determine the optimal human lambda light chain framework sequences to pair with a human VH3 sequences (the human heavy chain used for these antibodies). Based on these analyses, it was thought that human VH3 would pair well with human IGLV1 and IGLV2. See Glanville, J. et al. (2009) *Proc. Natl. Acad. Sci.* 106:20216-20221; Lloyd, C. et al. (2009) *Protein Eng. Des. Sel.* 22:159-168; and Jayaram, N. et al. (2012) *Protein Eng. Des. Sel.* 25:523-529.

(448) Therefore, six humanized light chains were created: Hum1 (AB25 HVRs+human IGLV3 framework), Hum2 (AB25 HVRs+human IGLV1 framework), Hum3 (AB66 HVRs+human IGLV3 framework), Hum4 (AB66 HVRs+human IGLV1 framework), Hum5 (AB25 HVRs+human IGLV2 framework), and Hum6 (AB21 HVRs+human IGLV1 framework). Sequences of the resulting light chain variable domains are provided in Table O1.

(449) TABLE-US-00031 TABLE O1 Sequences of humanized variable light chain domains tested. HVR sequences are bolded and underlined. LC Human LC Name HVRs framework Sequence Hum1 AB25 IGLV3

SYELTQPPSVSVSPGQTARITC**SGGSYSSYYYAWYQ**

QKPGQAPVTLIY**SDDKRPS**NIPERFSGSSSGTTVTTLTI

SGVQAEDEADYYC**G GYDQSSYTNP**FGGGTKLTVL (SEQ ID NO: 252) Hum2 AB25

IGLV1 QSVLTQPPSVSAAPGQKVTISC**SGGSYSSYYYAWYQ**

QLPGTAPKTLIY**SDDKRPS**NIPDRFSGSKSGTSATLGI

TGLQTGDEADYYC**G GYDQSSYTNP**FGTGTKVTVL (SEQ ID NO: 253) Hum3 AB66

IGLV3 SYELTQPPSVSVSPGQTARITC**SGGDYYSTYYAWYQ**

QKPGQAPVTVIH**SDDKRPS**DIPERFSGSSSGTTVTTLTI

SGVQAEADYYCGGYDGRTYINTFGGGTKLTVL (SEQ ID NO: 254) Hum4 AB66
 IGLV1 QSVLTQPPSVSAAPGQKVTISCSGGDYYSTYYAWY
 QQLPGTAPKTVIHSDDKRPSDIPDRFSGSKSGTSATL
 GITGLQTGDEADYYCGGYDGRTYINTFGTGTGKTVTV L (SEQ ID NO: 255) Hum5
 AB25 IGLV2 QSALTQPASVSGSPGQSITISCTGTSSDVGSYSSYYY
AWYQQHPGKAPKTLIYSDDKRPSNVSNRFSGSKSG
 NTASLTISGLQAEDEADYYCGGYDQSSYTNPFGGG TKLTVL (SEQ ID NO: 256)
 Hum6 AB21 IGLV1 QSVLTQPPSVSAAPGQKVTISCSGGDYYSYYYGWY
 QQLPGTAPKTVIYSDDKRPSDIPDRFSGSKSGTSATL
 GITGLQTGDEADYYCGGYDYSTYANAFGTGTGKTVTV L (SEQ ID NO: 257)

(450) Each of the 6 humanized light chains was paired with each of four heavy chains (derived from A1B21, A1B25, AB327, and AB366), generating 24 unique antibodies. Antibodies were expressed as described above. Surprisingly, human IGLV1 framework sequences resulted in decreased antibody expression regardless of the heavy chain. This refers to all the heavy chain pairings with Hum2, Hum4 and Hum6 (except when pairing was carried out with heavy chain from AB366). The results are summarized in FIG. 28 as “protein yield” (row 1). In contrast, antibodies with light chains including human IGLV2 and IGLV3 frameworks (Hum 1, Hum3, Hum5) showed higher levels of expression regardless of the heavy chain.

(451) Selected antibodies were next characterized for binding to a variety of SIRP proteins (e.g., to human SIRP- α v1, human SIRP- α v2, cynomolgus SIRP- α , mouse BALB/c SIRP- α , and human SIRP-7). These data are also summarized in FIG. 28. Selected humanized light chains caused a decrease in binding to one or more antigens. For instance, the human IGLV3 framework (represented by Hum1 and Hum3) was found to allow for superior levels of antibody production without perturbing binding affinity. For example, light chain variable domains with the IGLV3 frameworks and either the antibody 25 or antibody 66 HVR sequences (represented by Hum1 and Hum 3 respectively) combined well with a variety of heavy chains (e.g., heavy variable domains from antibodies 21, 25, 27, and 66) and showed similar binding to different SIRP- α and SIRP- γ proteins. In contrast, IGLV1 and IGLV2 frameworks (represented by Hum2, Hum4, Hum5 and Hum6) were found to either lower expression and/or decrease binding to SIRP when paired with heavy chains from antibodies 21, 25, and 27 and 66. Additional binding data from these experiments are provided in Table R infra. The human IGLV3 framework was selected for further testing.

(452) Another goal for humanization of these antibodies was antibody sequences having greater than or equal to 85% identity to human germline light chain/heavy chain sequences. Additional VL domain Hum9 and Hum8 was generated based on the Hum1 VL domain. Compared to Hum1, Hum9 contains 4 amino acid substitutions near or in HVR-L1 and -L2 that increase the humanness of the light chain to greater than or equal to 85% identity to human light chain sequence (FIG. 29). Compared to Hum1, Hum8 contains 5 amino acid substitutions respectively near or in HVR-L1 and -L2 that increase the humanness of the light chain to greater than or equal to 85% identity to human light chain sequence (FIG. 29). Hum1, Hum8 and Hum9 VLs when paired with heavy chain all_mut_AB21 (carrying germline mutations) produced anti-SIRP α antibodies that bind to human v1 with affinity equal or better than 10 pM (Table S). Similarly, when Hum1, Hum8 and Hum9 VLs were paired with heavy chain all_mut_AB25 (carrying germline mutations), the anti-SIRP α antibodies bind to human v1 with affinity equal or better than 10 pM. Additional binding data from these experiments are provided in Table S infra. These light chains can be combined interchangeably with antibody VH domains 21, 25, and 27 (as well as variants thereof, which were modified as described supra for antibodies 119, 135, and 136; FIG. 30). Without wishing to be bound to theory, it is thought that the humanization process described above can be applied to the light chain of any antibody of family 2 (bin 1).

Example 8: Induction of Phagocytosis and Dendritic Cell Activation by Anti-SIRP- α Antibodies

(453) Various anti-SIRP- α antibodies representing different modes of binding to SIRP- α (e.g., blocking, non-blocking, and “kick off” antibodies) were next examined in phagocytosis assays.

(454) CD47-Fc Binding Assays

(455) CD47-Fc was conjugated with Alexa Fluor 647 (AF647) using the Alexa Fluor 647 Microscale Protein Labeling Kit (Thermo Fisher Scientific). In 96 well polypropylene plates (Corning), 100,000 PBMCs were suspended in 100 μ l 0.25 μ M AF647-labeled CD47-Fc and 1 μ l anti-CD14 PE antibody (Biolegend) in FACS buffer. Cells were incubated on ice for 30 minutes, washed in FACS buffer, and incubated in 10-fold serial dilutions of anti-SIRP- α antibody from 1.25 μ M to 25 pM. Cells were incubated on ice for 30 minutes, washed in FACS buffer, and fixed in 75 μ l of 0.5 percent formaldehyde. Cells were analyzed on a FACS Canto™ II (BD Biosciences) flow cytometer, with subsequent data analysis by Flowjo 10.7 (Treestar). Geometric mean fluorescence intensity of the CD47-Fc signal was determined in the CD14-positive monocyte population.

(456) Dendritic Cell Activation Assays

(457) The dendritic cell activation assays were performed as described elsewhere herein. Briefly, Balb/c mice (n=3/group) were intravenously injected with a control rat anti-mouse anti-SIRP- α antagonistic antibody (clone p84), human IgG1 control, various anti-SIRP- α antibodies, mouse IgG control, or vehicle (PBS) at 10 mg/kg. Five hours post injection, spleens were harvested and processed into single cell suspension by mechanical dissociation. Activation marker CD86, MHCII and CCR7 level on CD4⁺ splenic dendritic cells was measured by flow cytometry.

(458) Cell Adhesion Assays

(459) The adhesion assay was performed using isogenic cells lacking or expressing human CD47. Hamster CD47 knockout CHO cells (CHO.sup.CD47 KO) were generated using CRISPR technology. Isogenic cells expressing human CD47 were generated by transiently transfecting human CD47 into CHO.sup.CD47 KO 24 hours post transfection, human CD47 transfected CHO.sup.CD47 KO cells (4 $\times$ 10⁵) were re-plated onto 24 well tissue culture treated plates and allowed to reattach and reach confluency overnight at 37° C.

(460) Human peripheral blood mononuclear cells (PBMCs) were separated from peripheral blood of healthy donors by Ficoll gradient centrifugation. CD3⁺ and CD14⁺ cells were isolated from PBMCs by negative selection using magnetic beads. Isolated CD3⁺ and CD14⁺ cells (5 $\times$ 10⁶) were pre-incubated with Hum1/AB21mutall, Hum9/AB21mutall or 136 wt/mut antibodies containing mutated human IgG1 (L234A L235A G237A, and N297A) (20 μ g/mL) at 37° C. for 20 minutes, before plating onto CHO.sup.CD47 KO and CHO.sup.hCD47⁺ cells and allowed to adhere for 1 hour at 37° C. Non-adherent cells were removed by five gentle washes with PBS. Adherent cells were detached with trypsin and neutralized with 10% FBS. Cells were transferred to 96 well plate and washed 2 times with PBS+0.5% BSA followed by cell surface labeling with fluorochrome conjugated human CD3, CD4, CD8 and CD14 antibodies. Counting beads were added and cell adhesion was quantified using a BD Canto™ II flow cytometer. Analysis of flow cytometry data was done using Flowjo.

(461) Results

(462) Five antibodies that “kick off” CD47 from binding SIRP- α were examined for their effects on phagocytosis of HER2(+) OE19 cells by M2 macrophages in combination with the anti-HER2 antibody trastuzumab (FIG. 31). All 5 “kick off” antibodies were found to enhance trastuzumab-induced phagocytosis.

(463) Five antibodies that do not block CD47 from binding SIRP- α were next examined for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 32). All non-blocking antibodies were found to enhance cetuximab-induced phagocytosis.

(464) Next, humanized antibodies described above were tested for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 33A). All humanized antibodies were found to enhance cetuximab-induced

phagocytosis. Additional variants of antibody 136 (described supra) were tested for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 33B). All variants of antibody 136 enhanced cetuximab-induced phagocytosis, but to varying degrees. Additional humanized antibodies described above (antibody 25 and 27 heavy chain variants combined with Hum1 or Hum9 light chain) were tested for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 33C). All humanized antibodies were found to enhance cetuximab-induced phagocytosis.

(465) To investigate the effect of anti-SIRP- α antibody Fc regions on phagocytosis, two non-blocking antibodies were tested either as full-length antibodies or F(ab).sub.2 fragments for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 34). These results indicated that non-blocking antibodies lose the ability to induce phagocytosis as F(ab).sub.2 fragments, suggesting that the Fc region of this class of antibody is required for induction of phagocytosis.

(466) Three variants of blocking antibody 119 (described supra) were tested for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 35). The results demonstrated that all three variants enhanced cetuximab-induced phagocytosis.

(467) Two variants of blocking antibody 135 (described supra) were tested for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 36). The results demonstrated that both variants enhanced cetuximab-induced phagocytosis.

(468) Additional non-blocking antibodies (described supra) were tested for their effects on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages in combination with the anti-EGFR antibody cetuximab (FIG. 37). All non-blocking antibodies were found to enhance cetuximab-induced phagocytosis.

(469) Next, various anti-SIRP- α antibodies were examined for their effects on in vivo dendritic cell activation (FIGS. 38A-38B), including known anti-SIRP- α antibody p84 (see, e.g., Tangsheng, Y. et al (2015) *Immunity* 43:1-12). Failure to engage mouse SIRP- α receptor on splenic dendritic cells via CD47 binding leads to splenic dendritic cell activation. Control anti-SIRP- α antagonist antibody p84 activated splenic dendritic cells when injected intravenously into mice. Non-blocking anti-SIRP- α antibodies (AB136b, AB3b and AB136 wt/mut) and blocking anti-SIRP- α antibodies (Hum1/AB21mutall, Hum8/AB21mutall, and Hum9/AB21mutall) were tested in vivo to determine if it leads to dendritic cell activation. As determined by CD86 and MHCII expression, both SIRP- α blockers and non-blockers induce activation of dendritic cells. These results suggest that blocking and non-blocking anti-SIRP- α antibodies induce activation of dendritic cells.

(470) Two variants of antibody 218, 218-Hum13/VH_wt and 218-Hum14/VH_wt, were generated by expressing humanized light chains (Hum13 and Hum14 respectively) with the wild-type heavy chain of antibody 218. Hum13 used the human IGLV2 framework, whereas Hum14 used the human IGLV3 framework (see SEQ ID NOs:333 and 334 for VL sequences of Hum13 and Hum14, respectively). Both clones showed lower affinity binding to v1 and v2 (K.sub.D is ~29.3 to 53.1 nM), as shown in Table O2.

(471) TABLE-US-00032 TABLE O2 Binding affinity (K.sub.D, M) of antibody 218 and 218 variants for human SIRP- α v1 and v2. v1 v2 K.sub.D (M) ALX135 ALX269 AB218a 1.60E-10 3.23E-10 218-Hum13/VH_wt 2.93E-08 3.38E-08 218-Hum14/VH_wt 3.45E-08 5.31E-08

(472) Both antibodies were tested in a phagocytosis assay in combination with cetuximab in DLD-1 cells. Interestingly, neither humanized variant was able to enhance phagocytosis over the parental AB218a antibody (FIG. 39A). Without wishing to be bound to theory, these results suggest that antibodies with K.sub.D at or below an approximate range of 30-50 nM may be more effective in inducing phagocytosis than antibodies that bind with weaker affinity.

(473) Exemplary blocking, non-blocking, and kick off anti-SIRP- α antibodies were next tested for induction of phagocytosis as single agents. First, three antibodies that block CD47 from binding SIRP- α , AB119a, AB120a, and AB122a, were examined for their effects as single agents on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages (as described above). All blocking antibodies were found to induce phagocytosis as single agents (FIG. 39B). Next, two antibodies that do not block CD47 from binding SIRP- α , AB136a and AB137a, were examined for their effects as single agents on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages. All non-blocking antibodies were found to induce phagocytosis as single agents (FIG. 39C). Finally, five “kick-off” anti-SIRP- α antibodies, AB115a, AB116a, AB117a, AB118a, and AB132a, were examined for their effects as single agents on phagocytosis of EGFR(+) DLD-1 cells by M2 macrophages. All “kick-off” antibodies were found to induce phagocytosis as single agents (FIG. 39D).

Example 9: Synergistic Anti-Tumor Effects of Combining Blocking or Non-Blocking Anti-SIRP- α Antibodies with Inhibition of the PD-L1/PD-1 Pathway

(474) Methods

(475) In Vivo Anti-Tumor Activity

(476) For the CT26 syngeneic mouse colon carcinoma model, CT26 cells were implanted subcutaneously in BALB/c mice and randomized into groups (8-9 mice/group). Treatment groups included vehicle (PBS), AB25b, anti-PD-L1, and AB25b/anti-PD-L1. Anti-PD-L1 is generated by fusing the VH and VL domain of Atezolizumab with mouse IgG1 Fc region bearing an N297A mutation. All anti-SIRP- α antibodies also have a mouse IgG1 Fc region bearing an N297A mutation. Treatment was initiated when tumors were an average of 75-80 mm^{sup.3}, day 7 or 8 post implant. Mice were dosed intraperitoneally (IP) at 3 mg/kg or 10 mg/kg twice a week for three weeks for anti-SIRP α antibodies and three doses at 3 mg/kg, five days apart for anti-PD-L1. Animals were sacrificed when tumors reached a volume of ~2000 mm^{sup.3}.

(477) For the MC38 syngeneic mouse colon carcinoma model, MC38 cells were implanted subcutaneously in C57BL/6 mice and randomized into groups (8-10 mice/group). Treatment groups included vehicle (PBS), AB25b, AB136b, anti-PD1 (clone RMP1-14, BioXCell), AB136b/anti-PD1, and AB25b/anti-PD1. All anti-SIRP α antibodies had a murine IgG1 Fc region bearing an N297A mutation except for AB25c. Treatment was initiated when tumors were an average of 60-65 mm^{sup.3}, day 7 post implant. Mice were dosed intraperitoneally (IP) at 10 mg/kg twice a week for three weeks for anti-SIRP α and three doses at 2 mg/kg for anti-PD1. Animals were sacrificed when tumors reached a volume of ~2000 mm^{sup.3}.

(478) Results

(479) Anti-tumor activity of the blocking AB25b anti-SIRP- α antibody was tested alone and in combination with an anti-PD-L1 antibody in the CT26 syngeneic mouse colon carcinoma model. As shown in FIG. 40, administration of AB25b at 10 mg/kg in combination with anti-PD-L1 at 3 mg/kg delayed tumor formation when compared to treatment with each single agent or vehicle control. On day 27 of the study, the combination treatment group had six mice with tumors below 600 mm^{sup.3} in size, as compared to two, two, and two mice with tumors below 600 mm^{sup.3} in size in the vehicle, anti-PD-L1 single agent, and anti-SIRP- α single agent treatment groups, respectively.

(480) Next, the anti-tumor activities of the blocking AB25b anti-SIRP- α antibody and non-blocking AB136b anti-SIRP- α antibody were tested alone and in combination with an anti-PD-1 antibody in the MC38 syngeneic mouse colon carcinoma model. As shown in FIG. 41, combining either AB25b or AB136b (at 10 mg/kg) with anti-PD-1 at 5 mg/kg delayed tumor formation when compared to treatment with each single agent or vehicle control. On day 27 of the study, the AB25b/PD-1 combination treatment group had seven mice with tumors below 600 mm^{sup.3} in size, and the AB136b/PD-1 combination treatment group had six mice with tumors below 600 mm^{sup.3} in size, as compared to one, five, two, and one mice with tumors below 600 mm^{sup.3} in

size in the vehicle, anti-PD-1 single agent, AB25b single agent, and AB136b single agent treatment groups, respectively.

(481) A summary of antibodies described herein and their properties is provided in Table P. Additional binding data are provided in Table T.

(482) Although the foregoing disclosure has been described in some detail by way of illustration and example for purposes of clarity of understanding, the descriptions and examples should not be construed as limiting the scope of the present disclosure. The disclosures of all patent and scientific literature cited herein are expressly incorporated in their entirety by reference.

(483) TABLE-US-00033 TABLE P Anti-SIRP- α antibody summary. Human Isoforms Type of In In Species Binding (+/-) (Koff) (+/-) (Koff) K.sub.D K.sub.D Heavy Light Binding vitro vivo

Human v1	Cyno	Mouse	129	Beta	Gamma	Human	V1	Human	V2	chain	Chain	Anti-	(NB/B/	Bin
phago	mouse	(SEQ ID	(SEQ ID	(SEQ ID	(SEQ ID	(SEQ ID	(SEQ ID	(SEQ ID	(SEQ ID	(Human/	(Human/	body	KO)	No.
(+/-)	(+/-)	NO: 5)	NO: 1)	NO: 7)	NO: 13)	NO: 15)	NO: 5)	NO: 6)	Chicken	Chicken)	Family	1	119	B
1	+	N.A.	+	+	+	+	+	1.83E-10	7.99E-11	Human	Human	120	B	1
+	+	+	+	+	+	+	+	2.14E-10	8.56E-11	Human	Human	121	B	1
+	+	N.A.	+	+	+	+	+	1.57E-10	5.01E-11	Human	Human	122	B	1
N.A.	+	+	+	+	+	+	+	2.14E-10	7.78E-11	Human	Human	135	b	1
+	+	+	+	+	+	+	+	1.51E-10	2.90E-11	Human	Human	Family	2	16
B	1	+	+	+	+	+	+	<1.0E-12	<1.0E-12	Human	Chicken	17	B	1
+	+	+	+	+	+	+	+	<1.0E-12	<1.0E-12	Human	Chicken	21	B	1
+	+	+	+	+	+	+	+	<1.0E-12	<1.0E-12	Human	Chicken	22	B	1
+	+	+	+	+	+	+	+	<1.0E-12	<1.0E-12	Human	Chicken	23	B	1
+	+	+	+	+	+	+	+	<1.0E-12	<1.0E-12	Human	Chicken	24	B	1
+	+	+	+	+	+	+	+	<1.0E-12	<1.0E-12	Human	Chicken	25	B	1
+	+	+	+	+	+	+	+	2.15E-11	Human	Chicken	26	B	1	
+	NT	+	+	+	+	+	+	2.15E-11	Human	Chicken	27	B	1	
NT	NT	+	+	+	+	+	+	Human	Chicken	28	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	29	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	30	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	55	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	56	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	59	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	60	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	65	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	66	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	69	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	70	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	71	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	73	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	74	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	76	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	201	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	202	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	206	B	1		
NT	NT	+	+	+	+	+	+	Human	Chicken	Family	3	136	NB	
2	+	+	+	+	+	+	+	4.58E-10	3.22E-09	Human	Human	137	NB	2
+	NT	+	+	+	+	+	+	7.74E-10	3.14E-09	Human	Human	175	NB	2
+	NT	+	+	+	+	+	+	1.37E-10	6.62E-10	Human	Human	177	NB	2
NT	NT	+	+	+	+	+	+	Human	Human	178	NB	2	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	180	NB	2	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	184	NB	2	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	185	NB	2	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	189	NB	2	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	190	NB	2	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	193	NB	2	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	Family	4	115	KO	3
+	NT	NT	+	+	+	+	+	Human	Human	116	KO	3	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	117	KO	3	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	118	KO	3	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	132	KO	3	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	191	KO	3	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	198	KO	3	NT	NT
+	NT	NT	+	+	+	+	+	Human	Human	Family	5	(additional	non-blockers)	3
NB	4	+	+	+	+	+	+	1.62E-10	7.67E-11	Chicken	Chicken	173	NB	4
+	NT	+	+	+	+	+	+	9.37E-10	9.28E-09	Human	Human	174	NB	4
NT	NT	+	+	+	+	+	+	1.71E-10	5.01E-09	Human	Human	209	NB	4
+	NT	+	+	+	+	+	+	6.05E-09	1.69E-09	Human	Chicken	214	NB	4
NT	NT	+	+	+	+	+	+	6.05E-10	NLB	Human	Human	149	NB	5
+	NT	+	+	+	+	+	+	8.73E-10	2.38E-10	Human	Human	161	NB	5
+	NT	+	+	+	+	+	+	1.03E-9	1.27E-10	Human	Human	162	NB	5
+	NT	+	+	+	+	+	+	4.50E-10	1.57E-08	Human	Human	163	NB	5
NT	NT	+	+	+	+	+	+	4.97E-10	NLB	Human	Human	218	NB	5
+	NT	+	+	+	+	+	+	1.23E-10	2.76E-10	Chicken	Human	S45	NB	6
+	NT	+	+	+	+	+	+	6.63E-11	1.34E-10	Chicken	Chicken	B	=	blocker;

NB = non-blocker; KO = knock off. NT or blank = not tested; NA = not applicable (antibodies do not cross-react). NLB = no binding

(484) TABLE-US-00034 TABLE Q Anti-SIRP- α antibody germline/liability mutation summary. Type of Bind- K.sub.D (M) Hu- Hu- ing SEQ ID SEQ ID SEQ SEQ SEQ SEQ ID SEQ ID In In man man (NB/ NO: 5 NO: 6 ID ID ID SEQ ID NO: 13 NO: 15 vitro vivo Anti- Light Heavy B/ Bin Human Human NO: 11 NO: 8 NO: 9 NO: 10 Human Human phago mouse body Chain Chain KO) No. v1 v2 Cyno NOD BL6 BALBc SIRPb SIRPg (+/-) (+/-) 119 wt wt B 1 1.83E-10 6.82E-11 1.12E-10 NLB NLB NLB 3.42E-10 2.67E-10 + N.A. 119 Mut wt B 1 2.14E-10 8.64E-11 1.38E-10 NLB NLB NLB NT 2.33E-10 NT N.A. 119 wt Mut B 1 2.54E-10 6.94E-11 1.55E-10 NLB NLB NLB NT NT + N.A. 119 Mut Mut B 1 3.17E-10 8.75E-11 1.95E-10 NLB NLB NLB 4.60E-10 3.36E-10 + N.A. 119 Mut Mut_ B 1 2.15E-10 6.88E-11 1.18E-10 NLB NLB NLB 3.37E-10 2.63E-10 NT N.A. V34M 135 wt wt B 1 1.51E-10 2.90E-11 9.69E-11 NLB NLB NLB 5.39E-10 + N.A. 135 Mut wt B 1 1.49E-10 2.66E-11 1.02E-10 NLB NLB NLB 5.22E-10 NT N.A. 135 Wt Mut B 1 1.84E-10 1.97E-11 1.25E-10 NLB NLB NLB 7.79E-10 7.33E-10 NT N.A. 135 Mut Mut B 1 1.92E-10 1.75E-11 1.27E-10 NLB NLB NLB 1.88E-10 7.33E-10 + N.A. 135 wt Mut_ B 1 1.50E-10 1.61E-11 7.94E-11 NLB NLB NLB 1.60E-10 5.33E-10 NT N.A. V34M 136 wt wt NB 2 4.58E-10 1.63E-09 2.15E-09 5.54E-10 1.27E-08 3.50E-10 4.35E-9 2.39E-08 + + 136 Mut wt NB 2 7.28E-09 1.74E-08 1.13E-08 4.12E-09 3.26E-08 2.80E-09 1.96E-08 - 136 wt Mut NB 2 5.58E-10 1.74E-09 2.26E-09 6.78E-10 2.31E-08 4.16E-10 3.54E-09 1.65E-08 NT 136 Mut Mut NB 2 7.29E-09 1.95E-08 1.30E-08 5.21E-09 3.17E-08 3.14E-09 1.68E-09 - 136 Mut_ Mut NB 2 7.17E-10 1.86E-09 2.33E-09 6.80E-10 1.53E-08 4.22E-10 3.16E-09 4.00E-08 + I2T 136 Mut_ Mut NB 2 4.88E-09 1.05E-08 7.99E-09 2.85E-09 1.63E-08 2.09E-09 6.67E-08 NT S12T 136 Mut_ Mut NB 2 4.99E-09 8.28E-09 6.84E-09 2.44E-09 1.33E-08 1.64E-09 6.17E-08 NT S22T 136 Mut_ Mut NB 2 6.18E-09 1.25E-08 1.04E-08 5.98E-09 3.31E-08 2.98E-09 1.93E-08 NT Q38E 136 Mut_ Mut_ NB 2 5.51E-10 1.74E-09 2.11E-09 5.96E-10 1.38E-08 3.61E-10 2.22E-09 3.35E-08 NT I2T V34M B = blocker; NB = non-blocker. NT or blank = not tested; NA = not applicable (antibodies do not cross-react); NLB = no binding 119 heavy chain mut = D1E, E43K, L112Q, M34V 119 light chain mut = F21L, R39K, E60A, T76S 135 heavy chain mut = D1E, R13Q, E16G, M34V, E43K, L112Q

(485) TABLE-US-00035 TABLE R Anti-SIRP- α antibody humanization summary (round 1). Koff (1/s) SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID In vitro In vivo Antibody NO: 5 NO: 6 NO: 11 NO: 10 NO: 15 phago mouse Designation VL VH Human V1 Human V2 Cyno BALBc SIRPg (+/-) (+/-) Parental Antibodies AB21 Chicken Human 7.07E-04 1.92E-03 2.29E-03 2.41E-03 9.02E-04 NT + (AB21_LC_wt)SEQ (AB21_HC_wt) ID NO: 136 SEQ ID NO: 135) AB25 Chicken Human 1.63E-04 3.53E-04 3.94E-04 1.78E-03 2.03E-04 + + (AB25_LC_wt) (AB25_HC_wt) SEQ ID NO: 138 SEQ ID NO: 137 AB27 Chicken Human 3.15E-04 5.79E-04 6.83E-04 5.00E-03 4.07E-04 + + (AB27_LC_wt) (AB27_HC_wt) SEQ ID NO: 140 SEQ ID NO: 139 AB66 Chicken Human 9.49E-04 2.81E-03 2.69E-03 3.46E-03 9.97E-04 NT + (AB66_LC_wt) (AB66_HC_wt) SEQ ID NO: 142 SEQ ID NO: 141 Humanization of chicken light chain of AB25, AB66 - replaced with human IGLV3 framework Hum1/ Hum1_Humanized Human 1.93E-04 3.03E-04 3.95E-04 2.91E-03 1.88E-04 + AB21_HC_wt (AB25_IGLV3) (AB21_HC_wt) SEQ ID NO: 252 SEQ ID NO: 135 Hum1/ Hum1_Humanized Human 1.33E-04 2.67E-04 3.30E-04 3.74E-03 2.03E-04 + AB25_HC_wt (AB25_IGLV3) (AB25_HC_wt) SEQ ID NO: 252 SEQ ID NO: 137 Hum1/ Hum1_Humanized Human 1.92E-04 2.92E-04 3.79E-04 3.70E-03 1.78E-04 + AB27_HC_wt (AB25_IGLV3) (AB27_HC_wt) SEQ ID NO: 252 SEQ ID NO: 139 Hum1/ Hum1_Humanized Human 1.24E-04 2.39E-04 3.03E-04 2.46E-03 1.36E-04 NT AB66_HC_wt (AB25_IGLV3) (AB66_HC_wt) SEQ ID NO: 252 SEQ ID NO: 141 Hum3/ Hum3_Humanized Human 1.37E-04 4.38E-04 4.32E-04 2.31E-03 2.10E-04 NT AB21_HC_wt (AB66_IGLV3) (AB21_HC_wt) SEQ ID NO: 254 SEQ ID NO: 135 Hum3/ Hum3_Humanized Human 5.94E-05 3.53E-04 3.75E-04 2.27E-04 1.69E-04 NT AB25_HC_wt (AB66_IGLV3)

(AB25_HC_wt) SEQ ID NO: 254 SEQ ID NO: 137 Hum3/ Hum3_Humanized Human 2.16E-04 3.96E-04 4.64E-04 2.02E-04 2.07E-04 NT AB27_HC_wt (AB66_IGLV3) (AB27_HC_wt) SEQ ID NO: 254 SEQ ID NO: 139 Hum3/ Hum3_Humanized Human 1.24E-04 2.87E-04 3.06E-04 2.05E-03 1.35E-04 NT AB66 HC wt (AB66_IGLV3) (AB66_HC_wt) SEQ ID NO: 254 SEQ ID NO: 141 NT or blank = not tested.

(486) TABLE-US-00036 TABLE S Anti-SIRP- α antibody humanization summary (round 2). KD (M) SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID In vitro Antibody NO: 5 NO: 6 NO: 11 NO: 8 NO: 9 NO: 10 NO: 13 NO: 15 phago Designation VL VH Human v1 Human V2 Cyno NOD BL6 BALBc SIRPb SIRPg (+/-) Pairing of humanized light chain with heavy chain (Germline mut) Hum1/ Hum1_ Human 5.32E-12 4.60E-12 2.91E-11 3.70E-09 9.50E-09 7.91E-09 6.7E-12 >1.0E-12 + AB21_HC_ Humanized (AB21_HC_ Mutall (AB25_IGL Mutall) V3) SEQ ID SEQ ID NO: 252 NO: 263 Hum1/ Hum1_ Human 5.19E-12 4.83E-09 1.03E-08 7.25E-09 + AB25_HC_ Humanized (AB25_HC_ Mutall (AB25_IGL Mutall) V3) SEQ ID SEQ ID NO: 252 NO: 265 Hum1/ Hum1_ Human 4.37E-12 2.92E-09 9.03E-09 5.77E-09 + AB27_HC_ Humanized (AB27_HC_ Mutall (AB25_IGL Mutall) V3) SEQ ID SEQ ID NO: 252 NO: 267 Mutation of humanized light chain to increase % humaness Hum8/ Hum8_ Human 2.01E-11 2.78E-08 4.15E-04 7.12E-08 + AB21_HC_ Humanized (AB21_HC_ Mutall (AB25_IGL Mutall) V3) + 5aa in SEQ ID CDR SEQ ID NO: 263 NO: 416 Hum8/ Hum8_ Human 2.62E-11 2.05E-08 4.97E-04 5.99E-08 NT AB25_HC_ Humanized (AB25_HC_ Mutall (AB25_IGL Mutall) V3) + 5aa in SEQ ID CDR SEQ ID NO: 265 NO: 416 Hum9/ Hum9_ Human 1.19E-11 1.19E-10 2.22E-10 2.41E-08 5.33E-04 1.36E-07 5.69E-11 3.45E-11 + AB21_HC_ Humanized (AB21_HC_ Mutall (AB25_IGL Mutall) V3) + 4aa in SEQ ID CDR SEQ ID NO: 263 NO: 262 Hum9/ Hum9_ Human 1.29E-11 6.12E-08 1.15E-08 3.83E-08 + AB25_HC_ Humanized (AB25_HC_ Mutall (AB25_IGL Mutall) V3) + 4aa in SEQ ID CDR SEQ ID NO: 265 NO: 262 NT or blank = not tested.

(487) TABLE-US-00037 TABLE T Anti-SIRP- α antibody binding data summary. Values indicated are tested by SPR (K.sub.off, 1/s). CV1-3 v1 v2 cyno1 cyno2 m129 NOD BL6 SIRPb SIRPg SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID CD47 Antibody NO: 18 NO: 5 NO: 6 NO: 11 NO: 12 NO: 7 NO: 8 NO: 9 NO: 13 NO: 15 blocking Family 1 S119 5.65E-04 4.15E-04 1.48E-04 2.34E-04 3.10E-04 NLB NLB NLB 4.28E-04 3.95E-04 block S120 6.26E-04 4.04E-04 1.49E-04 2.36E-04 3.14E-04 NLB NLB NLB 4.25E-04 3.93E-04 block S121 NT 4.79E-04 1.12E-04 3.02E-04 3.19E-04 NLB NLB NLB 5.64E-04 4.82E-04 block S122 6.63E-04 4.98E-04 2.26E-04 2.66E-04 3.27E-04 NLB NLB NLB 3.52E-04 3.61E-04 block S135 3.63E-03 6.58E-04 8.78E-05 4.73E-04 3.88E-04 NLB NLB NLB 5.21E-04 1.46E-03 block Family 2 S16 1.00E-04 6.22E-05 1.11E-04 1.32E-04 1.38E-04 1.76E-04 1.28E-03 2.35E-03 1.13E-04 8.17E-05 block S17 1.54E-04 1.24E-04 1.97E-04 1.93E-04 2.07E-04 2.34E-04 6.12E-04 1.13E-03 1.25E-04 1.02E-04 block S21 1.95E-04 1.80E-04 2.07E-04 2.33E-04 2.52E-04 2.81E-04 2.64E-03 8.06E-04 1.90E-04 1.84E-04 block S22 1.35E-04 1.21E-04 1.46E-04 1.77E-04 1.77E-04 1.53E-04 5.83E-04 7.36E-04 1.38E-04 1.08E-04 block S23 1.06E-04 8.35E-05 1.35E-04 1.60E-04 1.83E-04 1.34E-04 8.23E-04 1.32E-03 1.14E-04 1.04E-04 block S24 2.16E-04 2.57E-04 3.17E-04 3.82E-04 3.34E-04 4.02E-04 3.86E-03 1.26E-03 1.94E-04 1.97E-04 block S25 1.40E-04 1.12E-04 2.09E-04 2.19E-04 2.12E-04 1.33E-04 7.79E-04 2.90E-04 1.71E-04 1.41E-04 block S26 4.74E-05 5.81E-05 1.11E-04 9.67E-04 7.43E-04 1.08E-03 7.64E-04 NLB 1.29E-04 8.35E-05 block S27 8.94E-05 5.81E-05 9.97E-05 1.38E-04 1.61E-04 1.36E-04 1.30E-03 1.84E-03 8.97E-05 9.46E-05 block S28 6.59E-05 2.11E-05 1.47E-05 1.15E-04 1.32E-04 3.06E-04 6.16E-04 1.86E-03 6.29E-05 3.27E-05 block S29 3.43E-04 9.61E-05 1.78E-04 3.87E-04 3.69E-04 1.14E-03 NLB 1.86E-03 2.79E-04 2.16E-04 block S30 1.42E-04 1.32E-04 1.88E-04 3.35E-04 3.66E-04 4.76E-04 NLB 2.44E-03 2.37E-04 1.93E-04 block S55 6.00E-05 4.18E-05 7.16E-05 9.63E-05 1.16E-04 1.43E-04 1.19E-03 1.97E-03 6.48E-05 4.69E-05 block S56 2.03E-04

2.28E-04 2.86E-04 2.95E-04 2.80E-04 4.06E-04 1.06E-03 4.29E-03 3.62E-04 2.15E-04 block
S59 8.19E-05 4.98E-05 8.63E-05 1.06E-04 1.13E-04 2.09E-04 1.35E-03 2.39E-03 1.07E-04
4.28E-05 block S60 1.45E-04 1.40E-04 1.86E-04 1.92E-04 1.75E-04 2.21E-04 7.26E-04
1.50E-03 1.53E-04 1.00E-04 block S65 1.47E-04 1.45E-04 1.72E-04 1.88E-04 1.90E-04
1.59E-04 6.07E-04 7.86E-04 1.54E-04 1.34E-04 block S66 1.10E-04 1.18E-04 2.06E-04
2.50E-04 2.55E-04 1.27E-04 4.90E-04 1.02E-03 1.38E-04 1.26E-04 block S69 2.33E-04
2.05E-04 2.67E-04 2.55E-04 2.16E-04 2.46E-04 7.09E-04 7.65E-04 2.29E-04 1.99E-04 block
S70 2.60E-04 2.25E-04 2.90E-04 2.96E-04 2.45E-04 2.70E-04 6.72E-04 9.87E-04 2.25E-04
2.22E-04 block S71 3.79E-04 3.35E-04 4.03E-04 3.64E-04 3.13E-04 4.17E-04 1.12E-03
3.41E-03 3.88E-04 3.16E-04 block S73 5.28E-05 1.90E-05 7.49E-05 1.27E-04 1.19E-04
1.64E-04 1.48E-03 3.73E-04 1.08E-04 7.30E-05 block S74 1.66E-04 1.61E-04 2.78E-04
3.37E-04 3.20E-04 4.12E-04 NLB 1.37E-03 2.66E-04 2.25E-04 block S76 1.55E-04 1.40E-04
2.09E-04 2.77E-04 2.69E-04 1.79E-04 5.76E-04 9.98E-04 1.87E-04 1.77E-04 block S201
8.32E-05 3.46E-05 4.27E-05 6.36E-04 5.22E-04 8.18E-04 2.15E-03 1.11E-03 4.10E-03
1.85E-04 block S202 6.86E-05 3.75E-05 5.40E-05 1.13E-04 1.20E-04 4.58E-05 5.09E-04
6.30E-04 2.05E-03 4.44E-05 block S206 8.22E-05 3.50E-05 3.60E-05 6.24E-04 5.18E-04
9.01E-04 2.23E-03 1.57E-03 4.37E-03 1.92E-04 block Family 3 S136 1.59E-03 8.45E-04
1.61E-03 1.79E-03 1.32E-03 4.59E-04 3.86E-04 4.78E-03 3.60E-03 NLB Non-block S137
1.37E-03 1.65E-03 1.88E-03 2.04E-03 1.79E-03 5.15E-04 4.43E-04 4.63E-03 3.76E-03 NLB
Non-block S175 1.65E-03 8.39E-04 1.83E-03 1.77E-03 1.07E-03 6.05E-04 4.17E-04
3.20E-03 2.99E-03 NT Non-block S177 NLB 3.32E-03 5.37E-03 4.47E-03 2.69E-03 1.04E-03
7.36E-04 NLB NLB NT Non-block S178 4.26E-03 2.23E-03 3.17E-03 3.44E-03 1.77E-03
1.12E-03 7.62E-04 NLB 4.73E-03 NT Non-block S180 3.07E-03 1.60E-03 2.27E-03 2.50E-03
1.38E-03 8.06E-04 5.76E-04 4.66E-03 3.75E-03 NT Non-block S184 5.14E-03 2.53E-03
3.91E-03 3.90E-03 2.21E-03 8.21E-04 6.38E-04 4.22E-03 NLB NT Non-block S185 2.39E-03
1.22E-03 2.05E-03 1.78E-03 1.03E-03 7.17E-04 5.73E-04 5.14E-03 3.85E-03 NT Non-block
S189 2.28E-03 1.06E-03 2.81E-03 3.31E-03 1.87E-03 8.37E-04 5.04E-04 3.90E-03 2.78E-03
NT Non-block S190 3.08E-03 1.56E-03 1.99E-03 2.17E-03 1.24E-03 7.73E-04 5.63E-04 NLB
2.56E-03 NT Non-block S193 NLB 3.08E-03 5.17E-03 NLB 4.28E-03 2.09E-03 1.57E-03
3.89E-03 NLB NT Non-block Family 4 S115 4.84E-04 7.86E-06 1.95E-05 2.22E-05 6.91E-05
NLB NLB NLB 3.69E-05 3.34E-05 Kick-off S116 4.80E-04 2.84E-05 5.07E-05 7.01E-05
1.20E-04 NLB NLB NLB 3.09E-05 8.25E-05 Kick-off S117 2.75E-04 1.20E-05 3.40E-05
1.83E-05 5.92E-05 NLB NLB NLB 3.54E-05 4.30E-06 kick-off S118 2.47E-04 9.17E-06
4.12E-05 6.97E-05 8.06E-05 NLB NLB NLB 1.31E-05 1.13E-05 Kick-off S132 7.95E-05
3.34E-05 2.06E-05 4.28E-05 7.12E-05 NLB NLB NLB 1.01E-05 1.05E-05 Kick-off S191
2.95E-04 6.67E-05 3.53E-05 5.54E-05 1.38E-05 NLB NLB NLB 1.01E-05 NT Kick-off S198
7.87E-04 3.67E-05 4.95E-05 8.34E-05 7.73E-05 NLB NLB NLB 4.72E-05 NT kick-off Family
5 S3 3.94E-04 4.10E-04 2.76E-04 2.18E-03 1.74E-03 6.55E-04 1.19E-03 1.97E-03 1.92E-03
1.62E-04 Non-block S45 4.76E-05 7.24E-05 1.44E-05 NLB NLB NLB NLB NLB NLB NLB
Non-block S123 1.31E-03 1.55E-03 NLB 1.57E-03 1.54E-03 NLB NLB NLB NLB 1.43E-03
Non-block S149 NLB 4.15E-03 5.44E-04 NLB 2.58E-03 NLB NLB NLB 3.31E-04 NLB Non-
block S161 NLB 3.37E-03 3.84E-04 NLB 2.15E-03 NLB NLB NLB 3.21E-03 NLB Non-block
S162 1.52E-03 6.84E-04 NLB NLB 3.18E-04 NLB NLB NLB 1.98E-03 4.66E-03 Non-block
S163 1.74E-03 5.35E-04 6.64E-03 NLB 2.42E-04 NLB NLB NLB 1.70E-03 3.15E-03 Non-
block S164 1.34E-03 5.95E-04 9.89E-03 NLB 4.08E-04 NLB NLB NLB 1.67E-03 4.26E-03
Non-block S173 1.37E-03 1.12E-03 5.04E-03 NLB NLB NLB NLB NLB NLB NLB Non-block
S174 1.89E-03 1.43E-03 3.99E-03 NLB NLB NLB NLB NLB NLB NLB Non-block S194
1.11E-03 1.17E-03 NLB 1.31E-03 1.11E-03 NT NLB NLB NLB NT Non-block S209 2.45E-03
6.59E-04 4.01E-03 NLB NLB NT NLB NLB 6.54E-05 NLB Non-block S213 4.82E-03 NLB
6.85E-04 2.60E-04 3.13E-03 NT NLB NLB NLB NT Non-block S214 4.53E-03 NLB 6.77E-04

3.51E-03 2.71E-03 NT NLB NLB NLB NT Non-block S218 NT 7.06E-05 5.47E-05 2.49E-05
NT NT NLB NLB NT 2.12E-05 Non-block NLB = $K_{\text{sub.off}} > 5 \times 10^3$ and no binding; NT =
not tested.

Claims

1. An isolated antibody that binds an extracellular domain of a human SIRP- α polypeptide, wherein the antibody comprises: (a) a heavy chain variable (VH) domain comprising: (i) an HVR-H1 sequence comprising the amino acid sequence of SYDMN (SEQ ID NO: 270), SYDVN (SEQ ID NO:221), or SYDLN (SEQ ID NO:313); (ii) an HVR-H2 sequence comprising the amino acid sequence of LISGSGEIIYYADSVKG (SEQ ID NO: 156); and (iii) an HVR-H3 sequence comprising the amino acid sequence of ENNRYRFFDD (SEQ ID NO: 157); and (b) a light chain variable (VL) domain comprising: (i) an HVR-L1 sequence comprising the amino acid sequence of RASQSVYTYLA (SEQ ID NO:158); (ii) an HVR-L2 sequence comprising the amino acid sequence of GASSRAT (SEQ ID NO:159); and (iii) an HVR-L3 sequence comprising the amino acid sequence of QQYYDRPPLT (SEQ ID NO:160).
2. The antibody of claim 1, wherein the VH domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO: 249, 133, 260, or 329; and the VL domain comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO: 134, 250, or 251.
3. The antibody of claim 1, wherein: (a) the VH domain comprises the amino acid sequence of SEQ ID NO: 133, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; (b) the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; (c) the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; (d) the VH domain comprises the amino acid sequence of SEQ ID NO: 133, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; (e) the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; (f) the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; (g) the VH domain comprises the amino acid sequence of SEQ ID NO: 133, and the VL domain comprises the amino acid sequence of SEQ ID NO:251; (h) the VH domain comprises the amino acid sequence of SEQ ID NO:260, and the VL domain comprises the amino acid sequence of SEQ ID NO:251; (i) the VH domain comprises the amino acid sequence of SEQ ID NO:329, and the VL domain comprises the amino acid sequence of SEQ ID NO:251; (j) the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:134; (k) the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:250; or (l) the VH domain comprises the amino acid sequence of SEQ ID NO:249, and the VL domain comprises the amino acid sequence of SEQ ID NO:251.
4. The antibody of claim 1, wherein the antibody enhances phagocytosis by a macrophage expressing a human SIRP- α polypeptide.
5. The antibody of claim 1, wherein the antibody enhances activation of a dendritic cell expressing a human SIRP- α polypeptide.
6. The antibody of claim 1, wherein the antibody inhibits in vivo growth of a tumor that expresses CD47.
7. The antibody of claim 1, wherein the antibody does not prevent interactions between a CD47-expressing cell and a T cell.
8. The antibody of claim 1, wherein the antibody is a monoclonal antibody.
9. The antibody of claim 1, wherein the antibody is a scFv-Fc antibody.

10. The antibody of claim 1, wherein the antibody comprises a light chain constant region comprising the amino acid sequence of SEQ ID NO:325, 326, or 426.
11. The antibody of claim 1, wherein the antibody comprises an Fc region.
12. The antibody of claim 11, wherein the antibody comprises an antibody heavy chain polypeptide that comprises the VH domain and a constant region comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 320-324.
13. The antibody of claim 11, wherein the Fc region is a human Fc region selected from the group consisting of an IgG1 Fc region, an IgG2 Fc region, and an IgG4 Fc region.
14. The antibody of claim 11, wherein the Fc region comprises a human IgG1 Fc region comprising one or more mutations selected from the group consisting of L234A, L235A, G237A, and N297A, according to EU numbering.
15. The antibody of claim 11, wherein the Fc region comprises a human IgG2 Fc region comprising one or more mutations selected from the group consisting of A330S, P331S and N297A, according to EU numbering.
16. The antibody of claim 11, wherein the Fc region comprises a human IgG4 Fc region comprising one or more mutations selected from the group consisting of S228P, E233P, F234V, L235A, L235E, delG236, and N297A, according to EU numbering.
17. The antibody of claim 1, wherein the antibody is an antibody fragment selected from the group consisting of a Fab, F(ab')₂, Fab'-SH, Fv, and scFv fragment.
18. The antibody of claim 1, wherein the antibody is conjugated to a cytotoxic agent or label.
19. The antibody of claim 1, wherein the antibody is a bispecific antibody.
20. The antibody of claim 19, wherein the antibody comprises a first antigen binding domain comprising the VH and VL domain that bind an extracellular domain of a human SIRP- α polypeptide and a second antigen binding domain comprising a second VH domain and second VL domain that bind an antigen expressed by a cancer cell.
21. The antibody of claim 20, wherein the antigen expressed by the cancer cell is selected from the group consisting of CD19, CD20, CD22, CD30, CD33, CD38, CD52, CD56, CD70, CD74, CD79b, CD123, CD138, CS1/SLAMF7, Trop-2, 5T4, EphA4, BCMA, Mucin 1, Mucin 16, PD-L1, PTK7, STEAP1, Endothelin B Receptor, mesothelin, EGFRVIII, ENPP3, SLC44A4, GNMB, nectin 4, NaPi2b, LIV-1A, Guanylyl cyclase C, DLL3, EGFR, HER2, VEGF, VEGFR, integrin α V β 3, integrin α 5 β 1, MET, IGF1R, TRAILR1, TRAILR2, RANKL, FAP, Tenascin, Le.sup.y, EpCAM, CEA, gpA33, PSMA, TAG72, a mucin, CAIX, EPHA3, folate receptor α , GD2, GD3, and an MHC/peptide complex comprising a peptide from NY-ESO-1/LAGE, SSX-2, MAGE-A3, gp100/pmel17, Melan-A/MART-1, gp75/TRP1, tyrosinase, TRP2, CEA, PSA, TAG-72, immature laminin receptor, MOK/RAGE-1, WT-1, SAP-1, BING-4, EpCAM, MUC1, PRAME, survivin, BRCA1, BRCA2, CDK4, CML66, MART-2, p53, Ras, β -catenin, TGF- β RII, HPV E6, or HPV E7.
22. The antibody of claim 1, wherein the antibody is a human antibody.
23. A polynucleotide encoding the antibody of claim 1.
24. A vector comprising the polynucleotide of claim 23.
25. A host cell comprising the polynucleotide of claim 23.
26. A method of producing an antibody, the method comprising culturing the host cell of claim 25 such that the antibody that binds the extracellular domain of a human SIRP- α polypeptide is produced.
27. The method of claim 26, further comprising recovering the antibody from the host cell.
28. The antibody of claim 1, wherein the antibody binds an extracellular domain of a human SIRP- α v1 polypeptide comprising the amino acid sequence of TABLE-US-00038 (SEQ ID NO: 5)
EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELIY
NQKEGHFPRVTTVSDLTKRNNMDFSIRIGNITPADAGTYCYVKFRKGSPD
DVEFKSGAGTELSVRAKPS.
29. The antibody of claim 1, wherein the antibody binds an extracellular domain of a human SIRP-

α v2 polypeptide comprising the amino acid sequence of TABLE-US-00039 (SEQ ID NO: 6)
EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELI
YNQKEGHFPRVTTVSESTKRENMDFSISISNITPADAGTY YCVKFRKGS
PDTEFKSGAGTELSVRAKPS.

30. The antibody of claim 1, wherein the antibody is capable of binding an extracellular domain of a human SIRP- α v1 polypeptide comprising the amino acid sequence of
EEELQVIQPDKSVLVAAGETATLRCTATSLIPVGPIQWFRGAGPGRELIYNQKEGHFPRV
TTVSDLTKRNNMDFSIRIGNITPADAGTY YCVKFRKGSPDDVEFKSGAGTELSVRAKPS
(SEQ ID NO:5) and is capable of binding an extracellular domain of a human SIRP- α v2
polypeptide comprising the amino acid sequence of TABLE-US-00040 (SEQ ID NO: 6)
EEELQVIQPDKSVSVAAGESAILHCTVTSLIPVGPIQWFRGAGPARELIY
NQKEGHFPRVTTVSESTKRENMDFSISISNITPADAGTY YCVKFRKGSPD
TEFKSGAGTELSVRAKPS.

31. The antibody of claim 1, wherein the antibody binds an extracellular domain of a cynomolgus SIRP- α polypeptide comprising the amino acid sequence of SEQ ID NO:11.

32. The antibody of claim 1, wherein the antibody binds an extracellular domain of a murine SIRP- α polypeptide comprising the amino acid sequence of SEQ ID NO:7.

33. The antibody of claim 1, wherein the antibody binds the extracellular domain of a human SIRP- β polypeptide comprising the amino acid sequence of SEQ ID NO:13.

34. The antibody of claim 1, wherein the antibody does not block binding between an extracellular domain of a human SIRP- α polypeptide and an IgSF domain of a human CD47 polypeptide.

35. A host cell comprising the vector of claim 24.

36. A method of producing an antibody, the method comprising culturing the host cell of claim 35 such that the antibody that binds the extracellular domain of a human SIRP- α polypeptide is produced.

37. The method of claim 36, further comprising recovering the antibody from the host cell.
